

UWorld_Grouped_By_Passage_Export_2026-06-20_17-17

Passage groups: 8

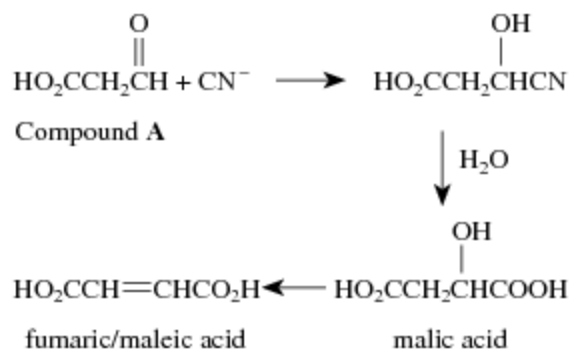
Passage 1 (Questions 1-5)

Passage

Passage 1 (Questions 1-5)

In order to understand the chemical evolution of life, scientists have studied the production of amino acids under conditions that simulate those of primitive Earth.

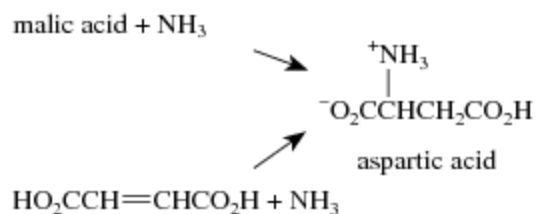
Malic acid is thought to have formed via hydrolysis of a cyanohydrin intermediate (Scheme I).



Scheme I

Subsequent dehydration of malic acid results in the formation of fumaric acid (mp 286°C) and its cis isomer, maleic acid (mp 130°C).

Ammonia was thought to be a constituent of the early atmosphere, and it was proposed that it reacted with fumaric, maleic, and malic acids to afford a prebiological synthesis of aspartic acid, an amino acid (Scheme II).



Scheme II

In an attempt to verify this hypothesis, chemists tried to reproduce the synthesis of aspartic acid under simulated primitive Earth conditions.

An open flask containing malic acid was treated with a continuous ammonia feed, and the mixture was heated at 150°C for 1.5 h. The experiment was repeated with a mixture of fumaric and maleic acids. The results were analyzed by paper chromatography; the R_f values of the reaction mixture were compared to the R_f value of a standard sample of aspartic acid. (Note: R_f = distance moved by compound/distance moved by solvent front.)

Question 1 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

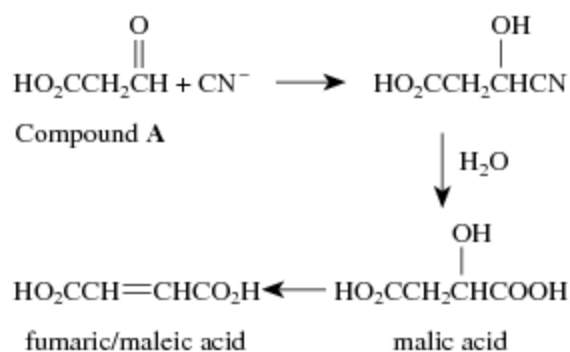
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Passage 1 (Questions 1-5)

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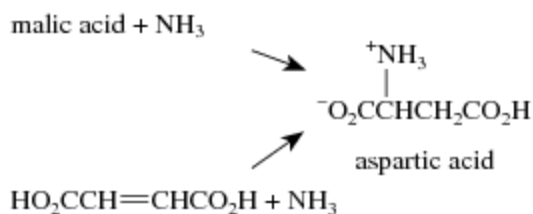
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In the chromatography of the reaction mixture, water absorbed on cellulose functioned as the stationary phase. What was the principal factor determining the migration of individual components in the sample?

- A.
Hydrogen bonding
- B.
Solute concentration
- C.
Stationary phase concentration
- D.
Thickness of paper

Solution: The correct answer is **A**.

1. **The immobilized water molecules bind to those components in the sample mixture capable of accepting or donating hydrogen bonds. Those components with stronger hydrogen bonding migrated more slowly.**
2. The solute concentration will affect the size of spots on a paper chromatogram, not the average migration rate.
3. The stationary phase does not have a concentration as it is an immobilized surface, not a solution.

4. The thickness of the paper will affect the amount of each component that can be fully separated, not the relative migration rates of the components.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Separations and purifications](#).

Correct

Your result is correct.

Your answer is A.

Correct answer is A.

Correct Answer Your Answer

A

✓

B

C

D

Time Spent:

1 min 12 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5C

FC5

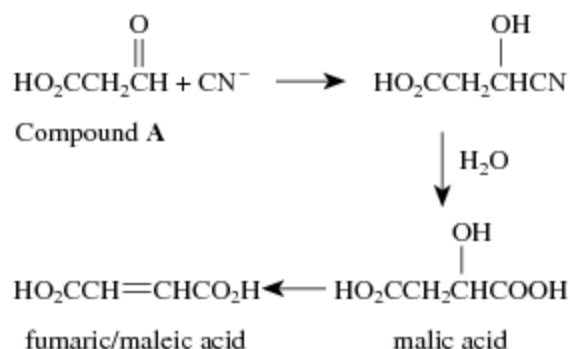
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RESULT / TIMING

Passage 1 (Questions 1-5)

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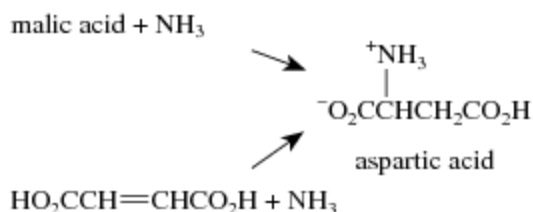
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- A.

- Hydrogen bonding
- B.
Solute concentration
- C.
Stationary phase concentration
- D.
Thickness of paper

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Question 2 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

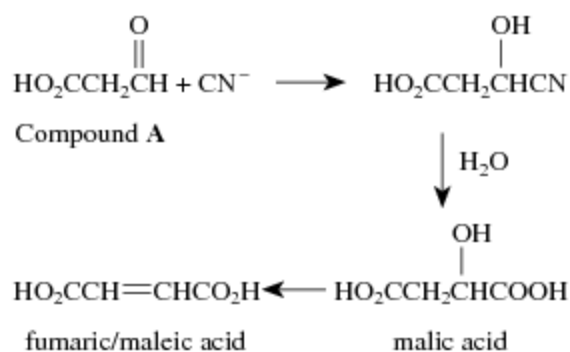
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Passage 1 (Questions 1-5)

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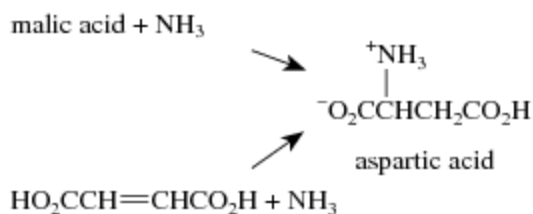
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Scheme II

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What assumption is being made if scientists conclude that aspartic acid was formed by the prebiological synthesis in the passage?

- A.
Aspartic acid is unstable at temperatures below 150°C.
- B.
All of the malic acid underwent the dehydration reaction to form fumaric/maleic acid.
- C.
Compound **A** and cyanide were available on primitive Earth.
- D.
The reaction between ammonia and fumaric acid was catalyzed by the presence of water.

Solution: The correct answer is **C**.

1. Because the prebiotic experiment was performed at 150°C in order to form aspartic acid, it is contradictory that the researchers would expect this temperature to degrade at lower temperatures. In addition, aspartic acid is found in living organisms at 37°C.
2. There is no requirement given that all of the malic acid dehydrate to fumaric/maleic acid, only that some does.

3. **Without Compound A and cyanide, no aspartic acid can form.**
4. There is no assumption in the passage that water acts as a catalyst in the reaction between ammonia and fumaric acid.

KHAN ACADEMY LESSONS

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Correct

Your result is correct.

Your answer is C.

Correct answer is C.

Correct Answer Your Answer

A

B

C

✓

D

Time Spent:

5 mins 0 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS3

CC5C

FC5

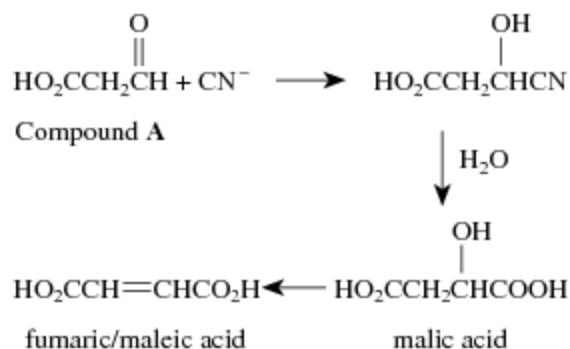
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RESULT / TIMING

Passage 1 (Questions 1-5)

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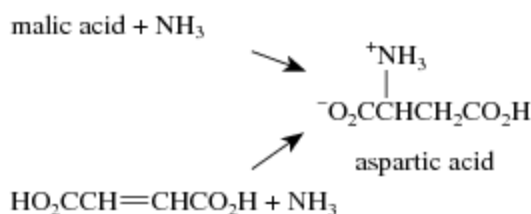
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Solution: The correct answer is **C**.

1. Because the prebiotic experiment was performed at 150°C in order to form aspartic acid, it is contradictory that the researchers would expect this temperature to degrade at lower temperatures. In addition, aspartic acid is found in living organisms at 37°C.
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KHAN ACADEMY LESSONS

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Question 3 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

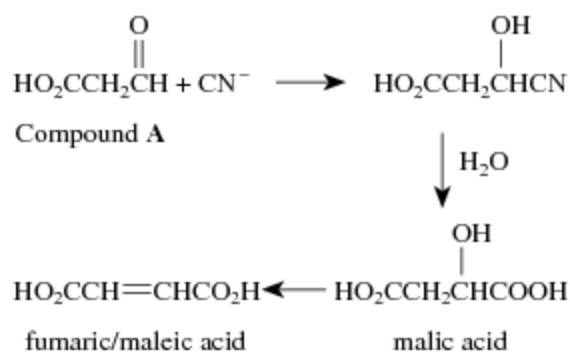
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Passage 1 (Questions 1-5)

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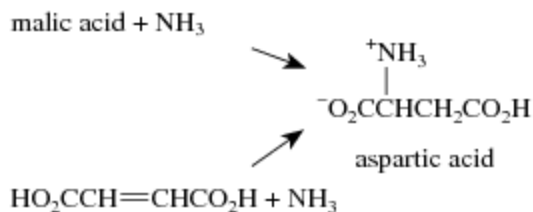
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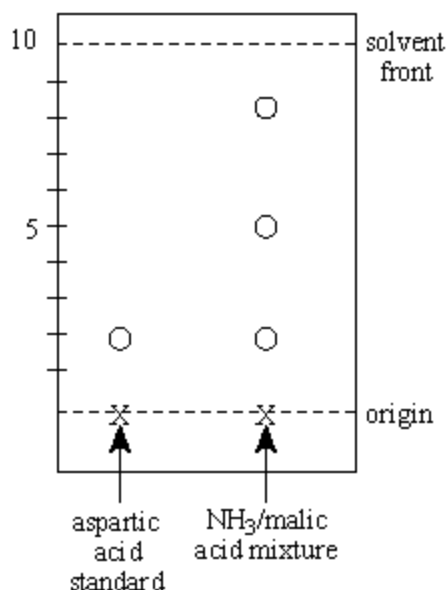


Scheme II

In an attempt to verify this hypothesis, chemists tried to reproduce the synthesis of aspartic acid under simulated primitive Earth conditions.

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According to the developed chromatography plate shown below, what is the approximate R_f value of aspartic acid?



- A.
0.20
- B.
0.50
- C.

- 5
- D.
- 10

Solution: The correct answer is **A**.

1. The R_f of aspartic acid = (distance traveled by aspartic acid)/(distance to the solvent front) = $2/10 = 0.20$
2. This value is the R_f of the middle spot of the reaction mixture, not aspartic acid.
3. This value is the distance traveled by the middle spot of the reaction mixture. It is not an R_f value.
4. This value is the distance to the solvent front.

KHAN ACADEMY LESSONS

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Correct

Your result is correct.

Your answer is A.

Correct answer is A.

Correct Answer Your Answer

A

✓

B

C

D

Time Spent:

1 min 15 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS4

CC5C

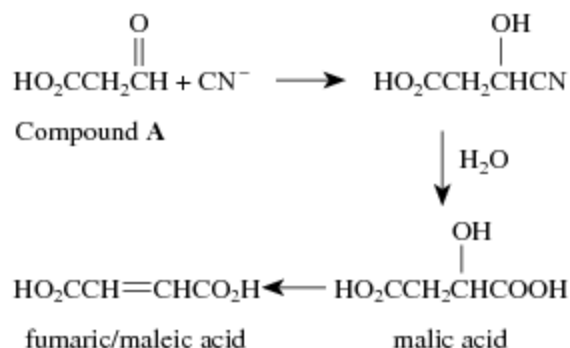
FC5

OCH

Passage 1 (Questions 1-5)

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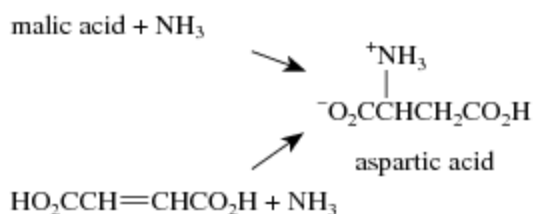
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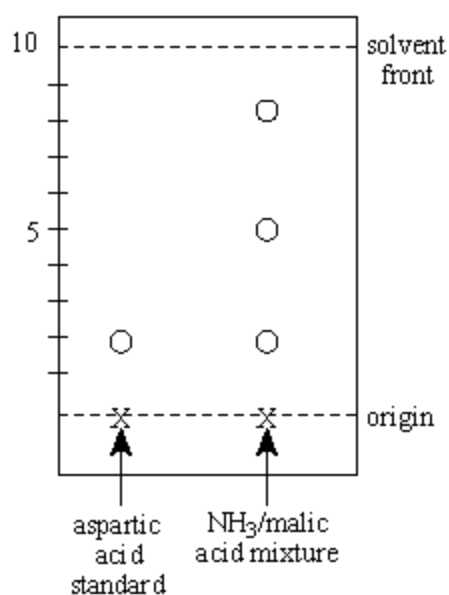
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4. This value is the distance to the solvent front.

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Question 4 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

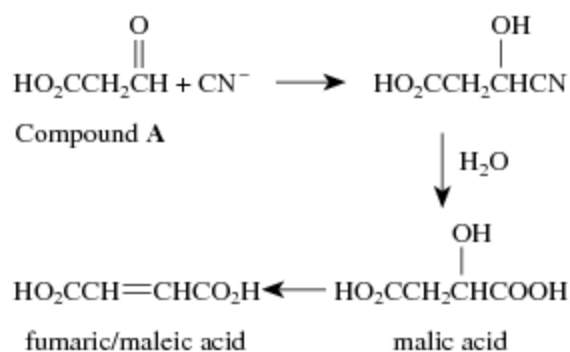
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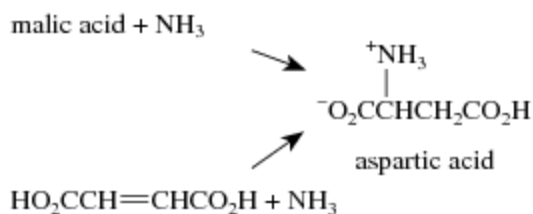
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Which of the following statements does NOT correctly describe the dehydration of malic acid to fumaric acid and maleic acid?

- A.
The reaction occurs most readily with tertiary alcohols.
- B.
The reaction involves the loss of a water molecule.
- C.
The reaction has a carbocation intermediate.
- D.
The reaction is stereospecific.

Solution: The correct answer is **D**.

1. Tertiary alcohols readily dehydrate.
2. Dehydration is loss of a water molecule by definition.
3. During dehydration, a carbocation forms.
4. **Because of the carbocation intermediate in dehydration, the subsequent loss of a proton will lead to a mixture of cis- and trans-alkenes. Thus, dehydration is NOT stereospecific.**

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Separations and purifications](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

2 mins 33 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS4

CC5C

FC5

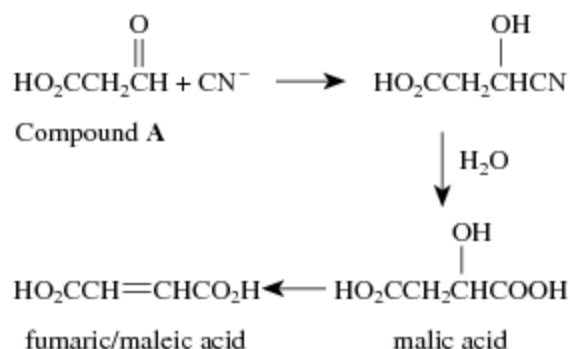
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RESULT / TIMING

Passage 1 (Questions 1-5)

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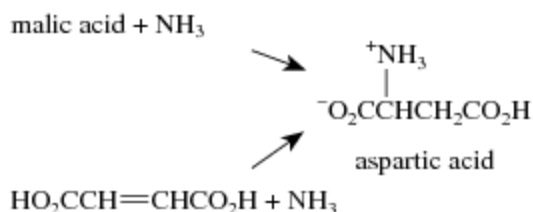
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- C.
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- D.
The reaction is stereospecific.

Solution: The correct answer is **D**.

1. Tertiary alcohols readily dehydrate.
2. Dehydration is loss of a water molecule by definition.
3. During dehydration, a carbocation forms.
4. **Because of the carbocation intermediate in dehydration, the subsequent loss of a proton will lead to a mixture of cis- and trans-alkenes. Thus, dehydration is NOT stereospecific.**

KHAN ACADEMY LESSONS

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Question 5 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

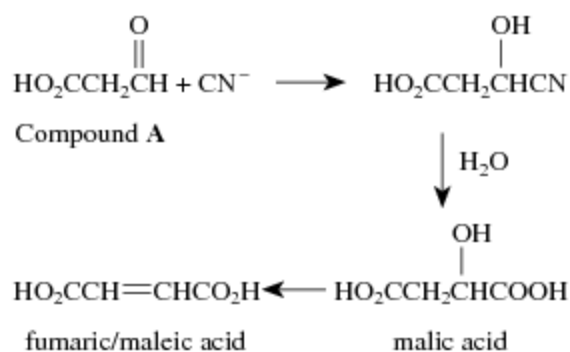
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Passage 1 (Questions 1-5)

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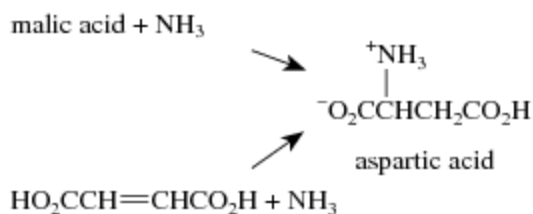
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Scheme II

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What type of functional group is formed when aspartic acid reacts with another amino acid to form a peptide bond?

- A.
An amine group
- B.
An aldehyde group
- C.
An amide group
- D.
A carboxyl group

Solution: The correct answer is **C**.

1. An amine is one of the functional groups that reacts to form a peptide bond, not the peptide bond itself.
2. An aldehyde is RCHO, whereas the product is an amide, RCONHR'.
3. **The functional group that represents a peptide bond is called an amide group.**
4. A carboxyl group reacts with an amine to form a peptide bond.

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Separations and purifications](#).

Correct

Your result is correct.

Your answer is C.

Correct answer is C.

Correct Answer Your Answer

A

B

C

✓

D

Time Spent:

1 min 7 secs

Difficulty Level:

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SIRS1

CC5C

FC5

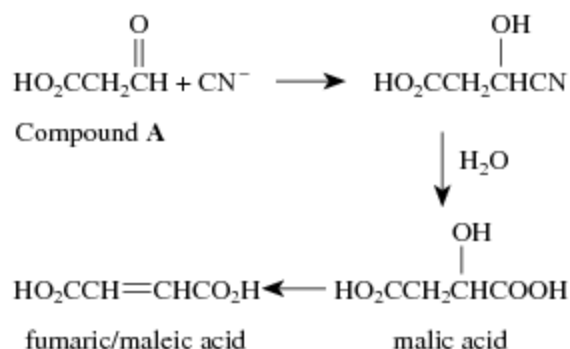
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RESULT / TIMING

Passage 1 (Questions 1-5)

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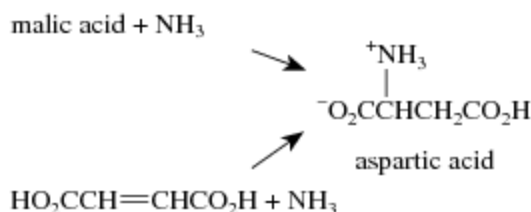
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Scheme I

Subsequent dehydration of malic acid results in the formation of fumaric acid (mp 286°C) and its cis isomer, maleic acid (mp 130°C).

Ammonia was thought to be a constituent of the early atmosphere, and it was proposed that it reacted with fumaric, maleic, and malic acids to afford a prebiological synthesis of aspartic acid, an amino acid (Scheme II).



Scheme II

In an attempt to verify this hypothesis, chemists tried to reproduce the synthesis of aspartic acid under simulated primitive Earth conditions.

An open flask containing malic acid was treated with a continuous ammonia feed, and the mixture was heated at 150°C for 1.5 h. The experiment was repeated with a mixture of fumaric and maleic acids. The results were analyzed by paper chromatography; the R_f values of the reaction mixture were compared to the R_f value of a standard sample of aspartic acid. (Note: R_f = distance moved by compound/distance moved by solvent front.)

What type of functional group is formed when aspartic acid reacts with another amino acid to form a peptide bond?

- A.
An amine group

- B.
An aldehyde group
- C.
An amide group
- D.
A carboxyl group

Solution: The correct answer is **C**.

1. An amine is one of the functional groups that reacts to form a peptide bond, not the peptide bond itself.
2. An aldehyde is RCHO , whereas the product is an amide, RCONHR' .
3. **The functional group that represents a peptide bond is called an amide group.**
4. A carboxyl group reacts with an amine to form a peptide bond.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Separations and purifications](#).

Passage 2 (Questions 6-9)

Passage

Passage 2 (Questions 6-9)

The nucleophilic alkyl substitution reactions are among the most thoroughly studied transformations in organic chemistry. A substitution by an S_N2 reaction mechanism is shown in Figure 1 and a substitution by an S_N1 mechanism is shown in Figure 2.

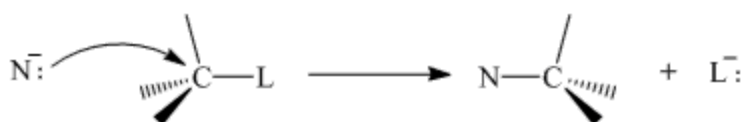


Figure 1	S_N2 Reaction mechanism
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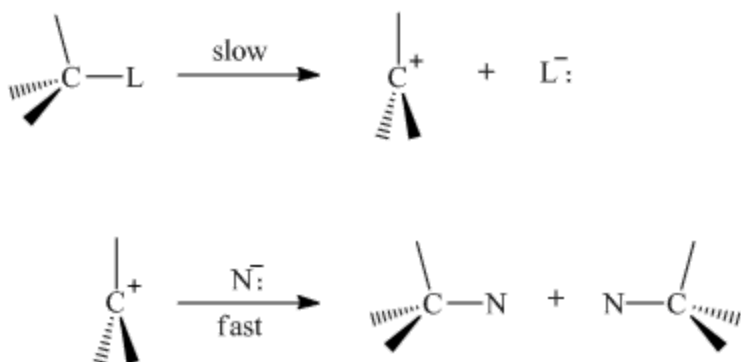


Figure 2	S_N1 Reaction mechanism
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The S_N2 mechanism involves a direct displacement of a leaving group (L) by a nucleophile (N). The reaction results in an inversion of configuration at the carbon bearing the leaving group, because the incoming nucleophile attacks it from the side opposite the leaving group. The S_N1 reaction mechanism involves an intermediate carbocation that is formed by loss of the leaving group. A nucleophile can react with the planar carbocation from either side, leading to the inversion or retention of the configuration at the site of the substitution.

The product distribution of two reactions is shown in figures 3 and 4.

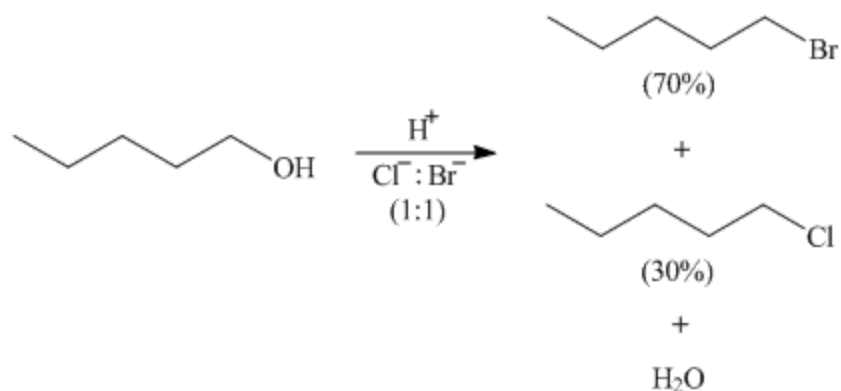


Figure 3

$\text{S}_{\text{N}}2$ Reaction of 1-pentanol

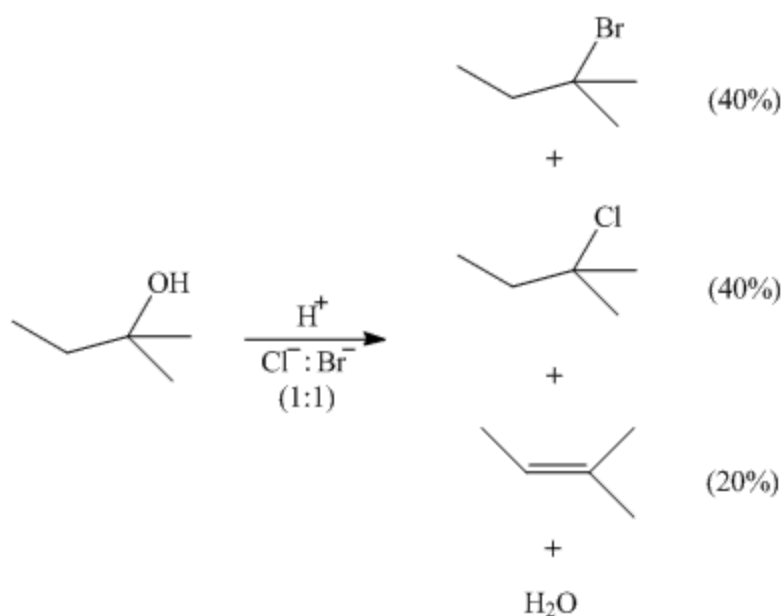


Figure 4

Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

Question 6 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 2 (Questions 6-9)

The nucleophilic alkyl substitution reactions are among the most thoroughly studied transformations in organic chemistry. A substitution by an S_N2 reaction mechanism is shown in Figure 1 and a substitution by an S_N1 mechanism is shown in Figure 2.



Figure 1	S_N2 Reaction mechanism
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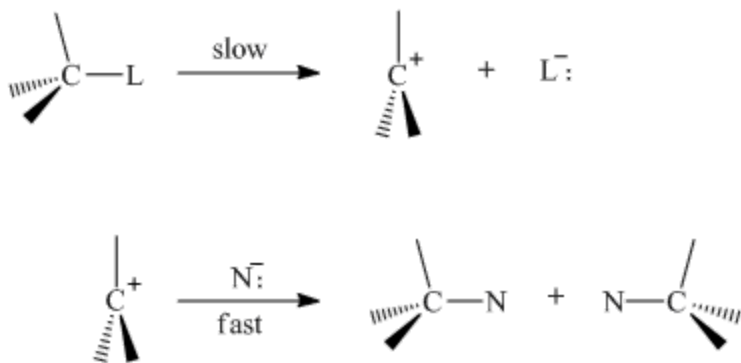


Figure 2	S_N1 Reaction mechanism
-----------------	---------------------------

The S_N2 mechanism involves a direct displacement of a leaving group (L) by a nucleophile (N). The reaction results in an inversion of configuration at the carbon bearing the leaving group, because the incoming nucleophile attacks it from the side opposite the leaving group. The S_N1 reaction mechanism involves an intermediate carbocation that is formed by loss of the leaving group. A nucleophile can react with the planar carbocation from either side, leading to the inversion or retention of the configuration at the site of the substitution.

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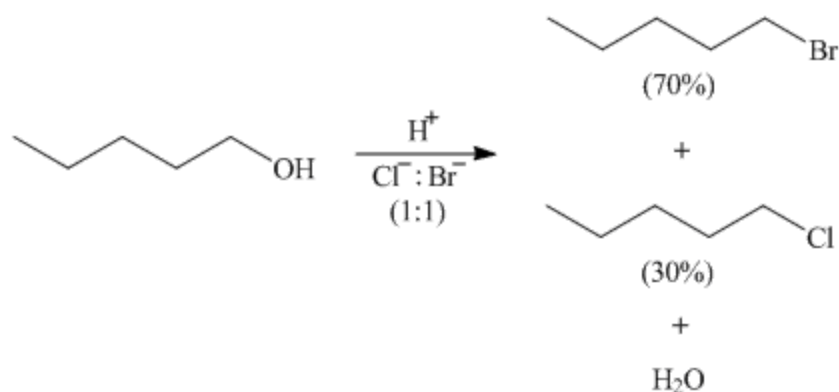


Figure 3

S_N2 Reaction of 1-pentanol

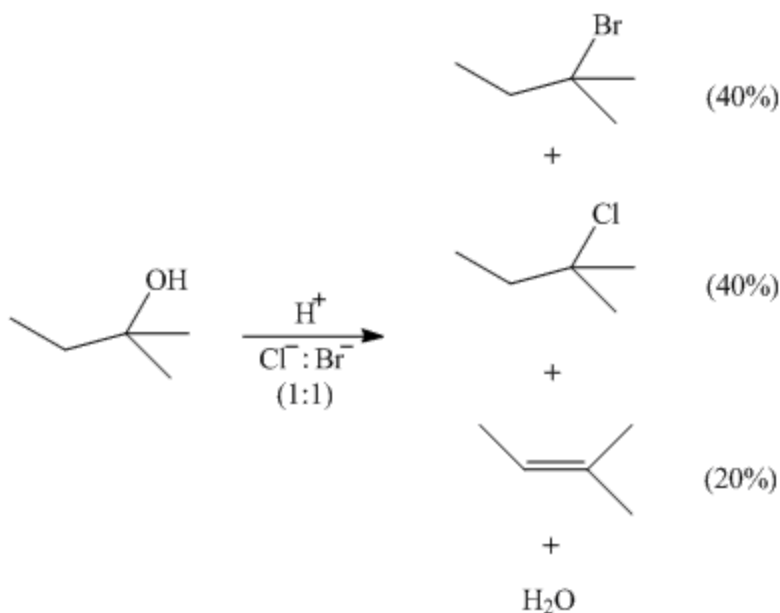


Figure 4

Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

If 2-pentanol replaces 1-pentanol in the reaction shown in Figure 3, the rate of substitution is less because:

- A.
the C–O bond in 2-pentanol is stronger than the C–O bond in 1-pentanol.
- B.
there is a competing elimination reaction that slows the rate of substitution.
- C.
there is more steric hindrance at the oxygen atom in 2-pentanol than in 1-pentanol, making protonation less likely.
- D.
there is more steric hindrance at the 2-position of 2-pentanol than at the 1-position of 1-pentanol.

Solution: The correct answer is **D**.

1. This statement is false. For both heterolytic and homolytic bond breaking, the more substituted C–O bond is weaker. Relative C–O bond strength is not the relevant parameter for this question.
2. Competing elimination does not slow the rate of substitution. Their rates are independent of each other.
3. Protonation at OH is a fast, pre-equilibrium step, thus it has no effect on the reaction rate.
4. **At a secondary carbon, there is more crowding from the adjacent carbon chains, which makes nucleophilic substitution slower.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Amino Acids, Peptides, Proteins](#); [Lipids](#); [Carbohydrates](#); [Aldehydes and Ketones](#); [Alcohols, Phenols, Polycyclic and Heterocyclic Aromatic Compounds](#); [Carboxylic Acids](#); and [Acid Derivatives \(Anhydrides, Amides, Esters\)](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

3 mins 18 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5D

FC5

OCH

RESULT / TIMING

Passage 2 (Questions 6-9)

The nucleophilic alkyl substitution reactions are among the most thoroughly studied transformations in organic chemistry. A substitution by an S_N2 reaction mechanism is shown in Figure 1 and a substitution by an S_N1 mechanism is shown in Figure 2.

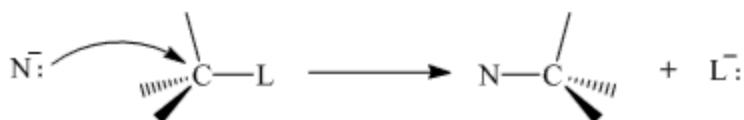


Figure 1

S_N2 Reaction mechanism

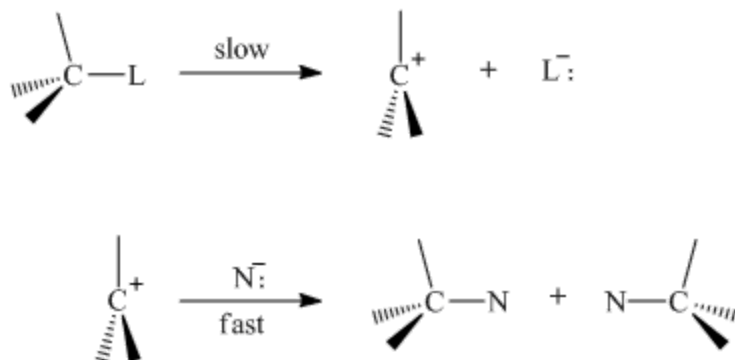


Figure 2	S _N 1 Reaction mechanism
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The S_N2 mechanism involves a direct displacement of a leaving group (L) by a nucleophile (N). The reaction results in an inversion of configuration at the carbon bearing the leaving group, because the incoming nucleophile attacks it from the side opposite the leaving group. The S_N1 reaction mechanism involves an intermediate carbocation that is formed by loss of the leaving group. A nucleophile can react with the planar carbocation from either side, leading to the inversion or retention of the configuration at the site of the substitution.

The product distribution of two reactions is shown in figures 3 and 4.

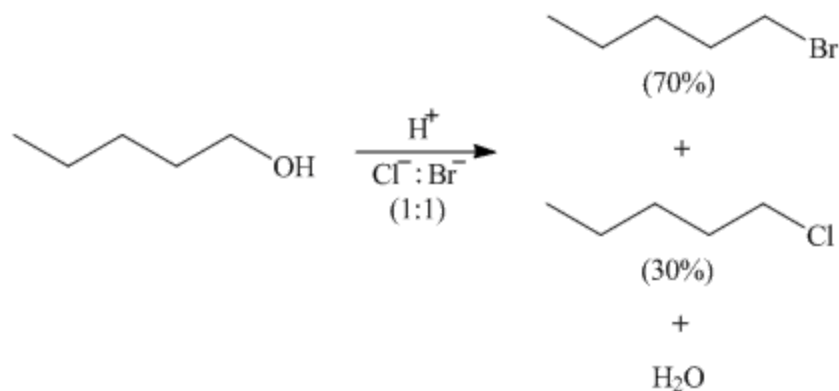


Figure 3	S _N 2 Reaction of 1-pentanol
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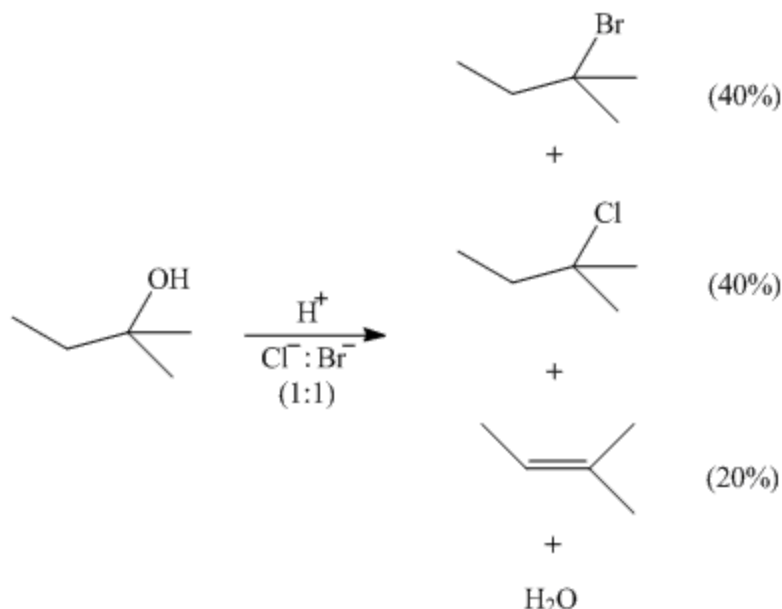


Figure 4 Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

If 2-pentanol replaces 1-pentanol in the reaction shown in Figure 3, the rate of substitution is less because:

- A.
the C–O bond in 2-pentanol is stronger than the C–O bond in 1-pentanol.
- B.
there is a competing elimination reaction that slows the rate of substitution.
- C.
there is more steric hindrance at the oxygen atom in 2-pentanol than in 1-pentanol, making protonation less likely.
- D.
there is more steric hindrance at the 2-position of 2-pentanol than at the 1-position of 1-pentanol.

Solution: The correct answer is **D**.

1. This statement is false. For both heterolytic and homolytic bond breaking, the more substituted C–O bond is weaker. Relative C–O bond strength is not the relevant parameter for this question.
2. Competing elimination does not slow the rate of substitution. Their rates are independent of each other.
3. Protonation at OH is a fast, pre-equilibrium step, thus it has no effect on the reaction rate.
4. **At a secondary carbon, there is more crowding from the adjacent carbon chains, which makes nucleophilic substitution slower.**

KHAN ACADEMY LESSONS

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Question 7 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 2 (Questions 6-9)

The nucleophilic alkyl substitution reactions are among the most thoroughly studied transformations in organic chemistry. A substitution by an S_N2 reaction mechanism is shown in Figure 1 and a substitution by an S_N1 mechanism is shown in Figure 2.



Figure 1	S_N2 Reaction mechanism
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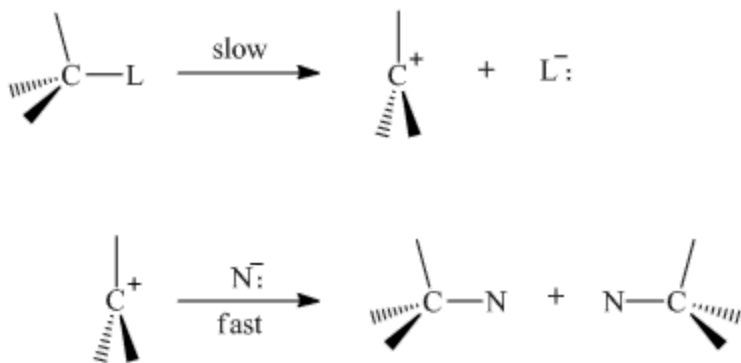


Figure 2	S_N1 Reaction mechanism
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The S_N2 mechanism involves a direct displacement of a leaving group (L) by a nucleophile (N). The reaction results in an inversion of configuration at the carbon bearing the leaving group, because the incoming nucleophile attacks it from the side opposite the leaving group. The S_N1 reaction mechanism involves an intermediate carbocation that is formed by loss of the leaving group. A nucleophile can react with the planar carbocation from either side, leading to the inversion or retention of the configuration at the site of the substitution.

The product distribution of two reactions is shown in figures 3 and 4.

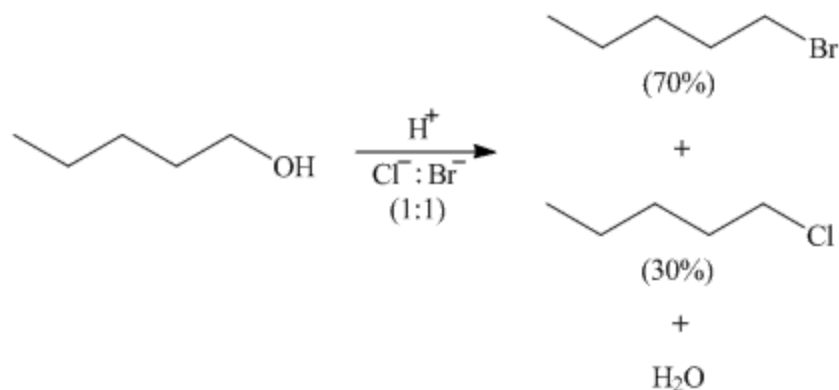


Figure 3

S_N2 Reaction of 1-pentanol

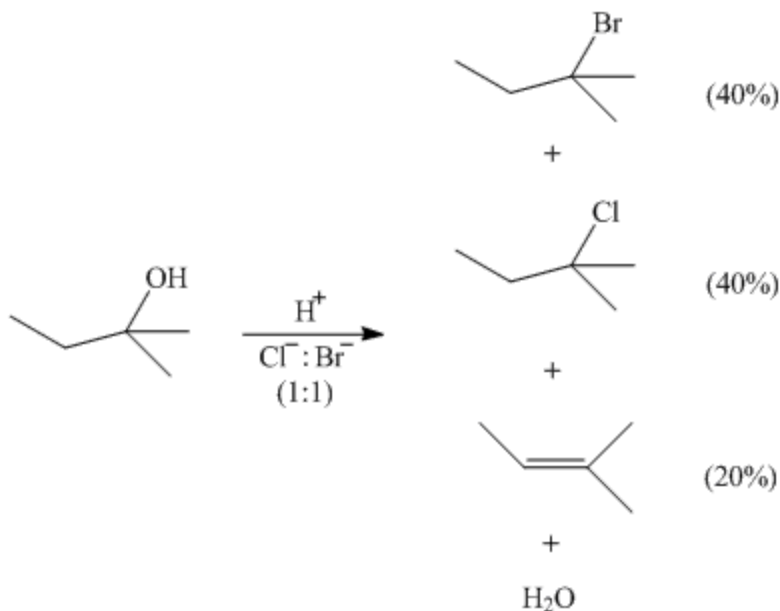
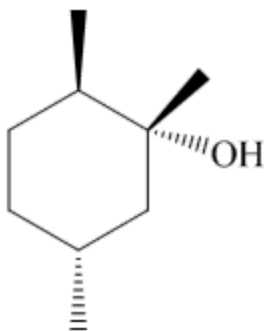


Figure 4

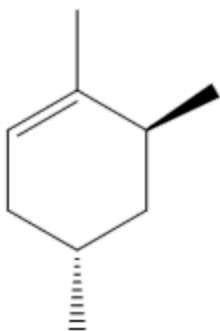
Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

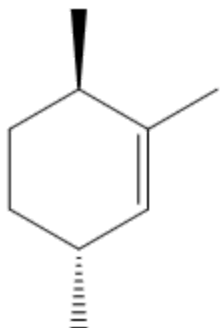
If Reaction 2 (Figure 4) is repeated with HCl and the compound shown below, which of the following compounds is NOT a direct product (without rearrangement)?



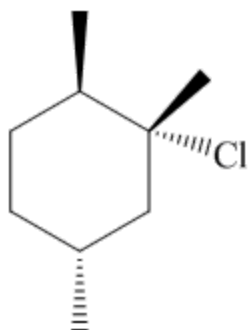
• A.



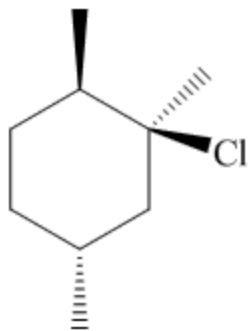
• B.



• C.



• D.



Solution: The correct answer is **A**.

1. **The only way that this product can form is by rearrangement, a 1,2-hydride shift following the initial formation of a tertiary carbocation.**
2. This product forms directly from deprotonation after the formation of the tertiary carbocation.
3. This product forms directly via chloride nucleophile covalently binding to the tertiary carbocation intermediate from below the plane of the carbocation.
4. This product forms directly via chloride nucleophile covalently binding to the tertiary carbocation intermediate from above the plane of the carbocation.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Amino Acids, Peptides, Proteins](#); [Lipids](#); [Carbohydrates](#); [Aldehydes and Ketones](#); [Alcohols, Phenols, Polycyclic and Heterocyclic Aromatic Compounds](#); [Carboxylic Acids](#); and [Acid Derivatives \(Anhydrides, Amides, Esters\)](#).

Correct

Your result is correct.

Your answer is A.

Correct answer is A.

Correct Answer Your Answer

A

✓

B

C

D

Time Spent:

3 mins 3 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5D

FC5

OCH

RESULT / TIMING

Passage 2 (Questions 6-9)

The nucleophilic alkyl substitution reactions are among the most thoroughly studied transformations in organic chemistry. A substitution by an S_N2 reaction mechanism is shown in Figure 1 and a substitution by an S_N1 mechanism is shown in Figure 2.

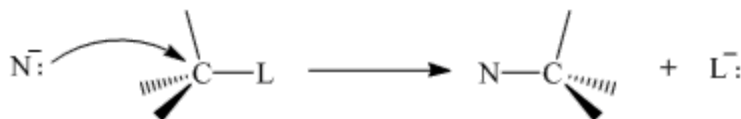
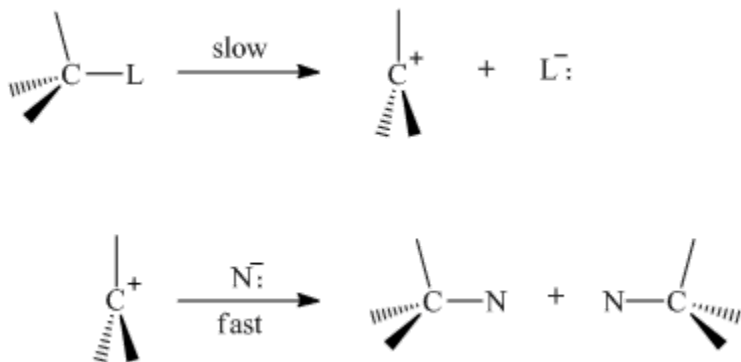
**Figure 1** S_N2 Reaction mechanism

Figure 2**S_N1 Reaction mechanism**

The S_N2 mechanism involves a direct displacement of a leaving group (L) by a nucleophile (N). The reaction results in an inversion of configuration at the carbon bearing the leaving group, because the incoming nucleophile attacks it from the side opposite the leaving group. The S_N1 reaction mechanism involves an intermediate carbocation that is formed by loss of the leaving group. A nucleophile can react with the planar carbocation from either side, leading to the inversion or retention of the configuration at the site of the substitution.

The product distribution of two reactions is shown in figures 3 and 4.

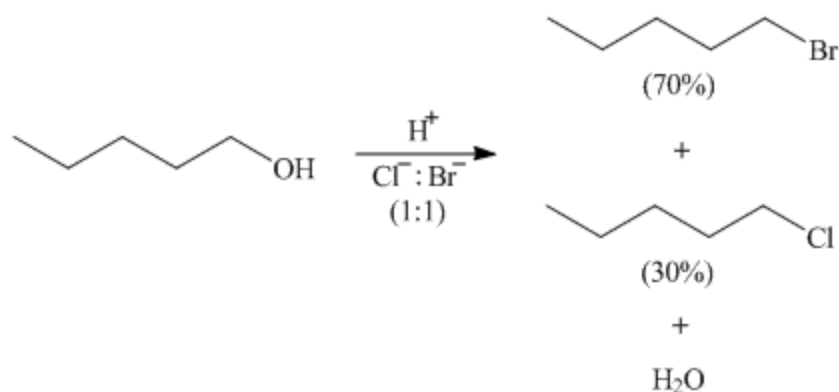
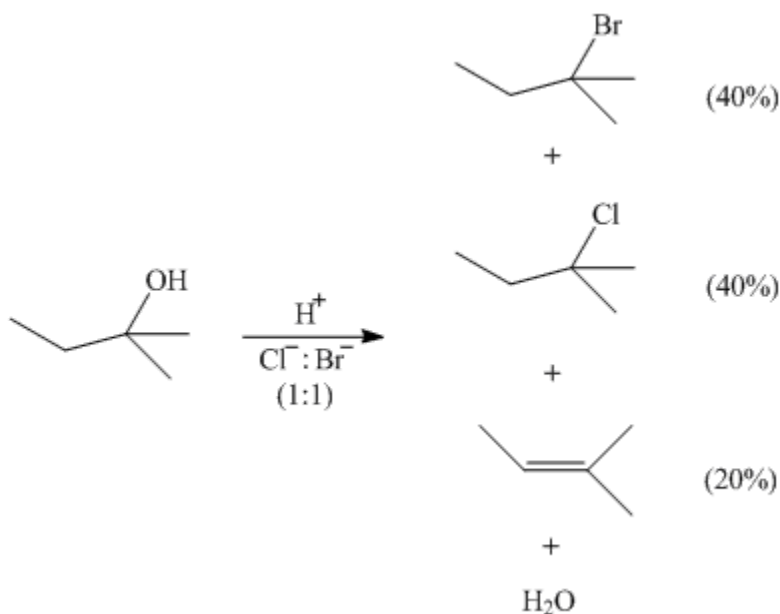
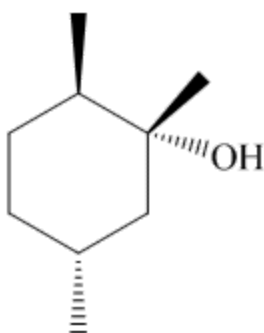
**Figure 3****S_N2 Reaction of 1-pentanol**

Figure 4

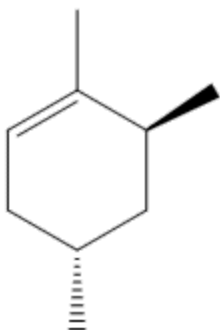
Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

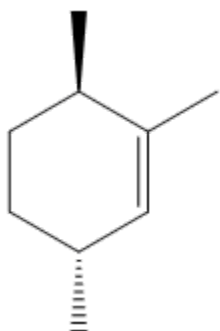
If Reaction 2 (Figure 4) is repeated with HCl and the compound shown below, which of the following compounds is NOT a direct product (without rearrangement)?



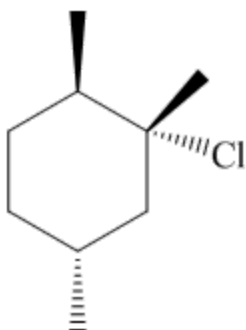
- A.



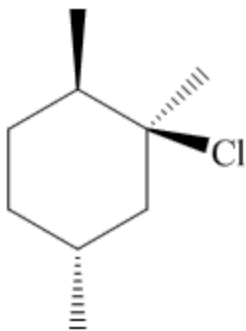
- B.



- C.



• D.



Solution: The correct answer is **A**.

1. **The only way that this product can form is by rearrangement, a 1,2-hydride shift following the initial formation of a tertiary carbocation.**
2. This product forms directly from deprotonation after the formation of the tertiary carbocation.
3. This product forms directly via chloride nucleophile covalently binding to the tertiary carbocation intermediate from below the plane of the carbocation.
4. This product forms directly via chloride nucleophile covalently binding to the tertiary carbocation intermediate from above the plane of the carbocation.

KHAN ACADEMY LESSONS

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Question 8 of 59

QUESTION

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MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 2 (Questions 6-9)

The nucleophilic alkyl substitution reactions are among the most thoroughly studied transformations in organic chemistry. A substitution by an S_N2 reaction mechanism is shown in Figure 1 and a substitution by an S_N1 mechanism is shown in Figure 2.



Figure 1	S_N2 Reaction mechanism
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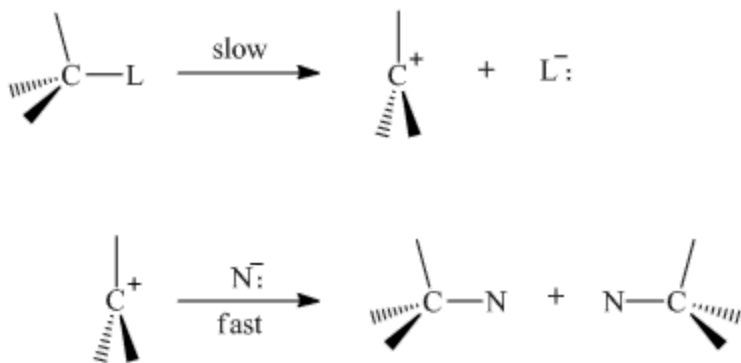


Figure 2	S_N1 Reaction mechanism
-----------------	---------------------------

The S_N2 mechanism involves a direct displacement of a leaving group (L) by a nucleophile (N). The reaction results in an inversion of configuration at the carbon bearing the leaving group, because the incoming nucleophile attacks it from the side opposite the leaving group. The S_N1 reaction mechanism involves an intermediate carbocation that is formed by loss of the leaving group. A nucleophile can react with the planar carbocation from either side, leading to the inversion or retention of the configuration at the site of the substitution.

The product distribution of two reactions is shown in figures 3 and 4.

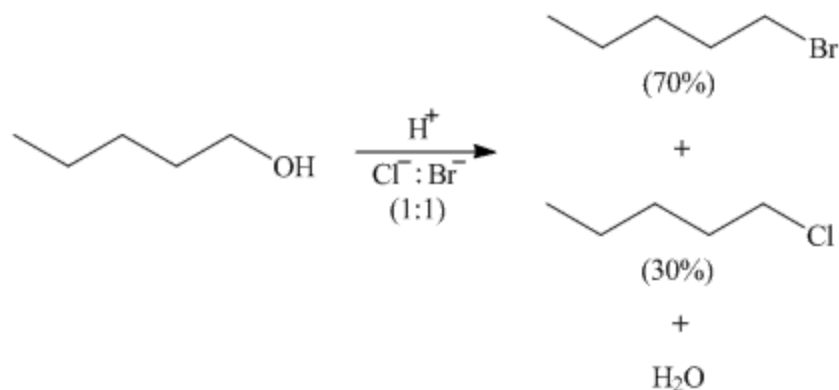


Figure 3

S_N2 Reaction of 1-pentanol

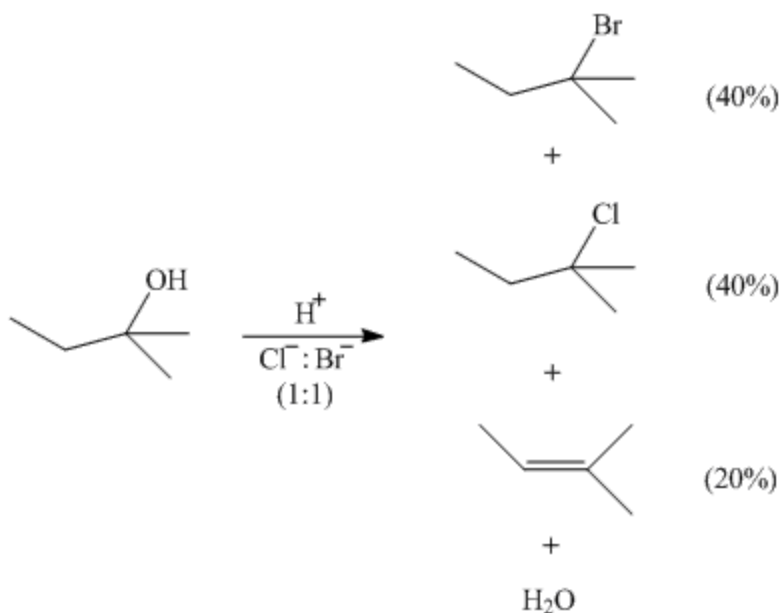


Figure 4

Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

If a solution containing the compounds shown in Figure 4, is injected into a gas-liquid chromatograph, the first peak observed in the gc trace is attributable to which compound?

- A.
2-Methyl-2-butanol
- B.
2-Methyl-2-butene
- C.
2-Chloro-2-methylbutane
- D.
2-Bromo-2-methylbutane

Solution: The correct answer is **B**.

1. 2-Methyl-2-butanol is highly polar and makes hydrogen bonds, so it is the least volatile of the options.
2. **In gas-liquid chromatography, the first peak to emerge will be from the least polar, most volatile compound. 2-Methyl-2-butene is nonpolar and the most volatile because it only has London forces for intermolecular attraction.**
3. Unlike 2-methyl-2-butene, 2-chloro-2-methylbutene possesses dipole-dipole forces in addition to greater London forces, so it will not be the most volatile.
4. Unlike 2-methyl-2-butene, 2-bromo-2-methylbutene possesses dipole-dipole forces in addition to greater London forces, so it will not be the most volatile.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Separations and purifications](#).

Incorrect

Your result is incorrect.

Your answer is C.

Correct answer is B.

Correct Answer Your Answer

A

B

C

x

D

Time Spent:

1 min 25 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5C

FC5

OCH

RESULT / TIMING

Passage 2 (Questions 6-9)

The nucleophilic alkyl substitution reactions are among the most thoroughly studied transformations in organic chemistry. A substitution by an S_N2 reaction mechanism is shown in Figure 1 and a substitution by an S_N1 mechanism is shown in Figure 2.

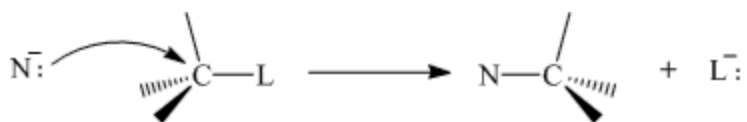


Figure 1

S_N2 Reaction mechanism

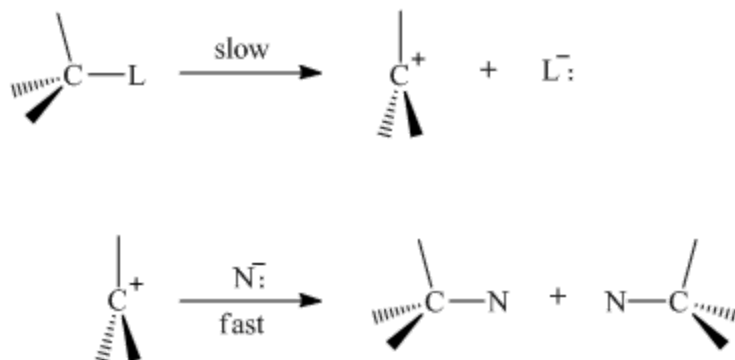


Figure 2	$\text{S}_{\text{N}}1$ Reaction mechanism
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The $\text{S}_{\text{N}}2$ mechanism involves a direct displacement of a leaving group (L) by a nucleophile (N). The reaction results in an inversion of configuration at the carbon bearing the leaving group, because the incoming nucleophile attacks it from the side opposite the leaving group. The $\text{S}_{\text{N}}1$ reaction mechanism involves an intermediate carbocation that is formed by loss of the leaving group. A nucleophile can react with the planar carbocation from either side, leading to the inversion or retention of the configuration at the site of the substitution.

The product distribution of two reactions is shown in figures 3 and 4.

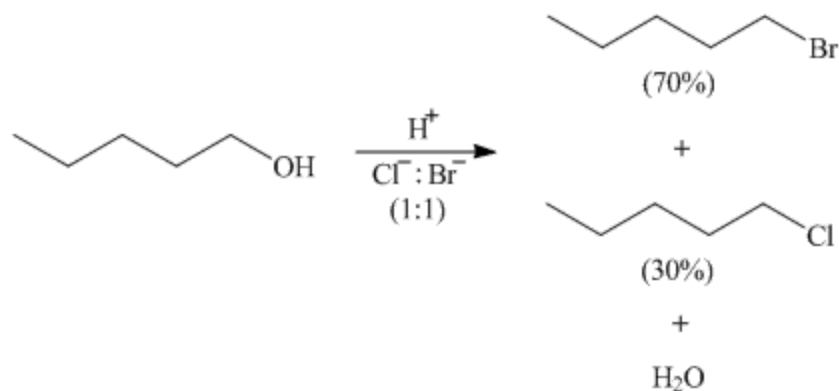


Figure 3	$\text{S}_{\text{N}}2$ Reaction of 1-pentanol
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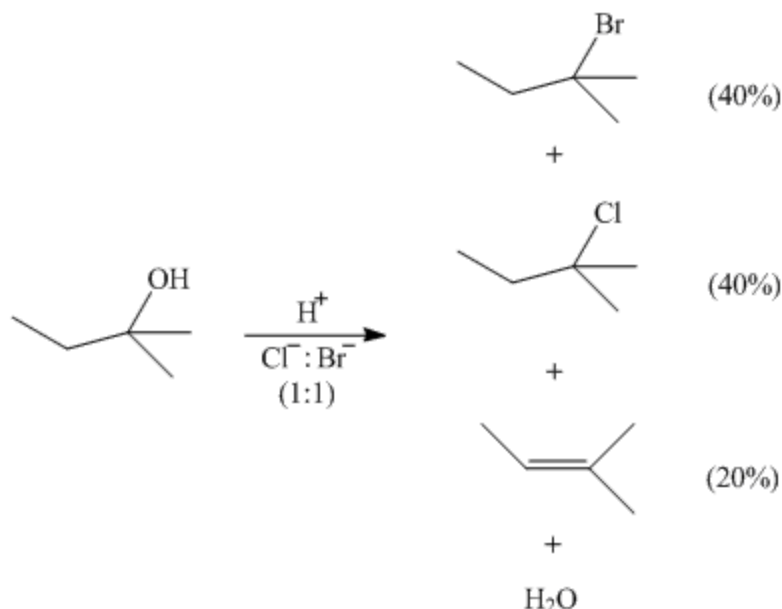


Figure 4 Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

If a solution containing the compounds shown in Figure 4, is injected into a gas-liquid chromatograph, the first peak observed in the gc trace is attributable to which compound?

- A.
2-Methyl-2-butanol
- B.
2-Methyl-2-butene
- C.
2-Chloro-2-methylbutane
- D.
2-Bromo-2-methylbutane

Solution: The correct answer is **B**.

1. 2-Methyl-2-butanol is highly polar and makes hydrogen bonds, so it is the least volatile of the options.
2. **In gas-liquid chromatography, the first peak to emerge will be from the least polar, most volatile compound. 2-Methyl-2-butene is nonpolar and the most volatile because it only has London forces for intermolecular attraction.**
3. Unlike 2-methyl-2-butene, 2-chloro-2-methylbutene possesses dipole-dipole forces in addition to greater London forces, so it will not be the most volatile.
4. Unlike 2-methyl-2-butene, 2-bromo-2-methylbutene possesses dipole-dipole forces in addition to greater London forces, so it will not be the most volatile.

KHAN ACADEMY LESSONS

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Question 9 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 2 (Questions 6-9)

The nucleophilic alkyl substitution reactions are among the most thoroughly studied transformations in organic chemistry. A substitution by an S_N2 reaction mechanism is shown in Figure 1 and a substitution by an S_N1 mechanism is shown in Figure 2.



Figure 1	S_N2 Reaction mechanism
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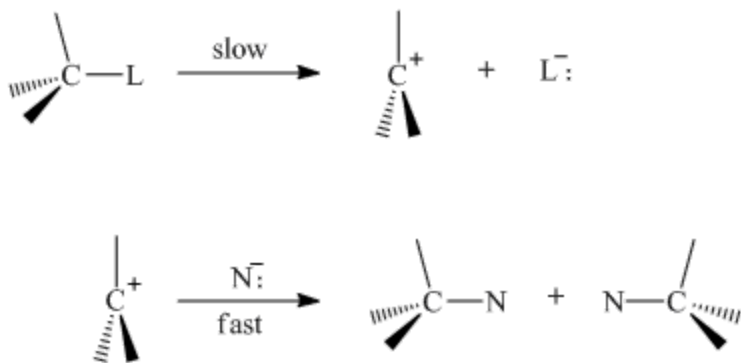


Figure 2	S_N1 Reaction mechanism
-----------------	---------------------------

The S_N2 mechanism involves a direct displacement of a leaving group (L) by a nucleophile (N). The reaction results in an inversion of configuration at the carbon bearing the leaving group, because the incoming nucleophile attacks it from the side opposite the leaving group. The S_N1 reaction mechanism involves an intermediate carbocation that is formed by loss of the leaving group. A nucleophile can react with the planar carbocation from either side, leading to the inversion or retention of the configuration at the site of the substitution.

The product distribution of two reactions is shown in figures 3 and 4.

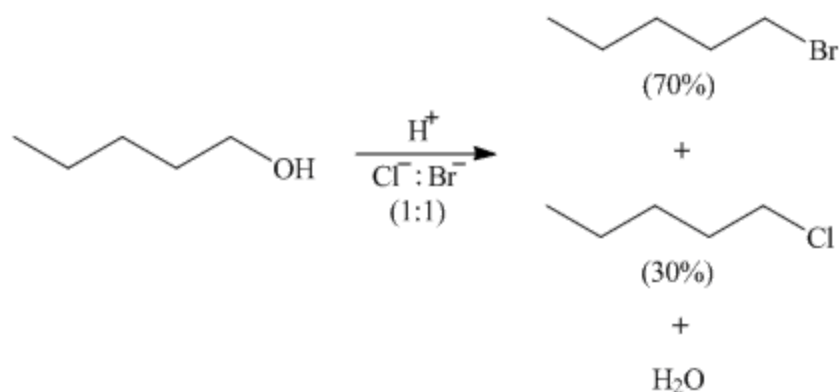


Figure 3

S_N2 Reaction of 1-pentanol

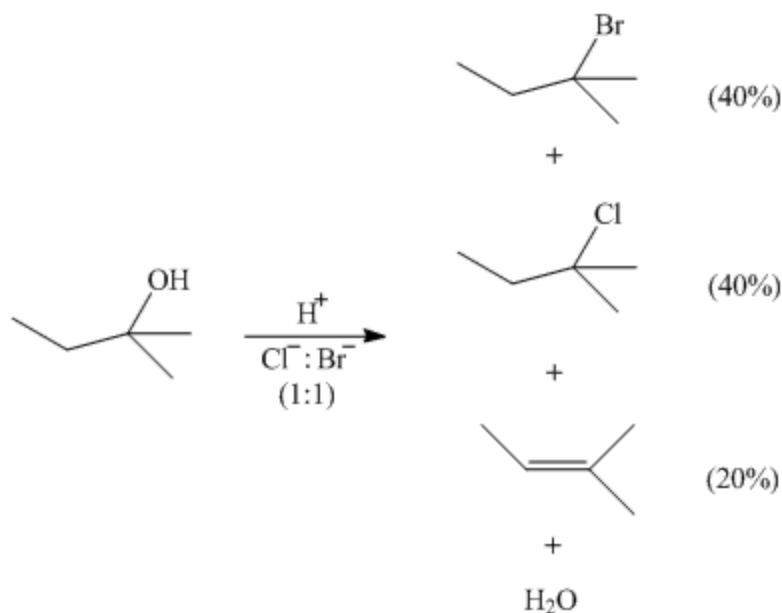
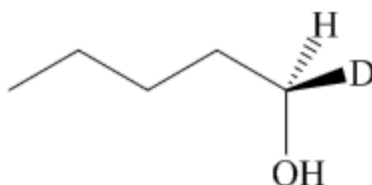


Figure 4

Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

What is the product of the reaction of Compound 1 (shown below) with HBr by the pathway shown in Figure 3?



Compound 1

- A.
(R)-1-bromo-1-deuteriohexane
- B.
(S)-1-bromo-1-deuteriohexane
- C.
(S)-1-bromo-1-deuteriopentane
- D.
(R)-1-bromo-1-deuteriopentane

Solution: The correct answer is **C**.

1. This product has both the wrong stereochemistry and the wrong number of carbon atoms.
2. Although this product has correct stereochemistry, it has the wrong number of carbon atoms.
3. **Because this is an $\text{S}_{\text{N}}2$ reaction, there will be inversion of configuration when bromide ion displaces water as a leaving group, changing the stereochemistry from R to S.**
4. Although this product has the correct number of carbons, the stereochemistry should be S, not R.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Stereochemistry](#).

Correct

Your result is correct.

Your answer is C.

Correct answer is C.

Correct Answer Your Answer

A

B

C

✓

D

Time Spent:

1 min 42 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5B

FC5

OCH

RESULT / TIMING

Passage 2 (Questions 6-9)

The nucleophilic alkyl substitution reactions are among the most thoroughly studied transformations in organic chemistry. A substitution by an S_N2 reaction mechanism is shown in Figure 1 and a substitution by an S_N1 mechanism is shown in Figure 2.



Figure 1

S_N2 Reaction mechanism

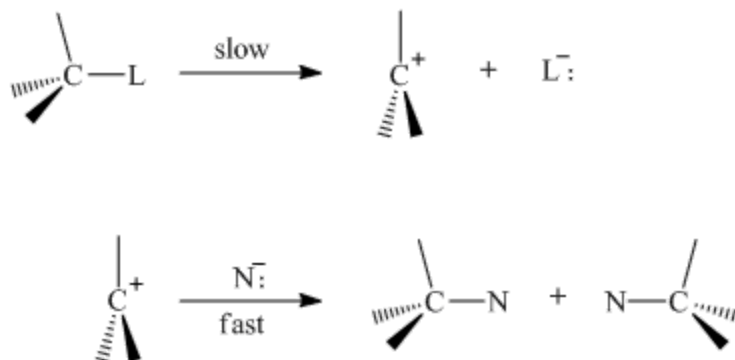


Figure 2	S _N 1 Reaction mechanism
-----------------	-------------------------------------

The S_N2 mechanism involves a direct displacement of a leaving group (L) by a nucleophile (N). The reaction results in an inversion of configuration at the carbon bearing the leaving group, because the incoming nucleophile attacks it from the side opposite the leaving group. The S_N1 reaction mechanism involves an intermediate carbocation that is formed by loss of the leaving group. A nucleophile can react with the planar carbocation from either side, leading to the inversion or retention of the configuration at the site of the substitution.

The product distribution of two reactions is shown in figures 3 and 4.

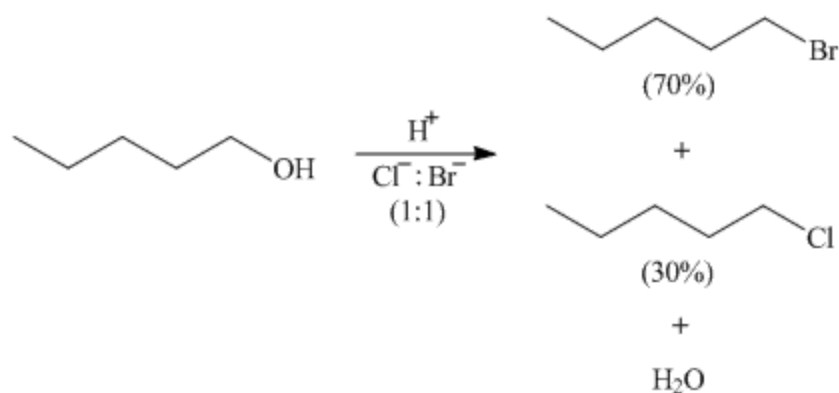


Figure 3	S _N 2 Reaction of 1-pentanol
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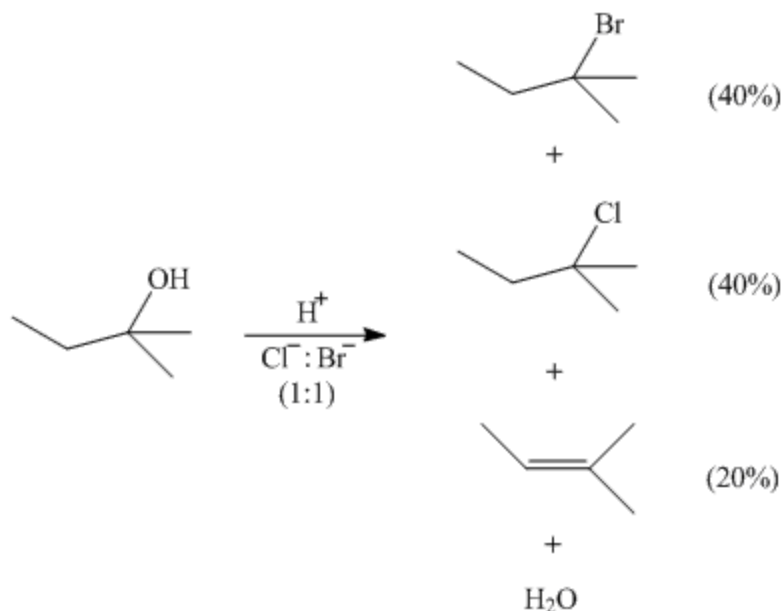
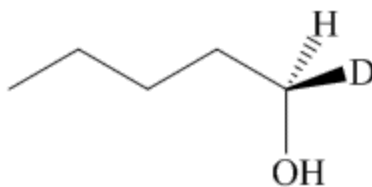


Figure 4 Reaction of 2-methyl-2-butanol

The rate of the reaction shown in Figure 3 increases as the concentration of alcohol, halide, or H^+ increases. The rate of the reaction shown in Figure 4 increases as the concentration of alcohol or H^+ increases. This reaction is independent of the halide concentration.

What is the product of the reaction of Compound **1** (shown below) with HBr by the pathway shown in Figure 3?



Compound **1**

- A.
(*R*)-1-bromo-1-deuteriohexane
- B.
(*S*)-1-bromo-1-deuteriohexane
- C.
(*S*)-1-bromo-1-deuteriopentane

- D.
(*R*)-1-bromo-1-deuteriopentane

Solution: The correct answer is **C**.

1. This product has both the wrong stereochemistry and the wrong number of carbon atoms.
2. Although this product has correct stereochemistry, it has the wrong number of carbon atoms.
3. **Because this is an S_N2 reaction, there will be inversion of configuration when bromide ion displaces water as a leaving group, changing the stereochemistry from R to S.**
4. Although this product has the correct number of carbons, the stereochemistry should be S, not R.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Stereochemistry](#).

Independent / Discrete Questions

Question 10 of 59

QUESTION

Done Reviewing

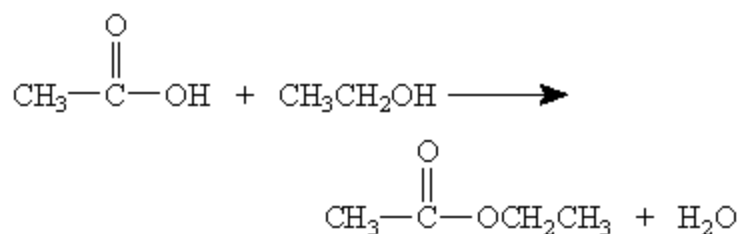
MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Acetic acid and ethanol react to form an ester product as shown below.



In determining which reactant loses the –OH group, which of the following isotopic substitutions would be most useful?

- A.
Replace the acidic H of acetic acid with D.
- B.
Replace the alcoholic H of ethanol with D.
- C.
Replace the carbonyl oxygen of acetic acid with O-18.
- D.
Replace the hydroxyl oxygen of ethanol with O-18.

Solution: The correct answer is **D**.

1. In order to determine which reactant loses the –OH group, an oxygen atom on ethanol must be radiolabeled.
2. In order to determine which reactant loses the –OH group, an oxygen atom on ethanol must be radiolabeled. Replacement the alcoholic H by D will not provide this information because of rapid H/D exchange between acetic acid and ethanol.

3. Replacing the carbonyl oxygen in acetic acid with O-18 will yield a mixture of O-18-labeled water and ester, further confusing the situation. There may be a mechanism of O-18 exchange with ethanol via O-18 labeled water that would obscure which reactant actually loses the –OH during ester formation.
4. **Labeling the oxygen in ethanol with O-18 definitely determines which –OH is lost because 100% of the O-18 in radiolabeled ethanol will be retained in the ester, not the water. Thus, only acetic acid supplies –OH that is lost to form water.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Amino Acids, Peptides, Proteins](#); [Lipids](#); [Carbohydrates](#); [Aldehydes and Ketones](#); [Alcohols, Phenols, Polycyclic and Heterocyclic Aromatic Compounds](#); [Carboxylic Acids](#); and [Acid Derivatives \(Anhydrides, Amides, Esters\)](#).

Incorrect

Your result is incorrect.

Your answer is A.

Correct answer is D.

Correct Answer Your Answer

A

x

B

C

D

Time Spent:

3 mins 23 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

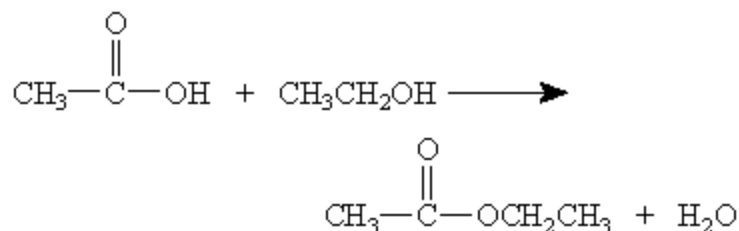
SIRS3

CC5D

FC5

OCH

Acetic acid and ethanol react to form an ester product as shown below.



In determining which reactant loses the –OH group, which of the following isotopic substitutions would be most useful?

- A.
Replace the acidic H of acetic acid with D.
- B.
Replace the alcoholic H of ethanol with D.
- C.
Replace the carbonyl oxygen of acetic acid with O-18.
- D.
Replace the hydroxyl oxygen of ethanol with O-18.

Solution: The correct answer is **D**.

1. In order to determine which reactant loses the –OH group, an oxygen atom on ethanol must be radiolabeled.
2. In order to determine which reactant loses the –OH group, an oxygen atom on ethanol must be radiolabeled. Replacement the alcoholic H by D will not provide this information because of rapid H/D exchange between acetic acid and ethanol.
3. Replacing the carbonyl oxygen in acetic acid with O-18 will yield a mixture of O-18-labeled water and ester, further confusing the situation. There may be a mechanism of O-18 exchange with ethanol via O-18 labeled water that would obscure which reactant actually loses the –OH during ester formation.
4. **Labeling the oxygen in ethanol with O-18 definitely determines which –OH is lost because 100% of the O-18 in radiolabeled ethanol will be retained in the ester, not the water. Thus, only acetic acid supplies –OH that is lost to form water.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Amino Acids, Peptides, Proteins](#); [Lipids](#); [Carbohydrates](#);

[Aldehydes and Ketones](#); [Alcohols](#), [Phenols](#), [Polycyclic and Heterocyclic Aromatic Compounds](#); [Carboxylic Acids](#); and [Acid Derivatives \(Anhydrides, Amides, Esters\)](#).

Question 11 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

A person, whose eye has a lens-to-retina distance of 2.0 cm, can only clearly see objects that are closer than 1.0 m away. What is the strength S of the person's eye lens? (Note: Use the thin lens formula $\frac{1}{O} + \frac{1}{I} = S$.)

- A.
–50 D
- B.
–10 D
- C.
51 D
- D.
55 D

Solution: The correct answer is **C**.

1. The negative sign of the lens strength is indicative of a diverging lens, whereas the eye lens is a converging lens.
2. The negative sign of the lens strength is indicative of a diverging lens, whereas the eye lens is a converging lens.
3. **According to the thin lens formula, the strength of the eye lens is equal to the inverse of the focal length of the eye lens. Its numerical value is given by $(1\text{ m})^{-1} + (0.02\text{ m})^{-1} = 1\text{ D} + 50\text{ D} = 51$**
4. This answer implies the person can only clearly see objects that are closer than 20 cm away.

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Sound](#); [Light, Electromagnetic Radiation](#); [Geometrical Optics](#); [Spherical mirrors](#); and [Reflection and refraction](#).

Incorrect

Your result is incorrect.

Your answer is A.

Correct answer is C.

Correct Answer Your Answer

A

x

B

C

D

Time Spent:

1 min 32 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC4D

FC4

PHY

RESULT / TIMING

A person, whose eye has a lens-to-retina distance of 2.0 cm, can only clearly see objects that are closer than 1.0 m away. What is the strength S of the person's eye lens? (Note: Use the thin lens formula $\frac{1}{O} + \frac{1}{I} = S$.)

- A.
-50 D
- B.
-10 D
- C.
51 D
- D.

Solution: The correct answer is **C**.

1. The negative sign of the lens strength is indicative of a diverging lens, whereas the eye lens is a converging lens.
2. The negative sign of the lens strength is indicative of a diverging lens, whereas the eye lens is a converging lens.
3. **According to the thin lens formula, the strength of the eye lens is equal to the inverse of the focal length of the eye lens. Its numerical value is given by $(1\text{ m})^{-1} + (0.02\text{ m})^{-1} = 1\text{ D} + 50\text{ D} = 51$**
4. This answer implies the person can only clearly see objects that are closer than 20 cm away.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Sound](#); [Light, Electromagnetic Radiation](#); [Geometrical Optics](#); [Spherical mirrors](#); and [Reflection and refraction](#).

Question 12 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Which statement correctly describes how enzymes affect chemical reactions?

Stabilization of:

- A.
the substrate changes the free energy of the reaction.
- B.
the transition state changes the free energy of the reaction.
- C.
the substrate changes the activation energy of the reaction.
- D.
the transition state changes the activation energy of the reaction.

Solution: The correct answer is **D**.

1. Although an enzyme binds to substrate, it does not affect the free energy of a reaction.
2. Although an enzyme does bind the transition state, it does not affect the free energy of a reaction.
3. An enzyme does not stabilize a substrate to change the activation energy of a reaction. Substrate stabilization would increase the activation energy and slow down the reaction.
4. **An enzyme stabilizes the transition state, thus lowering the activation energy of a reaction.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Enzymes](#); [Kinetics](#); and [Bioenergetics](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

28 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC5E

FC5

BCH

RESULT / TIMING

Which statement correctly describes how enzymes affect chemical reactions?

Stabilization of:

- A.
the substrate changes the free energy of the reaction.
- B.
the transition state changes the free energy of the reaction.
- C.
the substrate changes the activation energy of the reaction.
- D.
the transition state changes the activation energy of the reaction.

Solution: The correct answer is **D**.

1. Although an enzyme binds to substrate, it does not affect the free energy of a reaction.
2. Although an enzyme does bind the transition state, it does not affect the free energy of a reaction.
3. An enzyme does not stabilize a substrate to change the activation energy of a reaction. Substrate stabilization would increase the activation energy and slow down the reaction.
4. **An enzyme stabilizes the transition state, thus lowering the activation energy of a reaction.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Enzymes](#); [Kinetics](#); and [Bioenergetics](#).

Question 13 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Which of the following will *decrease* the percentage ionization of 1.0 M acetic acid, $\text{CH}_3\text{CO}_2\text{H}(aq)$?

- A.
Chlorinating the CH_3 group
- B.
Diluting the solution
- C.
Adding concentrated $\text{HCl}(aq)$
- D.
Adding a drop of basic indicator

Solution: The correct answer is **C**.

1. Replacing H with Cl will stabilize the formation of negative charge on the nearby carboxylic acid group. This is because chlorine is an electronegative atom. This change will therefore increase the percentage ionization of the acid.
2. Diluting the solution will increase the percentage ionization by Le Châtelier's Principle.
3. **HCl is a strong acid that will increase the amount of H^+ in solution and thus decrease the percentage of $\text{CH}_3\text{CO}_2\text{H}$ that ionizes.**
4. Adding a drop of basic indicator will slightly increase the amount of dissociated (ionized) acid due to the acid–base reaction that occurs between the two.

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Acid-Base Equilibria](#); [Solubility](#); and [Titration](#).

Incorrect

Your result is incorrect.

Your answer is B.

Correct answer is C.

Correct Answer Your Answer

A

B

x

C

D

Time Spent:

50 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5A

FC5

GCH

RESULT / TIMING

Which of the following will *decrease* the percentage ionization of 1.0 M acetic acid, $\text{CH}_3\text{CO}_2\text{H}(aq)$?

- A.
Chlorinating the CH_3 group
- B.
Diluting the solution
- C.
Adding concentrated $\text{HCl}(aq)$
- D.
Adding a drop of basic indicator

Solution: The correct answer is **C**.

1. Replacing H with Cl will stabilize the formation of negative charge on the nearby carboxylic acid group. This is because chlorine is an electronegative atom. This change will therefore increase the percentage ionization of the acid.
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4. Adding a drop of basic indicator will slightly increase the amount of dissociated (ionized) acid due to the acid–base reaction that occurs between the two.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Acid-Base Equilibria](#); [Solubility](#); and [Titration](#).

Question 22 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

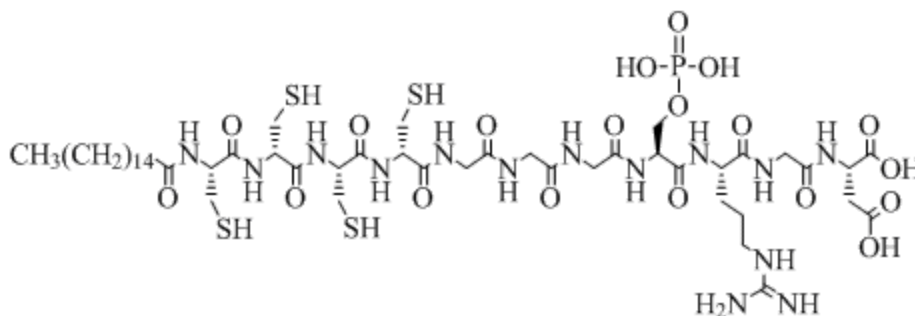
Please enter your note

Cancel Add

Passage 5 (Questions 22-25)

The natural development and repair of bone tissue provides a good example for the design of mineralized tissue fabrication schemes. Besides cells, bone tissue consists of both inorganic and organic components. The inorganic component is a carbonated, calcium-deficient, nano-crystalline hydroxyapatite [$\text{HA} = \text{Ca}_5(\text{PO}_4)_3(\text{OH})$]. The organic component is a matrix composed of type I collagen (95%) and non-collageneous proteins.

Researchers in tissue engineering have developed a peptide amphiphile (Compound 1, Figure 1) that can be used to make a nanostructured fibrous scaffold (composite) similar to an extracellular matrix found in bone.



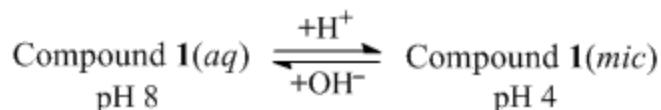
Compound 1

Figure 1

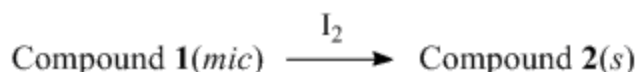
Structure of peptide amphiphile, Compound 1, used to build scaffolds for synthetic bone formation

Compound **1** contained five regions: (1) a hydrophobic region, (2) residues that could be cross-linked for additional structural integrity, (3) spacer residues, (4) a residue that was phosphorylated for future attachment of nanocrystals of HA, and (5) residues with affinity for cells.

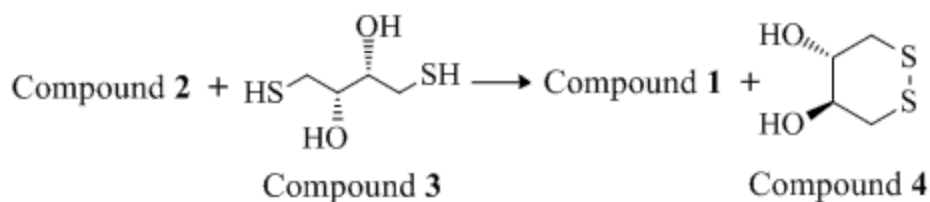
Molecules of Compound **1** spontaneously self-assemble into cylindrical micelles in water at pH 4 but freely redissolve at pH 8 (Equation 1). Oxidation of self-assembled, micellular Compound **1** with iodine resulted in a solid material (Compound **2**) that was stable to alkaline solutions (Equation 2). Reaction of Compound **2** with dithiol (Compound **3**) resulted in reformation of Compound **1** (Equation 3).



Equation 1



Equation 2



Equation 3

The researchers wanted to mimic the alignment that occurs between crystals of HA and collagen fibrils in bone tissue. They created a peptide amphiphile that assembles into cylindrical micelles as shown in Figure 2. Subsequent capture of nanocrystals of HA followed by mineralization was used to form synthetic bone.

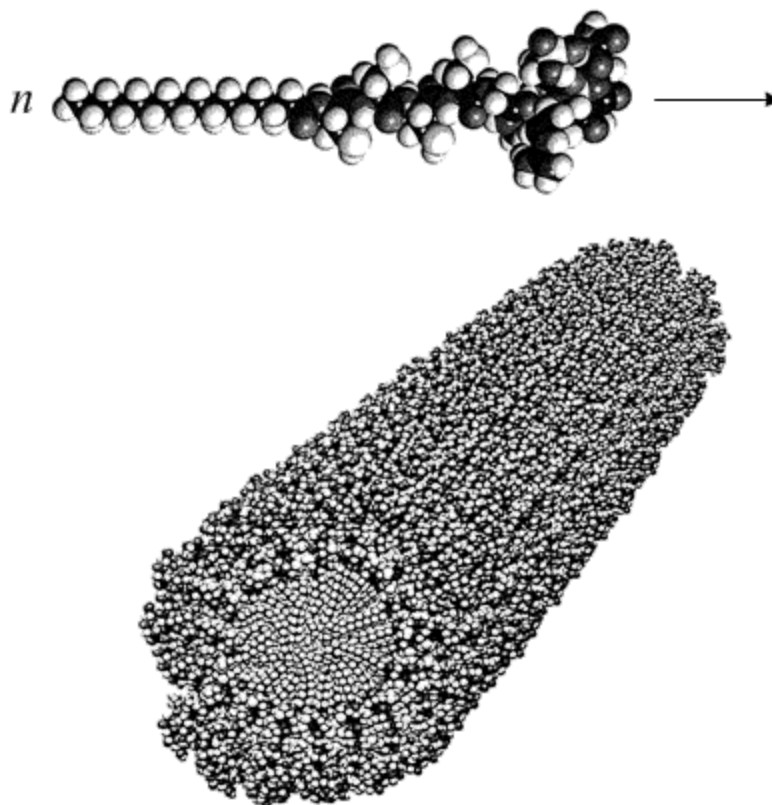


Figure 2

Space-filling molecular model illustrating cylindrical micelle formation by the peptide amphiphile, Compound **1**

Adapted from J. D. Hartgerink, E. Beniash, and S. I. Stupp, *Science*. ©2001 American Association for the Advancement of Science.

Compared to micellular Compound **1**, Compound **2** is structurally more rigid as a result of what type of interaction?

- A.
Intermolecular hydrogen bonding
- B.
Intermolecular covalent bonding
- C.
Intramolecular hydrogen bonding
- D.
Intramolecular covalent bonding

Solution: The correct answer is **B**.

1. Both compounds **1** and **2** form intermolecular hydrogen bonds. Only Compound **2** has disulfide bridges, which is a type of intermolecular covalent bonding.

2. **Only Compound 2 has disulfide bridges, which is a type of intermolecular covalent bonding.**
3. Although intramolecular hydrogen bonding would lead to structural rigidity, it is found in both compounds 1 and 2.
4. Intramolecular cross-linking could be found in a fraction of Compound 2 molecules, but these cross-links would not enhance the rigidity of a micelle as much as intermolecular cross-links would.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#); [Amino Acids](#), [Peptides](#), [Proteins](#); [The Three-Dimensional Protein Structure](#), [Nonenzymatic Protein Function](#); and [Phenols](#), [Polycyclic and Heterocyclic Aromatic Compounds](#).

Incorrect

Your result is incorrect.

Your answer is A.

Correct answer is B.

Correct Answer Your Answer

A

x

B

C

D

Time Spent:

2 mins 38 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5D

FC5

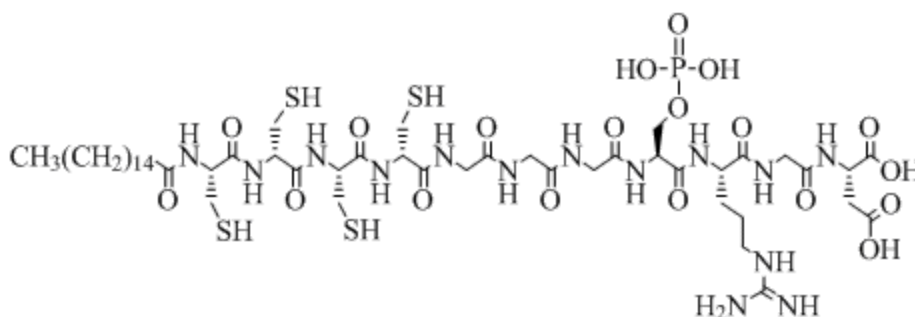
BCH

RESULT / TIMING

Passage 5 (Questions 22-25)

The natural development and repair of bone tissue provides a good example for the design of mineralized tissue fabrication schemes. Besides cells, bone tissue consists of both inorganic and organic components. The inorganic component is a carbonated, calcium-deficient, nano-crystalline hydroxyapatite [$\text{HA} = \text{Ca}_5(\text{PO}_4)_3(\text{OH})$]. The organic component is a matrix composed of type I collagen (95%) and non-collagenous proteins.

Researchers in tissue engineering have developed a peptide amphiphile (Compound 1, Figure 1) that can be used to make a nanostructured fibrous scaffold (composite) similar to an extracellular matrix found in bone.

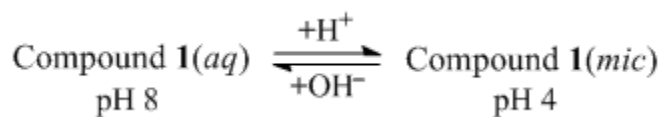


Compound 1

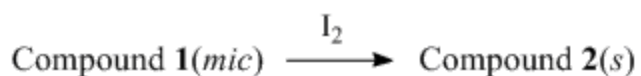
Figure 1	Structure of peptide amphiphile, Compound 1, used to build scaffolds for synthetic bone formation
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Compound 1 contained five regions: (1) a hydrophobic region, (2) residues that could be cross-linked for additional structural integrity, (3) spacer residues, (4) a residue that was phosphorylated for future attachment of nanocrystals of HA, and (5) residues with affinity for cells.

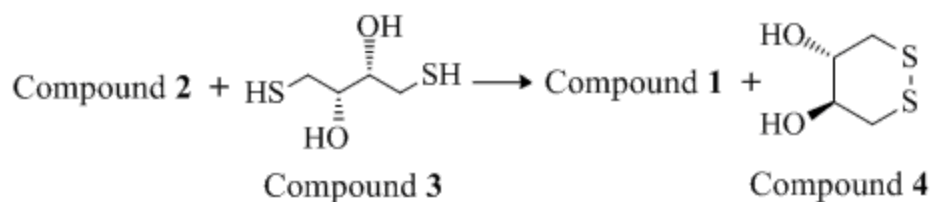
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Equation 1



Equation 2



Equation 3

The researchers wanted to mimic the alignment that occurs between crystals of HA and collagen fibrils in bone tissue. They created a peptide amphiphile that assembles into cylindrical micelles as shown in Figure 2. Subsequent capture of nanocrystals of HA followed by mineralization was used to form synthetic bone.

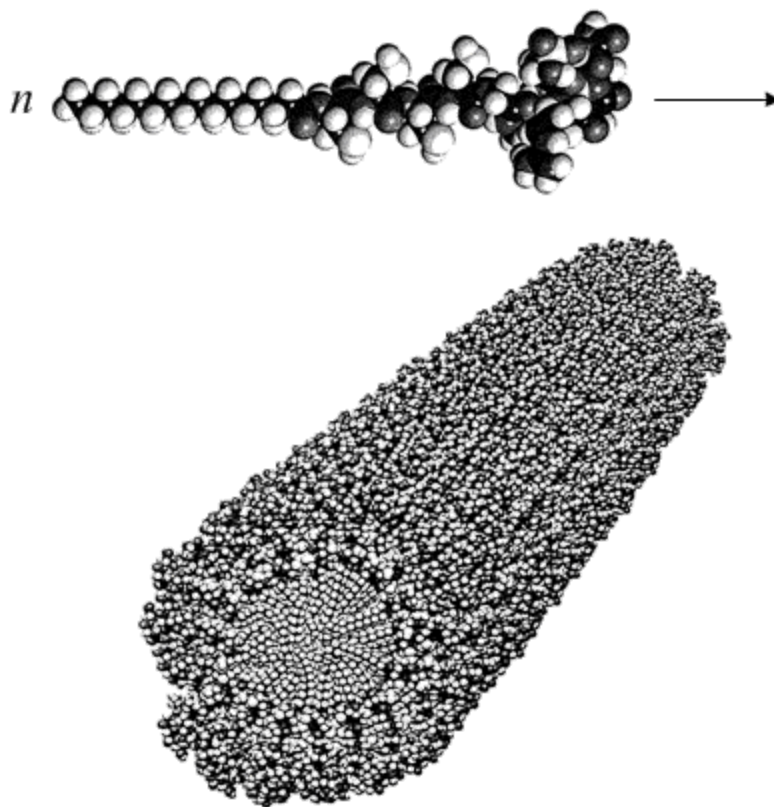


Figure 2

Space-filling molecular model illustrating cylindrical micelle formation by the peptide amphiphile, Compound 1

Compared to micellar Compound **1**, Compound **2** is structurally more rigid as a result of what type of interaction?

- A.
Intermolecular hydrogen bonding
- B.
Intermolecular covalent bonding
- C.
Intramolecular hydrogen bonding
- D.
Intramolecular covalent bonding

Solution: The correct answer is **B**.

1. Both compounds 1 and 2 form intermolecular hydrogen bonds. Only Compound 2 has disulfide bridges, which is a type of intermolecular covalent bonding.
2. **Only Compound 2 has disulfide bridges, which is a type of intermolecular covalent bonding.**
3. Although intramolecular hydrogen bonding would lead to structural rigidity, it is found in both compounds 1 and 2.
4. Intramolecular cross-linking could be found in a fraction of Compound 2 molecules, but these cross-links would not enhance the rigidity of a micelle as much as intermolecular cross-links would.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#); [Amino Acids](#), [Peptides](#), [Proteins](#); [The Three-Dimensional Protein Structure](#), [Nonenzymatic Protein Function](#); and [Phenols](#), [Polycyclic and Heterocyclic Aromatic Compounds](#).

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

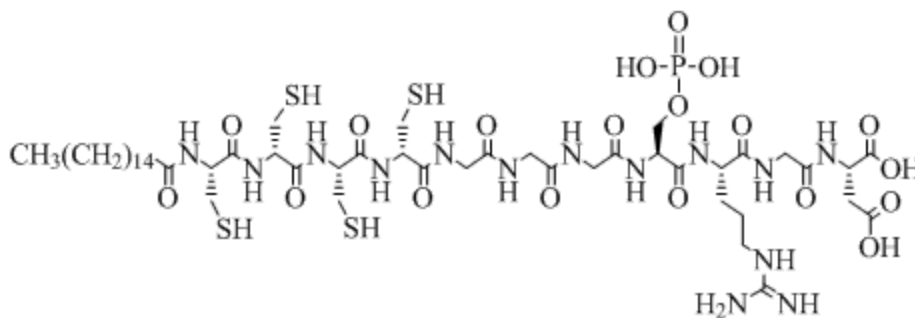
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Cancel Add

Passage 5 (Questions 22-25)

The natural development and repair of bone tissue provides a good example for the design of mineralized tissue fabrication schemes. Besides cells, bone tissue consists of both inorganic and organic components. The inorganic component is a carbonated, calcium-deficient, nano-crystalline hydroxyapatite [$\text{HA} = \text{Ca}_5(\text{PO}_4)_3(\text{OH})$]. The organic component is a matrix composed of type I collagen (95%) and non-collageneous proteins.

Researchers in tissue engineering have developed a peptide amphiphile (Compound 1, Figure 1) that can be used to make a nanostructured fibrous scaffold (composite) similar to an extracellular matrix found in bone.



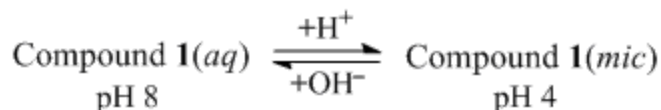
Compound 1

Figure 1

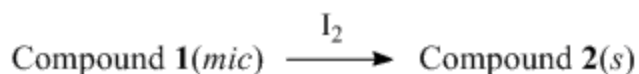
Structure of peptide amphiphile, Compound 1, used to build scaffolds for synthetic bone formation

Compound **1** contained five regions: (1) a hydrophobic region, (2) residues that could be cross-linked for additional structural integrity, (3) spacer residues, (4) a residue that was phosphorylated for future attachment of nanocrystals of HA, and (5) residues with affinity for cells.

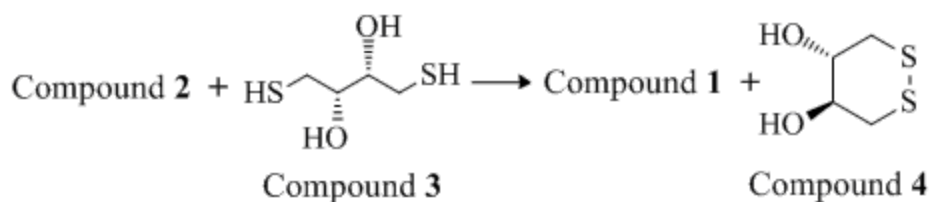
Molecules of Compound **1** spontaneously self-assemble into cylindrical micelles in water at pH 4 but freely redissolve at pH 8 (Equation 1). Oxidation of self-assembled, micellular Compound **1** with iodine resulted in a solid material (Compound **2**) that was stable to alkaline solutions (Equation 2). Reaction of Compound **2** with dithiol (Compound **3**) resulted in reformation of Compound **1** (Equation 3).



Equation 1



Equation 2



Equation 3

The researchers wanted to mimic the alignment that occurs between crystals of HA and collagen fibrils in bone tissue. They created a peptide amphiphile that assembles into cylindrical micelles as shown in Figure 2. Subsequent capture of nanocrystals of HA followed by mineralization was used to form synthetic bone.

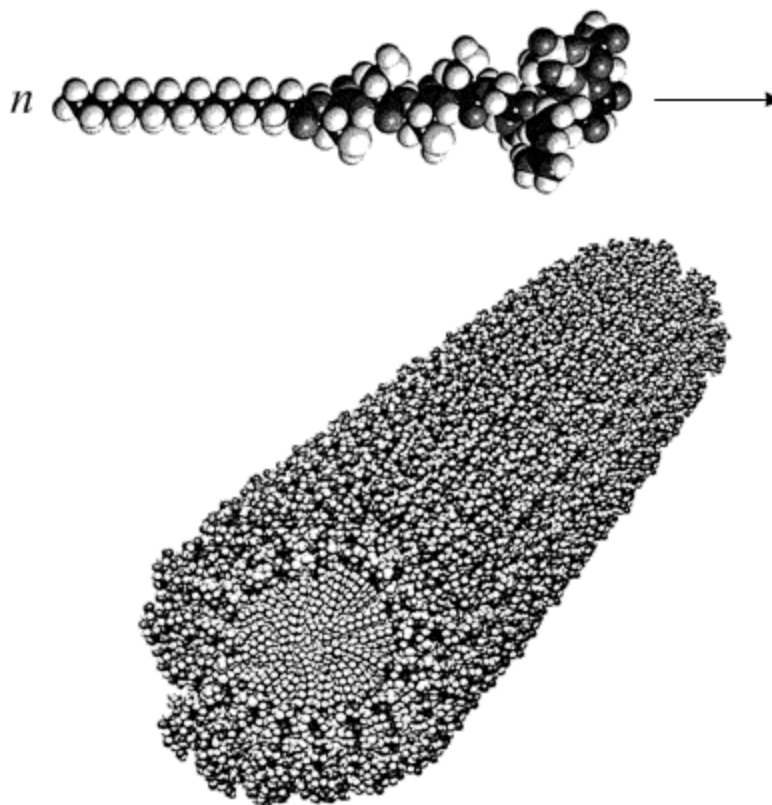


Figure 2

Space-filling molecular model illustrating cylindrical micelle formation by the peptide amphiphile, Compound **1**

Adapted from J. D. Hartgerink, E. Beniash, and S. I. Stupp, *Science*. ©2001 American Association for the Advancement of Science.

Which amino acid residues were incorporated into Compound **1** to promote the adhesion of cells on the scaffold surfaces?

- A.
Arg and Gly
- B.
Cys and Gly
- C.
Cys and Asp
- D.
Asp and Arg

Solution: The correct answer is **D**.

1. Arg has a side chain that can promote strong adhesion via electrostatic interaction, but Gly does not.

2. Cys is part of the cross-linking region of Compound 1, not the cell adhesion region, and Gly has no side chain that can form strong attractive forces of any type.
3. Although the side of Asp is in the cell adhesion region of Compound 1 and can form electrostatic interactions, Cys is part of the cross-linking region of Compound 1, not the cell adhesion region.
4. **Asp and Arg are part of the cell adhesion region of Compound 1. Both have side chains that can participate in electrostatic interactions.**

KHAN ACADEMY LESSONS

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Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

2 mins 53 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5D

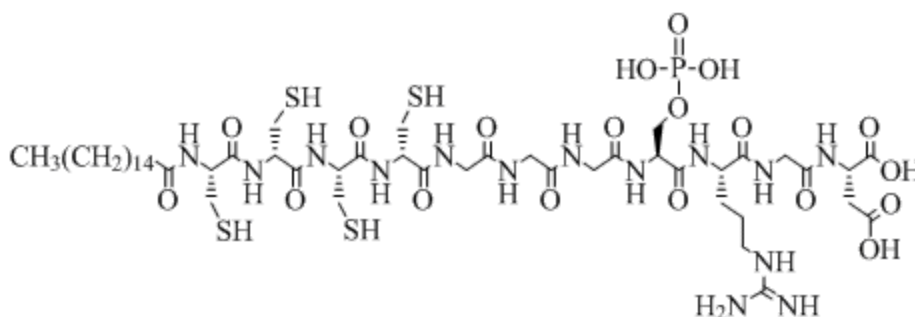
FC5

BCH

Passage 5 (Questions 22-25)

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Researchers in tissue engineering have developed a peptide amphiphile (Compound 1, Figure 1) that can be used to make a nanostructured fibrous scaffold (composite) similar to an extracellular matrix found in bone.

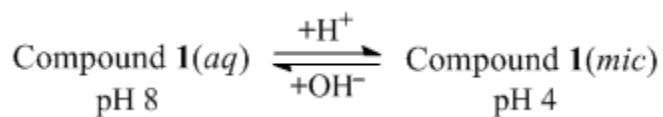


Compound 1

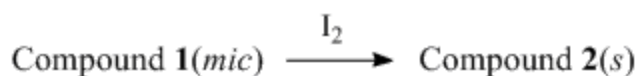
Figure 1	Structure of peptide amphiphile, Compound 1, used to build scaffolds for synthetic bone formation
-----------------	---

Compound 1 contained five regions: (1) a hydrophobic region, (2) residues that could be cross-linked for additional structural integrity, (3) spacer residues, (4) a residue that was phosphorylated for future attachment of nanocrystals of HA, and (5) residues with affinity for cells.

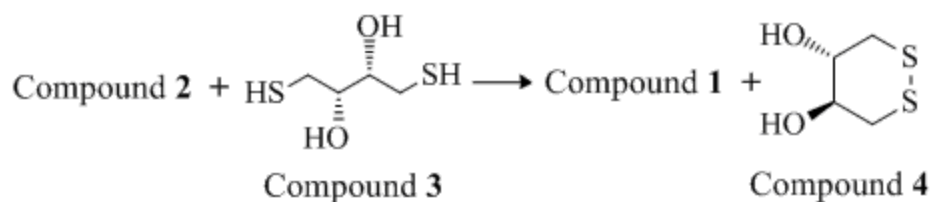
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Equation 1



Equation 2



Equation 3

The researchers wanted to mimic the alignment that occurs between crystals of HA and collagen fibrils in bone tissue. They created a peptide amphiphile that assembles into cylindrical micelles as shown in Figure 2. Subsequent capture of nanocrystals of HA followed by mineralization was used to form synthetic bone.

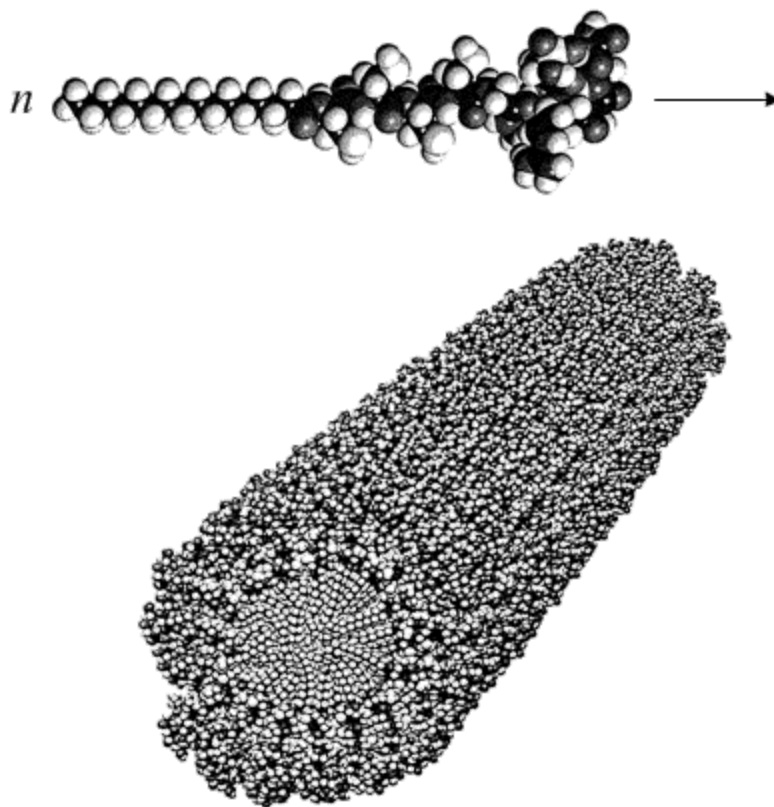


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- A.
Arg and Gly
- B.
Cys and Gly
- C.
Cys and Asp
- D.
Asp and Arg

Solution: The correct answer is **D**.

1. Arg has a side chain that can promote strong adhesion via electrostatic interaction, but Gly does not.
2. Cys is part of the cross-linking region of Compound 1, not the cell adhesion region, and Gly has no side chain that can form strong attractive forces of any type.
3. Although the side of Asp is in the cell adhesion region of Compound 1 and can form electrostatic interactions, Cys is part of the cross-linking region of Compound 1, not the cell adhesion region.
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KHAN ACADEMY LESSONS

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QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

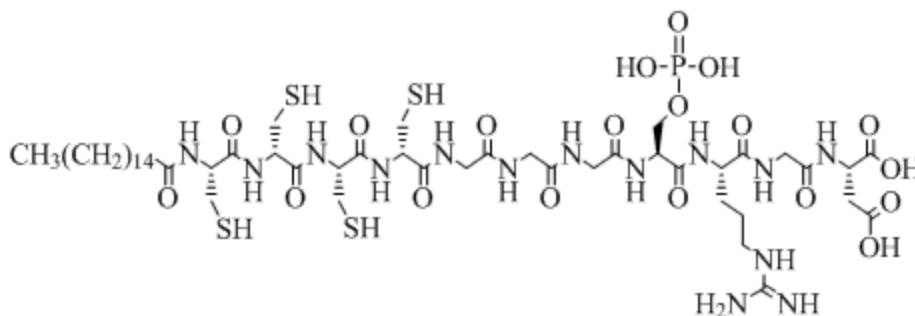
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Cancel Add

Passage 5 (Questions 22-25)

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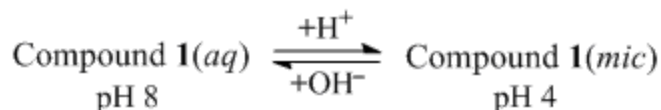
Compound 1

Figure 1

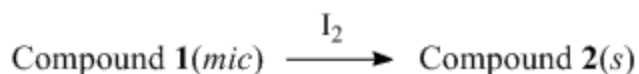
Structure of peptide amphiphile, Compound 1, used to build scaffolds for synthetic bone formation

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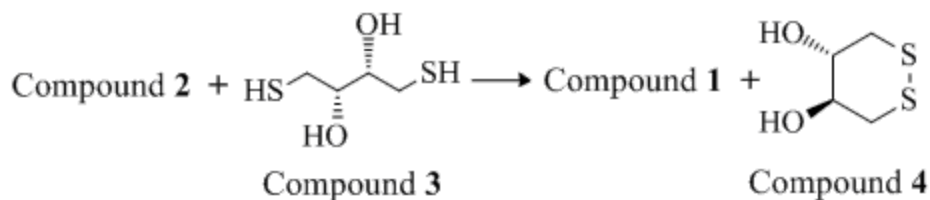
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Equation 1



Equation 2



Equation 3

The researchers wanted to mimic the alignment that occurs between crystals of HA and collagen fibrils in bone tissue. They created a peptide amphiphile that assembles into cylindrical micelles as shown in Figure 2. Subsequent capture of nanocrystals of HA followed by mineralization was used to form synthetic bone.

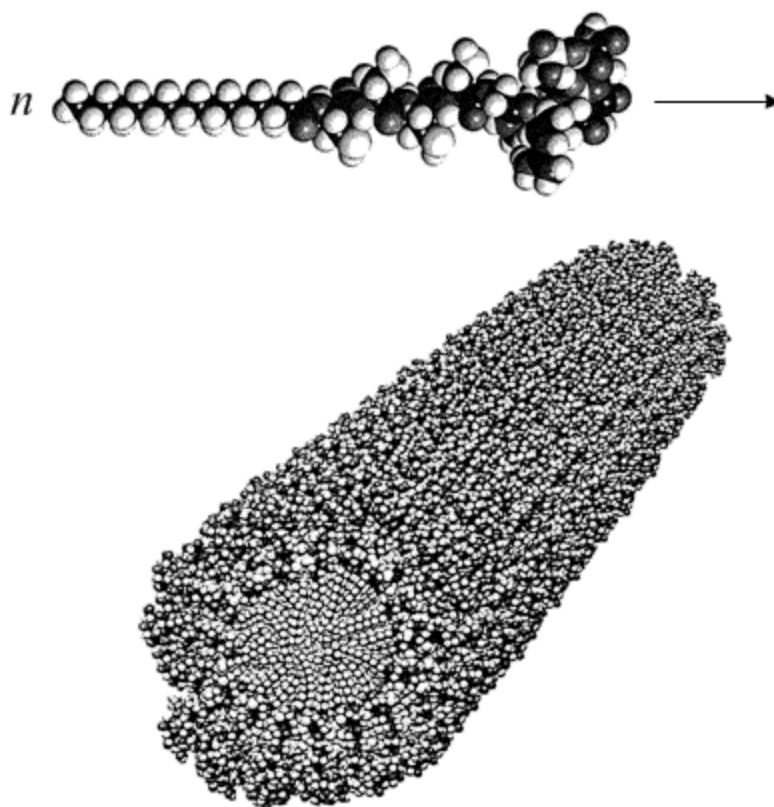


Figure 2

Space-filling molecular model illustrating cylindrical micelle formation by the peptide amphiphile, Compound **1**

Adapted from J. D. Hartgerink, E. Beniash, and S. I. Stupp, *Science*. ©2001 American Association for the Advancement of Science.

Which amino acid was incorporated into Compound **1** as a future site of covalent attachment to HA prior to mineralization?

- A.
Ser
- B.
Ala
- C.
Tyr
- D.
Thr

Solution: The correct answer is **A**.

1. **Ser is the residue that was phosphorylated at its primary alcohol side chain for future attachment of nanocrystals of HA**

2. Ala has a methyl side chain, not a primary alcohol.
3. The side chain of Tyr is a phenol, not a primary alcohol.
4. The side chain of Thr is a secondary alcohol, not a primary alcohol.

KHAN ACADEMY LESSONS

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Correct

Your result is correct.

Your answer is A.

Correct answer is A.

Correct Answer Your Answer

A

✓

B

C

D

Time Spent:

1 min 3 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5D

FC5

BCH

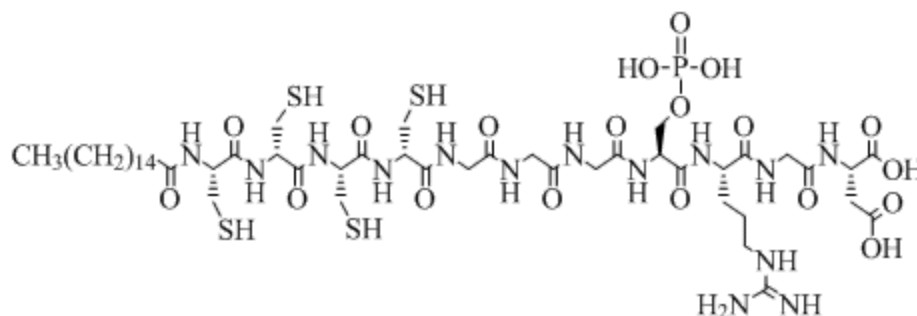
RESULT / TIMING

Passage 5 (Questions 22-25)

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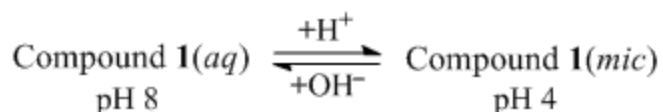


Compound **1**

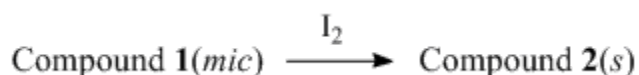
Figure 1	Structure of peptide amphiphile, Compound 1 , used to build scaffolds for synthetic bone formation
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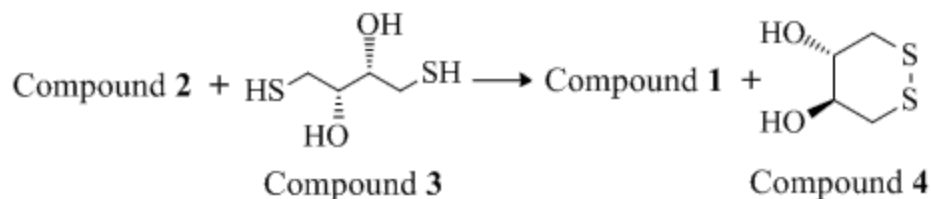
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Equation 1



Equation 2



Equation 3

The researchers wanted to mimic the alignment that occurs between crystals of HA and collagen fibrils in bone tissue. They created a peptide amphiphile that assembles into cylindrical micelles as shown in Figure 2. Subsequent capture of nanocrystals of HA followed by mineralization was used to form synthetic bone.

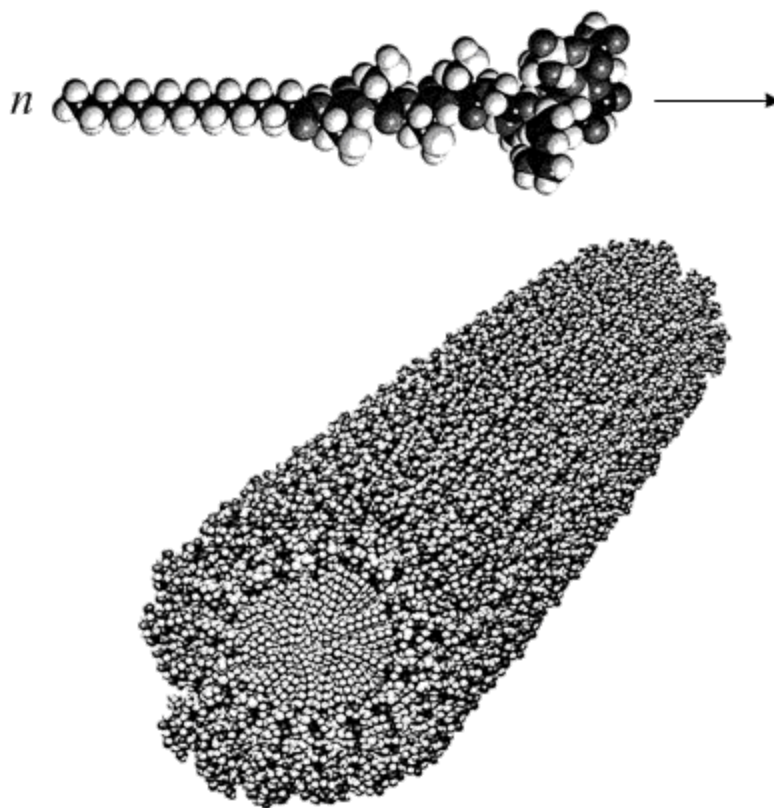


Figure 2

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Tyr
- D.
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QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

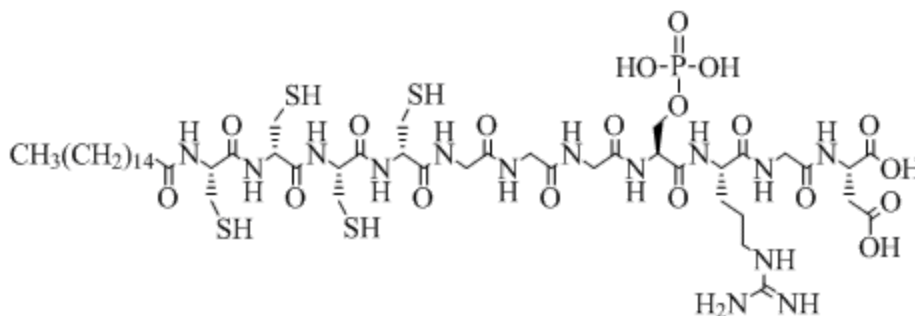
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Cancel Add

Passage 5 (Questions 22-25)

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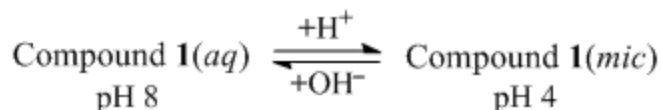
Compound 1

Figure 1

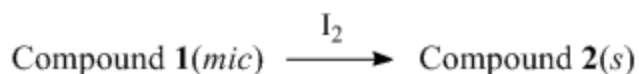
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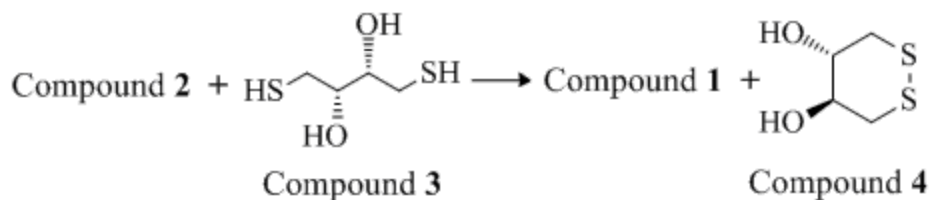
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Equation 1



Equation 2



Equation 3

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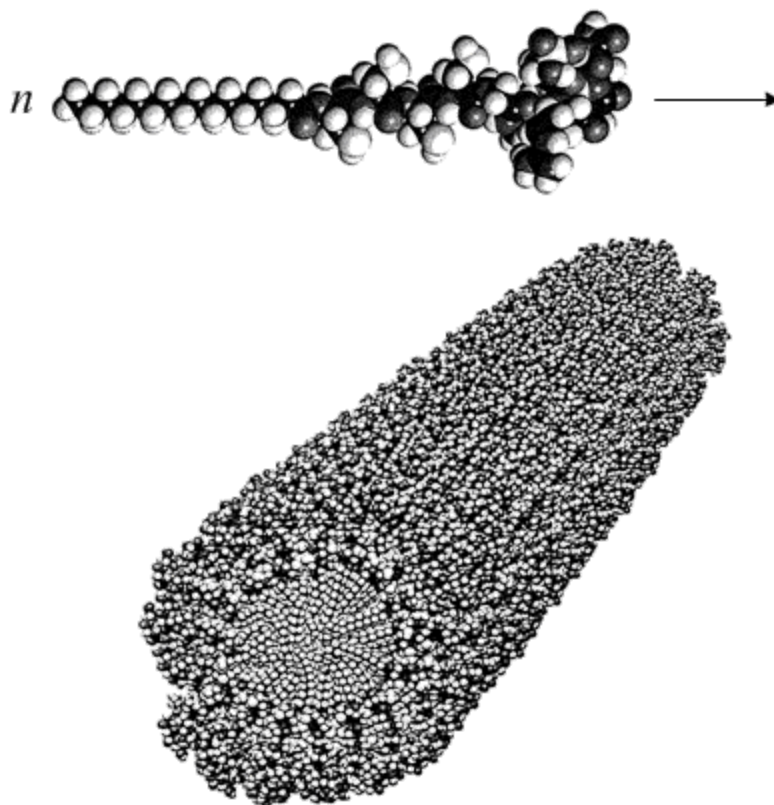


Figure 2

Space-filling molecular model illustrating cylindrical micelle formation by the peptide amphiphile, Compound **1**

Adapted from J. D. Hartgerink, E. Beniash, and S. I. Stupp, *Science*. ©2001 American Association for the Advancement of Science.

Compound **1** was designed to exhibit pH-dependent self-assembly. What feature(s) of the molecule is(are) responsible for the pH dependence of aggregation (Equation 1)?

- A.
Thiol side chains that can hydrogen bond
- B.
Long alkyl tail that exhibits predominantly London forces
- C.
Side chains whose net charge responds to pH
- D.
Covalent linkages that reversibly hydrolyze

Solution: The correct answer is **C**.

1. Charge development is the key to pH dependence of aggregation, not hydrogen bonding.

2. Long alkyl chains are not affected by pH.
3. **In order for pH to affect self-assembly of Compound 1, there must be side chains whose net charge respond to pH. At pH 4, there is a smaller net negative charge, self-assembly; at pH 8, the net negative charge is much higher, causing repulsion of Compound 1 molecules and blocking self-assembly.**
4. There are no pH-dependent, reversibly hydrolyzable covalent bonds formed when Compound 1 self-assembles.

KHAN ACADEMY LESSONS

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Correct

Your result is correct.

Your answer is C.

Correct answer is C.

Correct Answer Your Answer

A

B

C

✓

D

Time Spent:

33 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS3

CC5D

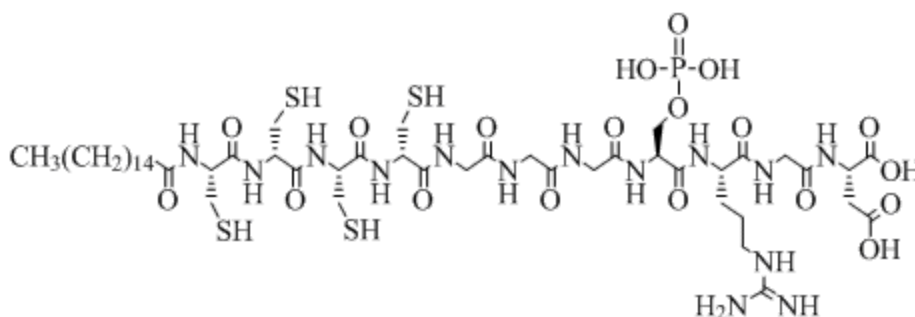
FC5

BCH

Passage 5 (Questions 22-25)

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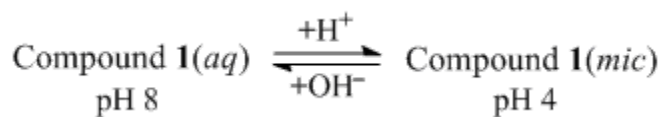


Compound **1**

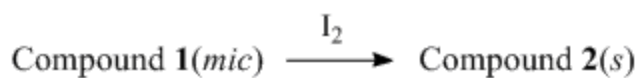
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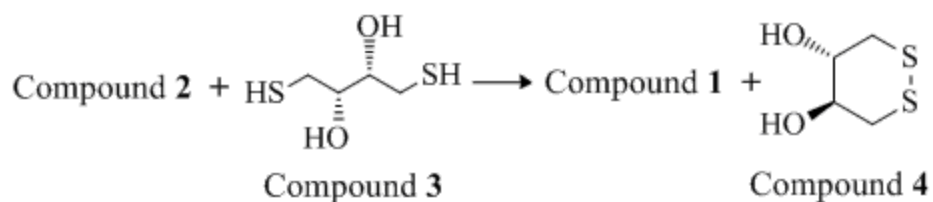
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Equation 1



Equation 2



Equation 3

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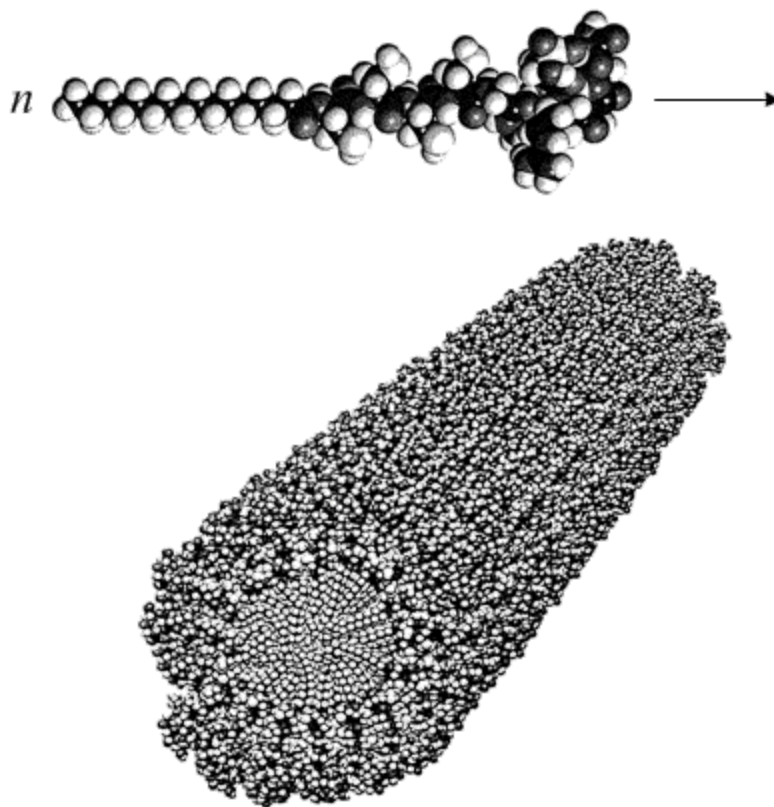


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Side chains whose net charge responds to pH
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Question 26 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

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Why is the velocity of blood flow slower in capillaries than in arteries?

- A.
Capillary walls are more elastic than arterial walls.
- B.
Capillaries have less resistance to blood flow than arteries.
- C.
The total cross-sectional area of capillaries exceeds that of arteries.
- D.
Blood pressure is higher in the capillaries than in arteries.

Solution: The correct answer is **C**.

1. The velocity of blood flow is primarily determined by the total cross-sectional area, not by elasticity of that level. The velocity of blood flow is inversely proportional to the total cross-sectional area.
2. The velocity of blood flow is primarily determined by the total cross-sectional area, not by resistance to blood flow of that level.
3. **The high number of capillaries in the body means that the total cross-sectional area of these vessels is larger than any other vessel type in the circulatory system. This causes the velocity of the blood to decrease.**
4. There is an increased ratio of surface area to blood volume in capillaries compared to arteries. This means that the blood is in contact with more blood vessel surface area. This slows down the velocity of the blood within capillaries and thus drops the blood pressure.

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Arterial and venous systems](#) and [Pressure and flow characteristics](#).

Incorrect

Your result is incorrect.

Your answer is A.

Correct answer is C.

Correct Answer Your Answer

A

x

B

C

D

Time Spent:

1 min 4 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC4B

FC4

BIO

RESULT / TIMING

Why is the velocity of blood flow slower in capillaries than in arteries?

- A.
Capillary walls are more elastic than arterial walls.
- B.
Capillaries have less resistance to blood flow than arteries.
- C.
The total cross-sectional area of capillaries exceeds that of arteries.
- D.
Blood pressure is higher in the capillaries than in arteries.

Solution: The correct answer is **C**.

1. The velocity of blood flow is primarily determined by the total cross-sectional area, not by elasticity of that level. The velocity of blood flow is inversely proportional to the total cross-sectional area.
2. The velocity of blood flow is primarily determined by the total cross-sectional area, not by resistance to blood flow of that level.
3. **The high number of capillaries in the body means that the total cross-sectional area of these vessels is larger than any other vessel type in the circulatory system. This causes the velocity of the blood to decrease.**
4. There is an increased ratio of surface area to blood volume in capillaries compared to arteries. This means that the blood is in contact with more blood vessel surface area. This slows down the velocity of the blood within capillaries and thus drops the blood pressure.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Arterial and venous systems](#) and [Pressure and flow characteristics](#).

Question 27 of 59

QUESTION

Done Reviewing

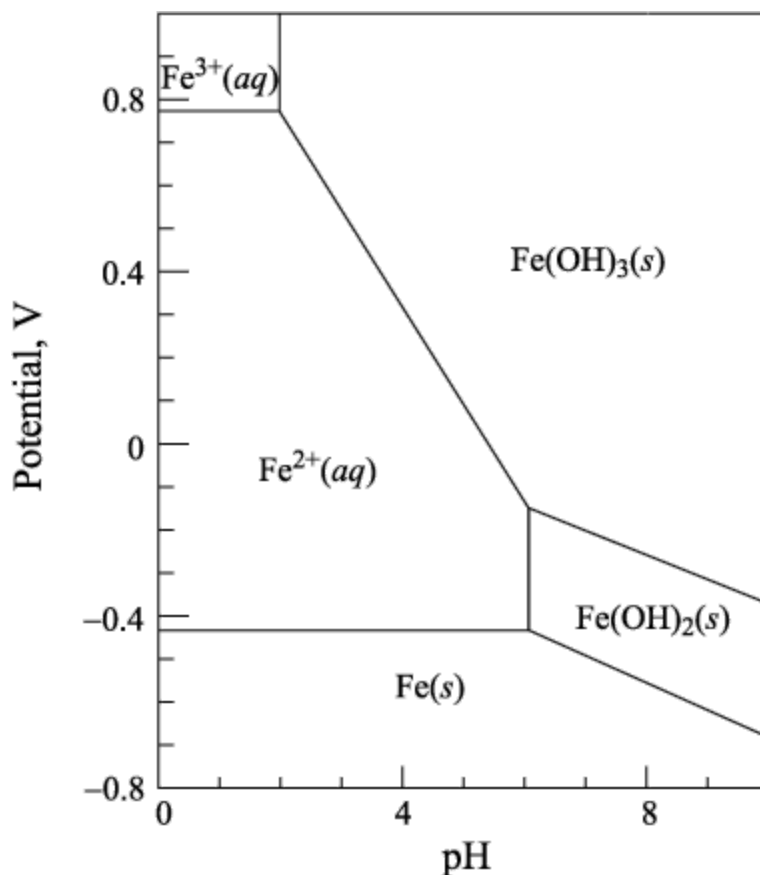
MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

The graph below shows the relationship between the predominant form of iron as a function of solution pH and applied potential. Based on the graph, which of the following statements is true?



- A.

At a potential of -0.4 V, as pH increases, Fe^{2+} is reduced and precipitates as $\text{Fe}(\text{OH})_3$.

- B.
At a potential of -0.44 V , the equilibrium between Fe and Fe^{2+} is independent of solution pH below pH 6.
- C.
At pH = 1, as the potential is changed from -0.2 to $+0.8$, Fe^{3+} is reduced to Fe^{2+} .
- D.
At pH = 8 and $V = -0.1\text{ V}$, $\text{Fe}(\text{OH})_2$ is the predominant form of iron.

Solution: The correct answer is **B**.

1. The conversion of Fe^{2+} to $\text{Fe}(\text{OH})_3(\text{s})$ represents an oxidation. From the graph, at a potential of -0.4 V , no oxidation or reduction occurs. $\text{Fe}(\text{OH})_2(\text{s})$ precipitates.
2. **The Pourbaix diagram for iron is given, but only one statement correctly conveys information presented by the diagram. At an applied potential of -0.44V and pH below 6, there is an equilibrium between $\text{Fe}(\text{s})$ and $\text{Fe}^{2+}(\text{aq})$, as shown by the line of demarcation. The fact that this line moves horizontally with increasing pH up until pH = 6 means that this equilibrium is unaffected by changing the pH.**
3. The opposite is true. At this pH as the potential is raised, Fe^{2+} is oxidized to Fe^{3+} .
4. At pH = 8 and $V = -0.1\text{ V}$, $\text{Fe}(\text{OH})_3(\text{s})$ is the predominant form of iron.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#); [Electronic Structure](#); [The Periodic Table - Classification of Elements Into Groups by Electronic Structure](#), [The Periodic Table - Variations of Chemical Properties with Group and Row](#); and [Stoichiometry](#).

Correct

Your result is correct.

Your answer is B.

Correct answer is B.

Correct Answer Your Answer

A

B

✓

C

D

Time Spent:

2 mins 5 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS4

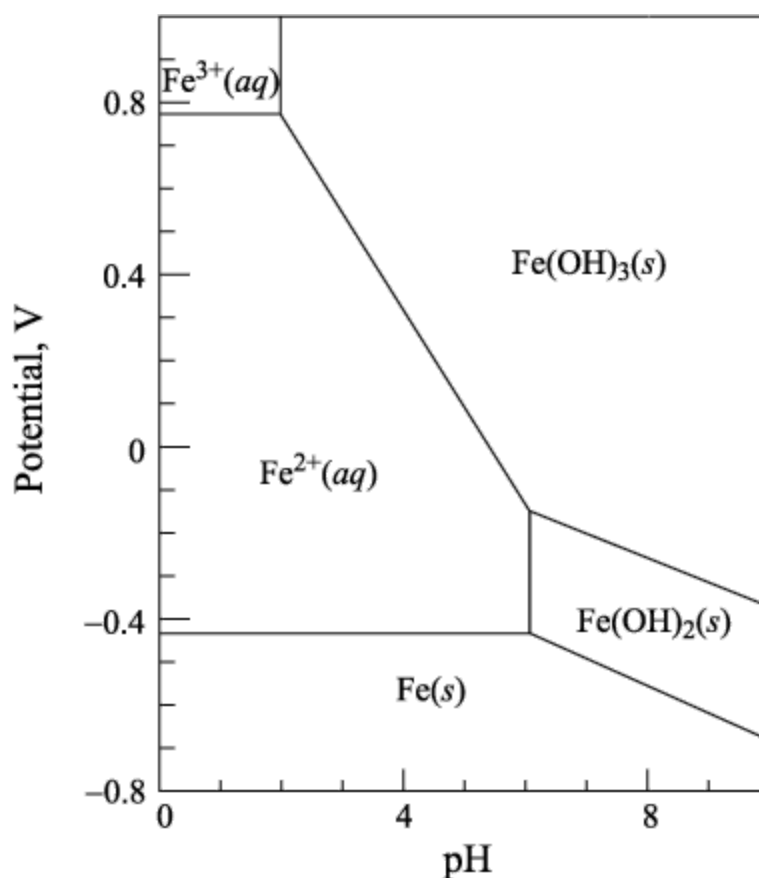
CC4E

FC4

GCH

RESULT / TIMING

The graph below shows the relationship between the predominant form of iron as a function of solution pH and applied potential. Based on the graph, which of the following statements is true?



- A.
At a potential of -0.4 V, as pH increases, Fe^{2+} is reduced and precipitates as Fe(OH)_3 .
- B.

At a potential of -0.44 V , the equilibrium between Fe and Fe^{2+} is independent of solution pH below pH 6.

- C.

At pH = 1, as the potential is changed from -0.2 to $+0.8$, Fe^{3+} is reduced to Fe^{2+} .

- D.

At pH = 8 and $V = -0.1\text{ V}$, $\text{Fe}(\text{OH})_2$ is the predominant form of iron.

Solution: The correct answer is **B**.

1. The conversion of Fe^{2+} to $\text{Fe}(\text{OH})_3(\text{s})$ represents an oxidation. From the graph, at a potential of -0.4 V , no oxidation or reduction occurs. $\text{Fe}(\text{OH})_2(\text{s})$ precipitates.
2. **The Pourbaix diagram for iron is given, but only one statement correctly conveys information presented by the diagram. At an applied potential of -0.44V and pH below 6, there is an equilibrium between $\text{Fe}(\text{s})$ and $\text{Fe}^{2+}(\text{aq})$, as shown by the line of demarcation. The fact that this line moves horizontally with increasing pH up until pH = 6 means that this equilibrium is unaffected by changing the pH.**
3. The opposite is true. At this pH as the potential is raised, Fe^{2+} is oxidized to Fe^{3+} .
4. At pH = 8 and $V = -0.1\text{ V}$, $\text{Fe}(\text{OH})_3(\text{s})$ is the predominant form of iron.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#); [Electronic Structure](#); [The Periodic Table - Classification of Elements Into Groups by Electronic Structure](#); [The Periodic Table - Variations of Chemical Properties with Group and Row](#); and [Stoichiometry](#).

Question 28 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Which of the following types of orbitals of the central atom are involved in bonding in octahedral compounds?

- A.
 sp
- B.
 sp^3
- C.
 p
- D.
 $d^2 sp^3$

Solution: The correct answer is **D**.

1. sp hybrid orbitals can be used to make linear AX_2E_n compounds.
2. Tetrahedral AX_4 compounds and other compounds based on a tetrahedral arrangement of electron pairs utilize sp^3 hybrid orbitals.
3. A pure p type orbital will typically be involved in pi bonding, or will represent a lone pair of electrons.
4. **Octahedral compounds have six σ bonds and no stereochemically active lone pairs. According to valence bond theory, the central atom requires the hybridization of six atomic orbitals, $d^2 sp^3$.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Covalent bonds](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

25 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC5B

FC5

GCH

RESULT / TIMING

Which of the following types of orbitals of the central atom are involved in bonding in octahedral compounds?

- A.
 sp
- B.
 sp^3
- C.
 p
- D.
 $d^2 sp^3$

Solution: The correct answer is **D**.

1. sp hybrid orbitals can be used to make linear AX_2E_n compounds.

2. Tetrahedral AX₄ compounds and other compounds based on a tetrahedral arrangement of electron pairs utilize sp^3 hybrid orbitals.
 3. A pure p type orbital will typically be involved in pi bonding, or will represent a lone pair of electrons.
 4. **Octahedral compounds have six σ bonds and no stereochemically active lone pairs. According to valence bond theory, the central atom requires the hybridization of six atomic orbitals, d^2sp^3 .**
-

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Covalent bonds](#).

Question 29 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

The term “ideal gas” refers to a gas for which certain assumptions have been made. Which of the following is such an assumption?

- A.
The law $PV = nRT^2$ is strictly obeyed.
- B.
Intermolecular molecular forces are infinitely large.
- C.
Individual molecular volume and intermolecular forces are negligible.
- D.
One gram-mole occupies a volume of 22.4 L at 25°C and one atmosphere pressure.

Solution: The correct answer is **C**.

1. The Ideal Gas Law is $PV = nRT$.
2. Large intermolecular forces are associated with condensed phases (solids and liquids), not gases.
3. **A property of an "ideal" gas is that it is composed of particles that have negligible volume and do not exert intermolecular forces.**
4. The gram-mole is an old way to refer to the molar mass of a substance. The molar volume of an ideal gas at 25°C is 24.4 L, not 22.4 L.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Gas Phase](#).

Correct

Your result is correct.

Your answer is C.

Correct answer is C.

Correct Answer Your Answer

A

B

C

✓

D

Time Spent:

34 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC4B

FC4

GCH

RESULT / TIMING

The term "ideal gas" refers to a gas for which certain assumptions have been made. Which of the following is such an assumption?

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1. The Ideal Gas Law is $PV = nRT$.
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KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Gas Phase](#).

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note
Cancel Add

Passage 7 (Questions 34-37)

The crystal structure of the protein lysosomal lipase (LipA) was determined. Crystal structure B factors, which are an indication of structural rigidity, were used to identify regions of high flexibility. Sites within these regions were varied in an attempt to create versions of LipA with higher thermal stability. All variants were then screened for retention of enzymatic activity. The three most thermally stable variants that retained enzymatic activity are shown in Table 1.

Table 1	Variants of LipA								
	<table><tr><th>Variant</th><th>Amino acid substitutions</th></tr><tr><td>IX</td><td>K112D, M134D, Y139C, I157M</td></tr><tr><td>X</td><td>R33Q, D34N, K35D, K112D, M134D, Y139C, I157M</td></tr><tr><td>XI</td><td>R33G, K112D, M134D, Y139C, I157M</td></tr></table>	Variant	Amino acid substitutions	IX	K112D, M134D, Y139C, I157M	X	R33Q, D34N, K35D, K112D, M134D, Y139C, I157M	XI	R33G, K112D, M134D, Y139C, I157M
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All proteins were expressed in *Escherichia coli* and purified chromatographically. After protein purity was verified via SDS-PAGE, wild-type and variant LipA proteins were characterized. The enzymatic activity of the proteins was measured at 37°C and then again after the protein was briefly exposed to higher temperatures and cooled down to 37°C (Figure 1). The thermal unfolding of the proteins was followed by loss of secondary structure (Table 2). Structural integrity was verified using NMR spectroscopy.

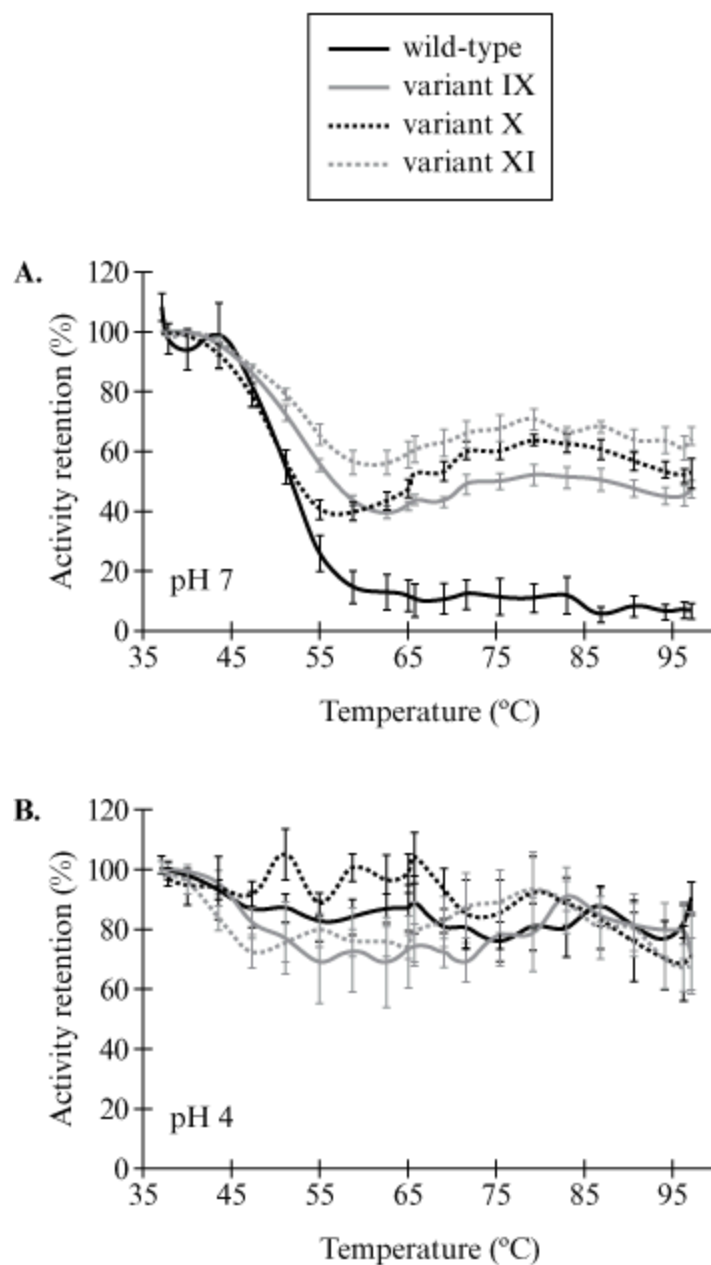


Figure 1 Enzymatic activity retention as a function of exposure temperature for wild-type and variant LipA proteins (Note: The solution contained either a pH 7 buffer (A) or a pH 4 buffer (B). The x-axis represents the temperature that the protein was heated to prior to the activity assay.)

Table 2 Parameters for Thermal Unfolding as Monitored by Loss of Secondary Structure

Parameter		Reference	2 M Urea	4 M Urea
T_i , (°C)	Wild-type	51.8	39.6	26.3
	Variant IX	45.6	32.0	24.9

Parameter		Reference	2 M Urea	4 M Urea
	Variant X	48.5	33.1	24.0
	Variant XI	49.3	34.3	25.4
Slope, deg cm ² /(mmol K)	Wild-type	8.2	3.9	3.5
	Variant IX	5.7	3.3	2.6
	Variant X	4.6	4.6	2.1
	Variant XI	6.0	3.9	3.0

Note: T_i is the temperature at the start of unfolding and the slope is measured for the unfolding transition. The experiments were performed in a pH 7 buffer (reference) and in the presence of 2 M and 4 M urea, which is a chemical denaturant.

Adapted from W. Augustyniak, A. Brzezinska, R. Pijning, H. Wienk, R. Boelens, et al., *Protein Science*. ©2012 The Protein Society.

Which reaction is catalyzed by LipA?

- A.
ATP hydrolysis
- B.
Peptide bond cleavage
- C.
Hydrolysis of triacylglycerides
- D.
Transfer of carboxyl groups

Solution: The correct answer is **C**.

1. LipA is a lipase, which does not hydrolyze ATP.
2. LipA is a lipase, not a protease.
3. **LipA is a lipase, which hydrolyzes triacylglycerides.**
4. LipA is a lipase, not a transferase.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Enzymes](#); [Kinetics](#); and [Bioenergetics](#).

Correct

Your result is correct.

Your answer is C.

Correct answer is C.

Correct Answer Your Answer

A

B

C

✓

D

Time Spent:

1 min 17 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC5E

FC5

BCH

RESULT / TIMING

Passage 7 (Questions 34-37)

The crystal structure of the protein lysosomal lipase (LipA) was determined. Crystal structure B factors, which are an indication of structural rigidity, were used to identify regions of high flexibility. Sites within these regions were varied in an attempt to create versions of LipA with higher thermal stability. All variants were then screened for retention of enzymatic activity. The three most thermally stable variants that retained enzymatic activity are shown in Table 1.

Table 1	Variants of LipA	
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	XI	R33G, K112D, M134D, Y139C, I157M

All proteins were expressed in *Escherichia coli* and purified chromatographically. After protein purity was verified via SDS-PAGE, wild-type and variant LipA proteins were characterized. The enzymatic activity of the proteins was measured at 37°C and then again after the protein was briefly exposed to higher temperatures and cooled down to 37°C (Figure 1). The thermal unfolding of the proteins was followed by loss of secondary structure (Table 2). Structural integrity was verified using NMR spectroscopy.

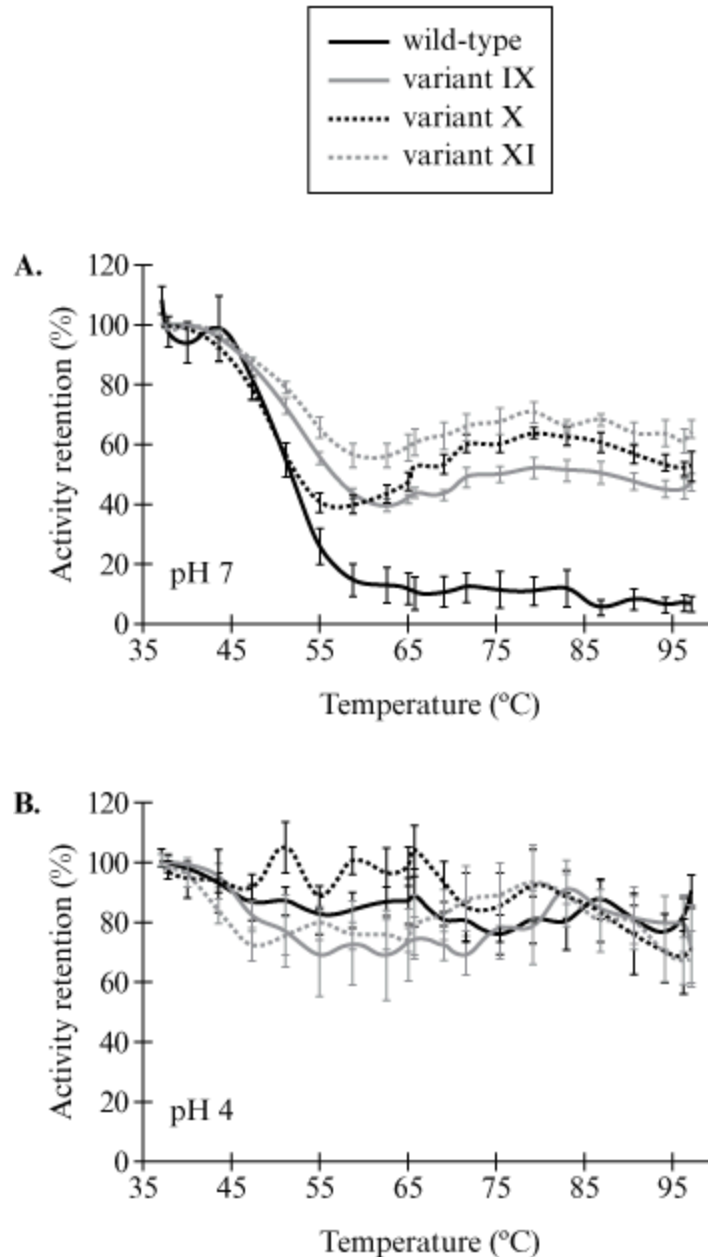


Figure 1

Enzymatic activity retention as a function of exposure temperature for wild-type and variant LipA proteins (Note: The solution contained either a pH 7 buffer (A) or a pH 4 buffer (B). The x-axis represents the temperature that the protein was heated to prior to the activity assay.)

Table 2 Parameters for Thermal Unfolding as Monitored by Loss of Secondary Structure

Parameter		Reference	2 M Urea	4 M Urea
T_i , (°C)	Wild-type	51.8	39.6	26.3
	Variant IX	45.6	32.0	24.9
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Note: T_i is the temperature at the start of unfolding and the slope is measured for the unfolding transition. The experiments were performed in a pH 7 buffer (reference) and in the presence of 2 M and 4 M urea, which is a chemical denaturant.

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Which reaction is catalyzed by LipA?

- A.
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Peptide bond cleavage
- C.
Hydrolysis of triacylglycerides
- D.
Transfer of carboxyl groups

Solution: The correct answer is **C**.

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2. LipA is a lipase, not a protease.
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4. LipA is a lipase, not a transferase.

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Enzymes](#); [Kinetics](#); and [Bioenergetics](#).

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note
Cancel Add

Passage 7 (Questions 34-37)

The crystal structure of the protein lysosomal lipase (LipA) was determined. Crystal structure B factors, which are an indication of structural rigidity, were used to identify regions of high flexibility. Sites within these regions were varied in an attempt to create versions of LipA with higher thermal stability. All variants were then screened for retention of enzymatic activity. The three most thermally stable variants that retained enzymatic activity are shown in Table 1.

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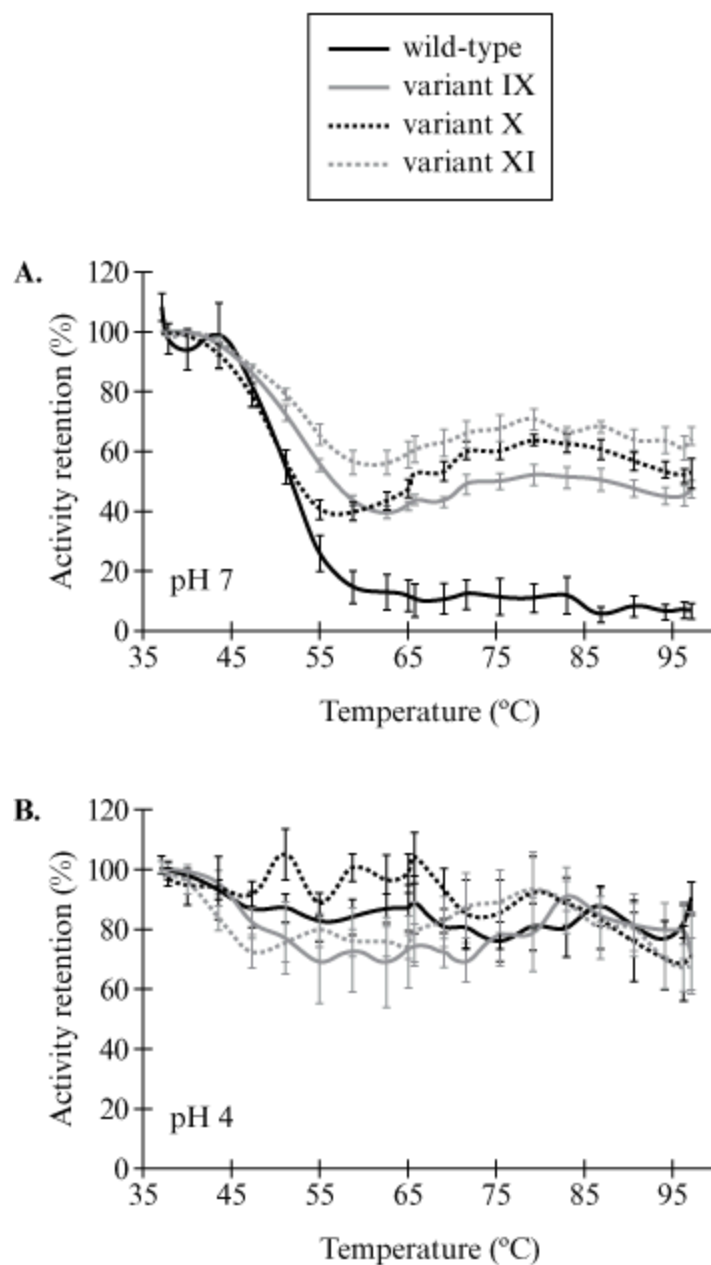


Figure 1

Enzymatic activity retention as a function of exposure temperature for wild-type and variant LipA proteins (Note: The solution contained either a pH 7 buffer (A) or a pH 4 buffer (B). The x-axis represents the temperature that the protein was heated to prior to the activity assay.)

Table 2

Parameters for Thermal Unfolding as Monitored by Loss of Secondary Structure

Parameter		Reference	2 M Urea	4 M Urea
T_i , (°C)	Wild-type	51.8	39.6	26.3
	Variant IX	45.6	32.0	24.9

Parameter		Reference	2 M Urea	4 M Urea
	Variant X	48.5	33.1	24.0
	Variant XI	49.3	34.3	25.4
Slope, deg cm ² /(mmol K)	Wild-type	8.2	3.9	3.5
	Variant IX	5.7	3.3	2.6
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	Variant XI	6.0	3.9	3.0

Note: T_i is the temperature at the start of unfolding and the slope is measured for the unfolding transition. The experiments were performed in a pH 7 buffer (reference) and in the presence of 2 M and 4 M urea, which is a chemical denaturant.

Adapted from W. Augustyniak, A. Brzezinska, R. Pijning, H. Wienk, R. Boelens, et al., *Protein Science*. ©2012 The Protein Society.

Compared to the wild-type LipA, what is the change in net charge in variant XI at pH 7?

- A.
+4
- B.
+3
- C.
-3
- D.
-4

Solution: The correct answer is **D**.

1. This change in net charge has the wrong sign but the correct magnitude.
2. Variant XI loses positive charge and gains negative charge.
3. This is an insufficient change in net negative charge.
4. **Variant XI has a change of -1 charge with R33G (+1 to 0), -2 charge with K112D (+1 to -1), and -1 charge with M134D (0 to -1) for a net change of -4.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#), [Amino Acids](#), [Peptides](#), [Proteins](#), [The Three-Dimensional Protein Structure](#), [Nonenzymatic Protein Function](#); and [Phenols](#), [Polycyclic and Heterocyclic Aromatic Compounds](#).

Incorrect

Your result is incorrect.

Your answer is C.

Correct answer is D.

Correct Answer Your Answer

A

B

C

x

D

Time Spent:

2 mins 59 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5D

FC5

BCH

RESULT / TIMING

Passage 7 (Questions 34-37)

The crystal structure of the protein lysosomal lipase (LipA) was determined. Crystal structure B factors, which are an indication of structural rigidity, were used to identify regions of high flexibility. Sites within these regions were varied in an attempt to create versions of LipA with higher thermal stability. All variants were then screened for retention of enzymatic activity. The three most thermally stable variants that retained enzymatic activity are shown in Table 1.

Table 1	Variants of LipA
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Variant	Amino acid substitutions
IX	K112D, M134D, Y139C, I157M
X	R33Q, D34N, K35D, K112D, M134D, Y139C, I157M

Variant	Amino acid substitutions
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All proteins were expressed in *Escherichia coli* and purified chromatographically. After protein purity was verified via SDS-PAGE, wild-type and variant LipA proteins were characterized. The enzymatic activity of the proteins was measured at 37°C and then again after the protein was briefly exposed to higher temperatures and cooled down to 37°C (Figure 1). The thermal unfolding of the proteins was followed by loss of secondary structure (Table 2). Structural integrity was verified using NMR spectroscopy.

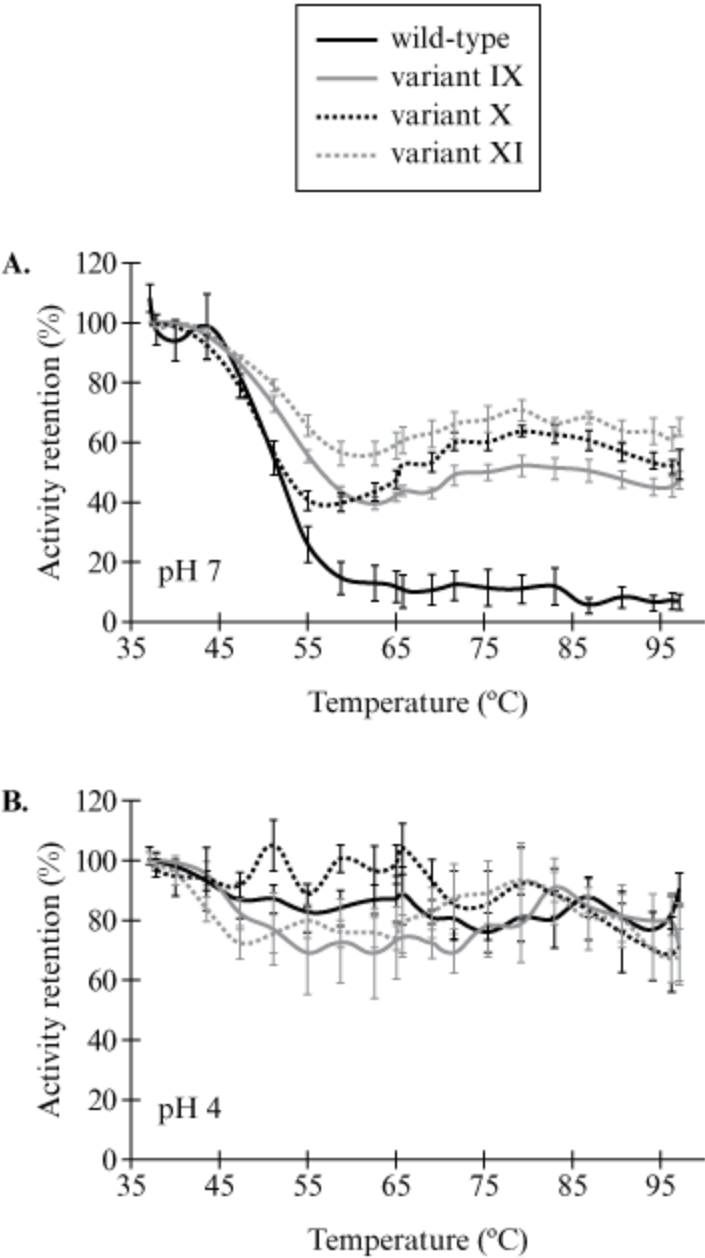


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a pH 7 buffer (A) or a pH 4 buffer (B). The x-axis represents the temperature that the protein was heated to prior to the activity assay.)

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	Variant IX	45.6	32.0	24.9
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	Variant IX	5.7	3.3	2.6
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	Variant XI	6.0	3.9	3.0

Note: T_i is the temperature at the start of unfolding and the slope is measured for the unfolding transition. The experiments were performed in a pH 7 buffer (reference) and in the presence of 2 M and 4 M urea, which is a chemical denaturant.

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- A.
+4
- B.
+3
- C.
-3
- D.
-4

Solution: The correct answer is **D**.

1. This change in net charge has the wrong sign but the correct magnitude.
2. Variant XI loses positive charge and gains negative charge.

3. This is an insufficient change in net negative charge.
4. **Variant XI has a change of -1 charge with R33G ($+1$ to 0), -2 charge with K112D ($+1$ to -1), and -1 charge with M134D (0 to -1) for a net change of -4 .**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#); [Amino Acids](#), [Peptides](#), [Proteins](#); [The Three-Dimensional Protein Structure](#), [Nonenzymatic Protein Function](#); and [Phenols](#), [Polycyclic and Heterocyclic Aromatic Compounds](#).

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note
Cancel Add

Passage 7 (Questions 34-37)

The crystal structure of the protein lysosomal lipase (LipA) was determined. Crystal structure B factors, which are an indication of structural rigidity, were used to identify regions of high flexibility. Sites within these regions were varied in an attempt to create versions of LipA with higher thermal stability. All variants were then screened for retention of enzymatic activity. The three most thermally stable variants that retained enzymatic activity are shown in Table 1.

Table 1	Variants of LipA								
	<table><tr><th>Variant</th><th>Amino acid substitutions</th></tr><tr><td>IX</td><td>K112D, M134D, Y139C, I157M</td></tr><tr><td>X</td><td>R33Q, D34N, K35D, K112D, M134D, Y139C, I157M</td></tr><tr><td>XI</td><td>R33G, K112D, M134D, Y139C, I157M</td></tr></table>	Variant	Amino acid substitutions	IX	K112D, M134D, Y139C, I157M	X	R33Q, D34N, K35D, K112D, M134D, Y139C, I157M	XI	R33G, K112D, M134D, Y139C, I157M
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All proteins were expressed in *Escherichia coli* and purified chromatographically. After protein purity was verified via SDS-PAGE, wild-type and variant LipA proteins were characterized. The enzymatic activity of the proteins was measured at 37°C and then again after the protein was briefly exposed to higher temperatures and cooled down to 37°C (Figure 1). The thermal unfolding of the proteins was followed by loss of secondary structure (Table 2). Structural integrity was verified using NMR spectroscopy.

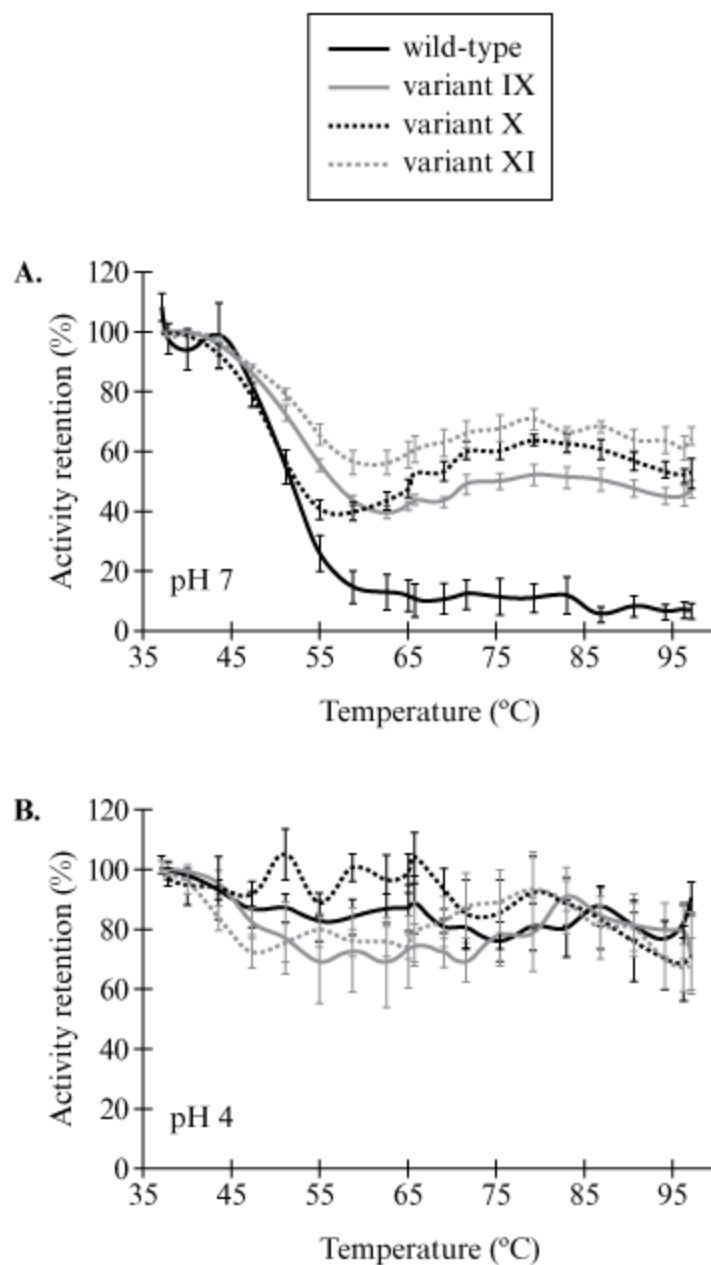


Figure 1

Enzymatic activity retention as a function of exposure temperature for wild-type and variant LipA proteins (Note: The solution contained either a pH 7 buffer (A) or a pH 4 buffer (B). The x-axis represents the temperature that the protein was heated to prior to the activity assay.)

Table 2

Parameters for Thermal Unfolding as Monitored by Loss of Secondary Structure

Parameter		Reference	2 M Urea	4 M Urea
T_i , (°C)	Wild-type	51.8	39.6	26.3
	Variant IX	45.6	32.0	24.9

Parameter		Reference	2 M Urea	4 M Urea
	Variant X	48.5	33.1	24.0
	Variant XI	49.3	34.3	25.4
Slope, deg cm ² /(mmol K)	Wild-type	8.2	3.9	3.5
	Variant IX	5.7	3.3	2.6
	Variant X	4.6	4.6	2.1
	Variant XI	6.0	3.9	3.0

Note: T_i is the temperature at the start of unfolding and the slope is measured for the unfolding transition. The experiments were performed in a pH 7 buffer (reference) and in the presence of 2 M and 4 M urea, which is a chemical denaturant.

Adapted from W. Augustyniak, A. Brzezinska, R. Pijning, H. Wienk, R. Boelens, et al., *Protein Science*. ©2012 The Protein Society.

The data in Figure 1 show that:

- A.
the unfolding of LipA proteins is a non-reversible process.
- B.
exposure to high temperature eliminates any variant protein activity.
- C.
variation in the flexible regions of LipA decreases the temperature sensitivity of activity.
- D.
wild-type LipA loses all activity when first heated to 50°C.

Solution: The correct answer is **C**.

1. At pH 4, the enzymatic activity of LipA proteins does not change even after heating and cooling.
2. In Figure 1, the highest temperatures used still show enzymatic activity for variants.
3. **Figure 1 shows that temperature has less effect on the enzymatic activity of the variants than wild-type LipA.**
4. At pH 7, wild-type LipA loses some but not all activity when first heated to 50°C.

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#); [Amino Acids](#), [Peptides](#), [Proteins](#); [The Three-Dimensional Protein Structure](#), [Nonenzymatic Protein Function](#); and [Phenols](#), [Polycyclic and Heterocyclic Aromatic Compounds](#).

Correct

Your result is correct.

Your answer is C.

Correct answer is C.

Correct Answer Your Answer

A

B

C

✓

D

Time Spent:

1 min 37 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS4

CC5D

FC5

BCH

RESULT / TIMING

Passage 7 (Questions 34-37)

The crystal structure of the protein lysosomal lipase (LipA) was determined. Crystal structure B factors, which are an indication of structural rigidity, were used to identify regions of high flexibility. Sites within these regions were varied in an attempt to create versions of LipA with higher thermal stability. All variants were then screened for retention of enzymatic activity. The three most thermally stable variants that retained enzymatic activity are shown in Table 1.

Table 1	Variants of LipA
----------------	------------------

Variant	Amino acid substitutions
IX	K112D, M134D, Y139C, I157M
X	R33Q, D34N, K35D, K112D, M134D, Y139C, I157M
XI	R33G, K112D, M134D, Y139C, I157M

All proteins were expressed in *Escherichia coli* and purified chromatographically. After protein purity was verified via SDS-PAGE, wild-type and variant LipA proteins were characterized. The enzymatic activity of the proteins was measured at 37°C and then again after the protein was briefly exposed to higher temperatures and cooled down to 37°C (Figure 1). The thermal unfolding of the proteins was followed by loss of secondary structure (Table 2). Structural integrity was verified using NMR spectroscopy.

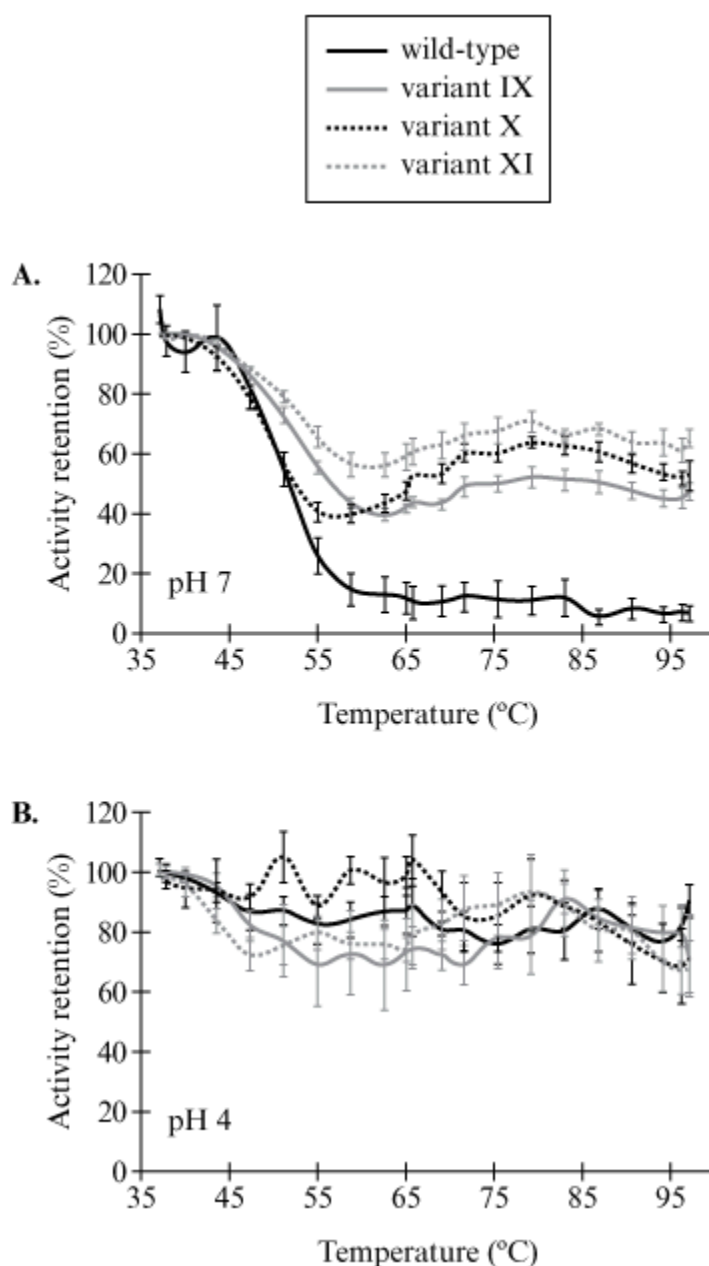


Figure 1 Enzymatic activity retention as a function of exposure temperature for wild-type and variant LipA proteins (Note: The solution contained either a pH 7 buffer (A) or a pH 4 buffer (B). The x-axis represents the temperature that the protein was heated to prior to the activity assay.)

Table 2 Parameters for Thermal Unfolding as Monitored by Loss of Secondary Structure

Parameter		Reference	2 M Urea	4 M Urea
T_i , (°C)	Wild-type	51.8	39.6	26.3
	Variant IX	45.6	32.0	24.9
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- A.
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- B.
exposure to high temperature eliminates any variant protein activity.
- C.
variation in the flexible regions of LipA decreases the temperature sensitivity of activity.
- D.
wild-type LipA loses all activity when first heated to 50°C.

Solution: The correct answer is **C**.

1. At pH 4, the enzymatic activity of LipA proteins does not change even after heating and cooling.
2. In Figure 1, the highest temperatures used still show enzymatic activity for variants.
3. **Figure 1 shows that temperature has less effect on the enzymatic activity of the variants than wild-type LipA.**
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KHAN ACADEMY LESSONS

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QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note
Cancel Add

Passage 7 (Questions 34-37)

The crystal structure of the protein lysosomal lipase (LipA) was determined. Crystal structure B factors, which are an indication of structural rigidity, were used to identify regions of high flexibility. Sites within these regions were varied in an attempt to create versions of LipA with higher thermal stability. All variants were then screened for retention of enzymatic activity. The three most thermally stable variants that retained enzymatic activity are shown in Table 1.

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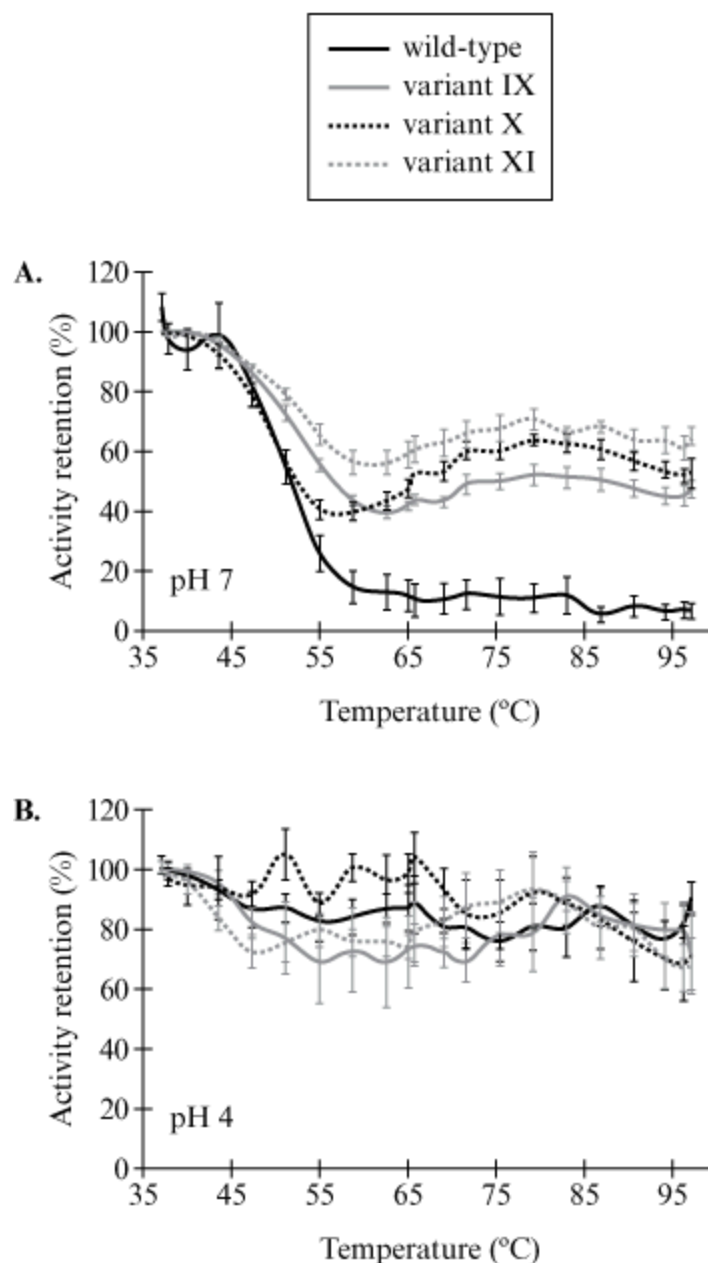


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Parameter		Reference	2 M Urea	4 M Urea
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	Variant IX	45.6	32.0	24.9

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Adapted from W. Augustyniak, A. Brzezinska, R. Pijning, H. Wienk, R. Boelens, et al., *Protein Science*. ©2012 The Protein Society.

By what factor is the proton concentration increased in the experiments shown in Figure 1B from those shown in Figure 1A?

- A.
2
- B.
10
- C.
200
- D.
1000

Solution: The correct answer is **D**.

1. Doubling the concentration of H_3O^+ involves decreasing the pH by only 0.3 units.
2. If the solution pH was lower by 1 pH unit, this would correspond to an increase in $[\text{H}_3\text{O}^+]$ by a factor of 10.
3. An increase of $[\text{H}_3\text{O}^+]$ by a factor of 200 corresponds to a decrease in 2.3 pH units.
4. **The pH scale is logarithmic ($\text{pH} = -\log([\text{H}^+])$). A difference of 3 pH units corresponds to a $10^3 = 1000$ -fold difference in proton concentrations.**

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Acid-Base Equilibria](#); [Solubility](#); and [Titration](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

47 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5A

FC5

GCH

RESULT / TIMING

Passage 7 (Questions 34-37)

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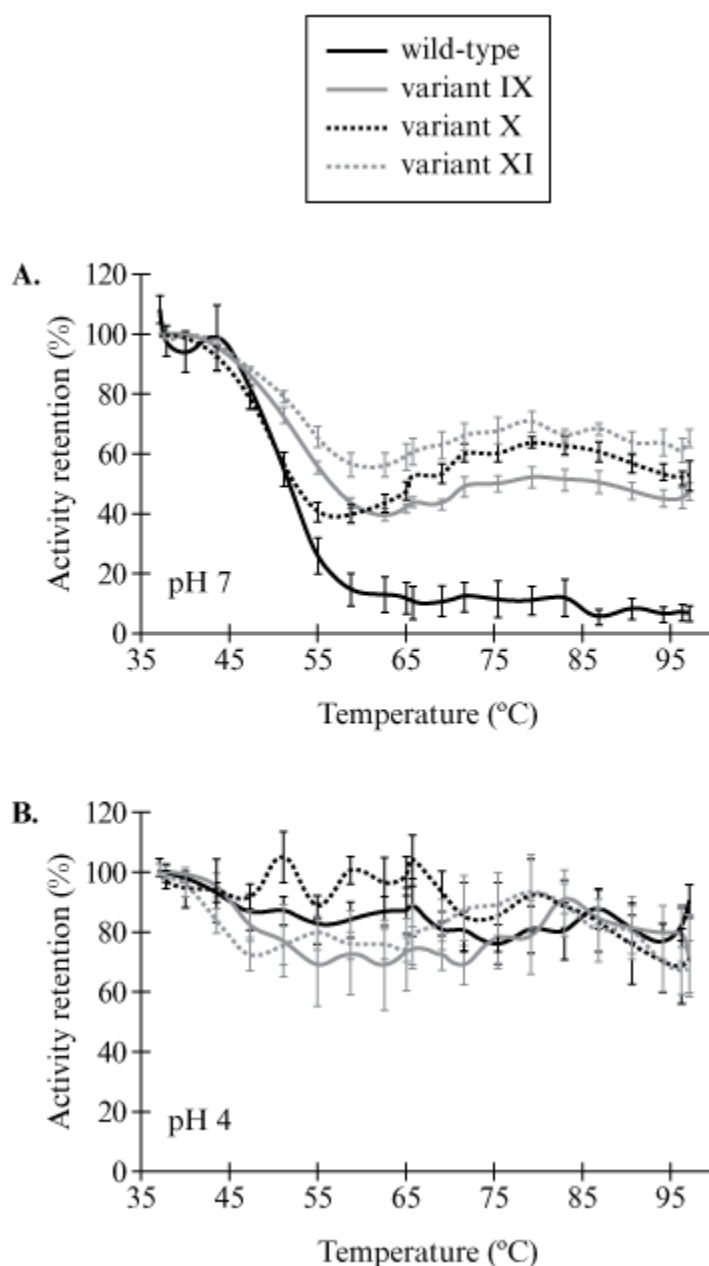


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Adapted from W. Augustyniak, A. Brzezinska, R. Pijning, H. Wienk, R. Boelens, et al., *Protein Science*. ©2012 The Protein Society.

By what factor is the proton concentration increased in the experiments shown in Figure 1B from those shown in Figure 1A?

- A.
2
- B.
10
- C.
200
- D.
1000

Solution: The correct answer is **D**.

1. Doubling the concentration of H_3O^+ involves decreasing the pH by only 0.3 units.
2. If the solution pH was lower by 1 pH unit, this would correspond to an increase in $[\text{H}_3\text{O}^+]$ by a factor of 10.
3. An increase of $[\text{H}_3\text{O}^+]$ by a factor of 200 corresponds to a decrease in 2.3 pH units.
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KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Acid-Base Equilibria](#); [Solubility](#); and [Titration](#).

Question 44 of 59

QUESTION

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MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Each of the following equations shows the dissociation of an acid in water. Which of the reactions occurs to the LEAST extent?

- A.
$$\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$$
- B.
$$\text{HPO}_4^{2-} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{PO}_4^{3-}$$
- C.
$$\text{H}_2\text{SO}_4 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{HSO}_4^-$$
- D.
$$\text{H}_3\text{PO}_4 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{H}_2\text{PO}_4^-$$

Solution: The correct answer is **B**.

1. Hydrochloric acid is a strong acid. This reaction will occur to completion.
2. **HPO_4^{2-} has a high negative charge and so dissociation of it will occur to the least extent.**
3. Sulfuric acid is a strong acid. This means that the dissociation shown will go to completion as water is a stronger base than the hydrogen sulfate anion.
4. Phosphoric acid is a much stronger Brønsted acid than the hydrogen phosphate anion. This is readily seen due to the increased charge on the anion which will make it much harder to lose another proton.

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Acid-Base Equilibria](#); [Solubility](#); and [Titration](#).

Correct

Your result is correct.

Your answer is B.

Correct answer is B.

Correct Answer Your Answer

A

B

✓

C

D

Time Spent:

1 min 2 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5A

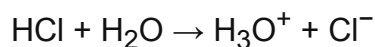
FC5

GCH

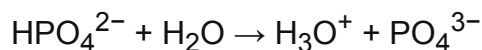
RESULT / TIMING

Each of the following equations shows the dissociation of an acid in water. Which of the reactions occurs to the LEAST extent?

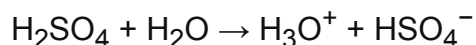
- A.



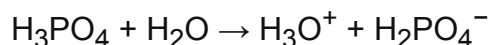
- B.



- C.



- D.



Solution: The correct answer is **B**.

1. Hydrochloric acid is a strong acid. This reaction will occur to completion.
2. HPO_4^{2-} has a high negative charge and so dissociation of it will occur to the least extent.
3. Sulfuric acid is a strong acid. This means that the dissociation shown will go to completion as water is a stronger base than the hydrogen sulfate anion.
4. Phosphoric acid is a much stronger Brønsted acid than the hydrogen phosphate anion. This is readily seen due to the increased charge on the anion which will make it much harder to lose another proton.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Acid-Base Equilibria](#); [Solubility](#); and [Titration](#).

Question 45 of 59

QUESTION

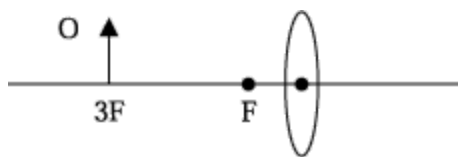
Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add



In the above figure, an object O is at a distance of three focal lengths from the center of a convex lens. What is the ratio of the height of the image to the height of the object?

- A.
1/3
- B.
1/2
- C.
2/3
- D.
3/2

Solution: The correct answer is **B**.

1. The ratio of the image height to the object height is equal to the ratio of the lens-image distance to the object-lens distance. According to the thin lens equation, the distance between the lens and the image is $\frac{3F \times 1F}{(3F - 1F)} = \left(\frac{3}{2}\right)F$. The ratio sought is then equal to $\frac{(3/2)F}{(3F)} = \frac{1}{2}$, and not $\frac{1}{3}$.
2. **This is correct because the ratio of the image height to the object height is equal to the ratio of the lens-image distance to the object-lens distance. According to the thin lens equation, the distance between the lens and the image is $\frac{3F \times 1F}{(3F - 1F)} = \left(\frac{3}{2}\right)F$. The ratio sought is then equal to $\frac{(3/2)F}{(3F)} = \frac{1}{2}$.**

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KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Sound](#); [Light, Electromagnetic Radiation](#); [Geometrical Optics](#); [Spherical mirrors](#); and [Reflection and refraction](#).

Correct

Your result is correct.

Your answer is B.

Correct answer is B.

Correct Answer Your Answer

A

B

✓

C

D

Time Spent:

1 min 16 secs

Difficulty Level:

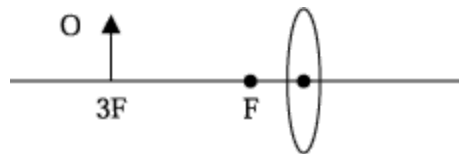
Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC4D

FC4

PHY



In the above figure, an object O is at a distance of three focal lengths from the center of a convex lens. What is the ratio of the height of the image to the height of the object?

- A.
1/3
- B.
1/2
- C.
2/3
- D.
3/2

Solution: The correct answer is **B**.

1. The ratio of the image height to the object height is equal to the ratio of the lens-image distance to the object-lens distance. According to the thin lens equation, the distance between the lens and the image is $\frac{3F \times 1F}{(3F - 1F)} = \left(\frac{3}{2}\right)F$. The ratio sought is then equal to $\frac{(3/2)F}{(3F)} = \frac{1}{2}$, and not $\frac{1}{3}$.
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For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Sound](#); [Light, Electromagnetic Radiation](#); [Geometrical Optics](#); [Spherical mirrors](#); and [Reflection and refraction](#).

Question 46 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Gamma decay occurs when a nucleus emits:

- A.
a photon.
- B.
a positron.
- C.
a neutron.
- D.
an electron.

Solution: The correct answer is **A**.

1. **By definition, gamma decay means the emission of photons by the nucleus.**
2. The emission of positrons by the nucleus is called beta-plus decay.
3. The emission of neutrons by the nucleus occurs only during stimulated nuclear fission.
4. The emission of an electron by the nucleus is called beta-minus decay.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#) and [Electronic Structure](#).

Correct

Your result is correct.

Your answer is A.

Correct answer is A.

Correct Answer Your Answer

A

✓

B

C

D

Time Spent:

15 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC4E

FC4

PHY

RESULT / TIMING

Gamma decay occurs when a nucleus emits:

- A.
a photon.
- B.
a positron.
- C.
a neutron.
- D.
an electron.

Solution: The correct answer is **A**.

1. **By definition, gamma decay means the emission of photons by the nucleus.**
2. The emission of positrons by the nucleus is called beta-plus decay.
3. The emission of neutrons by the nucleus occurs only during stimulated nuclear fission.
4. The emission of an electron by the nucleus is called beta-minus decay.

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#) and [Electronic Structure](#).

Question 47 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

When a 0.1 M H_2SO_4 solution is added to pulverized blackboard chalk, the following reaction takes place.



At 25°C, the reaction is spontaneous and has:

- A.
negative ΔG° and positive ΔS° .
- B.
negative ΔG° and negative ΔS° .
- C.
positive ΔG° and negative ΔS° .
- D.
positive ΔG° and positive ΔS° .

Solution: The correct answer is **A**.

1. **Spontaneous reactions must have a negative ΔG° , and reactions that create gaseous products from solids and liquids must have a positive ΔS° .**
2. The reaction does not have negative ΔS because it produces a gas.
3. The reaction must have negative ΔG because it is spontaneous. It is also apparent that the reaction has positive ΔS , because the reaction produces a gas.
4. Spontaneous reactions have negative ΔG .

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Thermodynamics](#); [Thermochemistry](#); and [Equilibrium](#).

Correct

Your result is correct.

Your answer is A.

Correct answer is A.

Correct Answer Your Answer

A

✓

B

C

D

Time Spent:

48 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC5E

FC5

GCH

RESULT / TIMING

When a 0.1 M H_2SO_4 solution is added to pulverized blackboard chalk, the following reaction takes place.



At 25°C, the reaction is spontaneous and has:

- A.
negative ΔG° and positive ΔS° .
- B.
negative ΔG° and negative ΔS° .
- C.

positive ΔG° and negative ΔS° .

- D.

positive ΔG° and positive ΔS° .

Solution: The correct answer is **A**.

1. **Spontaneous reactions must have a negative ΔG° , and reactions that create gaseous products from solids and liquids must have a positive ΔS° .**
2. The reaction does not have negative ΔS because it produces a gas.
3. The reaction must have negative ΔG because it is spontaneous. It is also apparent that the reaction has positive ΔS , because the reaction produces a gas.
4. Spontaneous reactions have negative ΔG .

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Thermodynamics](#); [Thermochemistry](#); and [Equilibrium](#).

Question 52 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, $X(s)$, into solutions containing different metallic ions, $Y^{m+}(aq)$, as shown in Figure 1.

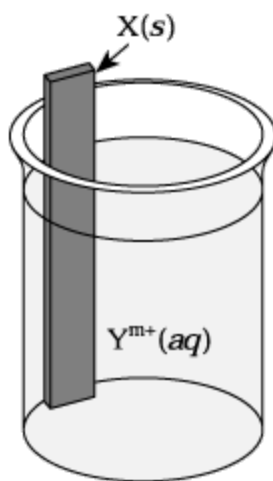


Figure 1

Metal strip in solution

After several hours, they observed whether or not $Y(s)$ formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
$\text{Al}^{3+}(\text{aq})$	*	no change	no change	no change	no change	no change
$\text{Cu}^{2+}(\text{aq})$	new solid	*	new solid	no change	no change	new solid
$\text{Zn}^{2+}(\text{aq})$	new solid	no change	*	no change	no change	no change
$\text{Ag}^{+}(\text{aq})$	new solid	new solid	new solid	*	no change	new solid
$\text{Au}^{3+}(\text{aq})$	*	*	*	new solid	*	*
$\text{Pb}^{2+}(\text{aq})$	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

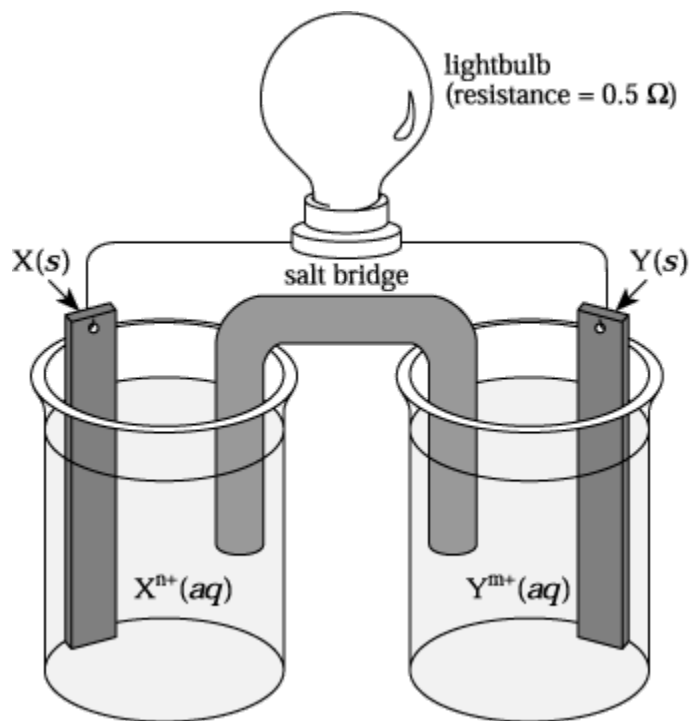
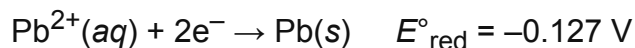
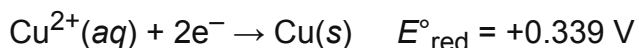


Figure 2

Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.

**Reaction 1****Reaction 2**

A student can most effectively increase the current passing through the circuit in Figure 2 by doing which of the following?

- A.
Using electrical wire with a smaller diameter
- B.
Increasing the temperature of the electrical wire
- C.
Decreasing the concentrations of $\text{X}^{\text{n}+}(\text{aq})$ and $\text{Y}^{\text{m}+}(\text{aq})$
- D.
Replacing the lightbulb with one that has a resistance of $0.2 \, \Omega$

Solution: The correct answer is **D**.

1. Using an electrical wire with a smaller diameter will increase the resistance of the wire. According to Ohm's law, this will decrease the current in the circuit.
2. Increasing the temperature of the electrical wire will increase the resistance of the wire. According to Ohm's law, this will decrease the current in the circuit.
3. Decreasing the concentrations will decrease the amount of charges that flow through the circuit that effectively decreases the current.
4. **This option is correct because according to Ohm's law, current is equal to voltage divided by resistance. In order to increase the current when the voltage is fixed, the resistance must be decreased, such as by replacing the lightbulb with one that has a resistance of $0.2 \, \Omega$ instead of $0.5 \, \Omega$.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Electrostatics](#); [Circuit Elements](#); and [Magnetism](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

43 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS3

CC4C

FC4

PHY

RESULT / TIMING

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, X(s), into solutions containing different metallic ions, $Y^{m+}(aq)$, as shown in Figure 1.

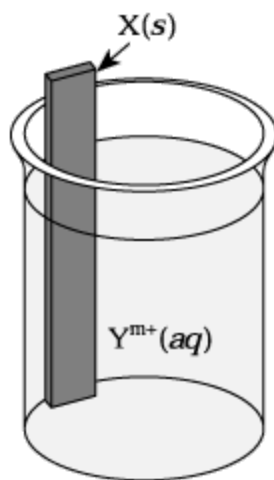


Figure 1 Metal strip in solution

After several hours, they observed whether or not $Y(s)$ formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
$Al^{3+}(aq)$	*	no change	no change	no change	no change	no change
$Cu^{2+}(aq)$	new solid	*	new solid	no change	no change	new solid
$Zn^{2+}(aq)$	new solid	no change	*	no change	no change	no change
$Ag^{+}(aq)$	new solid	new solid	new solid	*	no change	new solid
$Au^{3+}(aq)$	*	*	*	new solid	*	*
$Pb^{2+}(aq)$	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

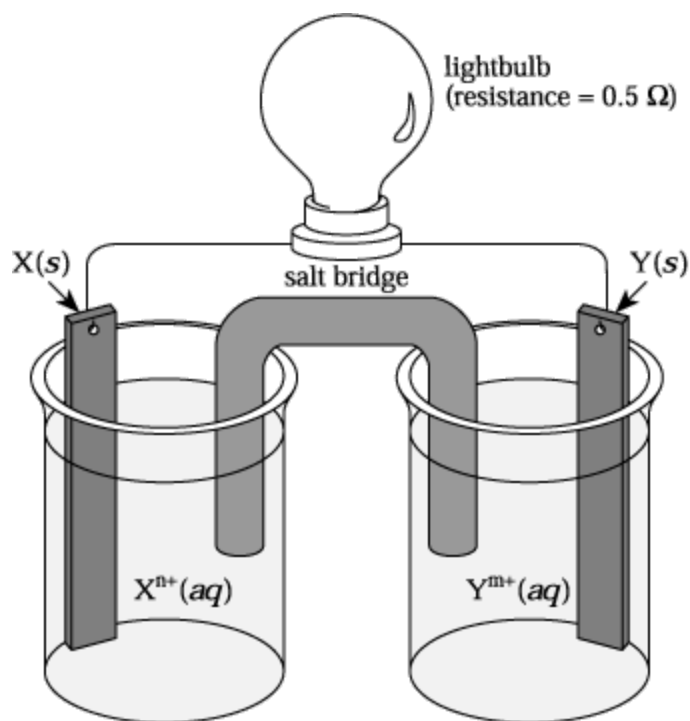
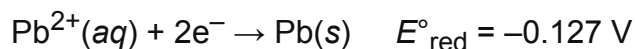
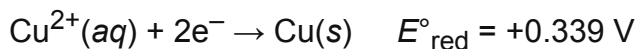


Figure 2	Complete circuit
-----------------	------------------

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.



Reaction 1



Reaction 2

A student can most effectively increase the current passing through the circuit in Figure 2 by doing which of the following?

- A.
Using electrical wire with a smaller diameter
- B.

Increasing the temperature of the electrical wire

- C.

Decreasing the concentrations of $X^{n+}(aq)$ and $Y^{m+}(aq)$

- D.

Replacing the lightbulb with one that has a resistance of $0.2\ \Omega$

Solution: The correct answer is **D**.

1. Using an electrical wire with a smaller diameter will increase the resistance of the wire. According to Ohm's law, this will decrease the current in the circuit.
2. Increasing the temperature of the electrical wire will increase the resistance of the wire. According to Ohm's law, this will decrease the current in the circuit.
3. Decreasing the concentrations will decrease the amount of charges that flow through the circuit that effectively decreases the current.
4. **This option is correct because according to Ohm's law, current is equal to voltage divided by resistance. In order to increase the current when the voltage is fixed, the resistance must be decreased, such as by replacing the lightbulb with one that has a resistance of $0.2\ \Omega$ instead of $0.5\ \Omega$.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Electrostatics](#); [Circuit Elements](#); and [Magnetism](#).

Question 53 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, $X(s)$, into solutions containing different metallic ions, $Y^{m+}(aq)$, as shown in Figure 1.

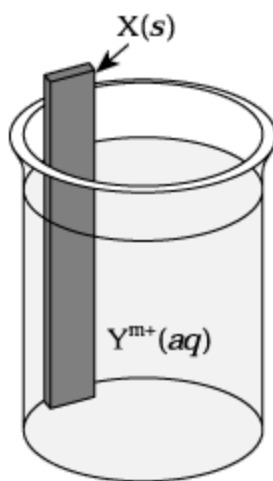


Figure 1

Metal strip in solution

After several hours, they observed whether or not $Y(s)$ formed on the surface of the metal strip (see Table 1).

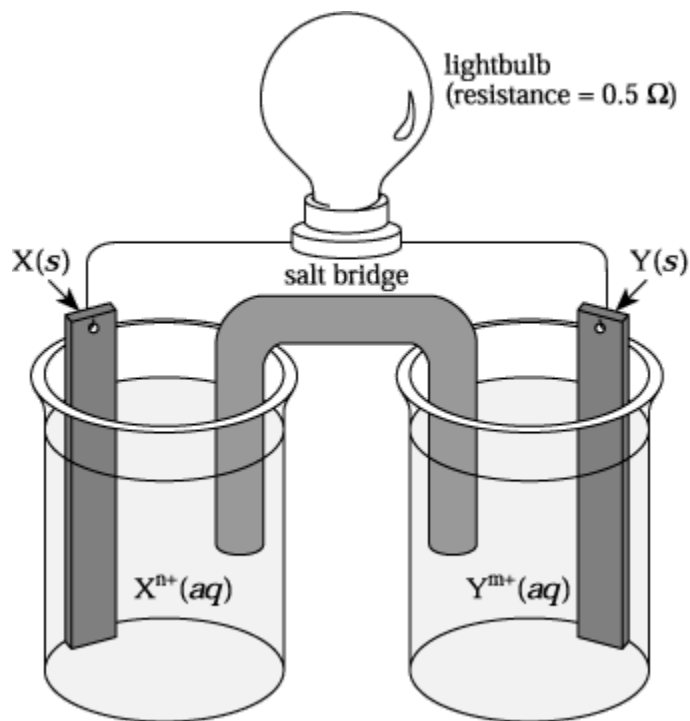
Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
$\text{Al}^{3+}(\text{aq})$	*	no change	no change	no change	no change	no change
$\text{Cu}^{2+}(\text{aq})$	new solid	*	new solid	no change	no change	new solid
$\text{Zn}^{2+}(\text{aq})$	new solid	no change	*	no change	no change	no change
$\text{Ag}^{+}(\text{aq})$	new solid	new solid	new solid	*	no change	new solid
$\text{Au}^{3+}(\text{aq})$	*	*	*	new solid	*	*
$\text{Pb}^{2+}(\text{aq})$	new solid	no change	new solid	no change	no change	*

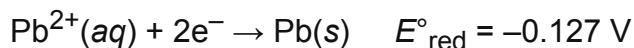
* = not studied

Experiment 2

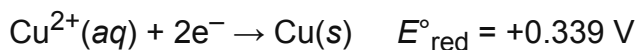
The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).



Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.



Reaction 1



Reaction 2

When a strip of Zn is placed in a beaker containing 0.1 M HCl, $\text{H}_2(\text{g})$ evolves. If a strip of Al is placed in a beaker containing 0.1 M HCl, does $\text{H}_2(\text{g})$ evolve?

- A.
Yes; Al is reduced and $\text{H}^{+}(\text{aq})$ is oxidized.
- B.
Yes; Al is oxidized and $\text{H}^{+}(\text{aq})$ is reduced.
- C.
No; Al is reduced and $\text{Cl}^{-}(\text{aq})$ is oxidized
- D.
No; Al is oxidized and $\text{H}_2\text{O}(\text{l})$ is produced.

Solution: The correct answer is **B**.

1. Al is an electropositive metal. It will not form anions. Formation of H_2 from H^{+} is a reduction.
2. **Since a new solid forms when $\text{Al}(\text{s})$ is mixed with $\text{Zn}^{2+}(\text{aq})$, it is reasonable to assume that $\text{Al}(\text{s})$ is more susceptible to oxidation than $\text{Zn}(\text{s})$.**
3. Electropositive metals are good reducing agents. Only exceptionally good oxidizing agents can oxidize chloride.
4. If Al is oxidized, something has to be reduced. Of the possible choices, reduction of H^{+} occurs at a potential of 0.0 V and is most likely.

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Electrochemistry](#).

Correct

Your result is correct.

Your answer is B.

Correct answer is B.

Correct Answer Your Answer

A

B

✓

C

D

Time Spent:

2 mins 22 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC4C

FC4

GCH

RESULT / TIMING

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, $X(s)$, into solutions containing different metallic ions, $Y^{m+}(aq)$, as shown in Figure 1.

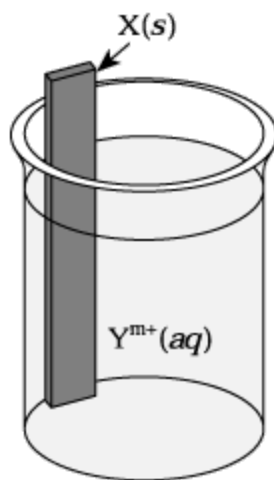


Figure 1 Metal strip in solution

After several hours, they observed whether or not $Y(s)$ formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
$Al^{3+}(aq)$	*	no change	no change	no change	no change	no change
$Cu^{2+}(aq)$	new solid	*	new solid	no change	no change	new solid
$Zn^{2+}(aq)$	new solid	no change	*	no change	no change	no change
$Ag^{+}(aq)$	new solid	new solid	new solid	*	no change	new solid
$Au^{3+}(aq)$	*	*	*	new solid	*	*
$Pb^{2+}(aq)$	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

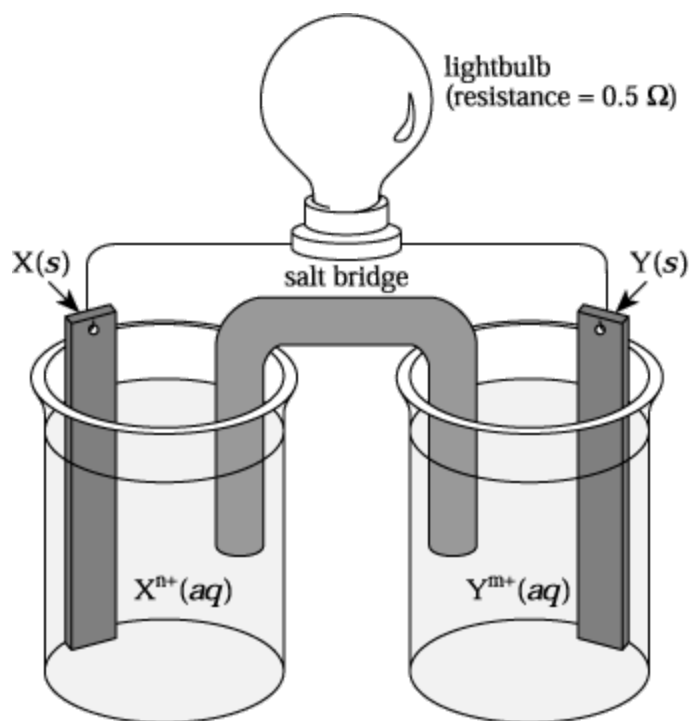
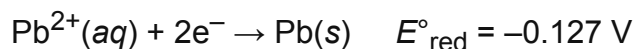
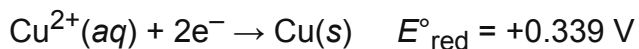


Figure 2 Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.



Reaction 1



Reaction 2

When a strip of Zn is placed in a beaker containing 0.1 M HCl, $\text{H}_2(\text{g})$ evolves. If a strip of Al is placed in a beaker containing 0.1 M HCl, does $\text{H}_2(\text{g})$ evolve?

- A.
Yes; Al is reduced and $\text{H}^{+}(\text{aq})$ is oxidized.
- B.

Yes; Al is oxidized and $\text{H}^+(\text{aq})$ is reduced.

- C.

No; Al is reduced and $\text{Cl}^-(\text{aq})$ is oxidized

- D.

No; Al is oxidized and $\text{H}_2\text{O}(\text{l})$ is produced.

Solution: The correct answer is **B**.

1. Al is an electropositive metal. It will not form anions. Formation of H_2 from H^+ is a reduction.
2. **Since a new solid forms when $\text{Al}(\text{s})$ is mixed with $\text{Zn}^{2+}(\text{aq})$, it is reasonable to assume that $\text{Al}(\text{s})$ is more susceptible to oxidation than $\text{Zn}(\text{s})$.**
3. Electropositive metals are good reducing agents. Only exceptionally good oxidizing agents can oxidize chloride.
4. If Al is oxidized, something has to be reduced. Of the possible choices, reduction of H^+ occurs at a potential of 0.0 V and is most likely.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Electrochemistry](#).

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, $X(s)$, into solutions containing different metallic ions, $Y^{m+}(aq)$, as shown in Figure 1.

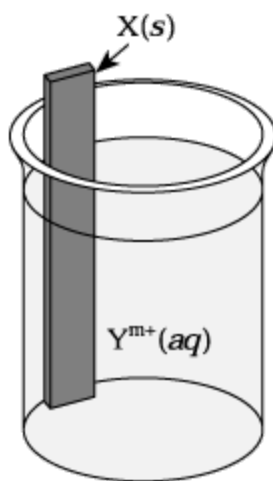


Figure 1

Metal strip in solution

After several hours, they observed whether or not $Y(s)$ formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
$\text{Al}^{3+}(\text{aq})$	*	no change	no change	no change	no change	no change
$\text{Cu}^{2+}(\text{aq})$	new solid	*	new solid	no change	no change	new solid
$\text{Zn}^{2+}(\text{aq})$	new solid	no change	*	no change	no change	no change
$\text{Ag}^{+}(\text{aq})$	new solid	new solid	new solid	*	no change	new solid
$\text{Au}^{3+}(\text{aq})$	*	*	*	new solid	*	*
$\text{Pb}^{2+}(\text{aq})$	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

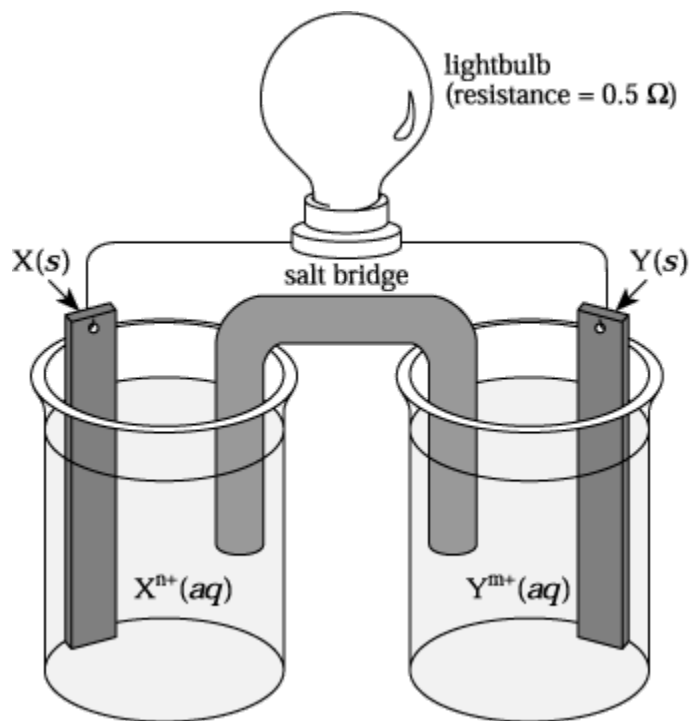
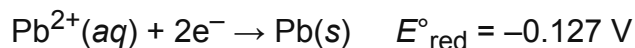
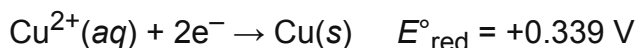


Figure 2

Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.

**Reaction 1****Reaction 2**

Assume that a circuit similar to that in Figure 2 is set up in which $\text{X} = \text{Al}$, $\text{X}^{n+} = \text{Al}^{3+}$, $\text{Y} = \text{Cu}$, and $\text{Y}^{m+} = \text{Cu}^{2+}$. Which of the following reactions will occur?

- A.
 $2\text{Al}^{3+}(\text{aq}) + 3\text{Cu}(\text{s}) \rightarrow 2\text{Al}(\text{s}) + 3\text{Cu}^{2+}(\text{aq})$
- B.
 $3\text{Al}^{3+}(\text{aq}) + 2\text{Cu}(\text{s}) \rightarrow 3\text{Al}(\text{s}) + 2\text{Cu}^{2+}(\text{aq})$
- C.
 $2\text{Al}(\text{s}) + 3\text{Cu}^{2+}(\text{aq}) \rightarrow 2\text{Al}^{3+}(\text{aq}) + 3\text{Cu}(\text{s})$
- D.
 $3\text{Al}(\text{s}) + 2\text{Cu}^{2+}(\text{aq}) \rightarrow 3\text{Al}^{3+}(\text{aq}) + 2\text{Cu}(\text{s})$

Solution: The correct answer is **C**.

1. While this reaction is balanced, it does not fit with the data in the table. $\text{Al}^{3+}(\text{aq})$ does not react with $\text{Cu}(\text{s})$
2. This reaction is not balanced for charge. Furthermore, it does not fit with the data in the table. $\text{Al}^{3+}(\text{aq})$ does not react with $\text{Cu}(\text{s})$.
3. **This response depicts the balanced chemical reaction that occurs when the four substances $[\text{Al}(\text{s})$, $\text{Al}^{3+}(\text{aq})$, $\text{Cu}(\text{s})$, and $\text{Cu}^{2+}(\text{aq})$] are mixed.**
4. This reaction is not balanced for charge. There is a net +9 on the product side, but only net +4 on the reactant side.

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#); [Electronic Structure](#); [The Periodic Table - Classification of Elements Into Groups by Electronic Structure](#); [The Periodic Table - Variations of Chemical Properties with Group and Row](#); and [Stoichiometry](#).

Incorrect

Your result is incorrect.

Your answer is A.

Correct answer is C.

Correct Answer Your Answer

A

x

B

C

D

Time Spent:

2 mins 20 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC4E

FC4

GCH

RESULT / TIMING

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, $X(s)$, into solutions containing different metallic ions, $Y^{m+}(aq)$, as shown in Figure 1.

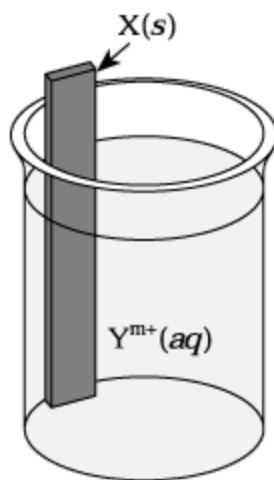


Figure 1 Metal strip in solution

After several hours, they observed whether or not $Y(s)$ formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
$Al^{3+}(aq)$	*	no change	no change	no change	no change	no change
$Cu^{2+}(aq)$	new solid	*	new solid	no change	no change	new solid
$Zn^{2+}(aq)$	new solid	no change	*	no change	no change	no change
$Ag^{+}(aq)$	new solid	new solid	new solid	*	no change	new solid
$Au^{3+}(aq)$	*	*	*	new solid	*	*
$Pb^{2+}(aq)$	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

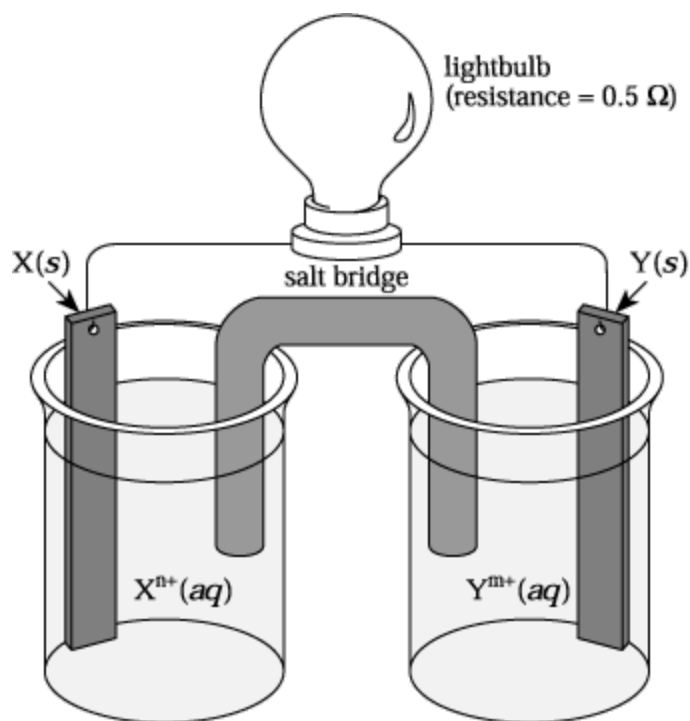
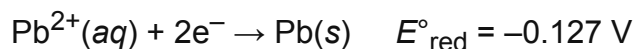
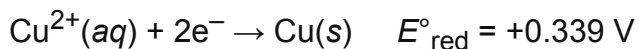


Figure 2 Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.



Reaction 1

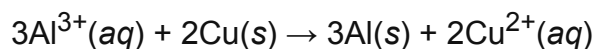


Reaction 2

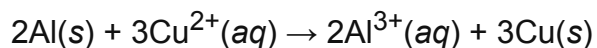
Assume that a circuit similar to that in Figure 2 is set up in which $\text{X} = \text{Al}$, $\text{X}^{n+} = \text{Al}^{3+}$, $\text{Y} = \text{Cu}$, and $\text{Y}^{m+} = \text{Cu}^{2+}$. Which of the following reactions will occur?

- A.

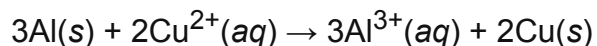
$$2\text{Al}^{3+}(\text{aq}) + 3\text{Cu}(\text{s}) \rightarrow 2\text{Al}(\text{s}) + 3\text{Cu}^{2+}(\text{aq})$$
- B.



- C.



- D.



Solution: The correct answer is **C**.

1. While this reaction is balanced, it does not fit with the data in the table. $\text{Al}^{3+}(\text{aq})$ does not react with $\text{Cu}(\text{s})$
2. This reaction is not balanced for charge. Furthermore, it does not fit with the data in the table. $\text{Al}^{3+}(\text{aq})$ does not react with $\text{Cu}(\text{s})$.
3. **This response depicts the balanced chemical reaction that occurs when the four substances $[\text{Al}(\text{s})$, $\text{Al}^{3+}(\text{aq})$, $\text{Cu}(\text{s})$, and $\text{Cu}^{2+}(\text{aq})$] are mixed.**
4. This reaction is not balanced for charge. There is a net +9 on the product side, but only net +4 on the reactant side.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#); [Electronic Structure](#); [The Periodic Table - Classification of Elements Into Groups by Electronic Structure](#), [The Periodic Table - Variations of Chemical Properties with Group and Row](#); and [Stoichiometry](#).

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, $X(s)$, into solutions containing different metallic ions, $Y^{m+}(aq)$, as shown in Figure 1.

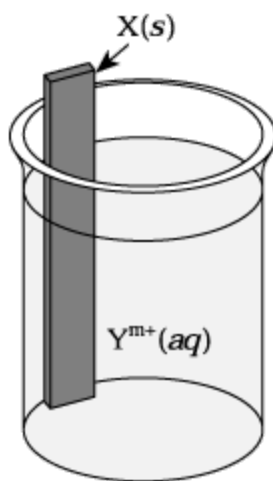


Figure 1

Metal strip in solution

After several hours, they observed whether or not $Y(s)$ formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
$\text{Al}^{3+}(\text{aq})$	*	no change	no change	no change	no change	no change
$\text{Cu}^{2+}(\text{aq})$	new solid	*	new solid	no change	no change	new solid
$\text{Zn}^{2+}(\text{aq})$	new solid	no change	*	no change	no change	no change
$\text{Ag}^{+}(\text{aq})$	new solid	new solid	new solid	*	no change	new solid
$\text{Au}^{3+}(\text{aq})$	*	*	*	new solid	*	*
$\text{Pb}^{2+}(\text{aq})$	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

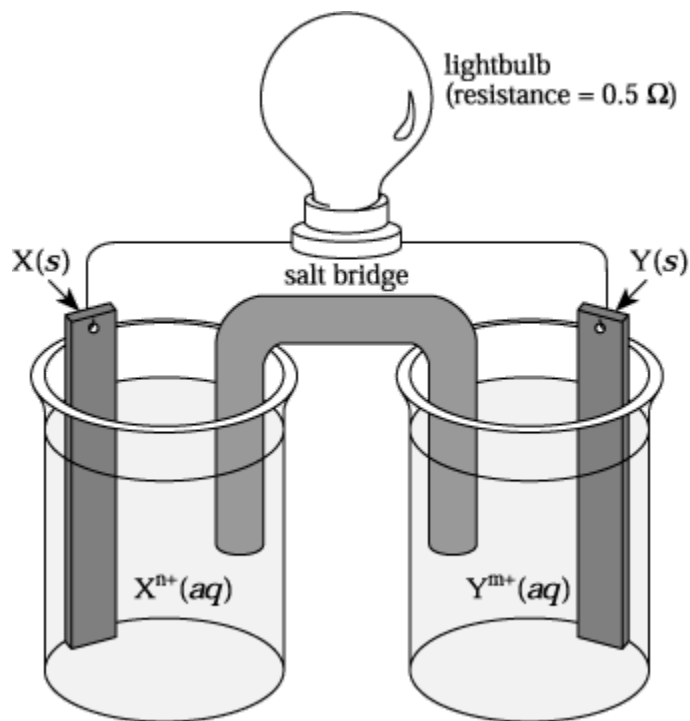
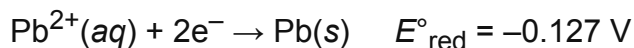
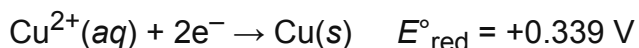


Figure 2

Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.

**Reaction 1****Reaction 2**

A circuit similar to that in Figure 2 is set up in which

$\text{X} = \text{Pb}$, $\text{X}^{\text{n}+} = \text{Pb}^{2+}$, $\text{Y} = \text{Cu}$, and $\text{Y}^{\text{m}+} = \text{Cu}^{2+}$. Will an oxidation–reduction reaction occur spontaneously?

- A.
No; $E^{\circ}_{\text{cell}} = -0.212$.
- B.
No; $E^{\circ}_{\text{cell}} = +0.212$.
- C.
Yes; $E^{\circ}_{\text{cell}} = -0.466$.
- D.
Yes; $E^{\circ}_{\text{cell}} = +0.466$.

Solution: The correct answer is **D**.

1. This is an incorrect calculation of E°_{cell} . In order to use the potentials provided, one of the half-reactions must be reversed. This changes the sign of the reduction potential. The calculated value is the result of reversing both half-reactions and adding the resulting potentials.
2. This is not the real value of E°_{cell} . It results from adding both reduction potentials listed. One of the reductions needs to be reversed in order to have a complete oxidation–reduction reaction.
3. This value of E°_{cell} is obtained by combining the oxidation of $\text{Cu}(\text{s})$ to $\text{Cu}^{2+}(\text{aq})$ with the reduction of $\text{Pb}^{2+}(\text{aq})$ to $\text{Pb}(\text{s})$. The fact that the value is negative

indicates that this reaction will not occur spontaneously.

4. **A spontaneous reaction occurs when E° is greater than zero, and this occurs if the oxidation of Pb(s) is combined with the reduction of $\text{Cu}^{2+}(\text{aq})$, resulting in a net E° of +0.466 V.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Electrochemistry](#).

Incorrect

Your result is incorrect.

Your answer is A.

Correct answer is D.

Correct Answer Your Answer

A

x

B

C

D

Time Spent:

55 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC4C

FC4

GCH

RESULT / TIMING

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, $X(s)$, into solutions containing different metallic ions, $Y^{m+}(aq)$, as shown in Figure 1.

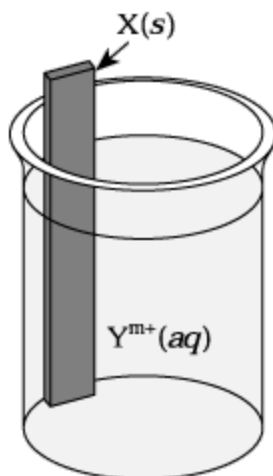


Figure 1 Metal strip in solution

After several hours, they observed whether or not $Y(s)$ formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
$Al^{3+}(aq)$	*	no change	no change	no change	no change	no change
$Cu^{2+}(aq)$	new solid	*	new solid	no change	no change	new solid
$Zn^{2+}(aq)$	new solid	no change	*	no change	no change	no change
$Ag^{+}(aq)$	new solid	new solid	new solid	*	no change	new solid
$Au^{3+}(aq)$	*	*	*	new solid	*	*
$Pb^{2+}(aq)$	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

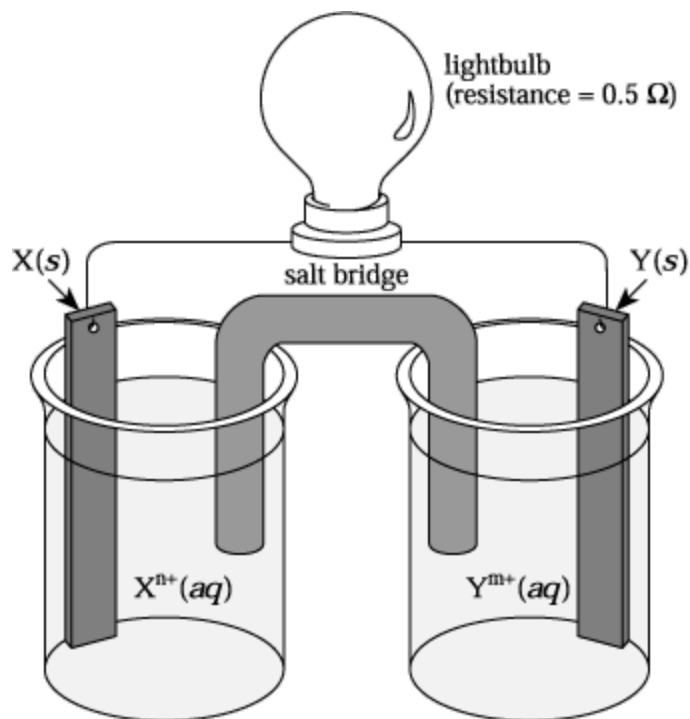
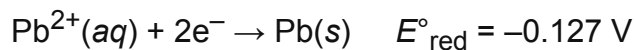
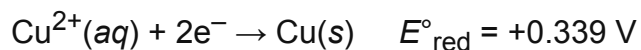


Figure 2	Complete circuit
-----------------	------------------

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.



Reaction 1



Reaction 2

A circuit similar to that in Figure 2 is set up in which

$\text{X} = \text{Pb}$, $\text{X}^{n+} = \text{Pb}^{2+}$, $\text{Y} = \text{Cu}$, and $\text{Y}^{m+} = \text{Cu}^{2+}$. Will an oxidation–reduction reaction

occur spontaneously?

- A.
No; $E^\circ_{\text{cell}} = -0.212$.
- B.
No; $E^\circ_{\text{cell}} = +0.212$.
- C.
Yes; $E^\circ_{\text{cell}} = -0.466$.
- D.
Yes; $E^\circ_{\text{cell}} = +0.466$.

Solution: The correct answer is **D**.

1. This is an incorrect calculation of E°_{cell} . In order to use the potentials provided, one of the half-reactions must be reversed. This changes the sign of the reduction potential. The calculated value is the result of reversing both half-reactions and adding the resulting potentials.
2. This is not the real value of E°_{cell} . It results from adding both reduction potentials listed. One of the reductions needs to be reversed in order to have a complete oxidation–reduction reaction.
3. This value of E°_{cell} is obtained by combining the oxidation of Cu(s) to $\text{Cu}^{2+}(\text{aq})$ with the reduction of $\text{Pb}^{2+}(\text{aq})$ to Pb(s) . The fact that the value is negative indicates that this reaction will not occur spontaneously.
4. **A spontaneous reaction occurs when E° is greater than zero, and this occurs if the oxidation of Pb(s) is combined with the reduction of $\text{Cu}^{2+}(\text{aq})$, resulting in a net E° of $+0.466 \text{ V}$.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Electrochemistry](#).

Question 56 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, $X(s)$, into solutions containing different metallic ions, $Y^{m+}(aq)$, as shown in Figure 1.

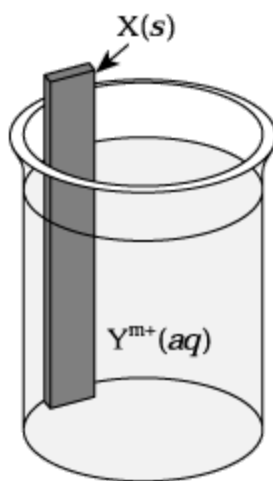


Figure 1

Metal strip in solution

After several hours, they observed whether or not $Y(s)$ formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
$\text{Al}^{3+}(\text{aq})$	*	no change	no change	no change	no change	no change
$\text{Cu}^{2+}(\text{aq})$	new solid	*	new solid	no change	no change	new solid
$\text{Zn}^{2+}(\text{aq})$	new solid	no change	*	no change	no change	no change
$\text{Ag}^{+}(\text{aq})$	new solid	new solid	new solid	*	no change	new solid
$\text{Au}^{3+}(\text{aq})$	*	*	*	new solid	*	*
$\text{Pb}^{2+}(\text{aq})$	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

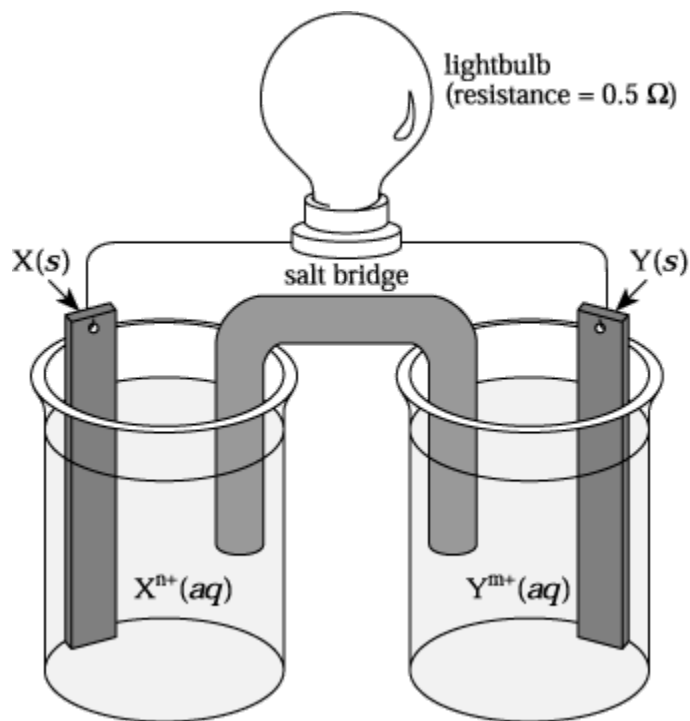
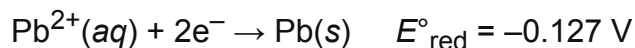
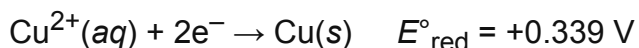


Figure 2

Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.

**Reaction 1****Reaction 2**

A circuit similar to that in Figure 2 is set up. It has a potential of 2.0 V. Assume that the resistance in the circuit is negligible compared to that of the lightbulb. Approximately how much current passes through the lightbulb?

- A.
0.5 A
- B.
1.0 A
- C.
2.0 A
- D.
4.0 A

Solution: The correct answer is **D**.

1. The current is equal to 2 V divided by 0.5 Ω equals 4.0 A, not 1.0 A.
2. The current is equal to 2 V divided by 0.5 Ω equals 4.0 A, not 1.0 A.
3. The current is equal to 2 V divided by 0.5 Ω equals 4.0 A, not 2.0 A.
4. **The current is equal to 2 V divided by 0.5 Ω equals 4.0 A.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Electrostatics](#); [Circuit Elements](#); and [Magnetism](#).

Incorrect

Your result is incorrect.

Your answer is C.

Correct answer is D.

Correct Answer Your Answer

A

B

C

x

D

Time Spent:

1 min 15 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC4C

FC4

PHY

RESULT / TIMING

Passage 10 (Questions 52-56)

A group of students investigated the reactions of several metals with solutions containing metallic ions.

Experiment 1

The students placed strips of a metal, $X(s)$, into solutions containing different metallic ions, $Y^{m+}(aq)$, as shown in Figure 1.

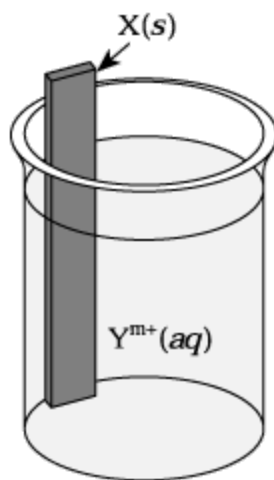


Figure 1 Metal strip in solution

After several hours, they observed whether or not $Y(s)$ formed on the surface of the metal strip (see Table 1).

Table 1 Results of Metal Strip Experiment

Identity of metal ions	Composition of metal strip					
	Al(s)	Cu(s)	Zn(s)	Ag(s)	Au(s)	Pb(s)
$Al^{3+}(aq)$	*	no change	no change	no change	no change	no change
$Cu^{2+}(aq)$	new solid	*	new solid	no change	no change	new solid
$Zn^{2+}(aq)$	new solid	no change	*	no change	no change	no change
$Ag^{+}(aq)$	new solid	new solid	new solid	*	no change	new solid
$Au^{3+}(aq)$	*	*	*	new solid	*	*
$Pb^{2+}(aq)$	new solid	no change	new solid	no change	no change	*

* = not studied

Experiment 2

The students placed a solid strip of each metal into an aqueous solution containing ions of the same metal. Complete circuits were established (Figure 2).

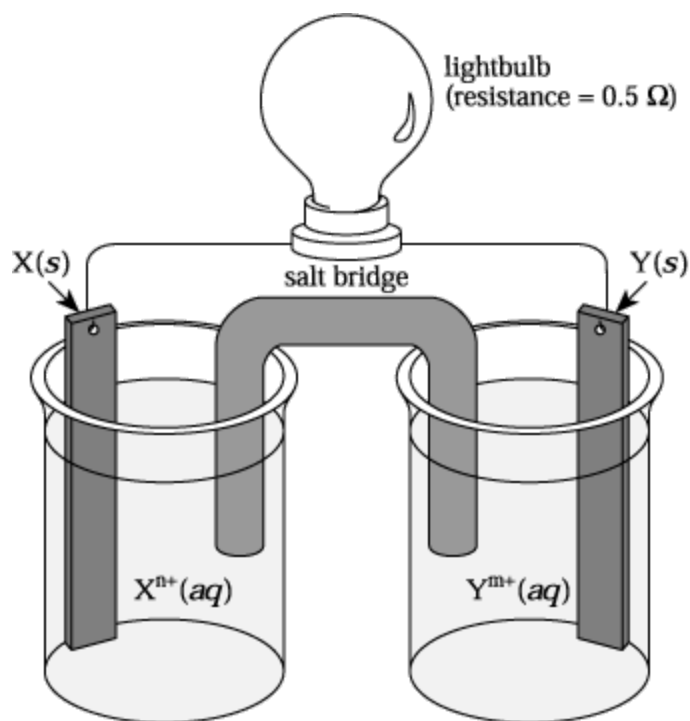
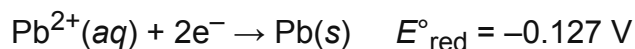
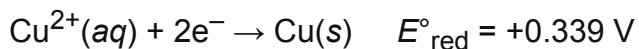


Figure 2 Complete circuit

Finally, the students checked a general chemistry text to determine the standard reduction potentials for the following reactions.



Reaction 1



Reaction 2

A circuit similar to that in Figure 2 is set up. It has a potential of 2.0 V. Assume that the resistance in the circuit is negligible compared to that of the lightbulb. Approximately how much current passes through the lightbulb?

- A.
0.5 A

- B.
1.0 A
- C.
2.0 A
- D.
4.0 A

Solution: The correct answer is **D**.

1. The current is equal to 2 V divided by 0.5 Ω equals 4.0 A, not 1.0 A.
2. The current is equal to 2 V divided by 0.5 Ω equals 4.0 A, not 1.0 A.
3. The current is equal to 2 V divided by 0.5 Ω equals 4.0 A, not 2.0 A.
4. **The current is equal to 2 V divided by 0.5 Ω equals 4.0 A.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Electrostatics](#); [Circuit Elements](#); and [Magnetism](#).

Question 57 of 59

QUESTION

Done Reviewing

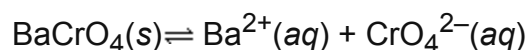
MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

The equilibrium



exists in a saturated aqueous solution of BaCrO_4 . Dissolution of Na_2CrO_4 in a saturated aqueous BaCrO_4 solution would:

- A.
have no effect on the position of this equilibrium.
- B.
shift this equilibrium left.
- C.
shift this equilibrium right.
- D.
shift this equilibrium first right and then left.

Solution: The correct answer is **B**.

1. Adding the chromate anion to solution will drive the indicated equilibrium to the left by Le Châtelier's Principle.
2. **Dissolution of Na_2CrO_4 would introduce the common ion, CrO_4^{2-} , which would reduce the solubility of BaCrO_4 due to the common ion effect.**
3. The reverse is true. Adding a product anion stresses the existing equilibrium. Upon addition of chromate, the reaction quotient Q will exceed K_{sp} and the equilibrium shifts left until Q and K_{sp} are equal.
4. This response does not follow the principles of chemical dynamics. While both reactions are constantly occurring (forward and reverse), the net reaction will only

go to the left to compensate for the added anion.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Acid-Base Equilibria](#); [Solubility](#); and [Titration](#).

Correct

Your result is correct.

Your answer is B.

Correct answer is B.

Correct Answer Your Answer

A

B

✓

C

D

Time Spent:

37 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

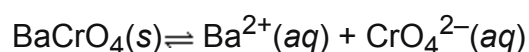
CC5A

FC5

GCH

RESULT / TIMING

The equilibrium



exists in a saturated aqueous solution of BaCrO_4 . Dissolution of Na_2CrO_4 in a saturated aqueous BaCrO_4 solution would:

- A.

have no effect on the position of this equilibrium.

- B.
shift this equilibrium left.
- C.
shift this equilibrium right.
- D.
shift this equilibrium first right and then left.

Solution: The correct answer is **B**.

1. Adding the chromate anion to solution will drive the indicated equilibrium to the left by Le Châtelier's Principle.
2. **Dissolution of Na_2CrO_4 would introduce the common ion, CrO_4^{2-} , which would reduce the solubility of BaCrO_4 due to the common ion effect.**
3. The reverse is true. Adding a product anion stresses the existing equilibrium. Upon addition of chromate, the reaction quotient Q will exceed K_{sp} and the equilibrium shifts left until Q and K_{sp} are equal.
4. This response does not follow the principles of chemical dynamics. While both reactions are constantly occurring (forward and reverse), the net reaction will only go to the left to compensate for the added anion.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Acid-Base Equilibria](#); [Solubility](#); and [Titration](#).

Question 58 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

What volume of a 0.120 M CaI_2 solution would contain 0.078 mol of the solute?

- A.
35.0 mL
- B.
65.0 mL
- C.
350 mL
- D.
650 mL

Solution: The correct answer is **D**.

1. A volume of 35 mL corresponds to only $0.035 \text{ L} \times 0.120 \text{ mol/L} = 0.0042 \text{ mol}$ of solute.
2. A volume of 65 mL corresponds to only $0.065 \text{ L} \times 0.120 \text{ mol/L} = 0.0078 \text{ mol}$ of solute.
3. A volume of 350 mL corresponds to only $0.35 \text{ L} \times 0.120 \text{ mol/L} = 0.042 \text{ mol}$ of solute.
4. **In order to obtain the volume of solution necessary to provide a given amount of solute in moles, one needs to take the number of moles and divide by the solution concentration in molarity: $0.078 \text{ mol} \times 1 \text{ L}/0.120 \text{ mol} = 0.65 \text{ L} = 650 \text{ mL}$.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Acid-Base Equilibria](#); [Solubility](#); and [Titration](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

2 mins 27 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5A

FC5

GCH

RESULT / TIMING

What volume of a 0.120 M CaI_2 solution would contain 0.078 mol of the solute?

- A.
35.0 mL
- B.
65.0 mL
- C.
350 mL
- D.
650 mL

Solution: The correct answer is **D**.

1. A volume of 35 mL corresponds to only $0.035 \text{ L} \times 0.120 \text{ mol/L} = 0.0042 \text{ mol}$ of solute.

2. A volume of 65 mL corresponds to only $0.065 \text{ L} \times 0.120 \text{ mol/L} = 0.0078 \text{ mol}$ of solute.
3. A volume of 350 mL corresponds to only $0.35 \text{ L} \times 0.120 \text{ mol/L} = 0.042 \text{ mol}$ of solute.
4. **In order to obtain the volume of solution necessary to provide a given amount of solute in moles, one needs to take the number of moles and divide by the solution concentration in molarity: $0.078 \text{ mol} \times 1 \text{ L}/0.120 \text{ mol} = 0.65 \text{ L} = 650 \text{ mL}$.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Acid-Base Equilibria](#); [Solubility](#); and [Titration](#).

Question 59 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Which measurement unit CANNOT be used to express power?

- A.
 $\text{kg} \cdot \text{m}^2 \cdot \text{s}^2$
- B.
 $\text{J} \cdot \text{s}^{-1}$
- C.
 $\text{ft} \cdot \text{lb} \cdot \text{s}^{-1}$
- D.
 W

Solution: The correct answer is **A**.

1. **This answer is correct because the measurement unit of power is watt, defined as $J/s = ft \cdot lb/s = kg \cdot m^2/s^3$.**
2. This unit is consistent with the definition of the watt, which is used to express power, because by definition $1W = J \cdot s^{-1}$
3. This unit is consistent with the definition of the watt, which is used to express power, because by definition $1W = J \cdot s^{-1} = 1 ft \cdot lb \cdot s^{-1}$
4. This unit is the watt, which is by definition is used to express power.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Vectors](#); [Newton's First Law](#); [Torque](#); and [Work and energy](#).

Incorrect

Your result is incorrect.

Your answer is C.

Correct answer is A.

Correct Answer Your Answer

A

B

C

x

D

Time Spent:

49 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC4A

FC4

PHY

RESULT / TIMING

Which measurement unit CANNOT be used to express power?

- A.
 $\text{kg} \cdot \text{m}^2 \cdot \text{s}^2$
- B.
 $\text{J} \cdot \text{s}^{-1}$
- C.
 $\text{ft} \cdot \text{lb} \cdot \text{s}^{-1}$
- D.
W

Solution: The correct answer is **A**.

1. **This answer is correct because the measurement unit of power is watt, defined as $J/s = ft \cdot lb/s = kg \cdot m^2/s^3$.**

2. This unit is consistent with the definition of the watt, which is used to express power, because by definition $1W = J \bullet s^{-1}$
3. This unit is consistent with the definition of the watt, which is used to express power, because by definition $1W = J \bullet s^{-1} = 1 \text{ ft} \bullet \text{lb} \bullet s^{-1}$
4. This unit is the watt, which is by definition is used to express power.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Vectors](#); [Newton's First Law](#); [Torque](#); and [Work and energy](#).

Passage 3 (Questions 14-17)

Passage

Passage 3 (Questions 14-17)

Researchers are designing a micro-robot equipped with an oxygen sensor for intraocular procedures. The micro-robot will be injected into the patient’s eye to determine if a blood clot is present. The micro-robot is powered by electromagnetic radiations that induce a current in its interior circuit. The current powers a motor that propels the micro-robot through the fluid of the vitreous humor. Inside the eye, the micro-robot experiences a viscous drag force $\mathbf{F}$ dependent on the vitreous humor viscosity η , and the micro-robot’s diameter d and velocity $\mathbf{v}$:

$$\mathbf{F} = -3\pi d\eta\mathbf{v}$$

Equation 1

The micro-robot uses a luminescent sensor film to determine the oxygen concentration, and potential blood clots, in the retina. In this method, a blue LED is shined into the eye and strikes the surface of the sensor. The sensor coating absorbs the photons from the LED and a short time later emits photons that are subsequently detected by a photodiode (Figure 1). Experiments in which oxygen parameters were controlled showed that oxygen, if present in the retina, affects both the time it takes the sensor to emit the photons and the intensity of the emitted radiation.

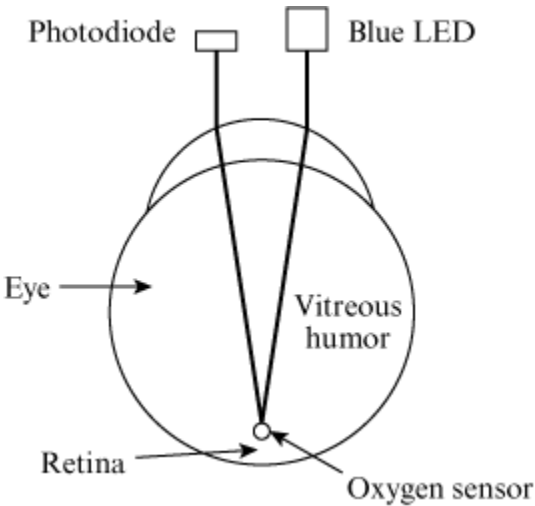


Figure 1	Micro-robot equipped with oxygen sensor used to detect blood clots
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If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

Question 14 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 3 (Questions 14-17)

Researchers are designing a micro-robot equipped with an oxygen sensor for intraocular procedures. The micro-robot will be injected into the patient's eye to determine if a blood clot is present. The micro-robot is powered by electromagnetic radiations that induce a current in its interior circuit. The current powers a motor that propels the micro-robot through the fluid of the vitreous humor. Inside the eye, the micro-robot experiences a viscous drag force $\mathbf{F}$ dependent on the vitreous humor viscosity η , and the micro-robot's diameter d and velocity $\mathbf{v}$:

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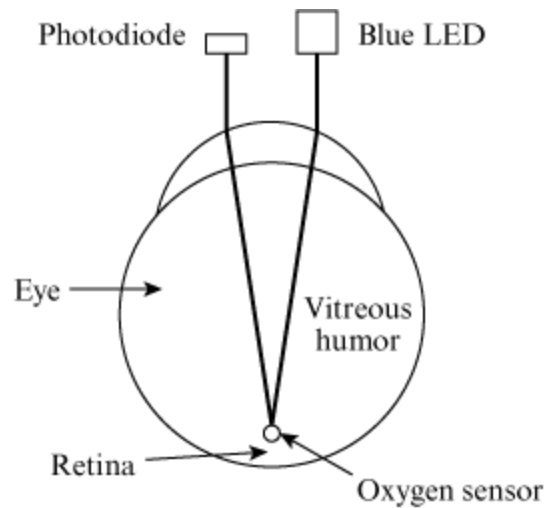


Figure 1 Micro-robot equipped with oxygen sensor used to detect blood clots

If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

When the current in the micro-robot's circuit increases and the resistance of the circuit remains constant, the voltage of the circuit:

- A.
decreases.
- B.
increases.
- C.
is constant.
- D.
is zero.

Solution: The correct answer is **B**.

1. This option contradicts Ohm's law which states that current is equal to voltage divided by resistance. If current increases and resistance is constant, then voltage cannot decrease.

2. This option is correct because according to Ohm's law, current is equal to voltage divided by resistance. If current increases and resistance is constant, then voltage increases as well.
3. This option contradicts Ohm's law which states that current is equal to voltage divided by resistance. If current increases and resistance is constant, then voltage cannot remain constant.
4. This option contradicts Ohm's law that states that current is equal to voltage divided by resistance. If current is not zero, then voltage cannot be zero, too.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Electrostatics](#); [Circuit Elements](#); and [Magnetism](#).

Correct

Your result is correct.

Your answer is B.

Correct answer is B.

Correct Answer Your Answer

A

B

✓

C

D

Time Spent:

47 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC4C

FC4

PHY

RESULT / TIMING

Passage 3 (Questions 14-17)

Researchers are designing a micro-robot equipped with an oxygen sensor for intraocular procedures. The micro-robot will be injected into the patient's eye to determine if a blood clot is present. The micro-robot is powered by electromagnetic radiations that induce a current in its interior circuit. The current powers a motor that propels the micro-robot through the fluid of the vitreous humor. Inside the eye, the micro-robot experiences a viscous drag force $\mathbf{F}$ dependent on the vitreous humor viscosity η , and the micro-robot's diameter d and velocity $\mathbf{v}$:

$$\mathbf{F} = -3\pi d\eta\mathbf{v}$$

Equation 1

The micro-robot uses a luminescent sensor film to determine the oxygen concentration, and potential blood clots, in the retina. In this method, a blue LED is shined into the eye and strikes the surface of the sensor. The sensor coating absorbs the photons from the LED and a short time later emits photons that are subsequently detected by a photodiode (Figure 1). Experiments in which oxygen parameters were controlled showed that oxygen, if present in the retina, affects both the time it takes the sensor to emit the photons and the intensity of the emitted radiation.

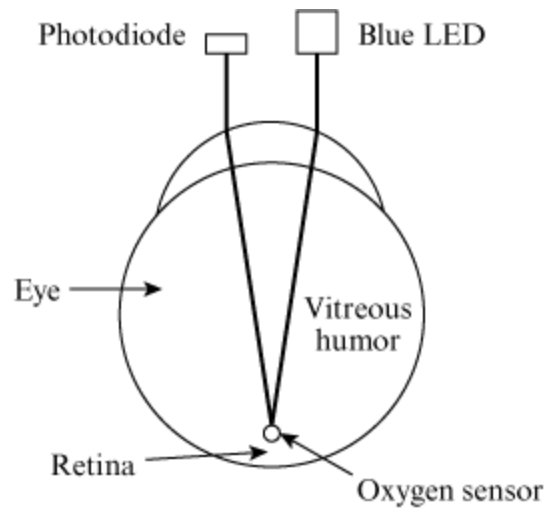


Figure 1

Micro-robot equipped with oxygen sensor used to detect blood clots

If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

When the current in the micro-robot's circuit increases and the resistance of the circuit remains constant, the voltage of the circuit:

- A.
decreases.
- B.
increases.
- C.
is constant.
- D.
is zero.

Solution: The correct answer is **B**.

1. This option contradicts Ohm's law which states that current is equal to voltage divided by resistance. If current increases and resistance is constant, then voltage cannot decrease.
2. **This option is correct because according to Ohm's law, current is equal to voltage divided by resistance. If current increases and resistance is constant, then voltage increases as well.**
3. This option contradicts Ohm's law which states that current is equal to voltage divided by resistance. If current increases and resistance is constant, then voltage cannot remain constant.
4. This option contradicts Ohm's law that states that current is equal to voltage divided by resistance. If current is not zero, then voltage cannot be zero, too.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Electrostatics](#); [Circuit Elements](#); and [Magnetism](#).

Question 15 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 3 (Questions 14-17)

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Equation 1

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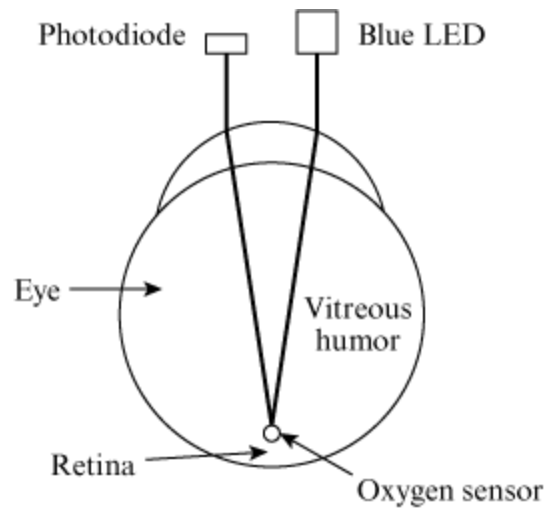


Figure 1	Micro-robot equipped with oxygen sensor used to detect blood clots
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If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

Knowing that the speed of light in the vitreous humor is 2.1×10^8 m/s, what is the index of refraction of the vitreous humor? (Note: The speed of light in a vacuum is 3.0×10^8 m/s.)

- A.
0.7
- B.
1.4
- C.
2.1
- D.
3.0

Solution: The correct answer is **B**.

1. This option implies light travels faster through the vitreous humor than through vacuum, which is a scientific impossibility.

2. The index of refraction of a medium is equal to the ratio of the speed of light in vacuum to the speed of light in the medium, thus in this case is equal to $(3.0 \times 10^8 \text{ m/s}) / (2.1 \times 10^8 \text{ m/s}) = 1.4$.
3. This is 50% larger than the index of refraction of the vitreous humor, which is equal to $(3.0 \times 10^8 \text{ m/s}) / (2.1 \times 10^8 \text{ m/s}) = 1.4$.
4. The index of refraction of the vitreous humor is equal to $(3.0 \times 10^8 \text{ m/s}) / (2.1 \times 10^8 \text{ m/s}) = 1.4$; however, this (3.0) number is 114% larger.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Sound](#); [Light, Electromagnetic Radiation](#); [Geometrical Optics](#); [Spherical mirrors](#); and [Reflection and refraction](#).

Correct

Your result is correct.

Your answer is B.

Correct answer is B.

Correct Answer Your Answer

A

B

✓

C

D

Time Spent:

2 mins 16 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC4D

FC4

PHY

RESULT / TIMING

Passage 3 (Questions 14-17)

Researchers are designing a micro-robot equipped with an oxygen sensor for intraocular procedures. The micro-robot will be injected into the patient's eye to determine if a blood clot is present. The micro-robot is powered by electromagnetic radiations that induce a current in its interior circuit. The current powers a motor that propels the micro-robot through the fluid of the vitreous humor. Inside the eye, the micro-robot experiences a viscous drag force $\mathbf{F}$ dependent on the vitreous humor viscosity η , and the micro-robot's diameter d and velocity $\mathbf{v}$:

$$\mathbf{F} = -3\pi d\eta\mathbf{v}$$

Equation 1

The micro-robot uses a luminescent sensor film to determine the oxygen concentration, and potential blood clots, in the retina. In this method, a blue LED is shined into the eye and strikes the surface of the sensor. The sensor coating absorbs the photons from the LED and a short time later emits photons that are subsequently detected by a photodiode (Figure 1). Experiments in which oxygen parameters were controlled showed that oxygen, if present in the retina, affects both the time it takes the sensor to emit the photons and the intensity of the emitted radiation.

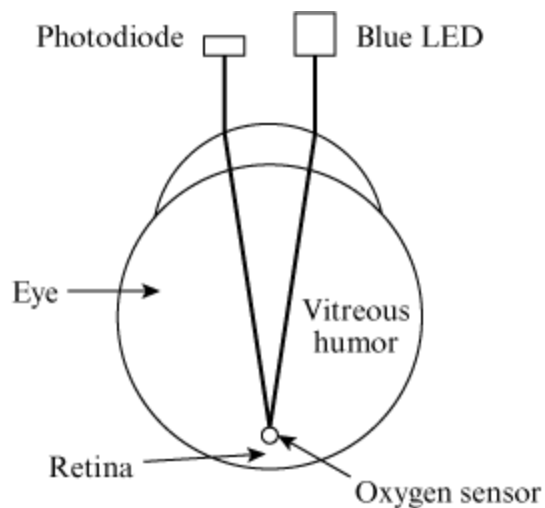


Figure 1	Micro-robot equipped with oxygen sensor used to detect blood clots
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If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA

is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

Knowing that the speed of light in the vitreous humor is 2.1×10^8 m/s, what is the index of refraction of the vitreous humor? (Note: The speed of light in a vacuum is 3.0×10^8 m/s.)

- A.
0.7
- B.
1.4
- C.
2.1
- D.
3.0

Solution: The correct answer is **B**.

1. This option implies light travels faster through the vitreous humor than through vacuum, which is a scientific impossibility.
2. **The index of refraction of a medium is equal to the ratio of the speed of light in vacuum to the speed of light in the medium, thus in this case is equal to $(3.0 \times 10^8 \text{ m/s}) / (2.1 \times 10^8 \text{ m/s}) = 1.4$.**
3. This is 50% larger than the index of refraction of the vitreous humor, which is equal to $(3.0 \times 10^8 \text{ m/s}) / (2.1 \times 10^8 \text{ m/s}) = 1.4$.
4. The index of refraction of the vitreous humor is equal to $(3.0 \times 10^8 \text{ m/s}) / (2.1 \times 10^8 \text{ m/s}) = 1.4$; however, this (3.0) number is 114% larger.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Sound](#); [Light, Electromagnetic Radiation](#); [Geometrical Optics](#); [Spherical mirrors](#); and [Reflection and refraction](#).

Question 16 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 3 (Questions 14-17)

Researchers are designing a micro-robot equipped with an oxygen sensor for intraocular procedures. The micro-robot will be injected into the patient's eye to determine if a blood clot is present. The micro-robot is powered by electromagnetic radiations that induce a current in its interior circuit. The current powers a motor that propels the micro-robot through the fluid of the vitreous humor. Inside the eye, the micro-robot experiences a viscous drag force $\mathbf{F}$ dependent on the vitreous humor viscosity η , and the micro-robot's diameter d and velocity $\mathbf{v}$:

$$\mathbf{F} = -3\pi d\eta\mathbf{v}$$

Equation 1

The micro-robot uses a luminescent sensor film to determine the oxygen concentration, and potential blood clots, in the retina. In this method, a blue LED is shined into the eye and strikes the surface of the sensor. The sensor coating absorbs the photons from the LED and a short time later emits photons that are subsequently detected by a photodiode (Figure 1). Experiments in which oxygen parameters were controlled showed that oxygen, if present in the retina, affects both the time it takes the sensor to emit the photons and the intensity of the emitted radiation.

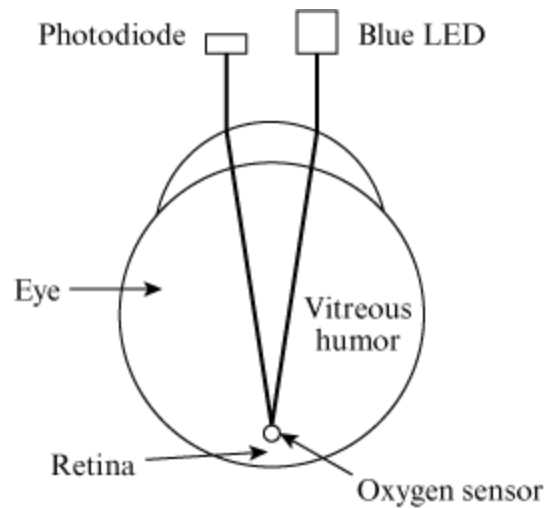


Figure 1 Micro-robot equipped with oxygen sensor used to detect blood clots

If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

The intensity of the radiation emitted by the oxygen sensor is directly proportional to the:

- A.
propagation speed of the radiation.
- B.
wavelength of the radiation.
- C.
polarization of photons emitted.
- D.
number of photons emitted.

Solution: The correct answer is **D**.

1. The intensity of radiation is not related to its propagation speed, but to the number of photons emitted per unit time and their respective frequency.
2. The intensity of radiation is not related to the wavelength of radiation, but to the number of photons emitted per unit time and their respective frequency.

3. The intensity of radiation is not related to the polarization state, but to the number of photons emitted per unit time and their respective frequency.
4. **The energy of electromagnetic radiation is directly proportional to the number of photons, and the intensity of electromagnetic radiation is defined as energy emitted per unit time. Thus, intensity is directly proportional to the number of photons emitted.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Sound](#); [Light, Electromagnetic Radiation](#); [Geometrical Optics](#); [Spherical mirrors](#); and [Reflection and refraction](#).

Incorrect

Your result is incorrect.

Your answer is A.

Correct answer is D.

Correct Answer Your Answer

A

x

B

C

D

Time Spent:

1 min 2 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC4D

FC4

PHY

RESULT / TIMING

Passage 3 (Questions 14-17)

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$$\mathbf{F} = -3\pi d\eta\mathbf{v}$$

Equation 1

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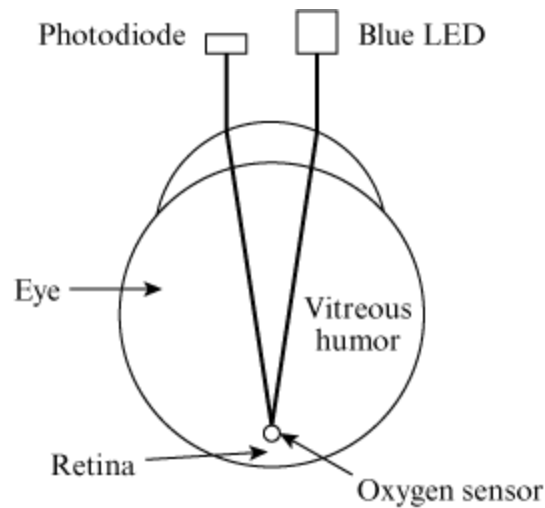


Figure 1

Micro-robot equipped with oxygen sensor used to detect blood clots

If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

The intensity of the radiation emitted by the oxygen sensor is directly proportional to the:

- A.
propagation speed of the radiation.
- B.
wavelength of the radiation.
- C.
polarization of photons emitted.
- D.
number of photons emitted.

Solution: The correct answer is **D**.

1. The intensity of radiation is not related to its propagation speed, but to the number of photons emitted per unit time and their respective frequency.
2. The intensity of radiation is not related to the wavelength of radiation, but to the number of photons emitted per unit time and their respective frequency.
3. The intensity of radiation is not related to the polarization state, but to the number of photons emitted per unit time and their respective frequency.
4. **The energy of electromagnetic radiation is directly proportional to the number of photons, and the intensity of electromagnetic radiation is defined as energy emitted per unit time. Thus, intensity is directly proportional to the number of photons emitted.**

KHAN ACADEMY LESSONS

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Question 17 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 3 (Questions 14-17)

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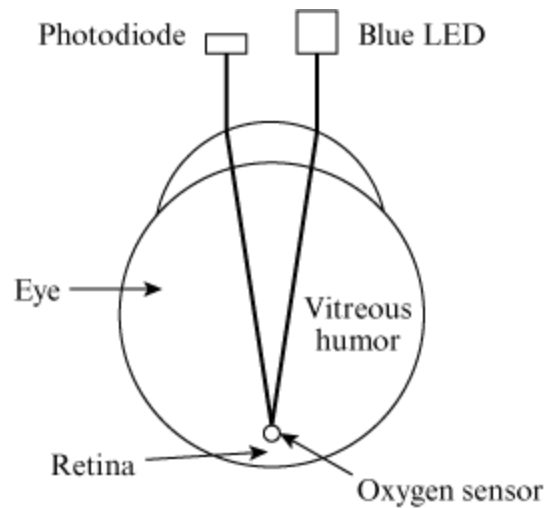


Figure 1 Micro-robot equipped with oxygen sensor used to detect blood clots

If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

What is the energy of the photons emitted by the LED at a frequency of 610 THz?
(Note: $h = 6.6 \times 10^{-34} \text{ J}\cdot\text{s}$)

- A.
 $9.2 \times 10^{-12} \text{ J}$
- B.
 $1.6 \times 10^{-16} \text{ J}$
- C.
 $1.1 \times 10^{-18} \text{ J}$
- D.
 $4.0 \times 10^{-19} \text{ J}$

Solution: The correct answer is **D**.

1. This is about seven orders of magnitude larger than the energy of a photon of frequency 610 THz which is $4 \times 10^{-19} \text{ J}$.

2. This is about three orders of magnitude larger than the energy of a photon of frequency 610 THz which is 4×10^{-19} J.
3. This is about three times larger than the energy of a photon of frequency 610 THz which is 4×10^{-19} J.
4. **The energy of a photon of frequency 610 THz is equal to 6.6×10^{-34} J•s $\times$ 610 x 10¹² Hz = 4×10^{-19} J.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Sound](#); [Light, Electromagnetic Radiation](#); [Geometrical Optics](#); [Spherical mirrors](#); and [Reflection and refraction](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

4 mins 3 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC4D

FC4

PHY

RESULT / TIMING

Passage 3 (Questions 14-17)

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$$\mathbf{F} = -3\pi d\eta\mathbf{v}$$

Equation 1

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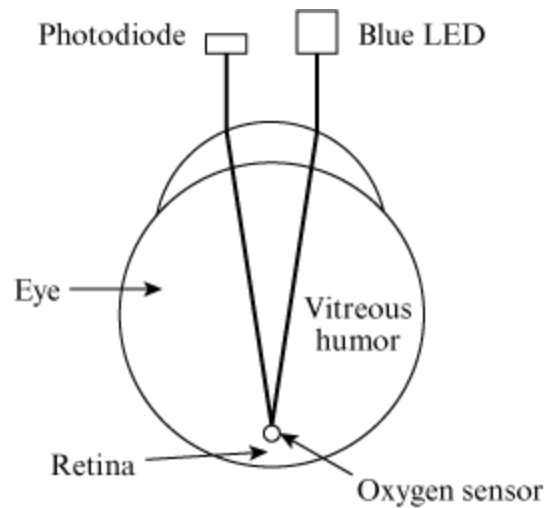


Figure 1

Micro-robot equipped with oxygen sensor used to detect blood clots

If the results of the test indicate that the patient has a clot that is obstructing blood flow, a second micro-robot, coated with tissue plasminogen activator (t-PA), is steered to the thrombus site where it punctures and docks with a retinal vein. The enzyme t-PA is a serine protease that converts plasminogen into plasmin, which subsequently degrades the clot by fibrinolysis.

What is the energy of the photons emitted by the LED at a frequency of 610 THz?

(Note: $h = 6.6 \times 10^{-34} \text{ J}\cdot\text{s}$)

- A.
 $9.2 \times 10^{-12} \text{ J}$
- B.
 $1.6 \times 10^{-16} \text{ J}$
- C.
 $1.1 \times 10^{-18} \text{ J}$
- D.
 $4.0 \times 10^{-19} \text{ J}$

Solution: The correct answer is **D**.

1. This is about seven orders of magnitude larger than the energy of a photon of frequency 610 THz which is $4 \times 10^{-19} \text{ J}$.
2. This is about three orders of magnitude larger than the energy of a photon of frequency 610 THz which is $4 \times 10^{-19} \text{ J}$.
3. This is about three times larger than the energy of a photon of frequency 610 THz which is $4 \times 10^{-19} \text{ J}$.
4. **The energy of a photon of frequency 610 THz is equal to $6.6 \times 10^{-34} \text{ J}\cdot\text{s} \times 610 \times 10^{12} \text{ Hz} = 4 \times 10^{-19} \text{ J}$.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Sound](#); [Light, Electromagnetic Radiation](#); [Geometrical Optics](#); [Spherical mirrors](#); and [Reflection and refraction](#).

Passage 4 (Questions 18-21)

Passage

Passage 4 (Questions 18-21)

Blood pressure can be measured in the brachial artery, which has a cross-sectional area of 25 mm^2 in a healthy adult. The blood flow rate is related to the power the person generates to move their body, as shown Figure 1.

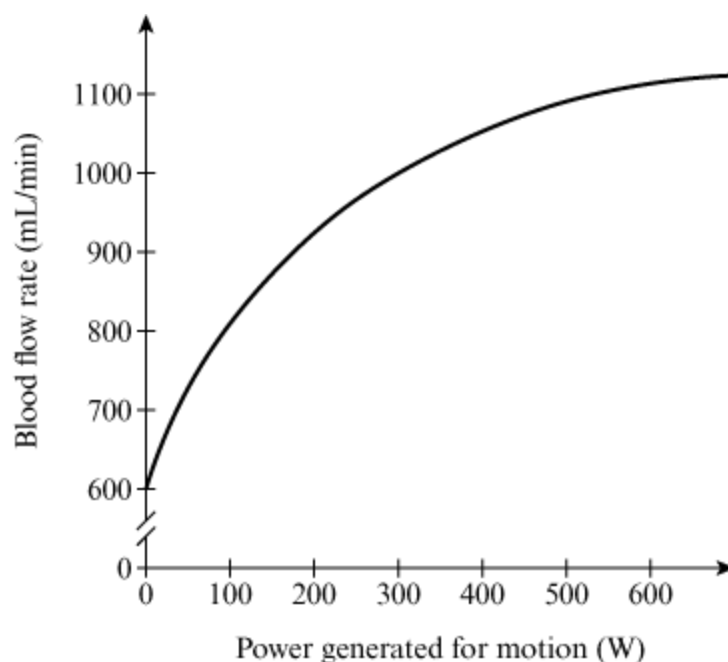


Figure 1	Relationship between the brachial artery blood flow rate of a healthy adult and the power generated for body motion
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An athlete running on a horizontal treadmill maintained a brachial artery blood flow rate of 1.1 L/min for 30 s . The treadmill had an electric circuit used to charge a parallel-plate capacitor of capacitance 1 mF .

The maximum and minimum blood pressures of a healthy adult (the systolic P_s and the diastolic P_d pressures, respectively) are about $P_s = 120 \text{ mmHg}$ and $P_d = 75 \text{ mmHg}$. The mean arterial pressure P_{MPA} is given by

$$P_{\text{MPA}} = P_d + (P_s - P_d)/3$$

Equation 1

and the pulse pressure P_p is defined as

$$P_p = P_s - P_d$$

Equation 2

Ca^{2+} ions are normally found in the blood at a concentration of 100 mg/L. The location of internal organs where changes in the calcium content occur can be found by placing positron-emitting isotopes such as ^{15}O (with a half-life $T_{1/2} = 2$ min) into the patient's bloodstream. When positrons annihilate with electrons, they produce pairs of gamma photons, each with an energy of 500 keV. The location where gamma photons originate gives the location of the organ of interest.

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note
Cancel Add

Passage 4 (Questions 18-21)

Blood pressure can be measured in the brachial artery, which has a cross-sectional area of 25 mm^2 in a healthy adult. The blood flow rate is related to the power the person generates to move their body, as shown Figure 1.

Power generated for motion (W)	Blood flow rate (mL/min)
0	600
100	800
200	920
300	1000
400	1060
500	1090
600	1100

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Equation 1

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What is the relationship between P_p , P_{MPA} , and P_d ?

- A.
 $P_p = P_{MPA} + 3P_d$
- B.
 $P_p = 3P_{MPA} + P_d$
- C.
 $P_p = 3P_{MPA} - 3P_d$
- D.
 $P_p = P_{MPA} - 3P_d$

Solution: The correct answer is **C**.

1. This option incorrectly implies that Equation 1 is $P_{\text{MPA}} = 3P_d$.
2. This option incorrectly implies that $P_d = 0$.
3. **Substituting Equation 2 in Equation 1 yields $P_{\text{PMA}} = P_d + \frac{P_p}{3}$ that can be arranged as $P_p = 3P_{\text{MPA}} - 3P_d$.**
4. This option incorrectly implies that $P_{\text{MPA}} = 0$.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Fluids](#) and [Gas Phase](#).

Incorrect

Your result is incorrect.

Your answer is D.

Correct answer is C.

Correct Answer Your Answer

A

B

C

D

x

Time Spent:

3 mins 54 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC4B

FC4

PHY

RESULT / TIMING

Passage 4 (Questions 18-21)

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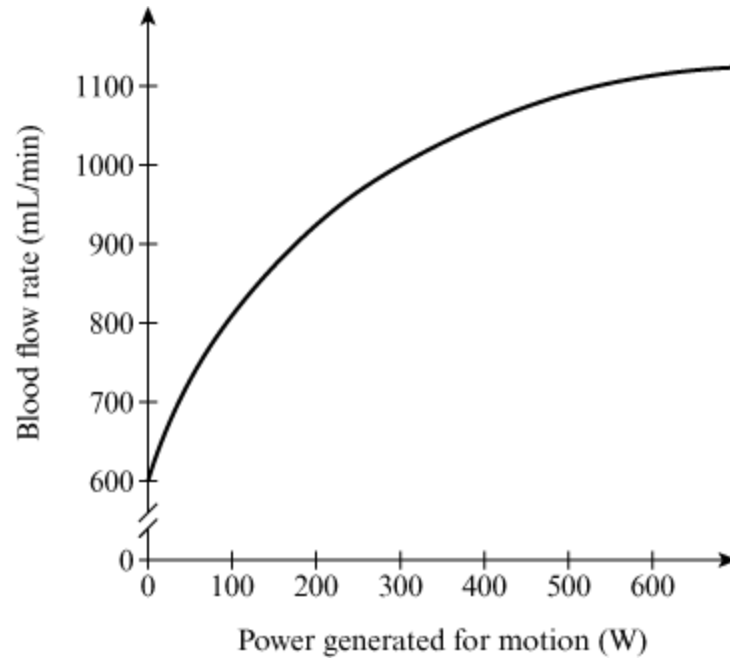


Figure 1	Relationship between the brachial artery blood flow rate of a healthy adult and the power generated for body motion
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An athlete running on a horizontal treadmill maintained a brachial artery blood flow rate of 1.1 L/min for 30 s . The treadmill had an electric circuit used to charge a parallel-plate capacitor of capacitance 1 mF .

The maximum and minimum blood pressures of a healthy adult (the systolic P_s and the diastolic P_d pressures, respectively) are about $P_s = 120 \text{ mmHg}$ and $P_d = 75 \text{ mmHg}$. The mean arterial pressure P_{MPA} is given by

$$P_{\text{MPA}} = P_d + (P_s - P_d)/3$$

Equation 1

and the pulse pressure P_p is defined as

$$P_p = P_s - P_d$$

Equation 2

Ca^{2+} ions are normally found in the blood at a concentration of 100 mg/L. The location of internal organs where changes in the calcium content occur can be found by placing positron-emitting isotopes such as ^{15}O (with a half-life $T_{1/2} = 2$ min) into the patient's bloodstream. When positrons annihilate with electrons, they produce pairs of gamma photons, each with an energy of 500 keV. The location where gamma photons originate gives the location of the organ of interest.

What is the relationship between P_p , P_{MPA} , and P_d ?

- A.
 $P_p = P_{\text{MPA}} + 3P_d$
- B.
 $P_p = 3P_{\text{MPA}} + P_d$
- C.
 $P_p = 3P_{\text{MPA}} - 3P_d$
- D.
 $P_p = P_{\text{MPA}} - 3P_d$

Solution: The correct answer is **C**.

1. This option incorrectly implies that Equation 1 is $P_{\text{MPA}} = 3P_d$.
2. This option incorrectly implies that $P_d = 0$.
3. **Substituting Equation 2 in Equation 1 yields $P_{\text{PMA}} = P_d + \frac{P_p}{3}$ that can be arranged as $P_p = 3P_{\text{MPA}} - 3P_d$.**
4. This option incorrectly implies that $P_{\text{MPA}} = 0$.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Fluids](#) and [Gas Phase](#).

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note
Cancel Add

Passage 4 (Questions 18-21)

Blood pressure can be measured in the brachial artery, which has a cross-sectional area of 25 mm^2 in a healthy adult. The blood flow rate is related to the power the person generates to move their body, as shown Figure 1.

Power generated for motion (W)	Blood flow rate (mL/min)
0	600
100	800
200	920
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400	1060
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What percentage of standard atmospheric pressure is the pulse pressure of a healthy adult?

- A.
10%
- B.
6%
- C.
2%
- D.
1%

Solution: The correct answer is **B**.

1. Atmospheric pressure is 760 mmHg. If pulse pressure was 10% of this value, it would be 76 mmHg.
2. **The pulse pressure in a healthy adult is $(120 - 75)$ mmHg = 45 mmHg, and so the percentage is $45 \text{ mmHg} / 760 \text{ mmHg} = 6\%$.**
3. A pulse pressure that is 2% of atmospheric pressure is $0.02 \times 760 \text{ mmHg} = 15.2 \text{ mmHg}$.
4. A pulse pressure that is 1% of atmospheric pressure would be $0.01 \times 760 \text{ mmHg} = 7.6 \text{ mmHg}$.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Gas Phase](#).

Incorrect

Your result is incorrect.

Your answer is C.

Correct answer is B.

Correct Answer Your Answer

A

B

C

x

D

Time Spent:

4 mins 4 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC4B

FC4

GCH

RESULT / TIMING

Passage 4 (Questions 18-21)

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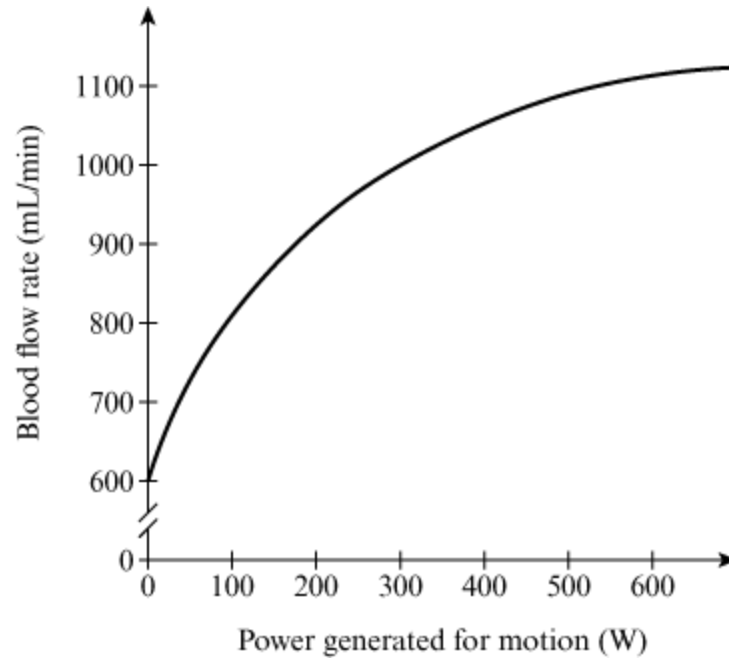


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Equation 1

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What percentage of standard atmospheric pressure is the pulse pressure of a healthy adult?

- A.
10%
- B.
6%
- C.
2%
- D.
1%

Solution: The correct answer is **B**.

1. Atmospheric pressure is 760 mmHg. If pulse pressure was 10% of this value, it would be 76 mmHg.
2. **The pulse pressure in a healthy adult is $(120 - 75)$ mmHg = 45 mmHg, and so the percentage is $45 \text{ mmHg} / 760 \text{ mmHg} = 6\%$.**
3. A pulse pressure that is 2% of atmospheric pressure is $0.02 \times 760 \text{ mmHg} = 15.2$ mmHg.
4. A pulse pressure that is 1% of atmospheric pressure would be $0.01 \times 760 \text{ mmHg} = 7.6$ mmHg.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Gas Phase](#).

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note
Cancel Add

Passage 4 (Questions 18-21)

Blood pressure can be measured in the brachial artery, which has a cross-sectional area of 25 mm^2 in a healthy adult. The blood flow rate is related to the power the person generates to move their body, as shown Figure 1.

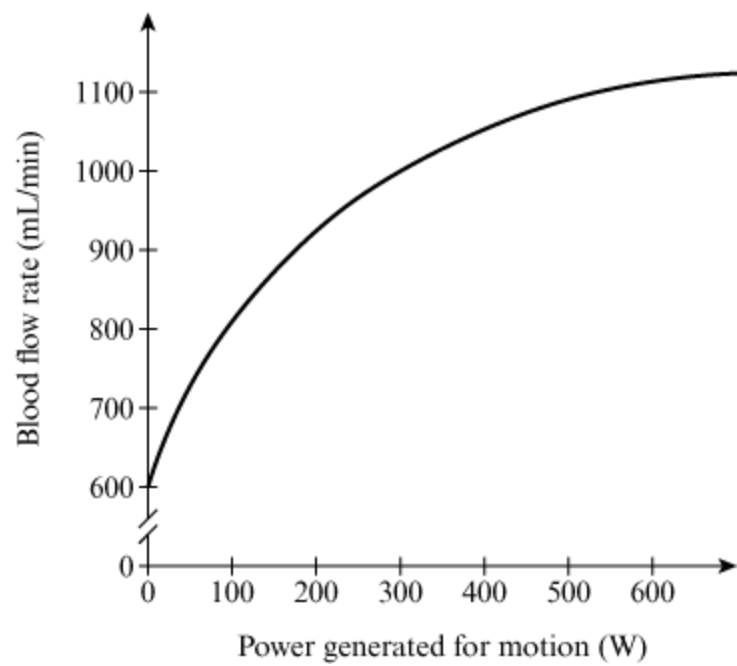


Figure 1	Relationship between the brachial artery blood flow rate of a healthy adult and the power generated for body motion
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Equation 1

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Equation 2

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What fraction of a ^{15}O sample decays in 10 min?

- A.
1/8
- B.
9/16
- C.
3/4
- D.
31/32

Solution: The correct answer is **D**.

1. Based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2 \text{ min} \times \ln(7/8) \div \ln(1/2) = 22$ seconds.
2. Based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2 \text{ min} \times \ln(7/16) \div \ln(1/2) = 2.4$ minutes.
3. Based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2 \text{ min} \times \ln(1/4) \div \ln(1/2) = 4$ minutes.
4. **This answer is correct because based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2 \text{ min} \times \ln(1/32) \div \ln(1/2) = 10$ minutes.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#) and [Electronic Structure](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

1 min 13 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC4E

FC4

PHY

Passage 4 (Questions 18-21)

Blood pressure can be measured in the brachial artery, which has a cross-sectional area of 25 mm^2 in a healthy adult. The blood flow rate is related to the power the person generates to move their body, as shown Figure 1.

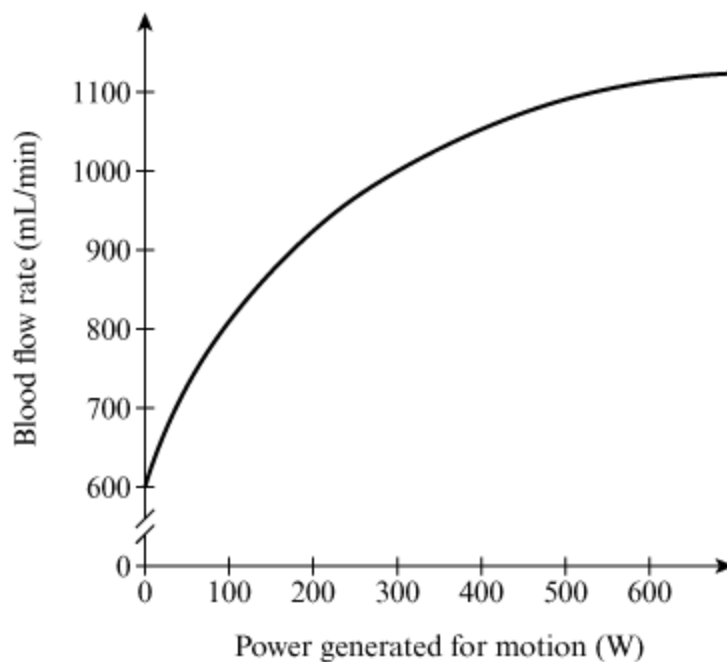


Figure 1	Relationship between the brachial artery blood flow rate of a healthy adult and the power generated for body motion
-----------------	---

An athlete running on a horizontal treadmill maintained a brachial artery blood flow rate of 1.1 L/min for 30 s . The treadmill had an electric circuit used to charge a parallel-plate capacitor of capacitance 1 mF .

The maximum and minimum blood pressures of a healthy adult (the systolic P_s and the diastolic P_d pressures, respectively) are about $P_s = 120 \text{ mmHg}$ and $P_d = 75 \text{ mmHg}$. The mean arterial pressure P_{MPA} is given by

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Equation 1

and the pulse pressure P_p is defined as

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Equation 2

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What fraction of a ^{15}O sample decays in 10 min?

- A.
1/8
- B.
9/16
- C.
3/4
- D.
31/32

Solution: The correct answer is **D**.

1. Based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2 \text{ min} \times \ln(7/8) \div \ln(1/2) = 22$ seconds.
2. Based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2 \text{ min} \times \ln(7/16) \div \ln(1/2) = 2.4$ minutes.
3. Based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2 \text{ min} \times \ln(1/4) \div \ln(1/2) = 4$ minutes.
4. **This answer is correct because based on the radioactive decay law $N(t) = N (1/2)^{t/T}$ where $T = 2$ min, this fraction of ^{15}O decays in $t = 2 \text{ min} \times \ln(1/32) \div \ln(1/2) = 10$ minutes.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#) and [Electronic Structure](#).

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note
Cancel Add

Passage 4 (Questions 18-21)

Blood pressure can be measured in the brachial artery, which has a cross-sectional area of 25 mm^2 in a healthy adult. The blood flow rate is related to the power the person generates to move their body, as shown Figure 1.

Power generated for motion (W)	Blood flow rate (mL/min)
0	600
100	800
200	920
300	1000
400	1060
500	1090
600	1100

Figure 1	Relationship between the brachial artery blood flow rate of a healthy adult and the power generated for body motion
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What is the work generated by a healthy adult who circulates 9 L of blood through the brachial artery in 10 min?

- A.
2 kJ
- B.
12 kJ
- C.
20 kJ
- D.
120 kJ

Solution: The correct answer is **D**.

1. This option is 60 times smaller than the correct work, because a volume flow rate of 9 liters in 10 minutes means 900 mL/min, and according to the figure, this flow rate correlates to a power of 200 W. The correct work is then $200 \text{ W} \times 600 \text{ s} = 120 \text{ kJ}$.
2. This option is 10 times smaller than the correct work, because a volume flow rate of 9 liters in 10 minutes means 900 mL/min and according to the figure, this flow rate correlates to a power of 200 W. The correct work is then $200 \text{ W} \times 600 \text{ s} = 120 \text{ kJ}$.
3. This option is one order of magnitude smaller than the correct work, because a volume flow rate of 9 liters in 10 minutes means 900 mL/min, and according to the figure, this flow rate correlates to a power of 200 W. The correct work is then $200 \text{ W} \times 600 \text{ s} = 120 \text{ kJ}$.
4. **A volume flow rate of 9 liters in 10 minutes means 900 mL/min. According to the figure, this flow rate correlates to a power of 200 W. The work is then $200 \text{ W} \times 600 \text{ s} = 120 \text{ kJ}$.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Vectors](#); [Newton's First Law](#); [Torque](#); and [Work and energy](#).

Incorrect

Your result is incorrect.

Your answer is C.

Correct answer is D.

Correct Answer Your Answer

A

B

C

x

D

Time Spent:

2 mins 7 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS4

CC4A

RESULT / TIMING

Passage 4 (Questions 18-21)

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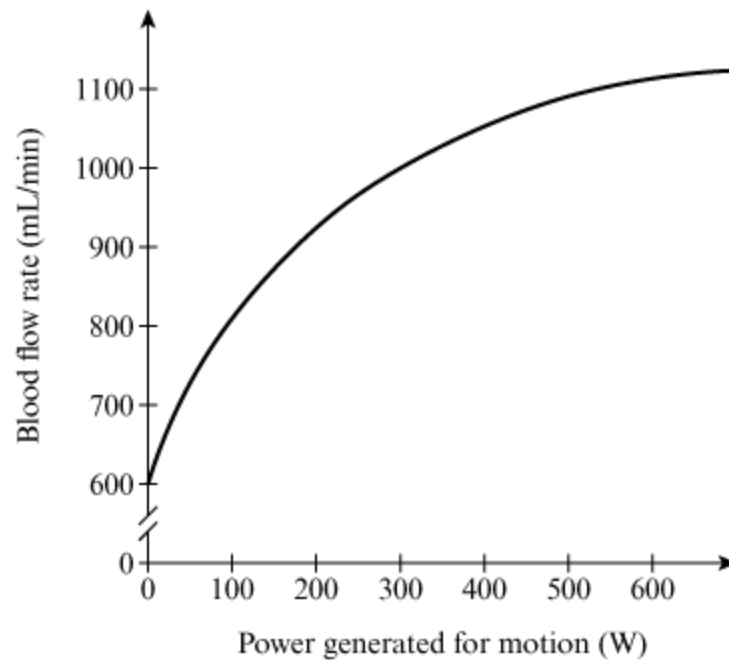


Figure 1 Relationship between the brachial artery blood flow rate of a healthy adult and the power generated for body motion

An athlete running on a horizontal treadmill maintained a brachial artery blood flow rate of 1.1 L/min for 30 s . The treadmill had an electric circuit used to charge a parallel-plate capacitor of capacitance 1 mF .

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Equation 1

and the pulse pressure P_p is defined as

$$P_p = P_s - P_d$$

Equation 2

Ca^{2+} ions are normally found in the blood at a concentration of 100 mg/L. The location of internal organs where changes in the calcium content occur can be found by placing positron-emitting isotopes such as ^{15}O (with a half-life $T_{1/2} = 2$ min) into the patient's bloodstream. When positrons annihilate with electrons, they produce pairs of gamma photons, each with an energy of 500 keV. The location where gamma photons originate gives the location of the organ of interest.

What is the work generated by a healthy adult who circulates 9 L of blood through the brachial artery in 10 min?

- A.
2 kJ
- B.
12 kJ
- C.
20 kJ
- D.
120 kJ

Solution: The correct answer is **D**.

1. This option is 60 times smaller than the correct work, because a volume flow rate of 9 liters in 10 minutes means 900 mL/min, and according to the figure, this flow rate correlates to a power of 200 W. The correct work is then $200 \text{ W} \times 600 \text{ s} = 120 \text{ kJ}$.
2. This option is 10 times smaller than the correct work, because a volume flow rate of 9 liters in 10 minutes means 900 mL/min and according to the figure, this flow rate correlates to a power of 200 W. The correct work is then $200 \text{ W} \times 600 \text{ s} = 120 \text{ kJ}$.
3. This option is one order of magnitude smaller than the correct work, because a volume flow rate of 9 liters in 10 minutes means 900 mL/min, and according to the

figure, this flow rate correlates to a power of 200 W. The correct work is then $200 \text{ W} \times 600 \text{ s} = 120 \text{ kJ}$.

4. **A volume flow rate of 9 liters in 10 minutes means 900 mL/min. According to the figure, this flow rate correlates to a power of 200 W. The work is then $200 \text{ W} \times 600 \text{ s} = 120 \text{ kJ}$.**

KHAN ACADEMY LESSONS

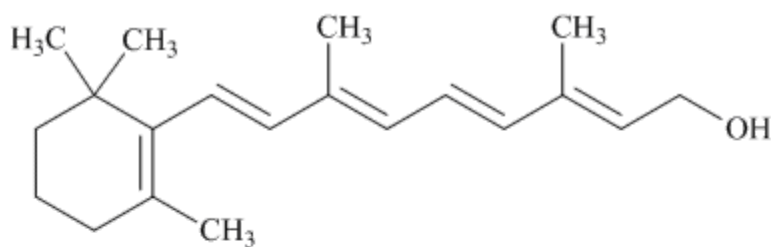
For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Vectors](#); [Newton's First Law](#); [Torque](#); and [Work and energy](#).

Passage 6 (Questions 30-33)

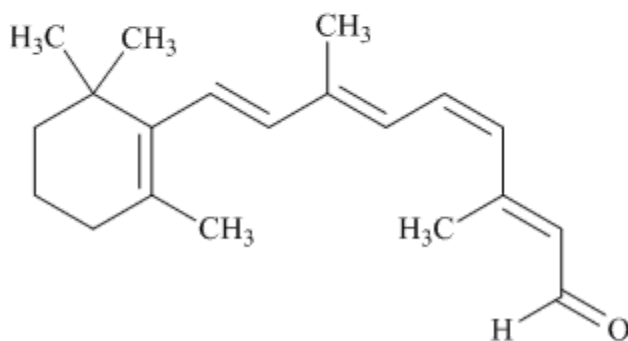
Passage

Passage 6 (Questions 30-33)

Human rhodopsin is a G protein-coupled receptor (GPCR) found in rod cells of the retina. Rhodopsin is composed of opsin, a family of seven helix transmembrane proteins, and a covalently attached coenzyme, retinal. Retinal is derived from all-*trans*-retinol, also known as vitamin A (Figure 1). When rhodopsin is exposed to light, the 11-*cis*-retinal (Figure 1) converts to all-*trans*-retinal. Subsequent structural changes in rhodopsin allow it to interact with and activate the intracellular G protein transducin. The signal is turned off via phosphorylation of rhodopsin by rhodopsin kinase and subsequent binding of the protein arrestin to rhodopsin. Rhodopsin has a maximum absorption at 500 nm. Additional opsin based receptors are expressed in specialized cone cells and have maximum absorptions at 420 nm, 530 nm, and 560 nm when reconstituted with 11-*cis*-retinal.



All-*trans*-retinol



11-*cis*-retinal

Figure 1

Structural representations of all-*trans*-retinol and 11-*cis*-retinal

Previous studies indicated that various intracellular loops of rhodopsin are important for interacting with arrestin. In order to determine which amino acids within the loops have the greatest impact on binding to arrestin, variants of rhodopsin containing one or more amino acid changes were prepared, and their binding to arrestin measured. Researchers focused on two loops, Loop I (Figure 2) and Loop II (Figure 3). Wild-type or variant rhodopsin was reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase in the presence of ATP before examining arrestin binding. In the absence of phosphorylation, arrestin binding to wild-type rhodopsin decreased by 90%.

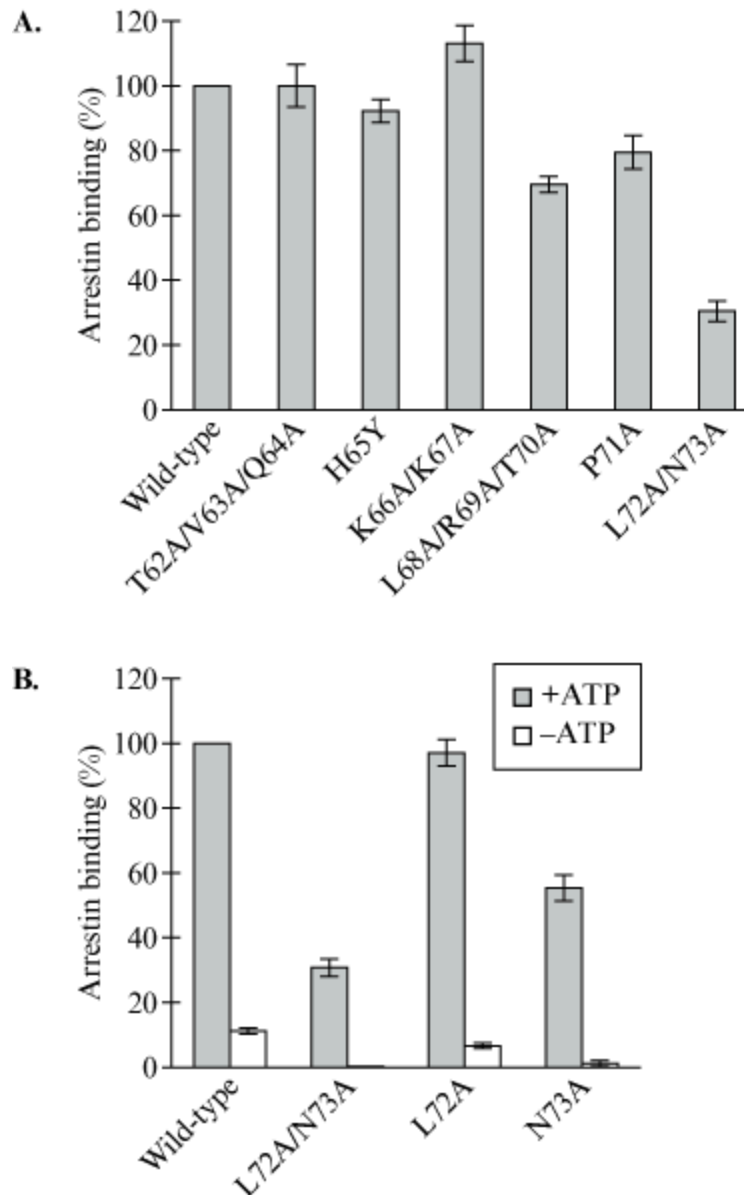


Figure 2

Arrestin binding to Loop I variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

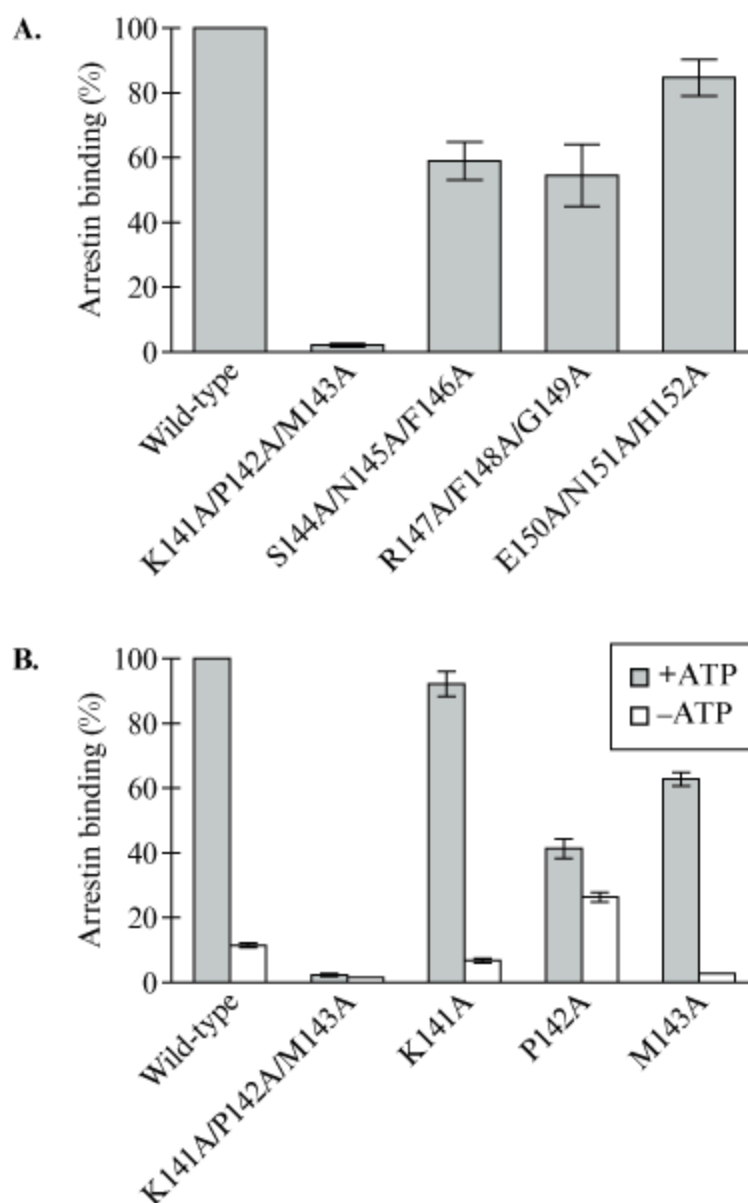


Figure 3 Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (–ATP)

Question 30 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

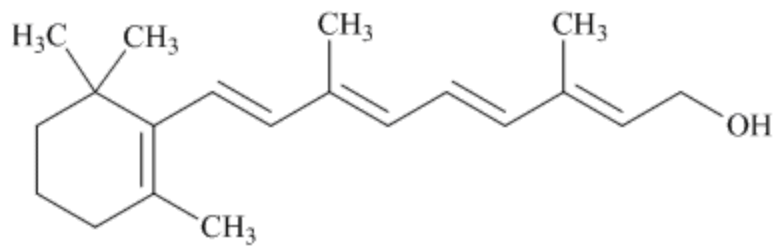
ANSWER CHOICES

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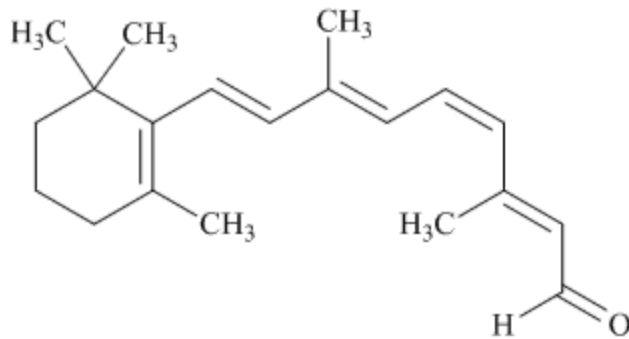
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Passage 6 (Questions 30-33)

Human rhodopsin is a G protein-coupled receptor (GPCR) found in rod cells of the retina. Rhodopsin is composed of opsin, a family of seven helix transmembrane proteins, and a covalently attached coenzyme, retinal. Retinal is derived from all-*trans*-retinol, also known as vitamin A (Figure 1). When rhodopsin is exposed to light, the 11-*cis*-retinal (Figure 1) converts to all-*trans*-retinal. Subsequent structural changes in rhodopsin allow it to interact with and activate the intracellular G protein transducin. The signal is turned off via phosphorylation of rhodopsin by rhodopsin kinase and subsequent binding of the protein arrestin to rhodopsin. Rhodopsin has a maximum absorption at 500 nm. Additional opsin based receptors are expressed in specialized cone cells and have maximum absorptions at 420 nm, 530 nm, and 560 nm when reconstituted with 11-*cis*-retinal.



All-trans-retinol



11-cis-retinal

Figure 1	Structural representations of <i>all-trans</i> -retinol and <i>11-cis</i> -retinal
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Previous studies indicated that various intracellular loops of rhodopsin are important for interacting with arrestin. In order to determine which amino acids within the loops have the greatest impact on binding to arrestin, variants of rhodopsin containing one or more amino acid changes were prepared, and their binding to arrestin measured. Researchers focused on two loops, Loop I (Figure 2) and Loop II (Figure 3). Wild-type or variant rhodopsin was reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase in the presence of ATP before examining arrestin binding. In the absence of phosphorylation, arrestin binding to wild-type rhodopsin decreased by 90%.

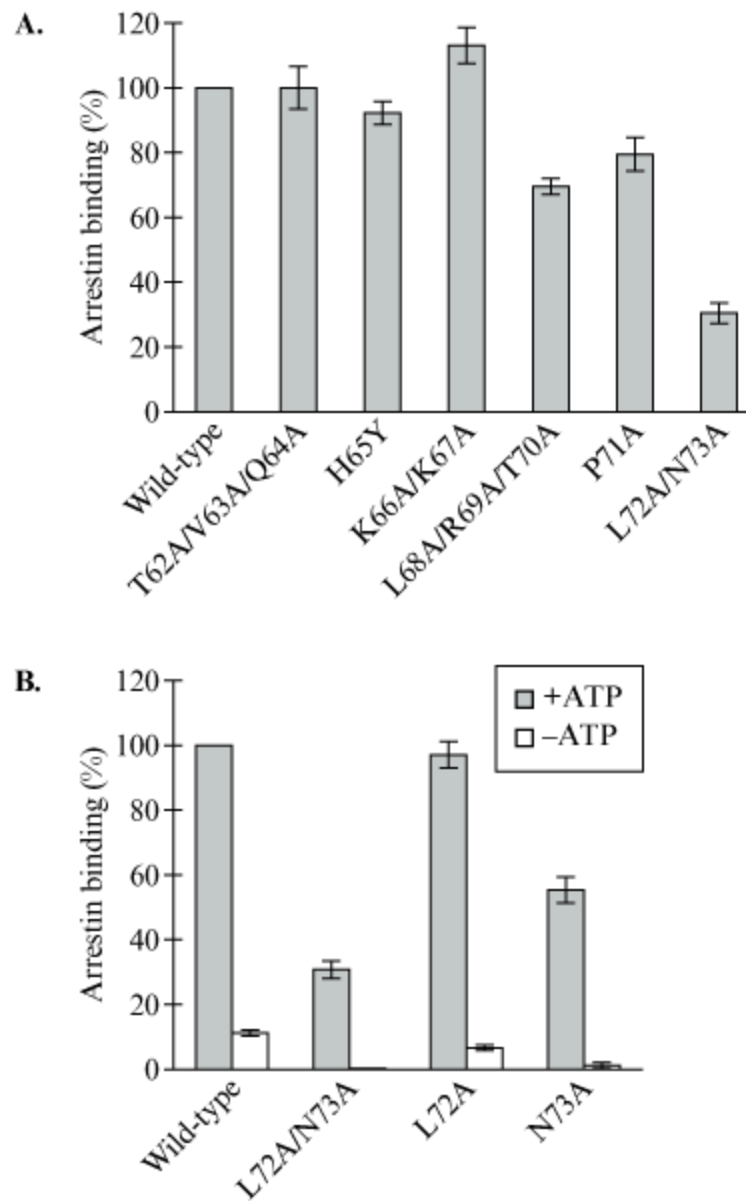


Figure 2 Arrestin binding to Loop I variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (–ATP)

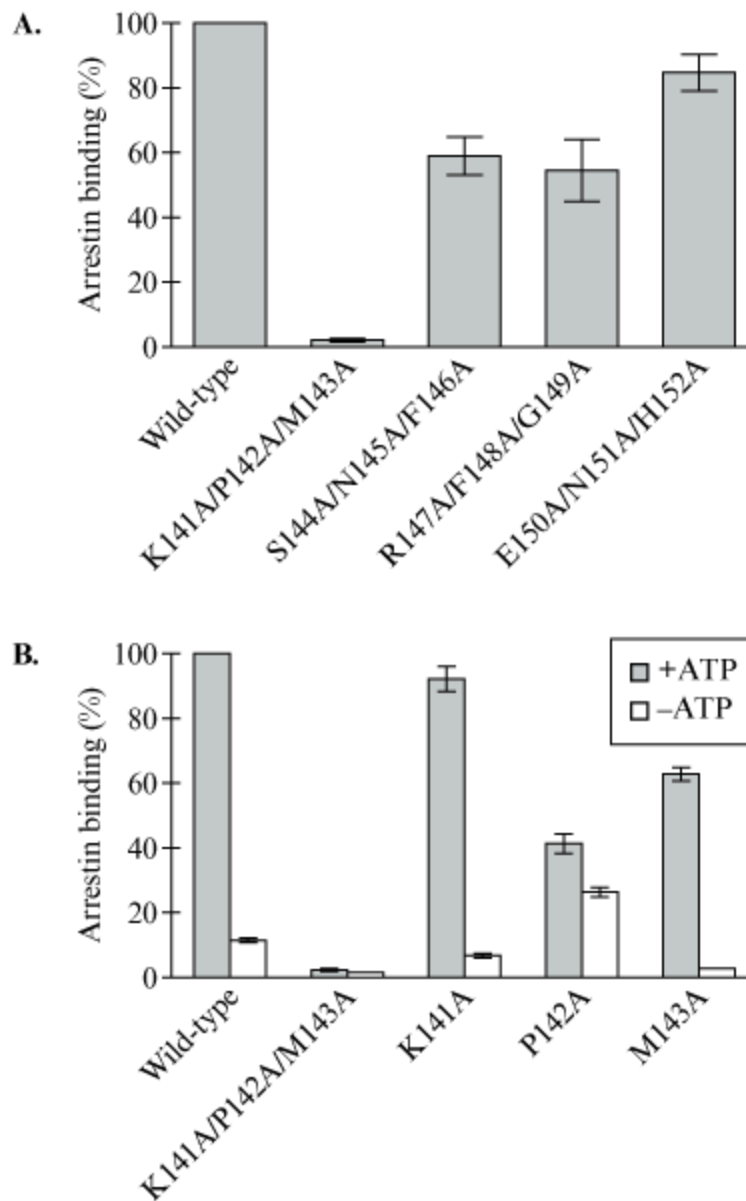


Figure 3 Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

Adapted from Raman, D.; Osawa, S.; and Weiss, E. ©1999 by American Chemical Society

The production of a variety of opsins functions to:

- A.
increase sensitivity to low light.
- B.

- enable the detection of different colors.
- C.
ensure fast recovery of 11-*cis*-retinal after exposure.
- D.
increase refractive index of the eye lens.

Solution: The correct answer is **B**.

1. There is no information in the passage indicating that some opsins are more sensitive to light than others.
2. **Because the wavelengths of maximum light absorption are significantly different for the specialized opsins in the cone cells, these opsins enable the detection of different colors of light.**
3. There is no information in the passage indicating that some opsins have faster recovery of 11-*cis*-retinal after light exposure than others.
4. There is no information in the passage indicating that opsins have any effect on the refractive index of the eye lens.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Molecular Structure and Absorption Spectra](#).

Incorrect

Your result is incorrect.

Your answer is A.

Correct answer is B.

Correct Answer Your Answer

A

x

B

C

D

Time Spent:

1 min 56 secs

Difficulty Level:

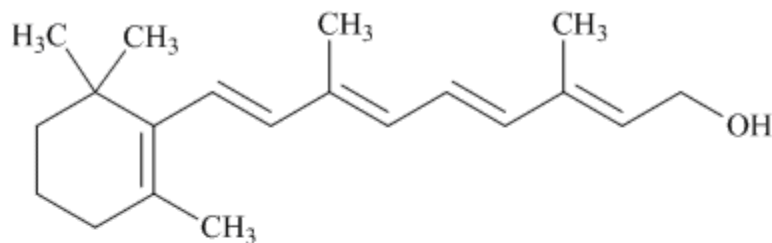
Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2
CC4D
FC4
OCH

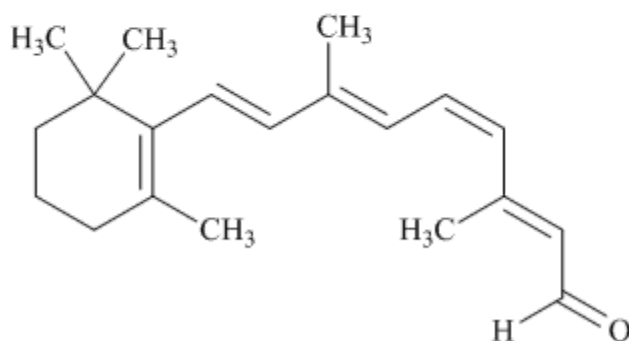
RESULT / TIMING

Passage 6 (Questions 30-33)

Human rhodopsin is a G protein-coupled receptor (GPCR) found in rod cells of the retina. Rhodopsin is composed of opsin, a family of seven helix transmembrane proteins, and a covalently attached coenzyme, retinal. Retinal is derived from all-*trans*-retinol, also known as vitamin A (Figure 1). When rhodopsin is exposed to light, the 11-*cis*-retinal (Figure 1) converts to all-*trans*-retinal. Subsequent structural changes in rhodopsin allow it to interact with and activate the intracellular G protein transducin. The signal is turned off via phosphorylation of rhodopsin by rhodopsin kinase and subsequent binding of the protein arrestin to rhodopsin. Rhodopsin has a maximum absorption at 500 nm. Additional opsin based receptors are expressed in specialized cone cells and have maximum absorptions at 420 nm, 530 nm, and 560 nm when reconstituted with 11-*cis*-retinal.



All-*trans*-retinol



11-*cis*-retinal

Figure 1

Structural representations of all-*trans*-retinol and 11-*cis*-retinal

Previous studies indicated that various intracellular loops of rhodopsin are important for interacting with arrestin. In order to determine which amino acids within the loops have the greatest impact on binding to arrestin, variants of rhodopsin containing one or more amino acid changes were prepared, and their binding to arrestin measured. Researchers focused on two loops, Loop I (Figure 2) and Loop II (Figure 3). Wild-type or variant rhodopsin was reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase in the presence of ATP before examining arrestin binding. In the absence of phosphorylation, arrestin binding to wild-type rhodopsin decreased by 90%.

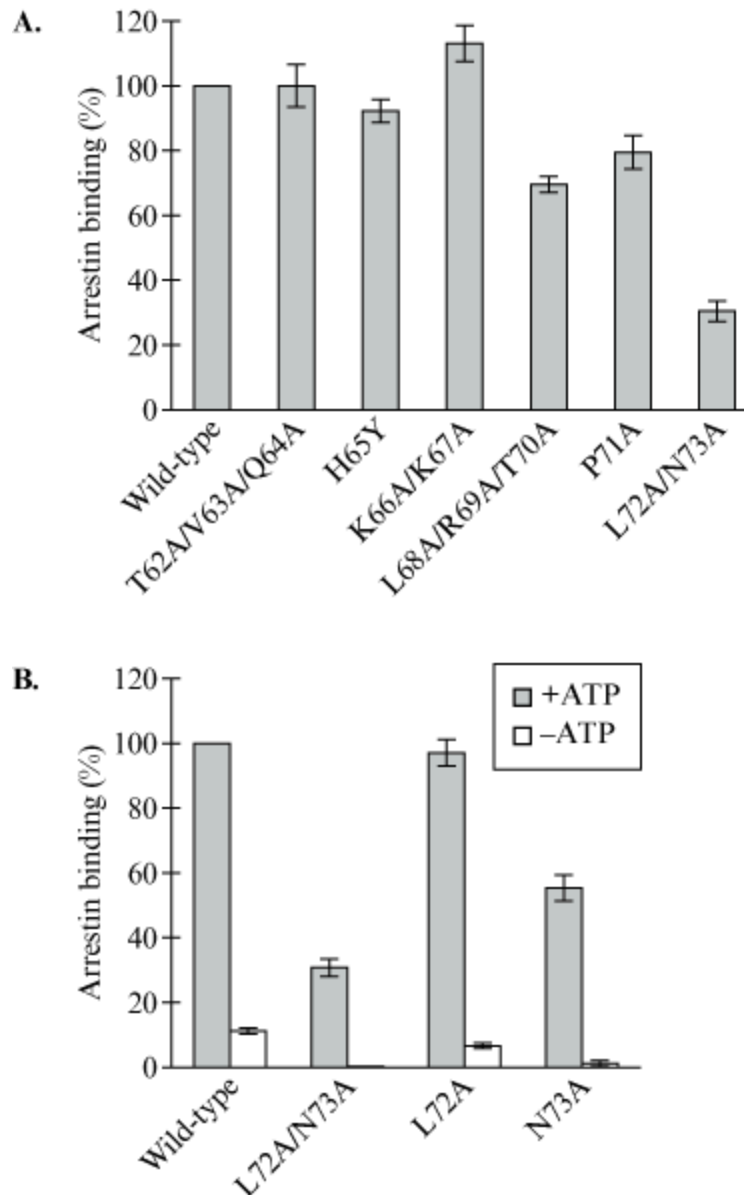


Figure 2

Arrestin binding to Loop I variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

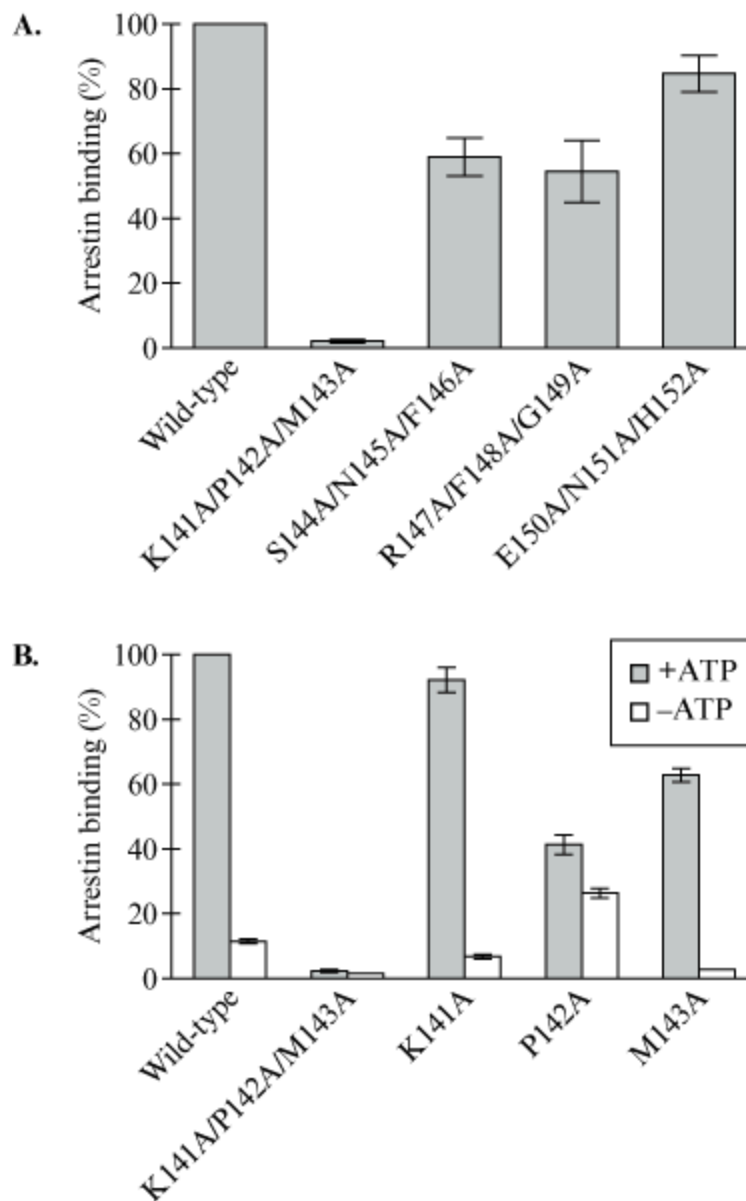


Figure 3 Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

Adapted from Raman, D.; Osawa, S.; and Weiss, E. ©1999 by American Chemical Society

The production of a variety of opsins functions to:

- A.
increase sensitivity to low light.

- B.
enable the detection of different colors.
- C.
ensure fast recovery of 11-*cis*-retinal after exposure.
- D.
increase refractive index of the eye lens.

Solution: The correct answer is **B**.

1. There is no information in the passage indicating that some opsins are more sensitive to light than others.
2. **Because the wavelengths of maximum light absorption are significantly different for the specialized opsins in the cone cells, these opsins enable the detection of different colors of light.**
3. There is no information in the passage indicating that some opsins have faster recovery of 11-*cis*-retinal after light exposure than others.
4. There is no information in the passage indicating that opsins have any effect on the refractive index of the eye lens.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Molecular Structure and Absorption Spectra](#).

Question 31 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

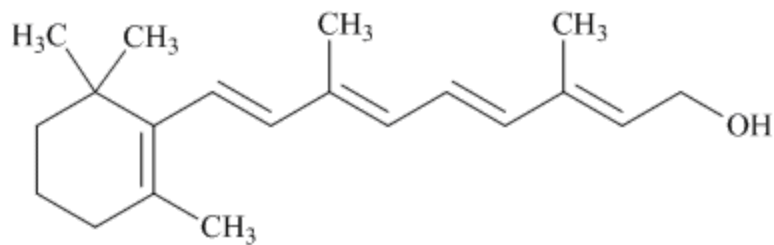
ANSWER CHOICES

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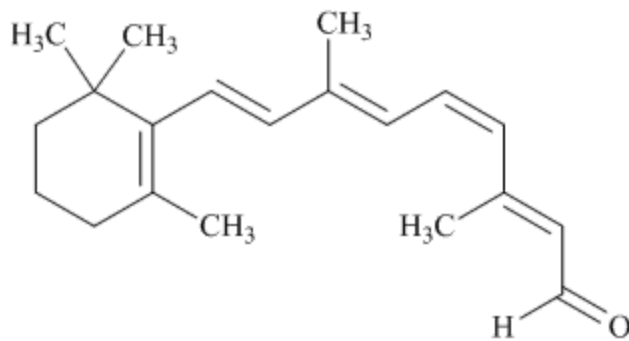
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Passage 6 (Questions 30-33)

Human rhodopsin is a G protein-coupled receptor (GPCR) found in rod cells of the retina. Rhodopsin is composed of opsin, a family of seven helix transmembrane proteins, and a covalently attached coenzyme, retinal. Retinal is derived from all-*trans*-retinol, also known as vitamin A (Figure 1). When rhodopsin is exposed to light, the 11-*cis*-retinal (Figure 1) converts to all-*trans*-retinal. Subsequent structural changes in rhodopsin allow it to interact with and activate the intracellular G protein transducin. The signal is turned off via phosphorylation of rhodopsin by rhodopsin kinase and subsequent binding of the protein arrestin to rhodopsin. Rhodopsin has a maximum absorption at 500 nm. Additional opsin based receptors are expressed in specialized cone cells and have maximum absorptions at 420 nm, 530 nm, and 560 nm when reconstituted with 11-*cis*-retinal.



All-trans-retinol



11-cis-retinal

Figure 1	Structural representations of <i>all-trans</i> -retinol and <i>11-cis</i> -retinal
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Previous studies indicated that various intracellular loops of rhodopsin are important for interacting with arrestin. In order to determine which amino acids within the loops have the greatest impact on binding to arrestin, variants of rhodopsin containing one or more amino acid changes were prepared, and their binding to arrestin measured. Researchers focused on two loops, Loop I (Figure 2) and Loop II (Figure 3). Wild-type or variant rhodopsin was reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase in the presence of ATP before examining arrestin binding. In the absence of phosphorylation, arrestin binding to wild-type rhodopsin decreased by 90%.

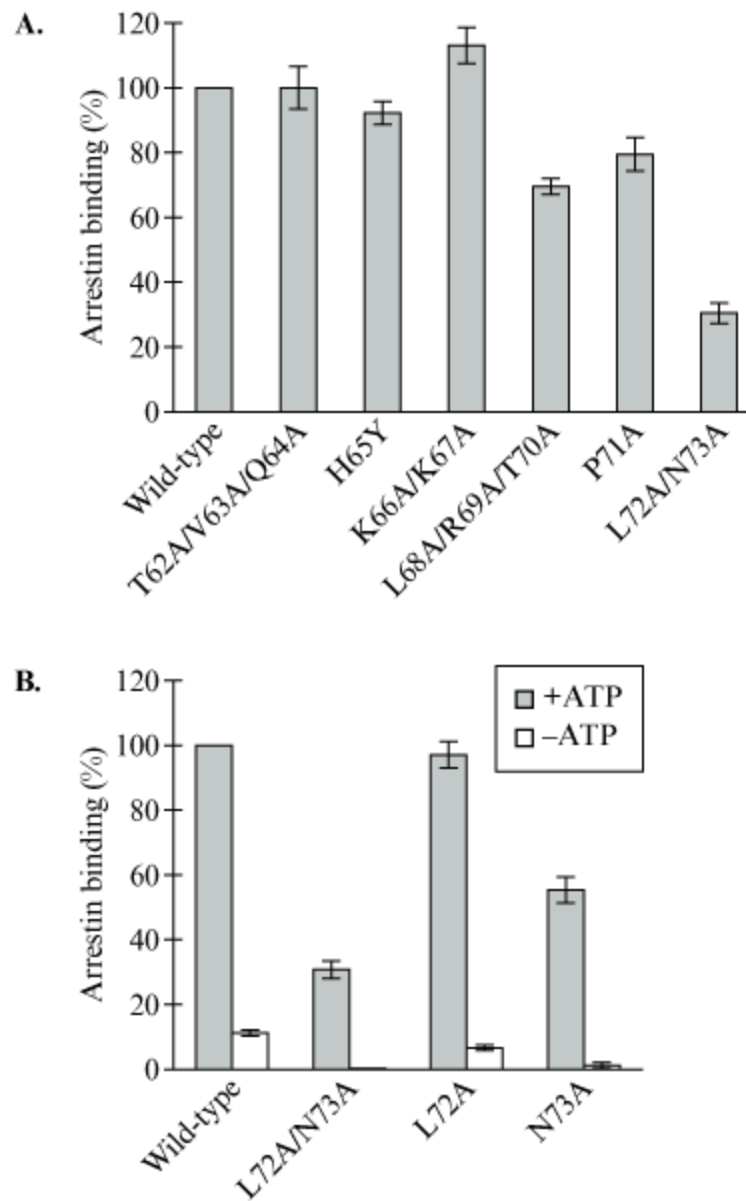


Figure 2

Arrestin binding to Loop I variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (–ATP)

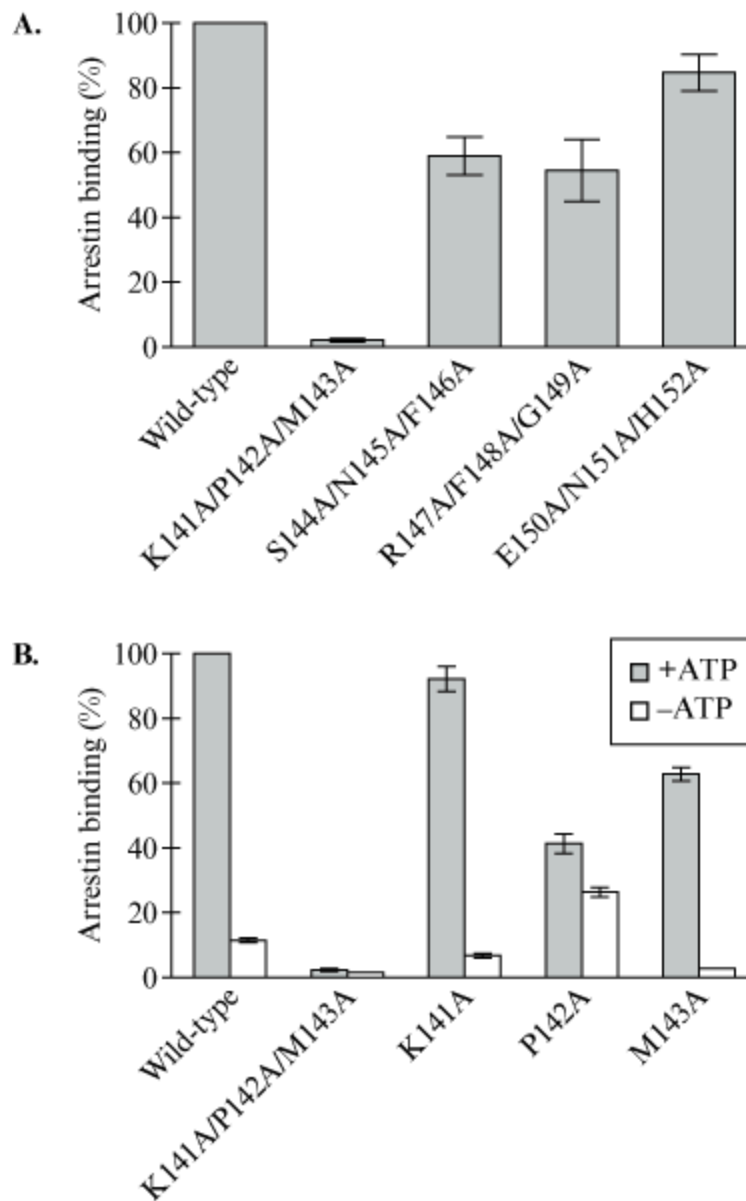


Figure 3

Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

Adapted from Raman, D.; Osawa, S.; and Weiss, E. ©1999 by American Chemical Society

The environment of the retinal binding site is most likely:

- A.
hydrophilic.
- B.

- positively charged.
- C.
negatively charged.
 - D.
hydrophobic.

Solution: The correct answer is **D**.

1. Retinal consists almost entirely of C-H bonds and is non-polar. Hydrophilic molecules have polar bonds and are especially capable of hydrogen-bonding.
2. A positively charged binding domain is well-suited for positioning substrates that have areas of negative charge. Retinal is non-polar.
3. A negatively charged binding domain is well-suited for positioning substrates that feature positive charge. Retinal is non-polar.
4. **Retinal is composed of mainly carbon and hydrogen, making it largely hydrophobic.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Covalent bonds](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

1 min 28 secs

Difficulty Level:

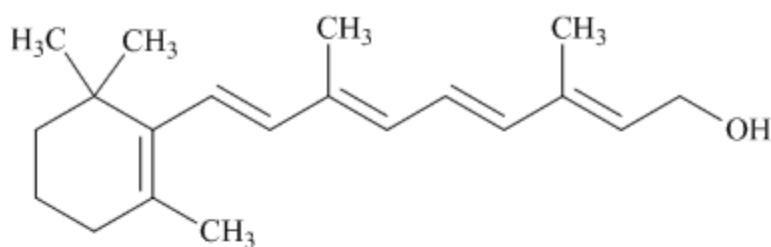
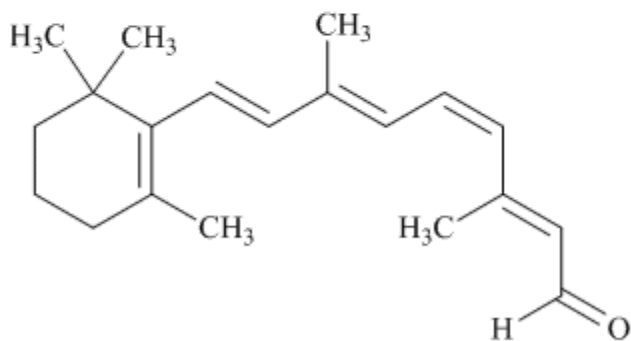
Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

RESULT / TIMING

Passage 6 (Questions 30-33)

Human rhodopsin is a G protein-coupled receptor (GPCR) found in rod cells of the retina. Rhodopsin is composed of opsin, a family of seven helix transmembrane proteins, and a covalently attached coenzyme, retinal. Retinal is derived from all-*trans*-retinol, also known as vitamin A (Figure 1). When rhodopsin is exposed to light, the 11-*cis*-retinal (Figure 1) converts to all-*trans*-retinal. Subsequent structural changes in rhodopsin allow it to interact with and activate the intracellular G protein transducin. The signal is turned off via phosphorylation of rhodopsin by rhodopsin kinase and subsequent binding of the protein arrestin to rhodopsin. Rhodopsin has a maximum absorption at 500 nm. Additional opsin based receptors are expressed in specialized cone cells and have maximum absorptions at 420 nm, 530 nm, and 560 nm when reconstituted with 11-*cis*-retinal.

*All-trans-retinol**11-cis-retinal***Figure 1**Structural representations of all-*trans*-retinol and 11-*cis*-retinal

Previous studies indicated that various intracellular loops of rhodopsin are important for interacting with arrestin. In order to determine which amino acids within the loops have the greatest impact on binding to arrestin, variants of rhodopsin containing one or more amino acid changes were prepared, and their binding to arrestin measured. Researchers focused on two loops, Loop I (Figure 2) and Loop II (Figure 3). Wild-type or variant rhodopsin was reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase in the presence of ATP before examining arrestin binding. In the absence of phosphorylation, arrestin binding to wild-type rhodopsin decreased by 90%.

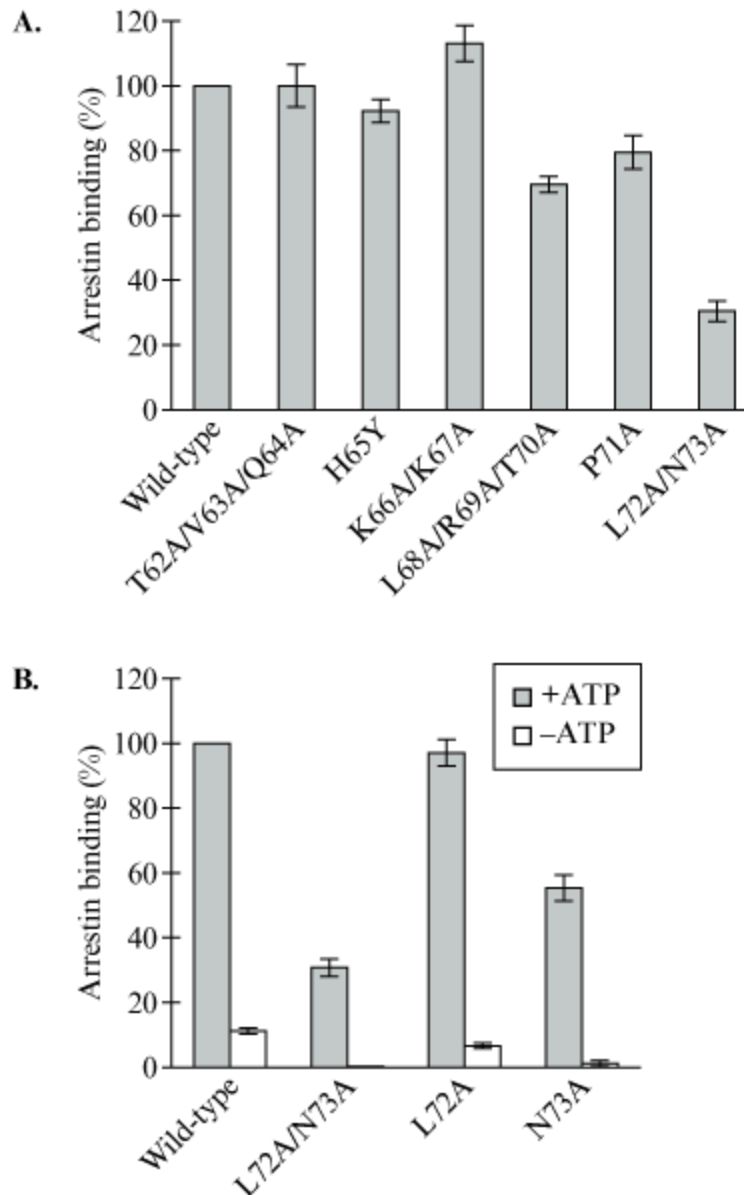


Figure 2

Arrestin binding to Loop I variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

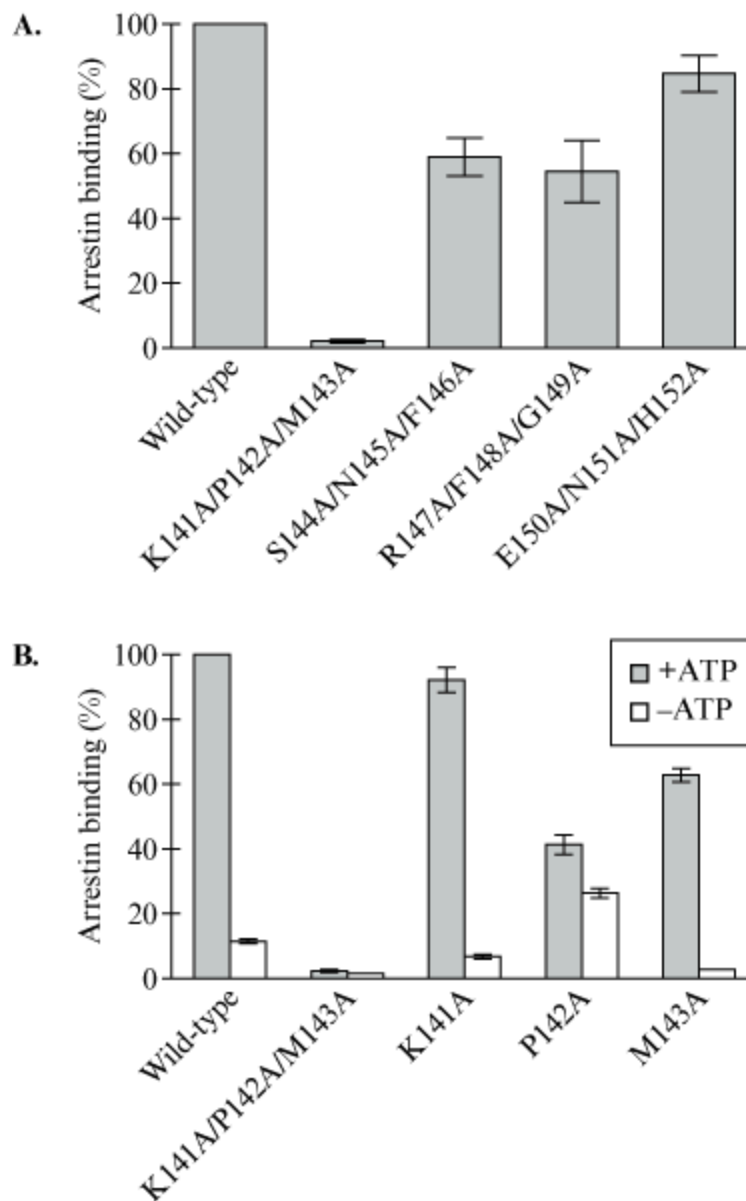


Figure 3 Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (-ATP)

Adapted from Raman, D.; Osawa, S.; and Weiss, E. ©1999 by American Chemical Society

The environment of the retinal binding site is most likely:

- A.
hydrophilic.

- B.
positively charged.
- C.
negatively charged.
- D.
hydrophobic.

Solution: The correct answer is **D**.

1. Retinal consists almost entirely of C-H bonds and is non-polar. Hydrophilic molecules have polar bonds and are especially capable of hydrogen-bonding.
2. A positively charged binding domain is well-suited for positioning substrates that have areas of negative charge. Retinal is non-polar.
3. A negatively charged binding domain is well-suited for positioning substrates that feature positive charge. Retinal is non-polar.
4. **Retinal is composed of mainly carbon and hydrogen, making it largely hydrophobic.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Covalent bonds](#).

Question 32 of 59

QUESTION

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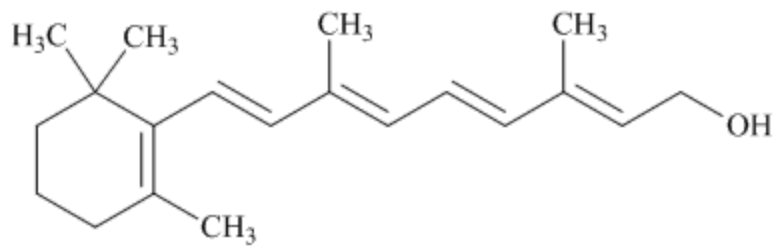
ANSWER CHOICES

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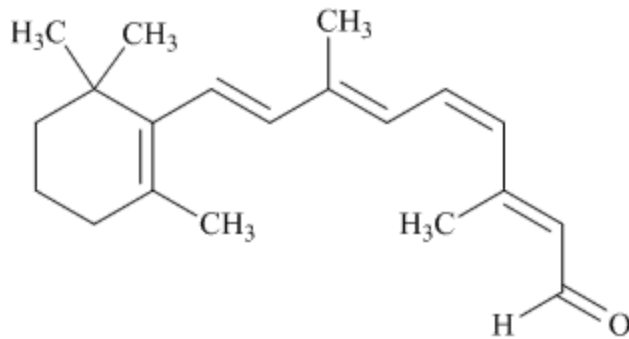
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Passage 6 (Questions 30-33)

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All-trans-retinol



11-cis-retinal

Figure 1	Structural representations of <i>all-trans</i> -retinol and <i>11-cis</i> -retinal
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Previous studies indicated that various intracellular loops of rhodopsin are important for interacting with arrestin. In order to determine which amino acids within the loops have the greatest impact on binding to arrestin, variants of rhodopsin containing one or more amino acid changes were prepared, and their binding to arrestin measured. Researchers focused on two loops, Loop I (Figure 2) and Loop II (Figure 3). Wild-type or variant rhodopsin was reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase in the presence of ATP before examining arrestin binding. In the absence of phosphorylation, arrestin binding to wild-type rhodopsin decreased by 90%.

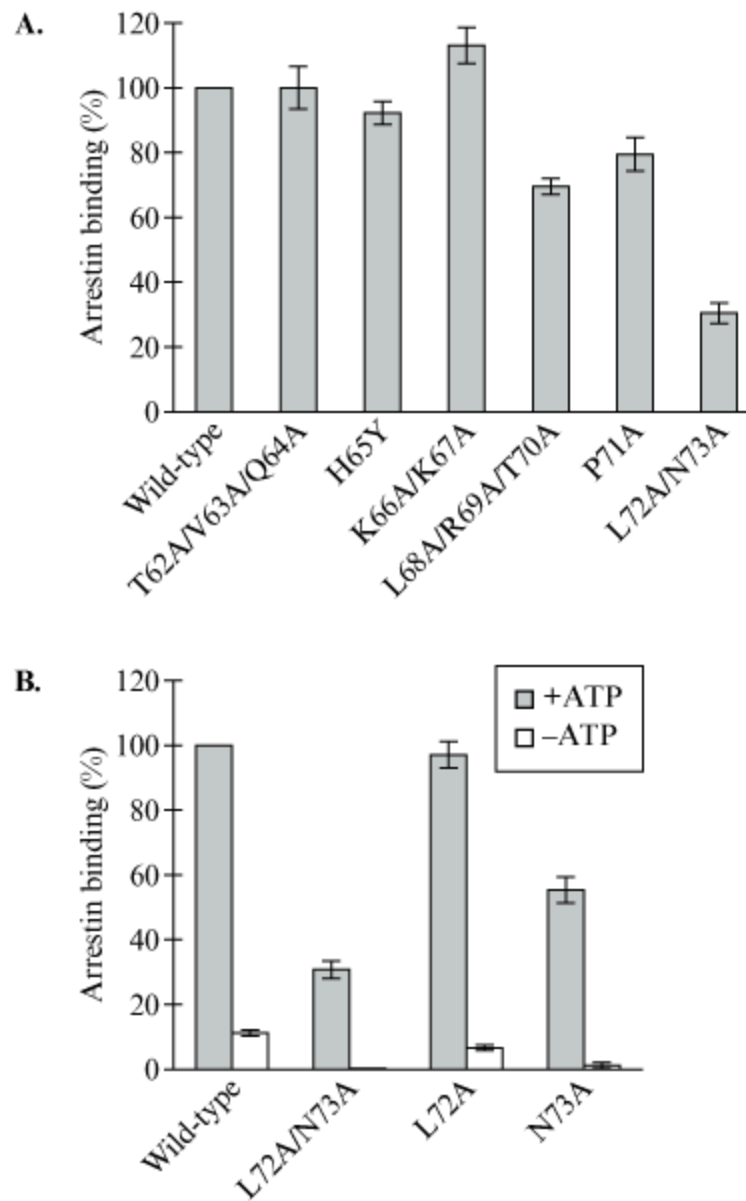


Figure 2 Arrestin binding to Loop I variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (–ATP)

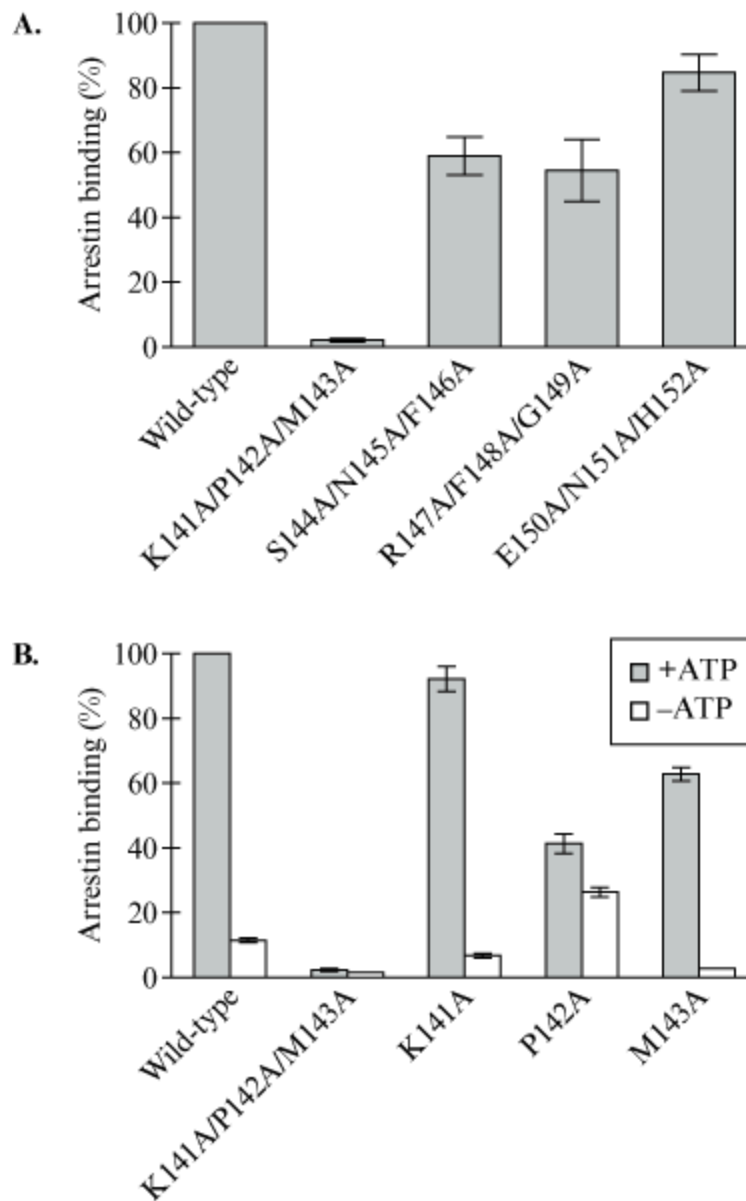


Figure 3

Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (–ATP)

Adapted from Raman, D.; Osawa, S.; and Weiss, E. ©1999 by American Chemical Society

The source of the phosphate groups that are added to rhodopsin is:

- A.
arrestin.
- B.

rhodopsin kinase.

- C.
ATP.
- D.
all-*trans*-retinol.

Solution: The correct answer is **C**.

1. Arrestin binds to rhodopsin after rhodopsin has been phosphorylated. Arrestin is not the source of the phosphate groups.
2. Rhodopsin kinase catalyzes the reaction between ATP and rhodopsin that causes rhodopsin to be phosphorylated, but ATP is the source of the phosphate.
3. **The substrate for rhodopsin kinase that the source of phosphate is ATP.**
4. Retinol contains no phosphate groups.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Enzymes](#); [Kinetics](#); and [Bioenergetics](#).

Correct

Your result is correct.

Your answer is C.

Correct answer is C.

Correct Answer Your Answer

A

B

C

✓

D

Time Spent:

28 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

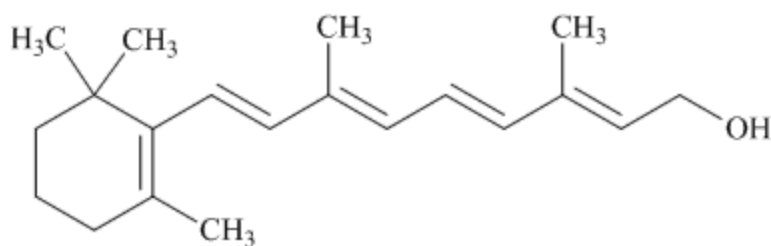
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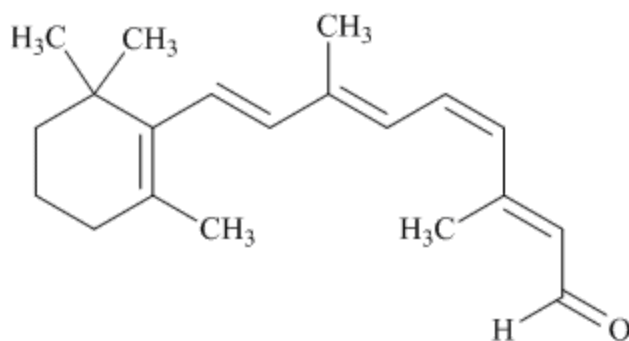
RESULT / TIMING

Passage 6 (Questions 30-33)

Human rhodopsin is a G protein-coupled receptor (GPCR) found in rod cells of the retina. Rhodopsin is composed of opsin, a family of seven helix transmembrane proteins, and a covalently attached coenzyme, retinal. Retinal is derived from all-*trans*-retinol, also known as vitamin A (Figure 1). When rhodopsin is exposed to light, the 11-*cis*-retinal (Figure 1) converts to all-*trans*-retinal. Subsequent structural changes in rhodopsin allow it to interact with and activate the intracellular G protein transducin. The signal is turned off via phosphorylation of rhodopsin by rhodopsin kinase and subsequent binding of the protein arrestin to rhodopsin. Rhodopsin has a maximum absorption at 500 nm. Additional opsin based receptors are expressed in specialized cone cells and have maximum absorptions at 420 nm, 530 nm, and 560 nm when reconstituted with 11-*cis*-retinal.



All-*trans*-retinol



11-*cis*-retinal

Figure 1

Structural representations of all-*trans*-retinol and 11-*cis*-retinal

Previous studies indicated that various intracellular loops of rhodopsin are important for interacting with arrestin. In order to determine which amino acids within the loops have the greatest impact on binding to arrestin, variants of rhodopsin containing one or more amino acid changes were prepared, and their binding to arrestin measured. Researchers focused on two loops, Loop I (Figure 2) and Loop II (Figure 3). Wild-type or variant rhodopsin was reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase in the presence of ATP before examining arrestin binding. In the absence of phosphorylation, arrestin binding to wild-type rhodopsin decreased by 90%.

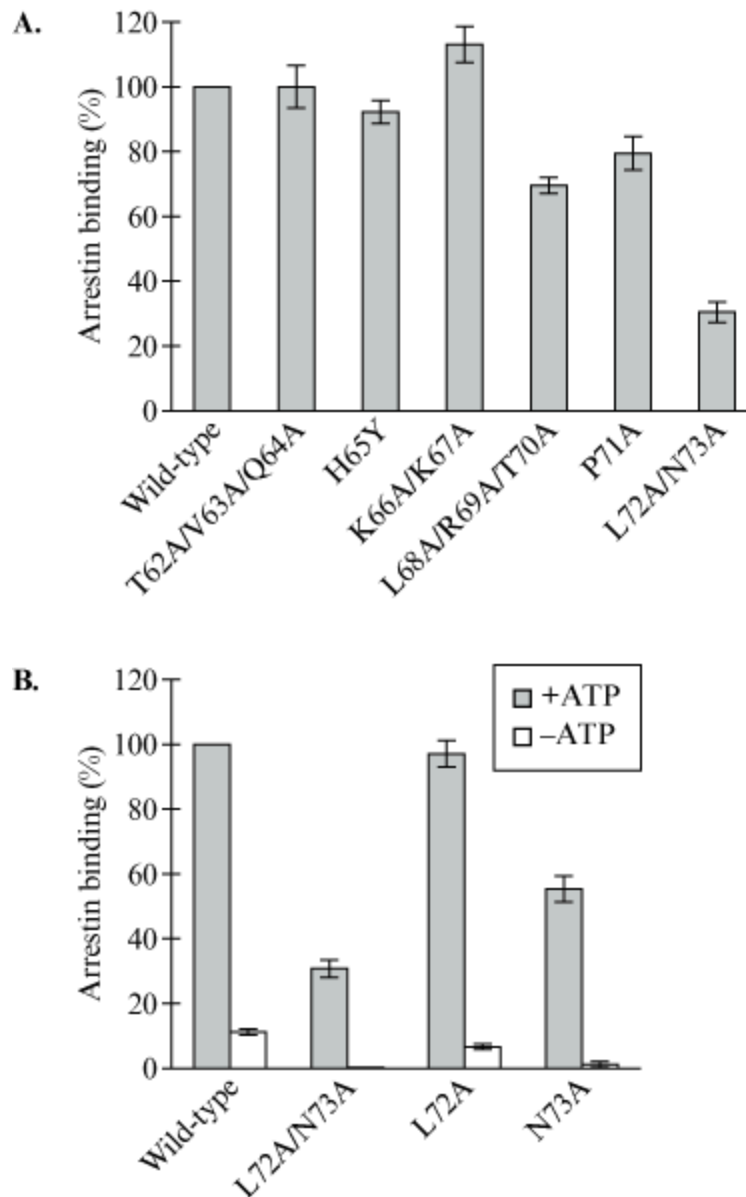


Figure 2 Arrestin binding to Loop I variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (–ATP)

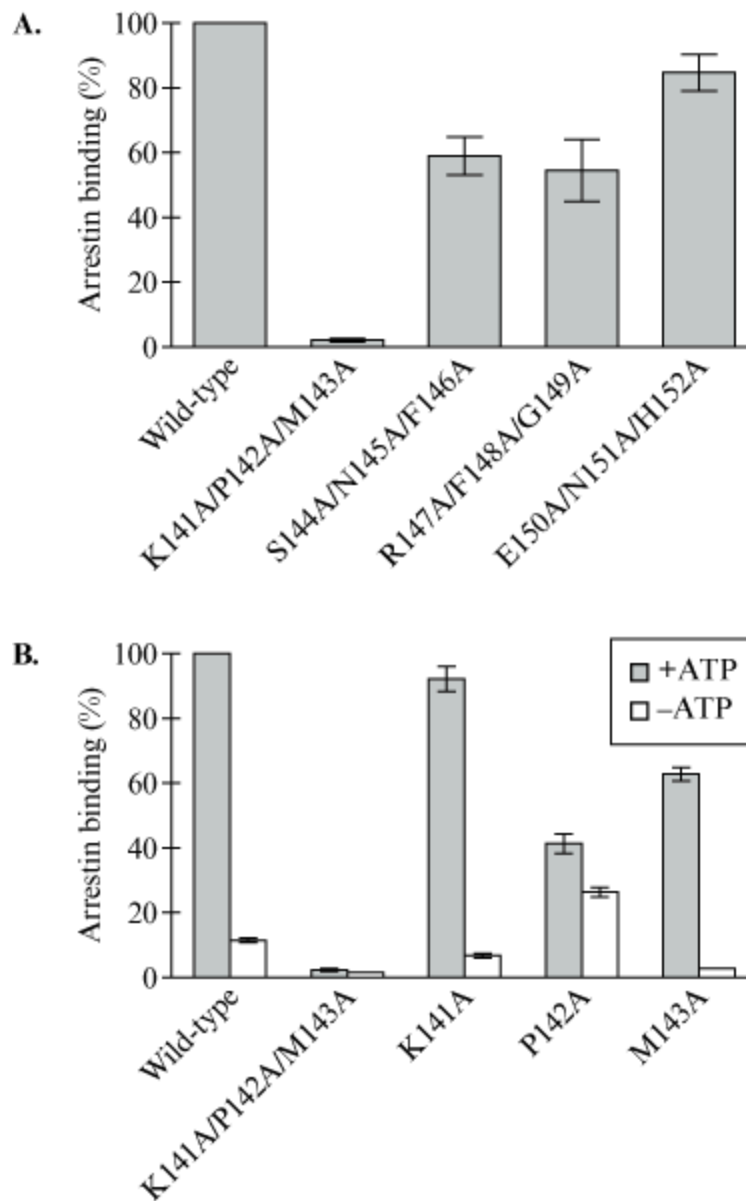


Figure 3

Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (–ATP)

Adapted from Raman, D.; Osawa, S.; and Weiss, E. ©1999 by American Chemical Society

The source of the phosphate groups that are added to rhodopsin is:

- A.
arrestin.
- B.

rhodopsin kinase.

- C.
ATP.
- D.
all-*trans*-retinol.

Solution: The correct answer is **C**.

1. Arrestin binds to rhodopsin after rhodopsin has been phosphorylated. Arrestin is not the source of the phosphate groups.
2. Rhodopsin kinase catalyzes the reaction between ATP and rhodopsin that causes rhodopsin to be phosphorylated, but ATP is the source of the phosphate.
3. **The substrate for rhodopsin kinase that the source of phosphate is ATP.**
4. Retinol contains no phosphate groups.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Enzymes](#); [Kinetics](#); and [Bioenergetics](#).

Question 33 of 59

QUESTION

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MCAT Official Prep Practice Exam 1

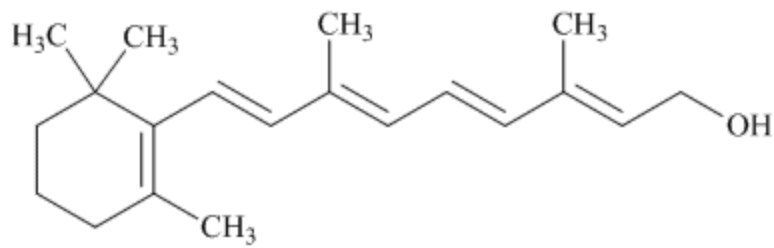
ANSWER CHOICES

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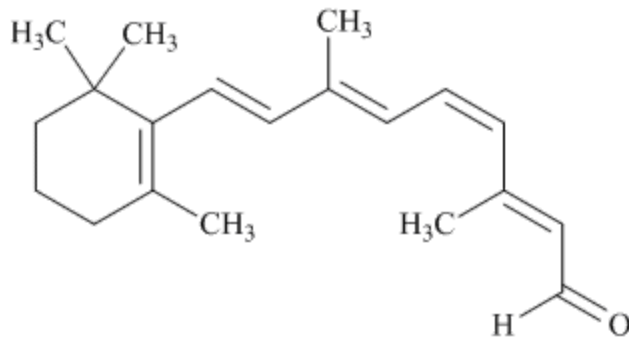
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Passage 6 (Questions 30-33)

Human rhodopsin is a G protein-coupled receptor (GPCR) found in rod cells of the retina. Rhodopsin is composed of opsin, a family of seven helix transmembrane proteins, and a covalently attached coenzyme, retinal. Retinal is derived from all-*trans*-retinol, also known as vitamin A (Figure 1). When rhodopsin is exposed to light, the 11-*cis*-retinal (Figure 1) converts to all-*trans*-retinal. Subsequent structural changes in rhodopsin allow it to interact with and activate the intracellular G protein transducin. The signal is turned off via phosphorylation of rhodopsin by rhodopsin kinase and subsequent binding of the protein arrestin to rhodopsin. Rhodopsin has a maximum absorption at 500 nm. Additional opsin based receptors are expressed in specialized cone cells and have maximum absorptions at 420 nm, 530 nm, and 560 nm when reconstituted with 11-*cis*-retinal.



All-trans-retinol



11-cis-retinal

Figure 1	Structural representations of <i>all-trans</i> -retinol and <i>11-cis</i> -retinal
-----------------	--

Previous studies indicated that various intracellular loops of rhodopsin are important for interacting with arrestin. In order to determine which amino acids within the loops have the greatest impact on binding to arrestin, variants of rhodopsin containing one or more amino acid changes were prepared, and their binding to arrestin measured. Researchers focused on two loops, Loop I (Figure 2) and Loop II (Figure 3). Wild-type or variant rhodopsin was reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase in the presence of ATP before examining arrestin binding. In the absence of phosphorylation, arrestin binding to wild-type rhodopsin decreased by 90%.

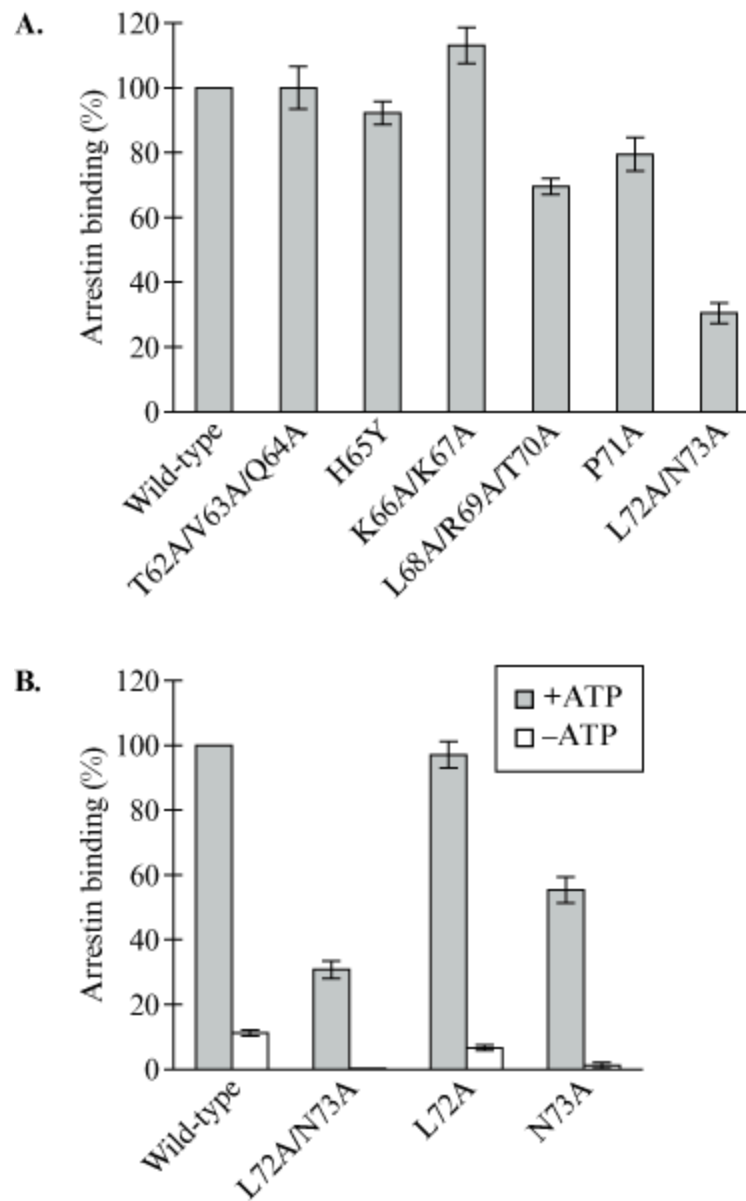


Figure 2

Arrestin binding to Loop I variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (–ATP)

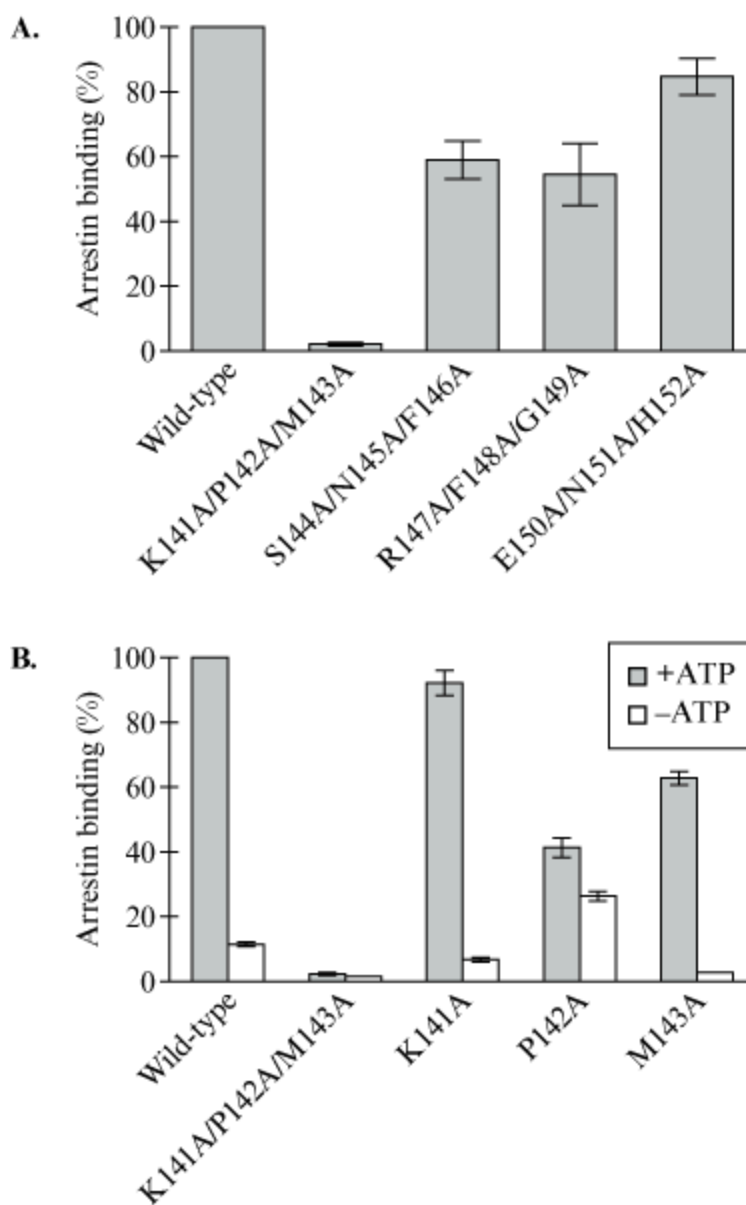


Figure 3

Arrestin binding to Loop II variants of rhodopsin: (A) Reconstituted with 11-*cis*-retinal and phosphorylated by rhodopsin kinase; (B) Reconstituted with 11-*cis*-retinal in the presence of rhodopsin kinase with ATP (+ATP) or without ATP (–ATP)

Adapted from Raman, D.; Osawa, S.; and Weiss, E. ©1999 by American Chemical Society

What are the hybridization states of the carbon atoms involved in the conversion of *trans* to *cis* retinal?

- A.
sp

- B.
 sp^2
- C.
 sp^3
- D.
 sp^3d

Solution: The correct answer is **B**.

1. A carbon atom that is sp hybridized is part of an AX_2 system. AX_2 systems are linear according to VSEPR.
2. **The reacting carbon atoms are both central to AX_3 systems with 3 bonded atoms and no lone pairs. The preferred geometry for such a system is trigonal planar and the hybridization scheme that facilitates this geometry is sp^2 .**
3. Tetrahedral AX_4 compounds and other compounds based on a tetrahedral arrangement of electron pairs utilize sp^3 hybrid orbitals. What is depicted is a geometric rearrangement at AX_3 systems, which are trigonal planar.
4. Transition metals are capable of using d orbitals in bonding. An sp^3d system has 5 electron pair domains and is based on a trigonal bipyramidal or other 5-coordinate geometry.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Covalent bonds](#).

Correct

Your result is correct.

Your answer is B.

Correct answer is B.

Correct Answer	Your Answer
A	
B	
✓	
C	
D	

Time Spent:

1 min 2 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC5B

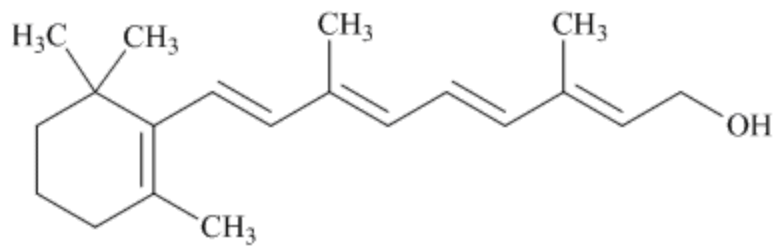
FC5

GCH

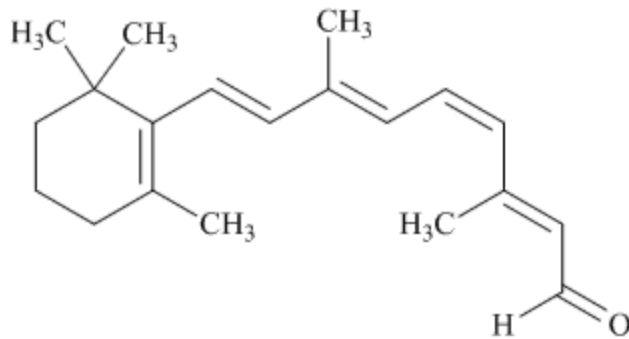
RESULT / TIMING

Passage 6 (Questions 30-33)

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All-trans-retinol



11-cis-retinal

Figure 1	Structural representations of <i>all-trans-retinol</i> and <i>11-cis-retinal</i>
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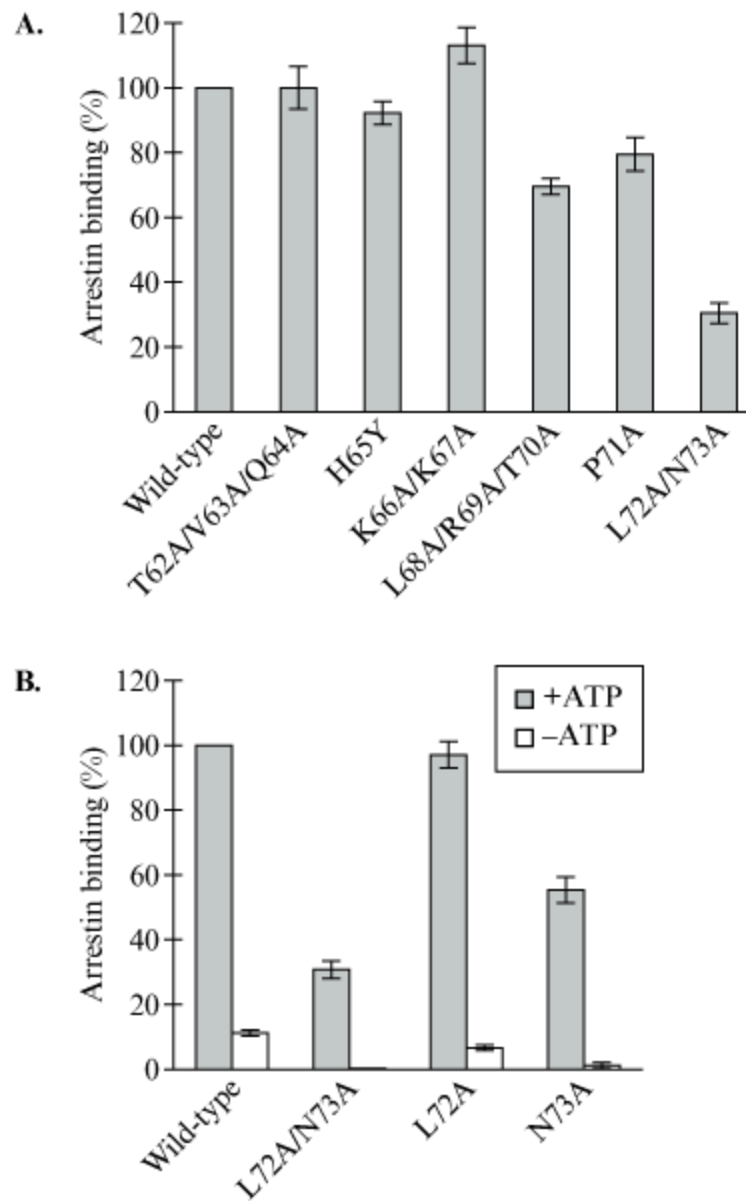


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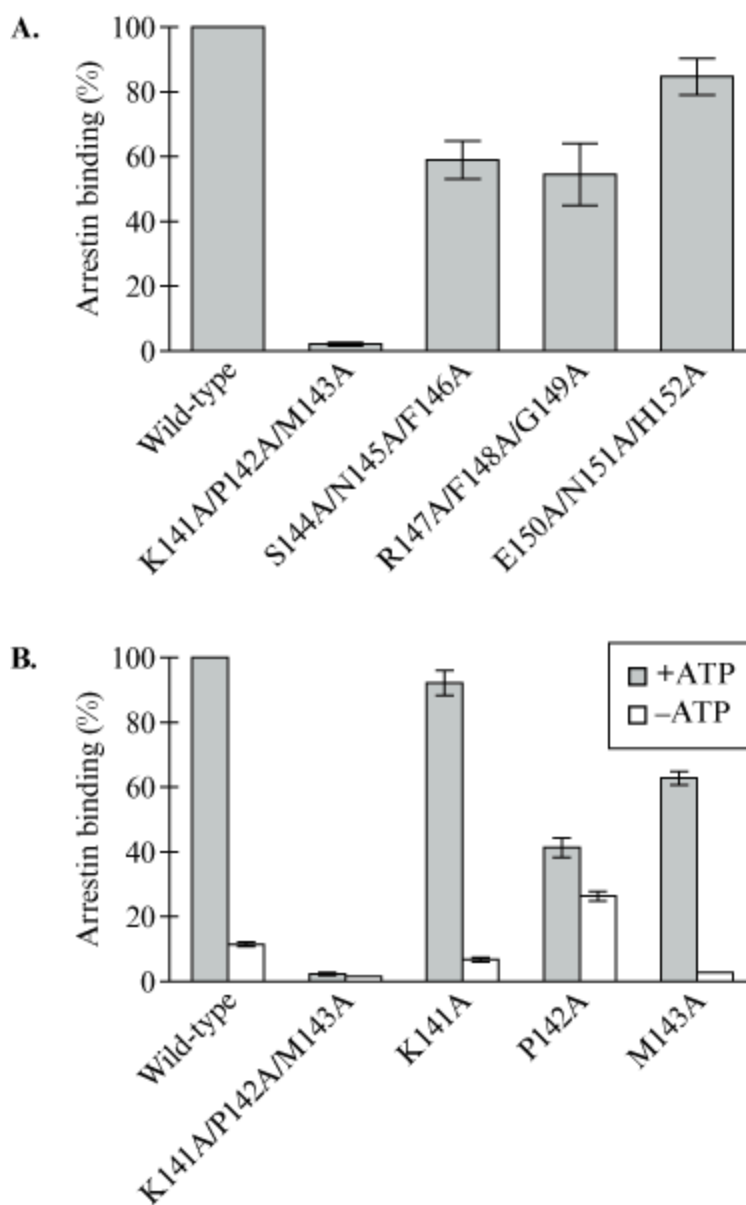


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Adapted from Raman, D.; Osawa, S.; and Weiss, E. ©1999 by American Chemical Society

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- A.
sp

- B.
 sp^2
- C.
 sp^3
- D.
 sp^3d

Solution: The correct answer is **B**.

1. A carbon atom that is sp hybridized is part of an AX_2 system. AX_2 systems are linear according to VSEPR.
2. **The reacting carbon atoms are both central to AX_3 systems with 3 bonded atoms and no lone pairs. The preferred geometry for such a system is trigonal planar and the hybridization scheme that facilitates this geometry is sp^2 .**
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KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Covalent bonds](#).

Passage 8 (Questions 38-43)

Passage

Passage 8 (Questions 38-43)

Glycogen synthase catalyzes the lengthening of glycogen polymers through the addition of glucose. The reaction consists of transferring a glucosyl moiety from uridine diphosphate glucose (UDP-glucose, Figure 1) to the nonreducing end of the glycogen molecule along with the release of UDP. UDP-glucose is formed from the enzymatically-catalyzed reaction between glucose 1-phosphate and UTP. It was discovered that approximately one out of every 600–1500 glucose molecules in muscle glycogen contains phosphate groups covalently linked to the C2 and C3 positions as monoesters. It is hypothesized that glycogen synthase is responsible for the phosphorylation.

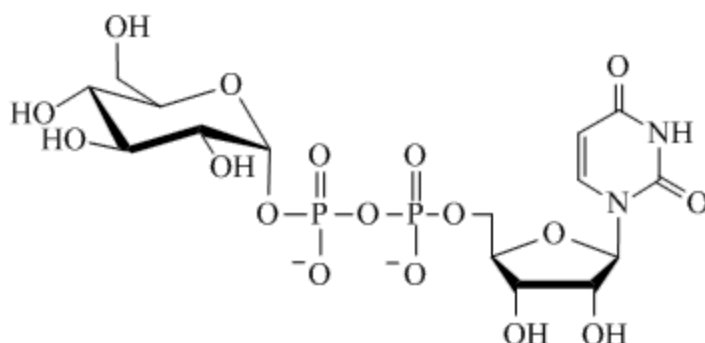


Figure 1	The structure of UDP-glucose
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Lafora disease is a fatal progressive epilepsy that is characterized by the formation of deposits of sparsely branched, insoluble glycogen-like polymers called Lafora bodies (LBs). LBs contain a much larger amount of covalently-linked phosphate compared to normal glycogen. The abundance of phosphate has been linked to a mutation in the gene for laforin, a phosphatase specific to carbohydrates.

In order to test if glycogen synthase is responsible for glycogen phosphorylation, both UDP-[U- ^{14}C]glucose and [β - ^{32}P]UDP-glucose were incubated with muscle extracts from wild-type (WT) mice and mice lacking muscle glycogen synthase (MGSKO). The reaction products were separated by SDS-PAGE and visualized by autoradiography that detects the products of β^- decay (Figure 2). The data were

normalized to account for the different decay rates. The samples were also treated with glucosidase, an enzyme that breaks down glycogen, and assayed in the same manner.

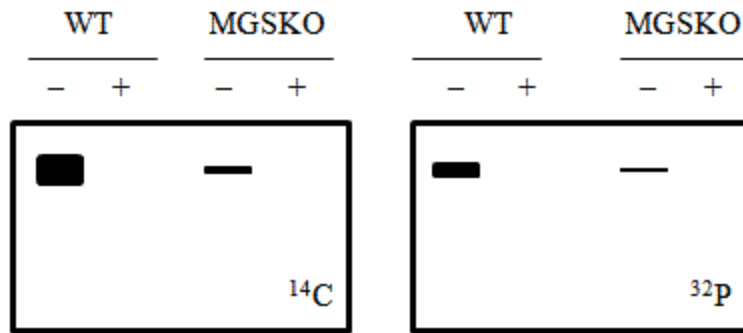


Figure 2

SDS-PAGE analysis from muscle extracts visualized with either ¹⁴C or ³²P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 8 (Questions 38-43)

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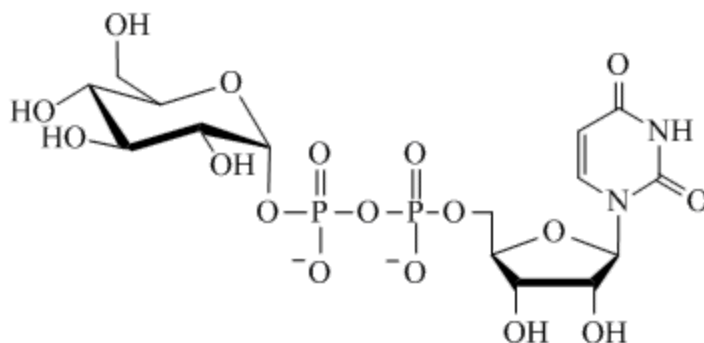


Figure 1	The structure of UDP-glucose
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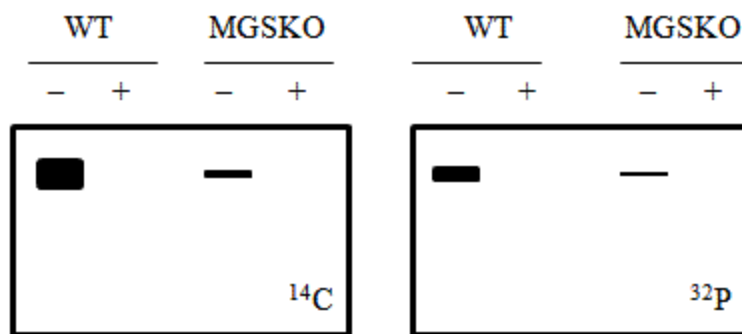


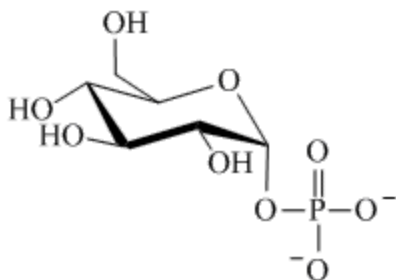
Figure 2

SDS-PAGE analysis from muscle extracts visualized with either ^{14}C or ^{32}P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

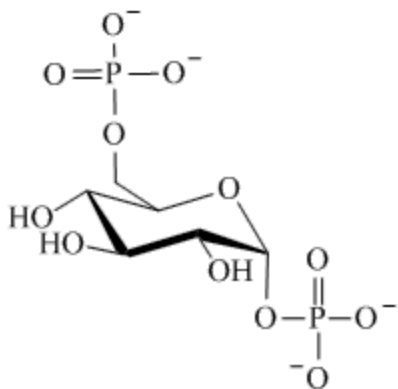
Adapted from Tagliabracci, V. et al. *Cell Metabolism*. © 2011 Elsevier Inc.

Which structure represents a phosphorylated subunit of glycogen as described in the passage?

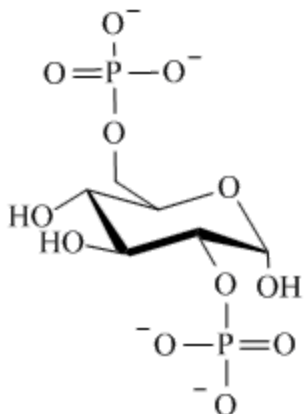
- A.



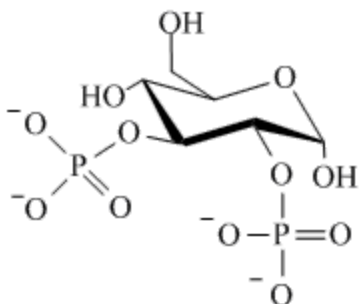
- B.



• C.



• D.



Solution: The correct answer is **D**.

1. This glucose molecule is phosphorylated at C1 instead of at C2 and C3.
2. This glucose molecule is phosphorylated at C1 and C6 instead of at C2 and C3.
3. This glucose molecule is phosphorylated at C2 and C6. The phosphorylation at C6 should be at C3.
4. **According to the passage, glycogen has phosphate monoesters at C2 and C3 in one out of every 600–1500 glucose molecules. This glucose molecule is phosphorylated at C2 and C3.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#); [Amino Acids](#),

[Peptides, Proteins; The Three-Dimensional Protein Structure, Nonenzymatic Protein Function](#); and [Phenols, Polycyclic and Heterocyclic Aromatic Compounds](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

1 min 27 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC5D

FC5

BCH

RESULT / TIMING

Passage 8 (Questions 38-43)

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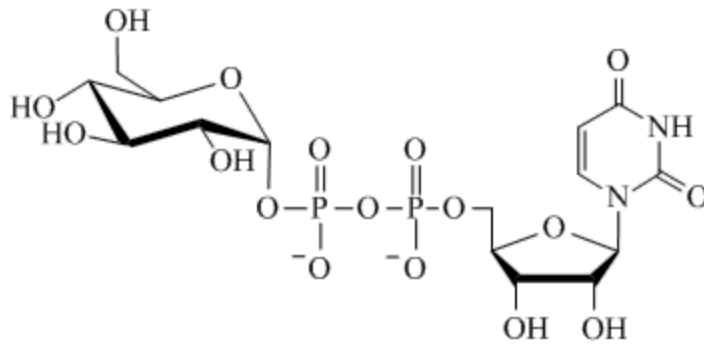


Figure 1 The structure of UDP-glucose

Lafora disease is a fatal progressive epilepsy that is characterized by the formation of deposits of sparsely branched, insoluble glycogen-like polymers called Lafora bodies (LBs). LBs contain a much larger amount of covalently-linked phosphate compared to normal glycogen. The abundance of phosphate has been linked to a mutation in the gene for laforin, a phosphatase specific to carbohydrates.

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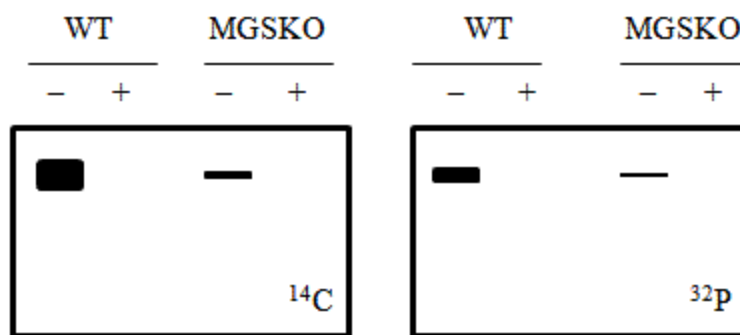
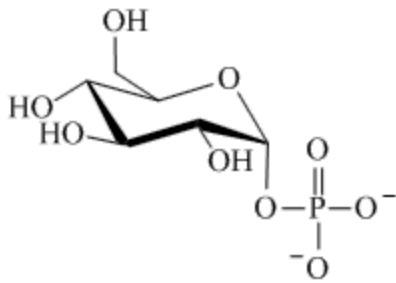


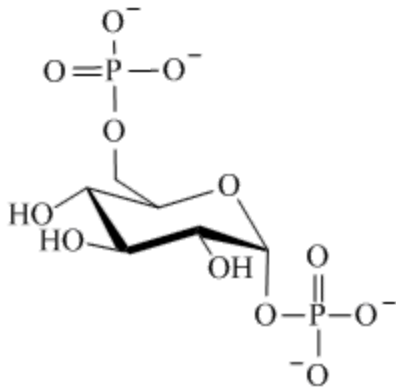
Figure 2 SDS-PAGE analysis from muscle extracts visualized with either ^{14}C or ^{32}P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

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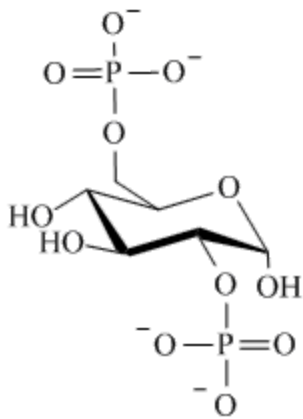
- A.



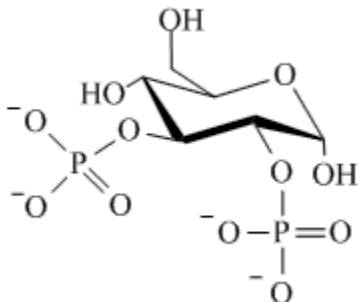
- B.



- C.



- D.



Solution: The correct answer is **D**.

1. This glucose molecule is phosphorylated at C1 instead of at C2 and C3.
2. This glucose molecule is phosphorylated at C1 and C6 instead of at C2 and C3.
3. This glucose molecule is phosphorylated at C2 and C6. The phosphorylation at C6 should be at C3.
4. **According to the passage, glycogen has phosphate monoesters at C2 and C3 in one out of every 600–1500 glucose molecules. This glucose molecule is phosphorylated at C2 and C3.**

KHAN ACADEMY LESSONS

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QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 8 (Questions 38-43)

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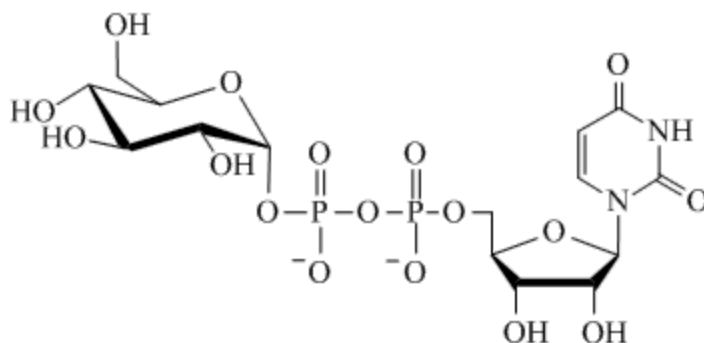


Figure 1	The structure of UDP-glucose
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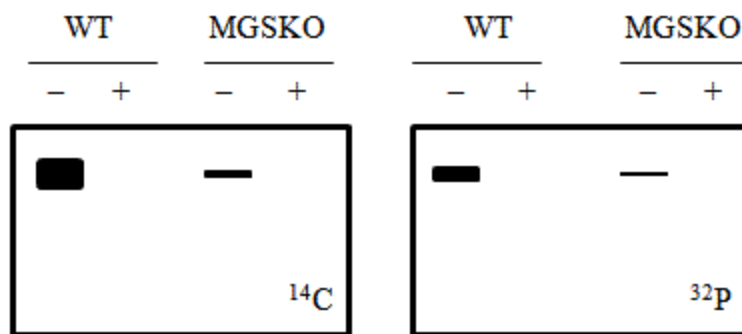


Figure 2 SDS-PAGE analysis from muscle extracts visualized with either ^{14}C or ^{32}P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

Adapted from Tagliabracci, V. et al. *Cell Metabolism*. © 2011 Elsevier Inc.

Which type of bond is formed by glycogen synthase upon release of UDP?

- A.
α-1,4-Glycosidic bond
- B.
α-1,6-Glycosidic bond
- C.
β-1,4-Glycosidic bond
- D.
β-1,6-Glycosidic bond

Solution: The correct answer is **A**.

1. In glycogen, each glucose subunit is connected to an adjacent glucose subunit by an α-1,4-glycosidic bond.

2. Glycogen contains glucose subunits connected by α -1,4-glycosidic bonds, not α -1,6-glycosidic bonds.
3. Glycogen contains glucose subunits connected by 1,4-glycosidic bonds that are α , not β .
4. Glycogen contains glucose subunits connected by α -1,4-glycosidic bonds, not β -1,6-glycosidic bonds.

KHAN ACADEMY LESSONS

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Incorrect

Your result is incorrect.

Your answer is C.

Correct answer is A.

Correct Answer Your Answer

A

B

C

x

D

Time Spent:

45 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC5D

FC5

BCH

RESULT / TIMING

Passage 8 (Questions 38-43)

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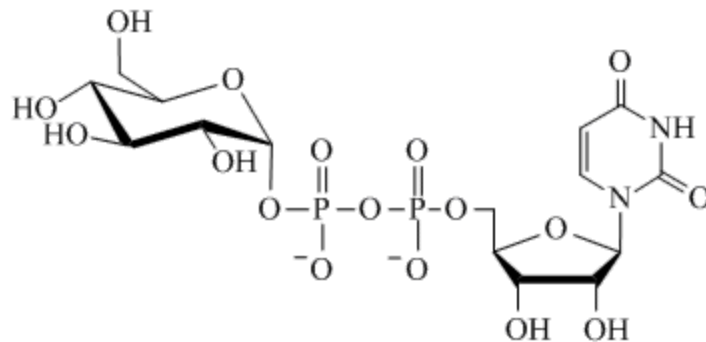


Figure 1	The structure of UDP-glucose
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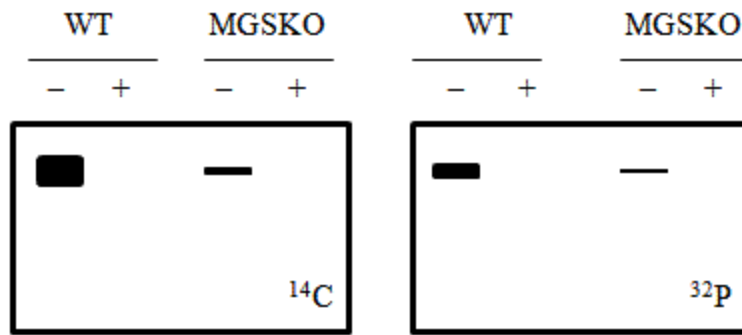


Figure 2

SDS-PAGE analysis from muscle extracts visualized with either ^{14}C or ^{32}P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

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- A.
α-1,4-Glycosidic bond
- B.
α-1,6-Glycosidic bond
- C.
β-1,4-Glycosidic bond
- D.
β-1,6-Glycosidic bond

Solution: The correct answer is **A**.

1. **In glycogen, each glucose subunit is connected to an adjacent glucose subunit by an α-1,4-glycosidic bond.**
2. Glycogen contains glucose subunits connected by α-1,4-glycosidic bonds, not α-1,6-glycosidic bonds.
3. Glycogen contains glucose subunits connected by 1,4-glycosidic bonds that are α, not β.
4. Glycogen contains glucose subunits connected by α-1,4-glycosidic bonds, not β-1,6-glycosidic bonds.

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QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 8 (Questions 38-43)

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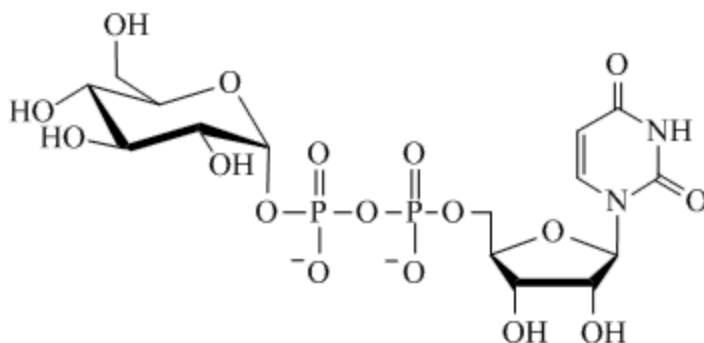


Figure 1	The structure of UDP-glucose
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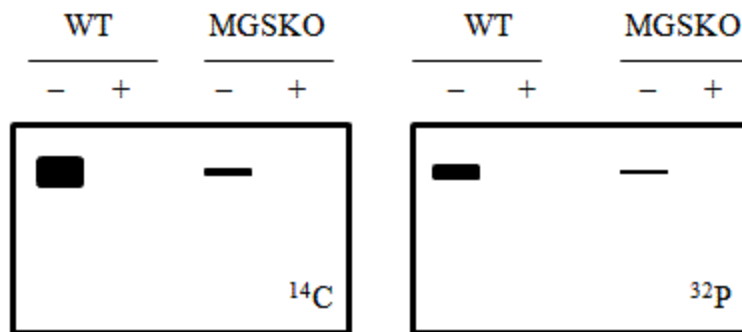


Figure 2

SDS-PAGE analysis from muscle extracts visualized with either ^{14}C or ^{32}P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

Adapted from Tagliabracci, V. et al. *Cell Metabolism*. © 2011 Elsevier Inc.

In addition to glucose, what other monosaccharide is part of the UDP-glucose structure?

- A.
Xylose
- B.
Fructose
- C.
Ribose
- D.
Arabinose

Solution: The correct answer is **C**.

1. Xylose is pentofuranose in which the C2 and C3 hydroxyl groups are *trans*, and the C3 hydroxyl group and the C4 hydroxymethyl group are *cis*. Although the other monosaccharide in UDP glucose is a pentofuranose, the C2 and C3 hydroxyl groups are *cis*, and the C3 hydroxyl group and the C4 hydroxymethyl group are *trans*.
2. Fructose is a hexose, not a pentose.
3. **The other monosaccharide in UDP-glucose is ribose, a pentofuranose in which the C2 and C3 hydroxyl groups are *cis*, and the C3 hydroxyl group and the C4 hydroxymethyl group are *trans*.**
4. Although arabinose can be a pentofuranose, the C2 and C3 hydroxyl groups would be *trans*, not *cis*.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#); [Amino Acids](#), [Peptides](#), [Proteins](#); [The Three-Dimensional Protein Structure](#), [Nonenzymatic Protein Function](#); and [Phenols](#), [Polycyclic and Heterocyclic Aromatic Compounds](#).

Incorrect

Your result is incorrect.

Your answer is B.

Correct answer is C.

Correct Answer Your Answer

A

B

x

C

D

Time Spent:

12 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC5D

FC5

BCH

Passage 8 (Questions 38-43)

Glycogen synthase catalyzes the lengthening of glycogen polymers through the addition of glucose. The reaction consists of transferring a glucosyl moiety from uridine diphosphate glucose (UDP-glucose, Figure 1) to the nonreducing end of the glycogen molecule along with the release of UDP. UDP-glucose is formed from the enzymatically-catalyzed reaction between glucose 1-phosphate and UTP. It was discovered that approximately one out of every 600–1500 glucose molecules in muscle glycogen contains phosphate groups covalently linked to the C2 and C3 positions as monoesters. It is hypothesized that glycogen synthase is responsible for the phosphorylation.

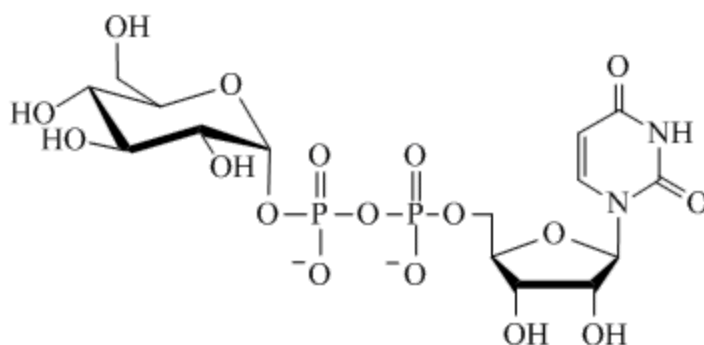


Figure 1	The structure of UDP-glucose
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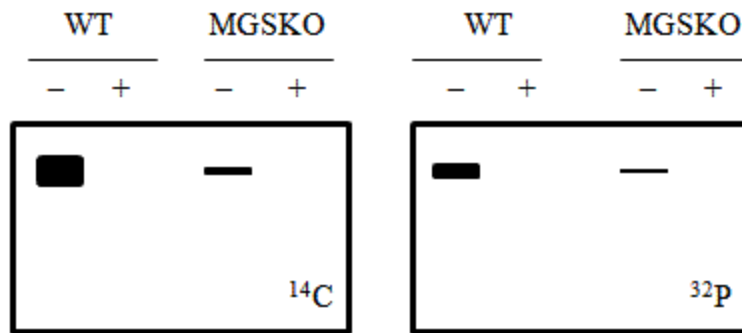


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- A.
Xylose
- B.
Fructose
- C.
Ribose
- D.
Arabinose

Solution: The correct answer is **C**.

1. Xylose is pentofuranose in which the C2 and C3 hydroxyl groups are *trans*, and the C3 hydroxyl group and the C4 hydroxymethyl group are *cis*. Although the other monosaccharide in UDP glucose is a pentofuranose, the C2 and C3 hydroxyl groups are *cis*, and the C3 hydroxyl group and the C4 hydroxymethyl group are *trans*.
2. Fructose is a hexose, not a pentose.
3. **The other monosaccharide in UDP-glucose is ribose, a pentofuranose in which the C2 and C3 hydroxyl groups are *cis*, and the C3 hydroxyl group and the C4 hydroxymethyl group are *trans*.**
4. Although arabinose can be a pentofuranose, the C2 and C3 hydroxyl groups would be *trans*, not *cis*.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#); [Amino Acids](#), [Peptides](#), [Proteins](#); [The Three-Dimensional Protein Structure](#), [Nonenzymatic Protein Function](#); and [Phenols](#), [Polycyclic and Heterocyclic Aromatic Compounds](#).

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 8 (Questions 38-43)

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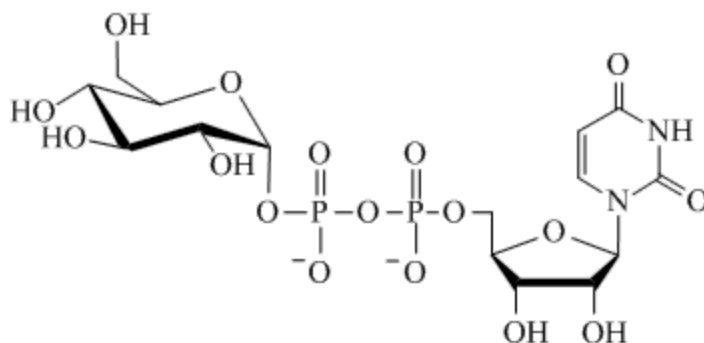


Figure 1	The structure of UDP-glucose
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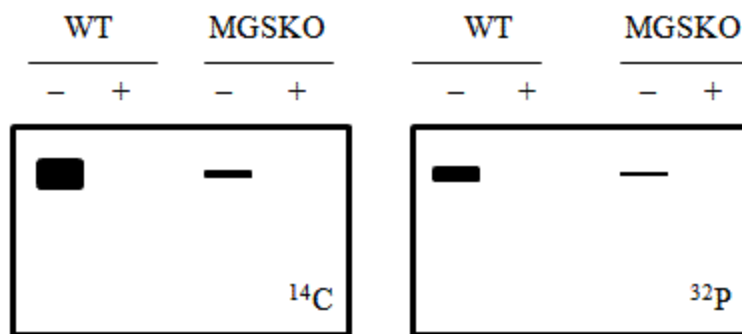


Figure 2

SDS-PAGE analysis from muscle extracts visualized with either ^{14}C or ^{32}P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

Adapted from Tagliabracci, V. et al. *Cell Metabolism*. © 2011 Elsevier Inc.

Addition of laforin to the reaction mixtures used to generate the data in Figure 2 will result in:

- A.
a decrease in the intensity of the band representing WT without glucosidase when visualized with ^{14}C decay.
- B.
an increase in the intensity of the band representing WT without glucosidase when visualized with ^{14}C decay.
- C.
a decrease in the intensity of the band representing WT without glucosidase when visualized with ^{32}P decay.
- D.

an increase in the intensity of the band representing WT without glucosidase when visualized with ^{32}P decay.

Solution: The correct answer is **C**.

1. A decrease in the intensity of this band will not be observed because laforin will not cleave off the ^{14}C -glucose residues from glycogen.
2. An increase in the intensity of this band will not be observed because laforin will not cleave off the ^{14}C -glucose residues from glycogen.
3. **Laforin will cleave off phosphate groups from glycogen, thus removing radioactive phosphate from the WT band and decreasing the intensity of the band.**
4. Laforin is a phosphatase, not a kinase. It will remove radioactive phosphate from glucose subunits in glycogen, not add them.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Enzymes](#); [Kinetics](#); and [Bioenergetics](#).

Incorrect

Your result is incorrect.

Your answer is A.

Correct answer is C.

Correct Answer Your Answer

A

x

B

C

D

Time Spent:

1 min 3 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS3

CC5E

FC5

RESULT / TIMING

Passage 8 (Questions 38-43)

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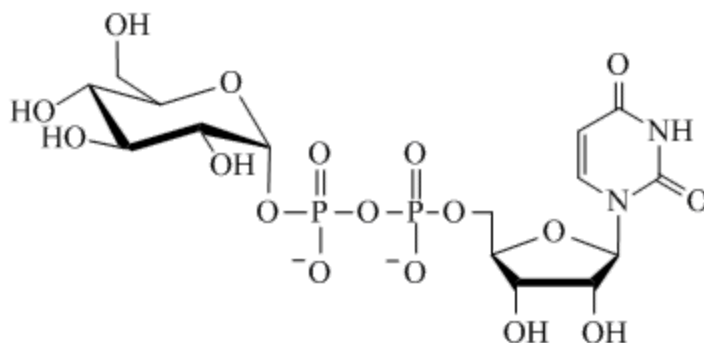


Figure 1	The structure of UDP-glucose
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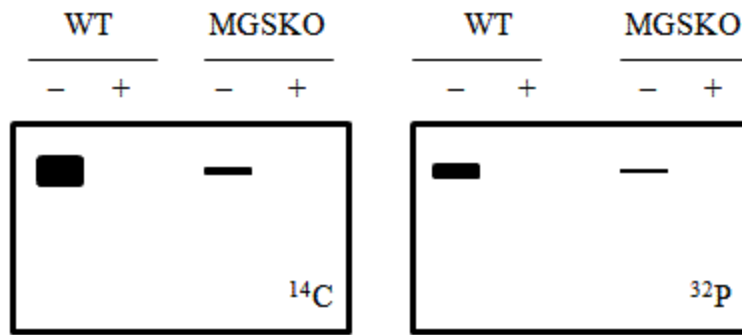


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Addition of laforin to the reaction mixtures used to generate the data in Figure 2 will result in:

- A.
a decrease in the intensity of the band representing WT without glucosidase when visualized with ^{14}C decay.
- B.
an increase in the intensity of the band representing WT without glucosidase when visualized with ^{14}C decay.
- C.
a decrease in the intensity of the band representing WT without glucosidase when visualized with ^{32}P decay.
- D.
an increase in the intensity of the band representing WT without glucosidase when visualized with ^{32}P decay.

Solution: The correct answer is **C**.

1. A decrease in the intensity of this band will not be observed because laforin will not cleave off the ^{14}C -glucose residues from glycogen.
2. An increase in the intensity of this band will not be observed because laforin will not cleave off the ^{14}C -glucose residues from glycogen.
3. **Laforin will cleave off phosphate groups from glycogen, thus removing radioactive phosphate from the WT band and decreasing the intensity of the**

band.

4. Laforin is a phosphatase, not a kinase. It will remove radioactive phosphate from glucose subunits in glycogen, not add them.

KHAN ACADEMY LESSONS

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QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 8 (Questions 38-43)

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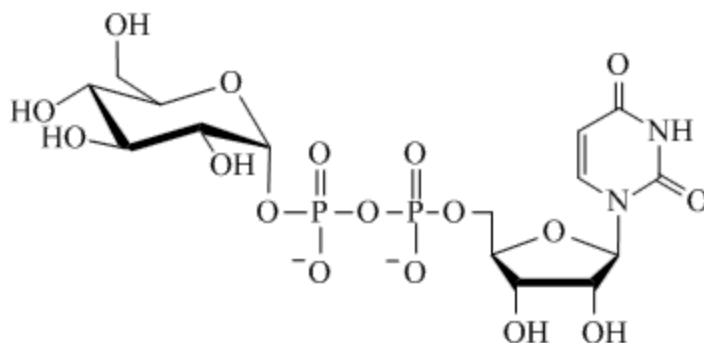


Figure 1	The structure of UDP-glucose
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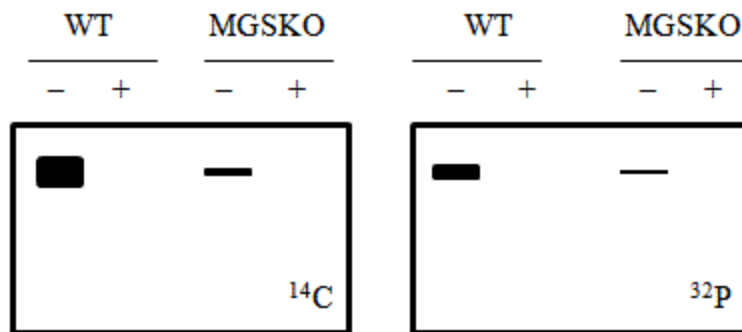


Figure 2

SDS-PAGE analysis from muscle extracts visualized with either ^{14}C or ^{32}P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

Adapted from Tagliabracci, V. et al. *Cell Metabolism*. © 2011 Elsevier Inc.

The radioactive decay described in the passage results in the formation of which two atoms?

- A.
N and S
- B.
B and Si
- C.
B and S
- D.
N and Si

Solution: The correct answer is **A**.

1. The β^- radioactive decay of ^{14}C occurs according to the reaction $^{14}\text{C} \rightarrow ^{14}\text{N} + e^- + \text{antineutrino}$, and that of ^{32}P according to $^{32}\text{P} \rightarrow ^{32}\text{S} + e^- + \text{antineutrino}$, so the resulting atoms are N and S.
2. The resulting atoms are neither B nor Si because the β^- radioactive decay of ^{14}C occurs according to the reaction $^{14}\text{C} \rightarrow ^{14}\text{N} + e^- + \text{antineutrino}$, and that of ^{32}P according to $^{32}\text{P} \rightarrow ^{32}\text{S} + e^- + \text{antineutrino}$.
3. B is not formed because the β^- radioactive decay of ^{14}C occurs according to the reaction $^{14}\text{C} \rightarrow ^{14}\text{N} + e^- + \text{antineutrino}$, and that of ^{32}P according to $^{32}\text{P} \rightarrow ^{32}\text{S} + e^- + \text{antineutrino}$.
4. Si is not formed because the β^- radioactive decay of ^{14}C occurs according to the reaction $^{14}\text{C} \rightarrow ^{14}\text{N} + e^- + \text{antineutrino}$, and that of ^{32}P according to $^{32}\text{P} \rightarrow ^{32}\text{S} + e^- + \text{antineutrino}$.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#) and [Electronic Structure](#).

Correct

Your result is correct.

Your answer is A.

Correct answer is A.

Correct Answer Your Answer

A

✓

B

C

D

Time Spent:

1 min 0 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC4E

FC4

RESULT / TIMING

Passage 8 (Questions 38-43)

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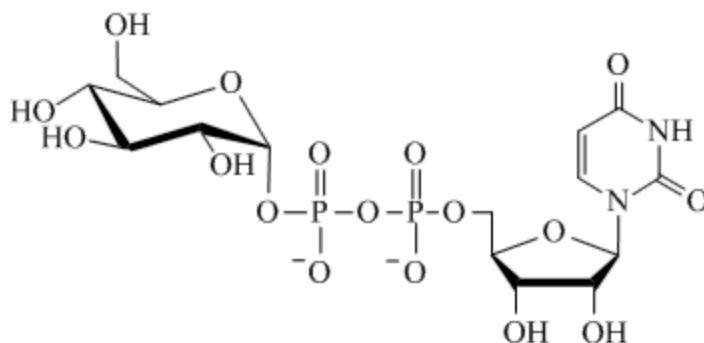


Figure 1	The structure of UDP-glucose
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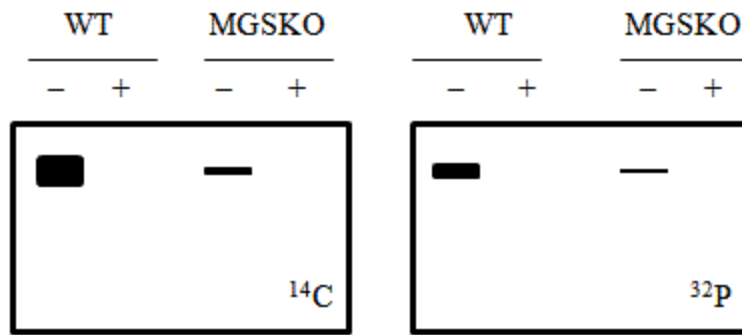


Figure 2

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The radioactive decay described in the passage results in the formation of which two atoms?

- A.
N and S
- B.
B and Si
- C.
B and S
- D.
N and Si

Solution: The correct answer is **A**.

1. The β^- radioactive decay of ^{14}C occurs according to the reaction $^{14}\text{C} \rightarrow ^{14}\text{N} + e^- + \text{antineutrino}$, and that of ^{32}P according to $^{32}\text{P} \rightarrow ^{32}\text{S} + e^- + \text{antineutrino}$, so the resulting atoms are N and S.
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3. B is not formed because the β^- radioactive decay of ^{14}C occurs according to the reaction $^{14}\text{C} \rightarrow ^{14}\text{N} + e^- + \text{antineutrino}$, and that of ^{32}P according to $^{32}\text{P} \rightarrow ^{32}\text{S} + e^- + \text{antineutrino}$.

4. Si is not formed because the β^- radioactive decay of ^{14}C occurs according to the reaction $^{14}\text{C} \rightarrow ^{14}\text{N} + e^- + \text{antineutrino}$, and that of ^{32}P according to $^{32}\text{P} \rightarrow ^{32}\text{S} + e^- + \text{antineutrino}$.
-

KHAN ACADEMY LESSONS

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QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

ANSWER CHOICES

Please enter your note

Cancel Add

Passage 8 (Questions 38-43)

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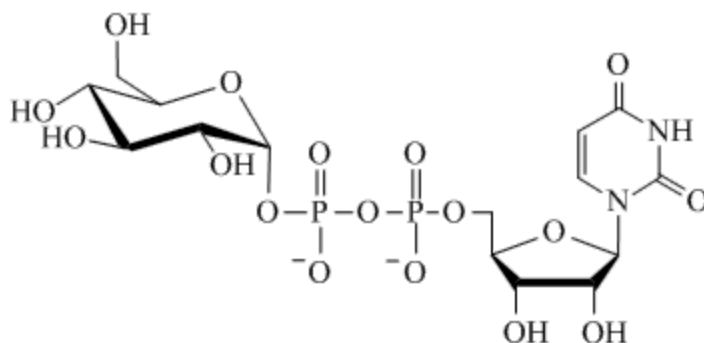


Figure 1	The structure of UDP-glucose
-----------------	------------------------------

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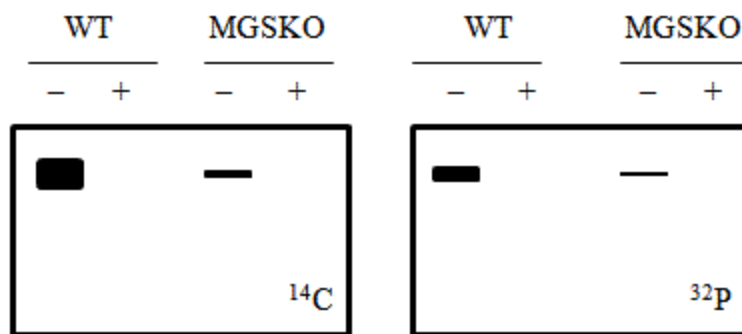


Figure 2

SDS-PAGE analysis from muscle extracts visualized with either ^{14}C or ^{32}P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

Adapted from Tagliabracci, V. et al. *Cell Metabolism*. © 2011 Elsevier Inc.

Compared to normal glycogen, the amount of what type of bond is decreased in Lafora bodies?

- A.
Phosphomonoester
- B.
Phosphodiester
- C.
 α -1,4-Glycosidic bond
- D.
 α -1,6-Glycosidic bond

Solution: The correct answer is **D**.

1. According to the passage, Lafora bodies should have more phosphomonoester bonds than normal glycogen.
2. There is no evidence from the passage that either Lafora bodies or normal glycogen have phosphodiester bonds.
3. The passage did not indicate that the polymerization of glucose subunits was less in Lafora bodies than normal glycogen.
4. **The passage stated that Lafora bodies have sparse branching compared to normal glycogen. In glycogen, branching is due to the formation of α -1, 6-glycosidic bonds. Thus, these bonds are decreased in Lafora bodies.**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#), [Amino Acids](#), [Peptides](#), [Proteins](#), [The Three-Dimensional Protein Structure](#), [Nonenzymatic Protein Function](#); and [Phenols, Polycyclic and Heterocyclic Aromatic Compounds](#).

Incorrect

Your result is incorrect.

Your answer is A.

Correct answer is D.

Correct Answer Your Answer

A

x

B

C

D

Time Spent:

1 min 7 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC5D

FC5

BCH

Passage 8 (Questions 38-43)

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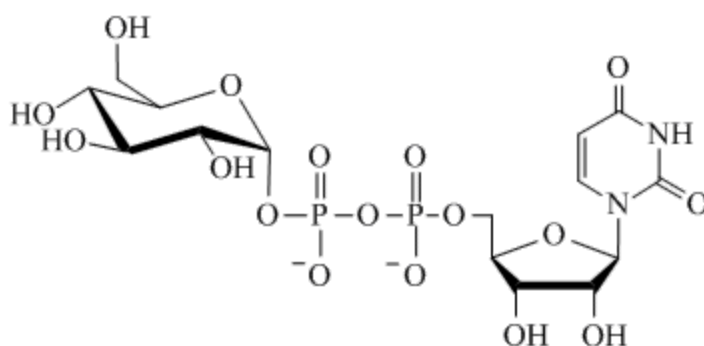


Figure 1	The structure of UDP-glucose
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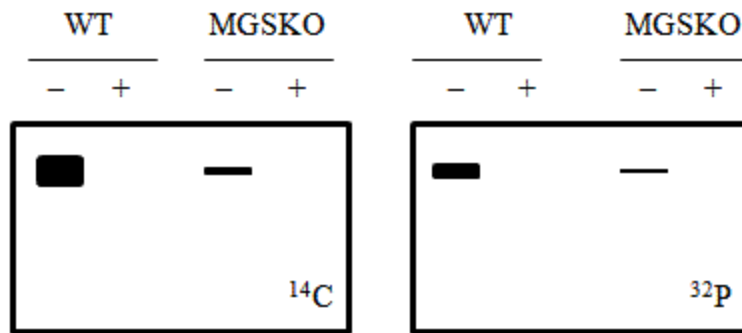


Figure 2

SDS-PAGE analysis from muscle extracts visualized with either ^{14}C or ^{32}P radioactive decay (Note: The + and – lanes represent the presence or absence of glucosidase treatment, respectively.)

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Compared to normal glycogen, the amount of what type of bond is decreased in Lafora bodies?

- A.
Phosphomonoester
- B.
Phosphodiester
- C.
 α -1,4-Glycosidic bond
- D.
 α -1,6-Glycosidic bond

Solution: The correct answer is **D**.

1. According to the passage, Lafora bodies should have more phosphomonoester bonds than normal glycogen.
2. There is no evidence from the passage that either Lafora bodies or normal glycogen have phosphodiester bonds.
3. The passage did not indicate that the polymerization of glucose subunits was less in Lafora bodies than normal glycogen.
4. **The passage stated that Lafora bodies have sparse branching compared to normal glycogen. In glycogen, branching is due to the formation of α -1, 6-glycosidic bonds. Thus, these bonds are decreased in Lafora bodies.**

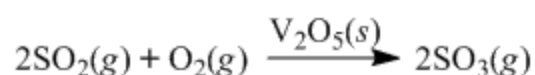
For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Nucleotides and Nucleic Acids](#), [Lipids](#); [Amino Acids](#), [Peptides](#), [Proteins](#); [The Three-Dimensional Protein Structure](#), [Nonenzymatic Protein Function](#); and [Phenols](#), [Polycyclic and Heterocyclic Aromatic Compounds](#).

Passage 9 (Questions 48-51)

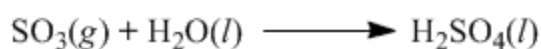
Passage

Passage 9 (Questions 48-51)

A catalyst is a substance that speeds up a reaction without being consumed. For example, V_2O_5 is used as a catalyst to oxidize SO_2 to SO_3 in the first step of the production of sulfuric acid (reactions 1 and 2).



Reaction 1



Reaction 2

Furthermore, H_2O_2 can be decomposed with the aid of a catalyst, as shown in reactions 3 and 4.



Reaction 3



Reaction 4

Catalysts are widely utilized in automobile catalytic converters, where the exhaust gases are passed through beads of a solid catalyst. The catalyst aids in the conversions of carbon monoxide to carbon dioxide, hydrocarbons to carbon dioxide and water, and nitric oxide to nitrogen gas. Unfortunately, catalytic converters also can catalyze the oxidation of SO_2 to SO_3 , which subsequently reacts with moisture to form unwanted sulfuric acid.

Catalysts are classified as either *homogeneous* or *heterogeneous*. A homogeneous catalyst exists in the same phase as the reactant molecules, whereas a heterogeneous catalyst exists in a different phase than the reactants.

Question 48 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

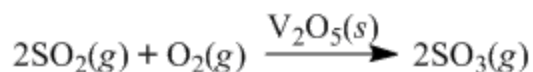
ANSWER CHOICES

Please enter your note

Cancel Add

Passage 9 (Questions 48-51)

A catalyst is a substance that speeds up a reaction without being consumed. For example, V_2O_5 is used as a catalyst to oxidize SO_2 to SO_3 in the first step of the production of sulfuric acid (reactions 1 and 2).



Reaction 1



Reaction 2

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Reaction 3



Reaction 4

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conversions of carbon monoxide to carbon dioxide, hydrocarbons to carbon dioxide and water, and nitric oxide to nitrogen gas. Unfortunately, catalytic converters also can catalyze the oxidation of SO_2 to SO_3 , which subsequently reacts with moisture to form unwanted sulfuric acid.

Catalysts are classified as either *homogeneous* or *heterogeneous*. A homogeneous catalyst exists in the same phase as the reactant molecules, whereas a heterogeneous catalyst exists in a different phase than the reactants.

Which of the following is the overall reaction for the decomposition of H_2O_2 that is shown in reactions 3 and 4?

- A.
$$2\text{H}_2\text{O}_2(aq) + \text{IO}^-(aq) \rightarrow 2\text{H}_2\text{O}(l) + \text{I}^-(aq) + \text{O}_2(g)$$
- B.
$$2\text{H}_2\text{O}_2(aq) + 2\text{I}^-(aq) \rightarrow 2\text{H}_2\text{O}(l) + \text{I}_2(g) + \text{O}_2(g)$$
- C.
$$\text{H}_2\text{O}_2(aq) \rightarrow \text{H}_2\text{O}(l) + \text{O}_2(g)$$
- D.
$$2\text{H}_2\text{O}_2(aq) \rightarrow 2\text{H}_2\text{O}(l) + \text{O}_2(g)$$

Solution: The correct answer is **D**.

1. Both of the anions shown in this equation are intermediates of the net of reactions 3 and 4. This reaction is also not balanced for O atoms.
2. Iodide is an intermediate in reactions 3 and 4. Furthermore, iodine does not appear in either reaction. The reaction shown in this response is not balanced for charge.
3. This reaction is not balanced for O atoms.
4. **This response depicts the net sum of reactions 3 and 4: $2\text{H}_2\text{O}_2(aq) \rightarrow 2\text{H}_2\text{O}(l) + \text{O}_2(g)$**

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#); [Electronic Structure](#); [The Periodic Table - Classification of Elements Into Groups by Electronic Structure](#); [The Periodic Table - Variations of Chemical Properties with Group and Row](#); and [Stoichiometry](#).

Correct

Your result is correct.

Your answer is D.

Correct answer is D.

Correct Answer Your Answer

A

B

C

D

✓

Time Spent:

1 min 18 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS1

CC4E

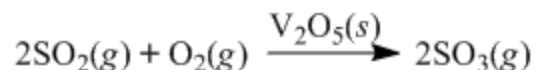
FC4

GCH

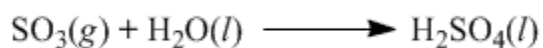
RESULT / TIMING

Passage 9 (Questions 48-51)

A catalyst is a substance that speeds up a reaction without being consumed. For example, V_2O_5 is used as a catalyst to oxidize SO_2 to SO_3 in the first step of the production of sulfuric acid (reactions 1 and 2).



Reaction 1



Reaction 2

Furthermore, H_2O_2 can be decomposed with the aid of a catalyst, as shown in reactions 3 and 4.



Reaction 3



Reaction 4

Catalysts are widely utilized in automobile catalytic converters, where the exhaust gases are passed through beads of a solid catalyst. The catalyst aids in the conversions of carbon monoxide to carbon dioxide, hydrocarbons to carbon dioxide and water, and nitric oxide to nitrogen gas. Unfortunately, catalytic converters also can catalyze the oxidation of SO_2 to SO_3 , which subsequently reacts with moisture to form unwanted sulfuric acid.

Catalysts are classified as either *homogeneous* or *heterogeneous*. A homogeneous catalyst exists in the same phase as the reactant molecules, whereas a heterogeneous catalyst exists in a different phase than the reactants.

Which of the following is the overall reaction for the decomposition of H_2O_2 that is shown in reactions 3 and 4?

- A.
 $2\text{H}_2\text{O}_2(aq) + \text{IO}^-(aq) \rightarrow 2\text{H}_2\text{O}(l) + \text{I}^-(aq) + \text{O}_2(g)$
- B.
 $2\text{H}_2\text{O}_2(aq) + 2\text{I}^-(aq) \rightarrow 2\text{H}_2\text{O}(l) + \text{I}_2(g) + \text{O}_2(g)$
- C.
 $\text{H}_2\text{O}_2(aq) \rightarrow \text{H}_2\text{O}(l) + \text{O}_2(g)$
- D.
 $2\text{H}_2\text{O}_2(aq) \rightarrow 2\text{H}_2\text{O}(l) + \text{O}_2(g)$

Solution: The correct answer is **D**.

1. Both of the anions shown in this equation are intermediates of the net of reactions 3 and 4. This reaction is also not balanced for O atoms.

2. Iodide is an intermediate in reactions 3 and 4. Furthermore, iodine does not appear in either reaction. The reaction shown in this response is not balanced for charge.
3. This reaction is not balanced for O atoms.
4. **This response depicts the net sum of reactions 3 and 4:** $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Atomic Nucleus](#); [Electronic Structure](#); [The Periodic Table - Classification of Elements Into Groups by Electronic Structure](#); [The Periodic Table - Variations of Chemical Properties with Group and Row](#); and [Stoichiometry](#).

Question 49 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

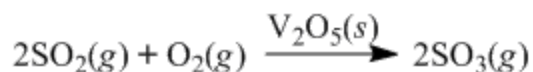
ANSWER CHOICES

Please enter your note

Cancel Add

Passage 9 (Questions 48-51)

A catalyst is a substance that speeds up a reaction without being consumed. For example, V_2O_5 is used as a catalyst to oxidize SO_2 to SO_3 in the first step of the production of sulfuric acid (reactions 1 and 2).



Reaction 1



Reaction 2

Furthermore, H_2O_2 can be decomposed with the aid of a catalyst, as shown in reactions 3 and 4.



Reaction 3



Reaction 4

Catalysts are widely utilized in automobile catalytic converters, where the exhaust gases are passed through beads of a solid catalyst. The catalyst aids in the

conversions of carbon monoxide to carbon dioxide, hydrocarbons to carbon dioxide and water, and nitric oxide to nitrogen gas. Unfortunately, catalytic converters also can catalyze the oxidation of SO_2 to SO_3 , which subsequently reacts with moisture to form unwanted sulfuric acid.

Catalysts are classified as either *homogeneous* or *heterogeneous*. A homogeneous catalyst exists in the same phase as the reactant molecules, whereas a heterogeneous catalyst exists in a different phase than the reactants.

Which of the following diagrams does NOT illustrate a function of an automobile catalytic converter?

- A.
 $\text{CH}_4 \rightarrow \text{converter} \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- B.
 $\text{CO} \rightarrow \text{converter} \rightarrow \text{CO}_2$
- C.
 $\text{SO}_3 \rightarrow \text{converter} \rightarrow \text{SO}_2 + \text{H}_2\text{O}$
- D.
 $\text{NO} \rightarrow \text{converter} \rightarrow \text{N}_2$

Solution: The correct answer is **C**.

1. Catalytic converters were stated to aid in the conversion of hydrocarbons to carbon dioxide and water.
 2. Catalytic converters were stated to aid in the conversion of carbon monoxide to carbon dioxide.
 3. **This response depicts a chemical transformation ($\text{SO}_3 \rightarrow \text{SO}_2 + \text{H}_2\text{O}$) that is NOT catalyzed by an automobile's catalytic converter.**
 4. Catalytic converters were stated to aid in the conversion of nitric oxide to nitrogen gas.
-

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Thermodynamics](#); [Thermochemistry](#); and [Equilibrium](#).

Correct

Your result is correct.

Your answer is C.

Correct answer is C.

Correct Answer Your Answer

A

B

C

✓

D

Time Spent:

1 min 28 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS2

CC5E

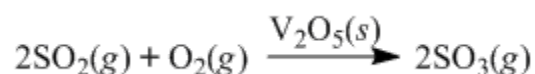
FC5

GCH

RESULT / TIMING

Passage 9 (Questions 48-51)

A catalyst is a substance that speeds up a reaction without being consumed. For example, V_2O_5 is used as a catalyst to oxidize SO_2 to SO_3 in the first step of the production of sulfuric acid (reactions 1 and 2).



Reaction 1



Reaction 2

Furthermore, H_2O_2 can be decomposed with the aid of a catalyst, as shown in reactions 3 and 4.



Reaction 3



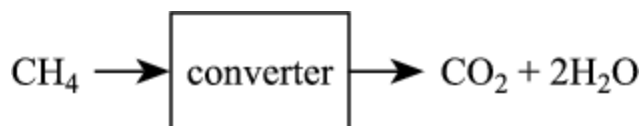
Reaction 4

Catalysts are widely utilized in automobile catalytic converters, where the exhaust gases are passed through beads of a solid catalyst. The catalyst aids in the conversions of carbon monoxide to carbon dioxide, hydrocarbons to carbon dioxide and water, and nitric oxide to nitrogen gas. Unfortunately, catalytic converters also can catalyze the oxidation of SO_2 to SO_3 , which subsequently reacts with moisture to form unwanted sulfuric acid.

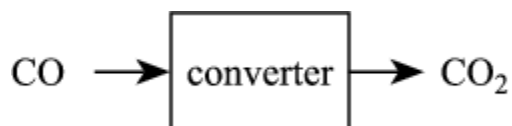
Catalysts are classified as either *homogeneous* or *heterogeneous*. A homogeneous catalyst exists in the same phase as the reactant molecules, whereas a heterogeneous catalyst exists in a different phase than the reactants.

Which of the following diagrams does NOT illustrate a function of an automobile catalytic converter?

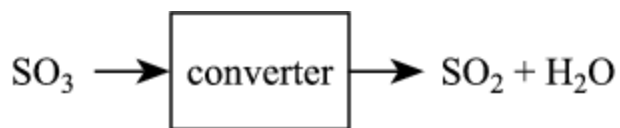
- A.



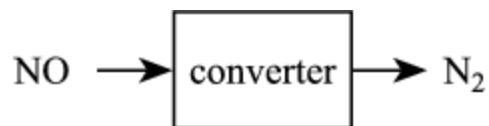
- B.



- C.



• D.



Solution: The correct answer is **C**.

1. Catalytic converters were stated to aid in the conversion of hydrocarbons to carbon dioxide and water.
2. Catalytic converters were stated to aid in the conversion of carbon monoxide to carbon dioxide.
3. **This response depicts a chemical transformation ($\text{SO}_3 \rightarrow \text{SO}_2 + \text{H}_2\text{O}$) that is NOT catalyzed by an automobile's catalytic converter.**
4. Catalytic converters were stated to aid in the conversion of nitric oxide to nitrogen gas.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Thermodynamics](#); [Thermochemistry](#); and [Equilibrium](#).

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QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

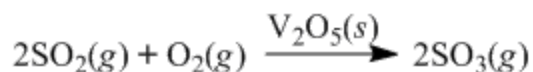
ANSWER CHOICES

Please enter your note

Cancel Add

Passage 9 (Questions 48-51)

A catalyst is a substance that speeds up a reaction without being consumed. For example, V_2O_5 is used as a catalyst to oxidize SO_2 to SO_3 in the first step of the production of sulfuric acid (reactions 1 and 2).



Reaction 1



Reaction 2

Furthermore, H_2O_2 can be decomposed with the aid of a catalyst, as shown in reactions 3 and 4.



Reaction 3



Reaction 4

Catalysts are widely utilized in automobile catalytic converters, where the exhaust gases are passed through beads of a solid catalyst. The catalyst aids in the

conversions of carbon monoxide to carbon dioxide, hydrocarbons to carbon dioxide and water, and nitric oxide to nitrogen gas. Unfortunately, catalytic converters also can catalyze the oxidation of SO_2 to SO_3 , which subsequently reacts with moisture to form unwanted sulfuric acid.

Catalysts are classified as either *homogeneous* or *heterogeneous*. A homogeneous catalyst exists in the same phase as the reactant molecules, whereas a heterogeneous catalyst exists in a different phase than the reactants.

Which of the following most likely will occur if a homogeneous catalyst CANNOT be separated from the products at the end of a reaction?

- A.
The catalyst will become heterogeneous.
- B.
The products will be contaminated.
- C.
The reaction will not occur.
- D.
The reaction rate will speed up.

Solution: The correct answer is **B**.

1. Of the many separation techniques available, only fractional distillation might cause the catalyst to become heterogeneous and that would facilitate separation by filtration.
2. **If a suitable separation technique is not found, the product will be contaminated by the catalyst.**
3. Separation of products from unreacted starting materials or catalysts does not affect the outcome of a reaction, which has already taken place.
4. The reaction rate is unaffected. The separation process will occur after the reaction is completed.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Thermodynamics](#); [Thermochemistry](#); and [Equilibrium](#).

Incorrect

Your result is incorrect.

Your answer is C.

Correct answer is B.

Correct Answer Your Answer

A

B

C

x

D

Time Spent:

55 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS3

CC5E

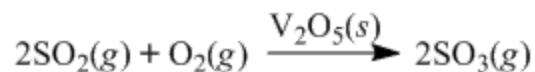
FC5

GCH

RESULT / TIMING

Passage 9 (Questions 48-51)

A catalyst is a substance that speeds up a reaction without being consumed. For example, V_2O_5 is used as a catalyst to oxidize SO_2 to SO_3 in the first step of the production of sulfuric acid (reactions 1 and 2).



Reaction 1



Reaction 2

Furthermore, H_2O_2 can be decomposed with the aid of a catalyst, as shown in reactions 3 and 4.



Reaction 3



Reaction 4

Catalysts are widely utilized in automobile catalytic converters, where the exhaust gases are passed through beads of a solid catalyst. The catalyst aids in the conversions of carbon monoxide to carbon dioxide, hydrocarbons to carbon dioxide and water, and nitric oxide to nitrogen gas. Unfortunately, catalytic converters also can catalyze the oxidation of SO_2 to SO_3 , which subsequently reacts with moisture to form unwanted sulfuric acid.

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- B.
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- C.
The reaction will not occur.
- D.
The reaction rate will speed up.

Solution: The correct answer is **B**.

1. Of the many separation techniques available, only fractional distillation might cause the catalyst to become heterogeneous and that would facilitate separation by filtration.
2. **If a suitable separation technique is not found, the product will be contaminated by the catalyst.**
3. Separation of products from unreacted starting materials or catalysts does not affect the outcome of a reaction, which has already taken place.

4. The reaction rate is unaffected. The separation process will occur after the reaction is completed.
-

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Thermodynamics](#); [Thermochemistry](#); and [Equilibrium](#).

Question 51 of 59

QUESTION

Done Reviewing

MCAT Official Prep Practice Exam 1

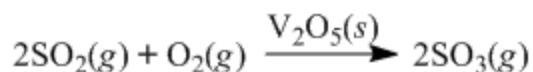
ANSWER CHOICES

Please enter your note

Cancel Add

Passage 9 (Questions 48-51)

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Reaction 1



Reaction 2

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Reaction 3



Reaction 4

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Catalysts are classified as either *homogeneous* or *heterogeneous*. A homogeneous catalyst exists in the same phase as the reactant molecules, whereas a heterogeneous catalyst exists in a different phase than the reactants.

How will the rate of a catalyzed reaction be affected if the solid catalyst is finely ground before it is added to the reaction mixture?

- A.
The rate will be faster because a greater mass of catalyst will be present.
- B.
The rate will be faster because a greater surface area of catalyst will be exposed.
- C.
The rate will be slower because the fine catalyst particles will interfere with product formation.
- D.
The rate will remain the same because the mass of catalyst will be the same.

Solution: The correct answer is **B**.

1. Grinding a solid does not change its mass.
2. **Grinding a heterogeneous catalyst increases the amount of catalyst available to the reaction and therefore increases its rate.**
3. Product formation occurs on a scale so small that the presence of more suspended solid cannot interfere with the course of reaction.
4. While it is true that the mass is the same, the rate will increase due to the greater amount of catalyst that is interacting with the liquid solution.

KHAN ACADEMY LESSONS

For more help on the skills, concepts, or content assessed by this question, see the Khan Academy lessons on [Thermodynamics](#); [Thermochemistry](#); and [Equilibrium](#).

Correct

Your result is correct.

Your answer is B.

Correct answer is B.

Correct Answer Your Answer

A

B

✓

C

D

Time Spent:

22 secs

Difficulty Level:

Content & Skills: "Content and Skills" shows the foundational concept, content category, skill, and discipline tested by each question. Your Score Report will show how you performed in each area. Learn more in the What's on the MCAT Exam? Content Outline.

SIRS3

CC5E

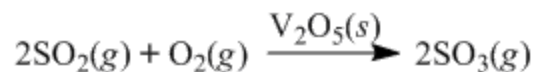
FC5

GCH

RESULT / TIMING

Passage 9 (Questions 48-51)

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Reaction 1



Reaction 2

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Reaction 3



Reaction 4

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-

KHAN ACADEMY LESSONS

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